

Bayesian Axiom Induction from Instances: Template Priors, Derivation Likelihoods and Time Penalties

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Abstract

Kaarel Hänni asked whether a Solomonoff-style inducer over axiom systems, with a prior over templates, a preference for short derivations and a time penalty, would pass from instances $\varphi(t)$ to $\forall x\varphi$, and whether it would robustly recover the axioms we actually have, or at least a deductively equivalent system. We make such an inducer precise. Theories are finite sets of templates with a proper prefix-code prior; the likelihood is a derivation grammar whose normaliser does not depend on the data, which gives the size principle; time is charged for checking axioms or for written derivation length. First, closed instances never favour $\forall x\varphi$ over its instance schema: the odds stay at the prior odds or fall by a constant factor per datum. “All future data are instances” is confirmed, while “the axioms prove $\forall x\varphi$ ” tends to a prior share or to zero, a Bayesian ω -gap. Second, with well-specified data the posterior concentrates on the theories with the data’s law and identifies their theorems, not the axioms. Third, human-written data are not drawn from the model; the posterior then goes to the best-fitting generator, which can be weaker than the axioms behind the data, or unsound; so the actual axioms are not robustly recovered. Fourth, penalising the time to check or generate axioms does not stop Hänni’s collapse of axiom induction into function induction; derivation length in written symbols prices it. All positive guarantees need well-specified data. Adversarial referees with independent code checked the results; no human has.

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1 Introduction

Kaarel Hänni asked the following question on 8 October 2026 (verbatim):

let’s consider the case of going from a bunch of $\phi(x)$ statements to $\forall x \phi(x)$. i wonder if this should happen basically like so:

- $\forall x \phi(x)$ is a simple law which implies all the other statements. so once we see all these other statements, we should up the probability on $\forall x \phi(x)$. i guess we should still be entertaining other hypotheses as well
- i guess this is some sort of solomonoff induction idea. see my writing on solomonoff axiom induction here: <https://github.com/kaarelh/notes>
- could you investigate whether there is a good version of sth like this here? some further ideas:
 - we might want to not allow just any enumerable axioms, but ones with templates
 - we might want to have a derivation length prior (like, statements given without proof in our data set should be fairly easily derived from axioms)
 - we might want to have a time complexity penalty. like, it should not take too long to generate axioms / to check whether a statement is an axiom
- i’m interested in whether you can come up with something that robustly picks up the axioms we actually have, or at least something that is equivalent in terms of theorems derived. or if this will at least get a lot of posterior mass

This paper answers the question within a precise model. It is a study of an idea: what a Bayesian inducer over axiom systems would conclude, and under which conditions. It is not the design of a stronger prover. The code of Section 8 is a small exact laboratory for checking claims, not a prover.

1.1 Hänni’s note and his collapse argument

Hänni’s working notes (Hänni, 2026)¹ describe “solomonoff axiom induction/abduction”. One is told that some mathematical statements are true and some false, and weighs the axiom systems that prove the givens “and not their negations”, or, in a second variant, those “that do not contradict the given statements”. The weight may depend “both on the complexity of specifying the axioms [...] and also on how many of the given statements can be proven”, and “one could also penalize longer proofs”. Every other statement then gets “a $p(\text{true/false/independent})$ ”, and he asks whether to renormalise away the independent part, split it 50/50, or keep the triple.

He then argues that “solomonoff function induction with consistent assigners only is equivalent to the version of solomonoff axiom induction that requires proofs”. One direction runs proof search from the axioms. For the other, a consistent assigner T becomes the schema “ $T(\text{quoted-}\phi) = \text{accept} \Rightarrow \phi$ ” (and likewise for rejection), where “being an axiom of the right form is in fact decidable”; the schema is consistent because “a model of the ϕ s alone would be a model of the added sentences also”. So axiom induction over arbitrary decidable axiom sets collapses into function induction, and he turns to “a version where axioms are not just printed by a program, where instead you have substitution schemas”. His “other ideas for how to fix axiom induction” are to be “fine with only generating a very small subset of observations”, to be “fine with making some mistakes”, and to grade near misses. A second note ([hanni-polytime-solomonoff.md](#))

¹File `research/prior/hanni-solomonoff-axiom-induction.md` of the accompanying repository. These are working notes; we quote them verbatim and do not hold them to the standard of a paper.

gives, for each polynomial p , a predictor with constant regret against every predictor running in time $O(p(n))$. The question adds three refinements: templates, a derivation-length prior and a time penalty.

1.2 The question as we read it, and the model

We read the question as asking for three things: (1) *a good version*, a precise Bayesian inducer over axiom systems with a template prior, a likelihood that makes statements with short derivations probable, and a time penalty; (2) what it does with instances $\varphi(t)$ of a law $\forall x\varphi$; (3) *guarantees*: when the posterior concentrates on the actual axioms, on a deductively equivalent system, or at least gives them much mass, and when and why it fails.

A *theory* is a finite set of templates of the class DT° of the earlier report AS (Claude (Anthropic), 2026a): formulas with metavariables applied to metavariable-free arguments, such as the instance schema $\varphi(z)$, the induction template T_{Ind} , or a single sentence (Definition 2.1); matching against them is unique and linear (AS Thm 2.5). The prior is a prefix code over theories (Definition 2.4); mixture weights get a Dirichlet prior. Data are sentences drawn i.i.d. from a theory's law P_T . Under L0 a datum cites a template whose metavariables a fixed stochastic grammar $\mathcal{Q}$ fills in (Definition 2.7). Under L1 a stochastic derivation grammar grows a tree whose nodes cite axioms or apply modus ponens, generalisation or $\forall$ -elimination; $P_T(s)$ is the probability of a valid tree concluding s , divided by a normaliser that depends on T and not on the data (Definition 2.9). That normaliser gives the *size principle*: a theory that spends probability on sentences never observed loses a fixed amount per datum (Lemma 2.21). Variants, including a penalty on written symbol size (L1^σ) and selection-aware forms (L1^{sel} , $\text{L1}_{\text{cit}}^{\text{sel}}$), are in Definition 2.11; Hänni's own proposals are kept as scores (Definition 2.12). Time is charged either for checking or generating axioms or through the written size of derivations (Section 6).

1.3 Answers in brief

Each answer holds within the hypotheses of the results it points to.

- A1. A good version exists, in a precise and limited sense.** Theories are finite sets of DT° templates with a proper prefix-code prior and Dirichlet weights; the likelihood is a derivation grammar with a data-independent normaliser, which gives the size principle (Lemmas 2.5 and 2.10 and Proposition 2.22). Hänni's 0/1 scores lack the size principle: a consistent strengthening keeps its prior odds forever. His graded score, divided by its normaliser, is a two-part code that charges written derivation length (Remark 2.23). The likelihood is a computable real whose positivity is undecidable (Proposition 2.24). For $\forall x\varphi$ a good version also models which theorems get reported and includes noise (Section 3.9).
- A2. From closed instances, the data never favour $\forall x\varphi$ over its instance schema.** Between $H_\forall = \{\forall x\varphi\}$ and $H_{\text{sch}} = \{\varphi(z)\}$ the posterior odds stay at the prior odds (citation with the closure reading, L1^{sel} , Hänni's scores) or fall by a constant factor per datum under a derivation likelihood in the minimal calculus C_{min} (Theorems 3.3 and 3.4). The reason is the size principle: $\forall x\varphi$ spends probability on outputs never observed. The direction survives noise, penalties monotone in derivation length or time, and a learned $\forall$ -elimination rate (Proposition B.1); with $\wedge$ -rules on quantified formulas it is open (Remark 3.5). Over-general templates lose mass exponentially; the memoriser class, under a geometric numeral law, only like $\exp(-O(\ln^2 n))$, which refutes the brief's "exponentially" for memorisation (Theorem B.4 and Proposition B.6).

- A3. “All future data are instances” is confirmed; “the axioms prove $\forall x\varphi$ ” is not.** The first gets posterior probability $\rightarrow 1$ (Theorem 3.8). The second tends to a prior share strictly between 0 and 1 (between 0.03 and 0.48 for $\varphi = 0 + x = x$, depending on the prior code), to 0, or, when the likelihood ignores how the data were selected, to a class-dependent limit (Theorem 3.9, Table 16, Proposition 3.12): a Bayesian form of the ω -gap of AS. One instance at a parameter licenses $\forall x\varphi$, and so does one quantified datum that entails it over the background (Propositions 3.10 and B.14). Without a closedness guard and with the selection unmodelled, the posterior does not take the ω -step in pure logic (Remark 3.11).
- A4. Well specified, the posterior identifies the generator: the theorem set, not the axioms.** With fixed weights it concentrates on the generator class $C^* = \{T : P_T = P_{T^*}\}$ and inside C^* equals the prior; deductive beliefs converge to the theorems of T^* under L0, which identifies even the instance union, and under L1 with parameters admissible (Theorem 4.2 and Corollaries 4.3 and 4.4). With Dirichlet weights under L0 the instance union is identified for almost every weight vector (Theorem 4.7). No bound on the number of templates is needed, and Gold’s impossibility applies only on a null set of texts (Proposition 4.9). So a deductively equivalent system is found, while the actual axioms keep the mass $\pi(T^*)/\pi(C^*)$. Under L1 equivalent axiomatisations can have different laws, and the posterior picks the one that generated the data (Proposition 4.5). As far as proved, a derivation likelihood does not change the MDL finding of AS: a schema and its root split tie at matched weights and differ by Occam terms with learned weights (Remark 4.13). Spare templates are never refuted by positive data (Proposition 4.14).
- A5. Robustly picking up the actual axioms: no.** Human data are theorems of the actual axioms but not draws from a law of the model. The posterior then goes to the best-fitting generator (Theorem 4.16), which can be strictly weaker than the axioms behind the data or unsound (Table 23). Much posterior mass on PA-equivalents is obtained for data that are direct uses of axioms, but only within the hand-picked candidate sets and pools scored (Section 7.7.1); with atomic induction motives the posterior moves to a strictly weaker theory (Proposition 7.29).
- A6. The thresholded verifier is sound only for well-specified data.** With fixed weights no adaptive prover ever gets a non-theorem accepted, with probability at least $1 - \delta/\pi(C_d^*)$, uniformly in time (Theorem 5.2). With Dirichlet weights this holds on average over the weights or with a threshold shrinking in n , not with a constant threshold (Theorems 5.6 and 5.7 and Example 5.8). Misspecified, a waiting prover wins with probability 1 (Proposition 5.4). Hänni’s triple is a belief function with its plausibility, and both point estimates he considers are incoherent in general (Section 2.7).
- A7. A time penalty on checking or generating axioms does not stop Hänni’s collapse; derivation length in written symbols prices it.** His equivalence holds up to a constant via Craig’s trick, while his own consistency argument fails in single-sorted arithmetic over PA (Theorem 6.3 and Proposition 6.5). Membership in the collapse sets is polynomial, which refutes the brief’s H6 (Proposition 6.8). Templates block his schema but not the collapse (Propositions 6.6 and 6.7). Symbol-size derivation penalties charge a hard assigner polynomially, between its nondeterministic time and its deterministic time (the upper bound a proof sketch; Proposition 6.13 and Remark 6.1); plain L1 is proved to charge only logarithmically (Corollary 6.12).
- A8. “Statements given without proof should be easily derivable” does not by itself**

	the brief’s hypothesis	verdict	where
H1	template prior; derivation grammar; size principle	confirmed with qualification ($\mathcal{Q}$ uniformly subcritical; Hänni’s scores lack the size principle)	Section 2
H2	concentration on $\{T : P_T = P_{T^*}\}$; misspecified limits at KL-minimisers	confirmed for fixed weights; a.e. weight vector with Dirichlet weights; KL-minimisers for finite classes only	Theorems 4.2 and 4.16
H3	verifier sound with probability $\geq 1 - \delta/\pi(T^*)$	confirmed for fixed weights; refuted as stated for Dirichlet weights, then repaired	Theorems 5.2 and 5.7
H4(a)	memorisation and over-general templates lose exponentially	over-general: confirmed; memorisation: refuted	Proposition B.6
H4(b)	instances do not favour $H_\forall$ over H_{sch}	confirmed, exact factors; qualified for learned weights and $\wedge$ -detours	Theorem 3.4
H4(c), (d)	“all future data are instances” $\rightarrow 1$; “the axioms prove $\forall x\varphi$ ” need not	confirmed: the Bayesian ω -gap	Theorem 3.9
H4(e)	open instances and Gen make $H_{\text{open}} \vdash \forall x\varphi$	confirmed for parameter data; without a guard the ω -step is not taken under L1	Proposition 3.10
H4(f)	quantified data favour provers of $\forall x\varphi$	confirmed	Proposition B.14
H5	positive data and the size principle get around Gold; spares cost $\approx \frac{1}{2} \log n$	confirmed with qualification: some spares never vanish	Propositions 4.9 and 4.14
H6	a membership-time penalty prices the collapse	refuted; written derivation size prices it	Proposition 6.8
H7	PA and ZF: equivalent forms, $\text{Th}(\mathbb{N})$, mixtures, failures	usage decides; no r.e. theory survives $\text{Th}(\mathbb{N})$; narrow practice can end weaker; ten robust failures	Table 6

Table 1: The hypotheses of the research brief (`research/00-brief.md`), which were to be checked, not assumed, and their verdicts (model §9.1, universal §0, pa Summary, experiments §0).

find the axioms. Under track pa’s two-part code, memorising a short theorem is 6 to 24 times cheaper than deriving it, and on theorem data without direct uses of induction the MAP is strictly weaker than PA (Proposition 7.15 and Remark 7.16). The generator wins when the data come from its own derivation process or use its schemas directly (Propositions 7.17 and 7.18); frequently used lemmas become axioms.

A9. Positive data suffice in the limit, but some errors are never corrected by them. Both ends of Gold’s limit point are identified (Proposition 4.9). But a spare slot holding a false sentence decays only like $n^{-1/2}$ (Proposition 4.14), under the two-part likelihood inconsistency is invisible to the size principle (Proposition 7.36), and for data from all of $\text{Th}(\mathbb{N})$ what can be promised is regret against every theory plus its exceptions (Propositions 7.20 and 7.23).

1.4 Contributions, and what is not achieved

We contribute a precise model that sets four tracks’ calculi, priors and likelihoods side by side (Section 2); an exact analysis of $\forall x\varphi$ from instances (Section 3); identification, split, spare-slot and misspecification results (Section 4); soundness of the thresholded verifier for fixed and Dirichlet weights, and the trichotomy as a belief function (Section 5); an analysis of Hänni’s collapse and

of which penalties price it (Section 6); an application to PA and ZF with a list of robust failures (Section 7); exact posteriors over finite pools in a small Python package, with eight experiments (Section 8); and a record of every claim refuted or weakened while the results were checked (Appendix H). Table 1 gives the verdicts on the hypotheses the research started from.

What is not achieved. (i) Posteriors over the full template class are not computed; computations use hand-picked candidate sets and finite pools, and the winners changed when the pools changed (Remark 8.3). (ii) Misspecified limits are known only for specific families; in the full class even the KL-projection picture may fail (Remark C.5). (iii) Several rates are conjectures, among them the Occam rate under L1 (Proposition 4.11(d)), identification for every weight vector (Theorem 4.7(c)) and a polynomial charge by plain L1 (Conjecture 6.14). (iv) No human-written theorem corpus was used. (v) PA under a derivation likelihood was not run in code; track pa computes with a two-part code or with L0, never with the generative L1. (vi) Nothing is checked in a proof assistant, no human has checked the mathematics, and several references were not verified against their sources (Appendix H).

1.5 Earlier work, method and reader’s guide

Two earlier reports prepared for Hänni are background, cited by their own numbering. AS (Claude (Anthropic), 2026a) is the non-Bayesian treatment of the same questions (templates, matching, the cautious verifier, the ω -gap, the MDL finding, the learner DTRC); its code is reused here. “IL” (Claude (Anthropic), 2026b) supplies the Ville argument behind Section 5. The wider literature is in Section 9.4.

The results come from four research tracks: *model* (general theory), *universal* ($\forall x\varphi$), *pa* (PA and ZF) and *experiments* (code). An adversarial referee checked each track’s notes and wrote independent code; the track then revised its notes and kept a verification log. Every theorem-like statement carries a status in brackets (proved; computed, by a seeded script; known, with a reference; proof sketch; conjecture; refuted, kept with its counterexample) and, as a grey superscript, its item in the track’s final notes (e.g. “model Thm 4.1”). All roles, including the writers of this paper, were separate Claude instances; no human has checked the mathematics (Appendix H).

Reader’s guide. Section 2: theories, calculi, priors, likelihoods, Hänni’s scores. Section 3: $\forall x\varphi$ from instances. Section 4: what the posterior identifies, well specified and not. Section 5: the verifier against adaptive provers; the trichotomy. Section 6: Hänni’s collapse and the time penalty. Section 7: PA and ZF. Section 8: the code and experiments E1–E8. Section 9: conclusions, literature, open problems. The appendices give proofs, computed tables and the record of verification.

2 The model

This section fixes theories, calculi, priors, the instantiation grammar, likelihoods, Hänni’s scores and his trichotomy. Proofs are in Appendix A, with Tables 11 and 12, which list the calculi and the likelihoods (normalisers, size principle, names in the research notes).

2.1 Syntax and theories

We use the syntax of AS §2 without change. The languages are $L_A = \{0, S, +, \cdot, <, =\}$ and $L_\in$. Formulas may contain *parameters* $p, p_1, \dots$ (free names) and are read under their universal closure; leading universal quantifiers are not stripped, so $\forall x (x + 0 = x)$ and $p + 0 = p$ are different data with equivalent closures (closure-normal form, AS §2.1). $\mathcal{S}$ is the set of these formulas, called *sentences*.

A *template* (AS Def 2.1) may contain metavariables applied to metavariable-free arguments, $M(t_1, \dots, t_n)$. Term metavariables are lower case (z, z' if 0-ary; f, g), formula metavariables capital (P, F). An occurrence is a *pattern occurrence* if its arguments are distinct bound variables; in the class DT° every metavariable has one. $\text{FO} \subseteq \text{PAT} \subseteq \text{DT}^\circ \subseteq \text{SO}^\circ$, and DT_F° (0-ary term metavariables) is the class implemented in code. A *ground* template is a single sentence. A substitution θ maps metavariables to bodies, λ -terms whose only free names are parameters, and $\text{inst}(\tau) = \{\tau\theta\}$ (AS Def 2.2); whether parameters are admissible in bodies is said at each use. *Matching* (AS Thm 2.5): for $\tau \in \text{DT}^\circ$ and $s \in \mathcal{S}$ at most one θ has $\tau\theta = s$, found in time $O(|\tau| + |s|)$. The chains of Sections 3 and 8 give each metavariable a *guard*: *closed* (no parameter in the body; the default) or *open*.

Robinson arithmetic **Q** has the axioms Q1 $\forall x \neg Sx = 0$; Q2 $\forall x \forall y (Sx = Sy \rightarrow x = y)$; Q3 $\forall x (\neg x = 0 \rightarrow \exists y x = Sy)$; Q4 $\forall x (x + 0 = x)$; Q5 $\forall x \forall y (x + Sy = S(x + y))$; Q6 $\forall x (x \cdot 0 = 0)$; Q7 $\forall x \forall y (x \cdot Sy = x \cdot y + x)$, numbered as in AS. **Q** has no axiom about $<$; it is Σ_1 -complete for $<$ -free Σ_1 sentences. Section 7 writes Q3 as $x = 0 \vee \exists y x = Sy$. Induction is the template $T_{\text{Ind}} = (P(0) \wedge \forall x (P(x) \rightarrow P(Sx))) \rightarrow \forall x P(x) \in \text{DT}^\circ \setminus \text{PAT}$.

Definition 2.1 (Theories). ^{model Def 1.1} A *theory* is a finite set $T = \{\tau_1, \dots, \tau_k\}$ of DT° templates, taken up to renaming of metavariables; $\bigcup \text{inst}(T) := \bigcup_i \text{inst}(\tau_i)$. $\mathcal{H}$ is the class of all theories. A *weighted theory* is a pair (T, w) with w in the open simplex over T .

2.2 Calculi

Definition 2.2 (Background calculus; theorems). ^{model §1.2} The background calculus is Mendelson's first-order calculus with equality **K** (Mendelson, 1997, Ch. 2; chapter and numbering not checked): axiom schemas A1 $B \rightarrow (C \rightarrow B)$; A2 $(B \rightarrow (C \rightarrow D)) \rightarrow ((B \rightarrow C) \rightarrow (B \rightarrow D))$; A3 $(\neg C \rightarrow \neg B) \rightarrow ((\neg C \rightarrow B) \rightarrow C)$; A4 $\forall x B \rightarrow B[t/x]$, t free for x ; A5 $\forall x (B \rightarrow C) \rightarrow (B \rightarrow \forall x C)$, x not free in B ; reflexivity and substitutivity $x = y \rightarrow (B(x, x) \rightarrow B(x, y))$ of equality; rules MP and Gen. Mendelson's free variables are our parameters; Gen infers $\forall x \varphi[x/p]$ from φ . $\text{Th}(T)$ is the set of formulas derivable in **K** from $\bigcup \text{inst}(T)$, with Gen on every parameter; $\text{Th}_d(T)$ the set of those with a derivation of symbol size at most d , axioms included.

Since parameters are read under closure, $\varphi(p)$ and $\forall x \varphi$ are interderivable (AS Prop 4.10), while a closed instance $\varphi(t)$ is not (AS Ex 4.9). **K** is complete (known). $\text{Th}(T)$ and $\text{Th}_d(T)$ depend on T only through $\bigcup \text{inst}(T)$. The unit of d is stated at each use: symbol size here and in Section 5, chain steps in Section 8, derivation size in Section 7.

$\forall$ -elimination is not a template (Lemma A.5, in Appendix A.3), so it belongs to the fixed background (compare AS Rem A.3); the other logical axioms are templates. Derivation grammars use $\forall E$ as a primitive rule; it is derivable in **K**, so Th is unchanged.

Definition 2.3 (The restricted calculi). ^{universal §1.4; experiments §1.6; pa Def 0.3; model Ex 3.6} Later sections use stochastic grammars over derivations in restricted calculi (rules in Appendix A.3, comparison in Table 11): the chains $\mathcal{C}_{\min}$ (cite with probability $1 - c$, $\forall E$ with probability c) and

$C_{\text{open}}(c, g)$ (plus Gen on the parameter, probability g ; $p_{\text{cite}} = 1 - c - g$) and the trees C_{U3} , $C_{\wedge}$ of Section 3; the forward chain Ch_J of Section 8 (stop probability c_{stop} while a rule applies; $c_{\text{ch}} := 1 - c_{\text{stop}} = \frac{1}{2}$); natural deduction ND with the two-part code L1^{sch} (Section 7); and the equational grammar of L1^{eq} .

2.3 Prior

Definition 2.4 (Template code and prior). ^{model Def 1.3} Templates are coded by a prefix code on their preorder token sequences, one token per node (details in Appendix A.1). Renamings get the same code, argument permutations $(M(x, y), M(y, x))$ different ones. A theory is coded as its number of templates followed by their codes in lexicographic order; $\ell(T)$ is the length in bits, and

$$\pi_{\lambda}(T) := 2^{-\lambda \ell(T)} / Z_{\lambda}, \quad \lambda \geq 1, \quad Z_{\lambda} := \sum_{T \in \mathcal{H}} 2^{-\lambda \ell(T)}.$$

Mixture weights get the prior $\text{Dir}(\alpha, \dots, \alpha)$, by default $\alpha = \frac{1}{2}$ (Krichevsky and Trofimov, 1981). Statements that need fixed weights say so.

Lemma 2.5 (Kraft). ^{[proved]model Lemma 1.4} *Theory codes are prefix-free, so $\sum_{T \in \mathcal{H}} 2^{-\ell(T)} \leq 1$ and π_{λ} is a probability for $\lambda \geq 1$; and $\sum_T \pi_{\lambda}(T)^{1/2} < \infty$ for $\lambda \geq 2$.*

Other priors (Remark A.1, in Appendix A.1): the experiments' prior is proper for $\lambda \geq 1$; those of Sections 3 and 7 are improper, used only on finite or summable families; a Levin-style time factor changes $\ln \pi$ only by $O(\ln \ell(T))$ inside DT° .

2.4 Instantiation grammar

Definition 2.6 (Instantiation grammar $\mathcal{Q}$). ^{model Def 1.5} $\mathcal{Q}$ is a probabilistic context-free grammar on metavariable bodies with one production law per *type* (s, k) : the sort $s \in \{\iota, o\}$ of the node and the number k of variables available there (holes and bound indices alike). A body of type $\iota^n \rightarrow s$ starts at type (s, n) ; a node chooses a function or relation symbol or connective (children of type $(\cdot, k)$), a quantifier (child of type $(o, k + 1)$), one of the k variables, a constant, or a parameter (from a geometric law). $\mathcal{Q}(\theta) := \prod_M \mathcal{Q}_{\text{type}(M)}(\theta(M))$. We require *full support* (all productions have positive probability) and *uniform subcriticality*: at most $b < \infty$ children per production, and type weights $h \in [h_{\min}, h_{\max}]$, $h_{\min} > 0$, and $\rho < 1$ with $\mathbb{E}_i[\sum_{\text{children } c} h(\text{type}(c))] \leq \rho h(i)$ for every type i .

Under these conditions bodies are finite a.s. with exponential tails, and $\mathcal{Q}_{\tau}(s) := \mathcal{Q}(\theta_s)$ if $\tau\theta_s = s$, else 0, is a probability on $\mathcal{S}$, with support $\text{inst}(\tau)$ under full support (Lemma A.2, in Appendix A.2). The uniform condition replaces a per-type condition that does not give finiteness (Remark A.3); it holds for a grammar for L_A with $\rho = 0.75$ (Example A.4), for the term laws of Definition 3.1 and for the experiments' grammar. The code of Section 7 instead *learns* a positional grammar (a sequential KT model of the body), not a fixed PCFG, so results that assume a fixed $\mathcal{Q}$, such as the exact split ties of Proposition 4.6, do not apply to it verbatim (Proposition 7.12).

2.5 Likelihoods

Definition 2.7 (L0: axiom citation). ^{model §1.5} For data $D = (X_1, \dots, X_n)$, a sequence of sentences, and fixed weights, $P_{T,w}(s) := \sum_{\tau \in T} w_{\tau} \mathcal{Q}_{\tau}(s)$ and $P_{T,w}(D) := \prod_i P_{T,w}(X_i)$. With Dirichlet weights, $P_T^{\text{Dir}}(D) := \int \prod_i P_{T,w}(X_i) \text{Dir}(dw; \alpha)$.

Lemma 2.8 (Dirichlet labelling sum). *[proved; computed]^{model §1.5 (c8); experiments Prop X2} With $A := \sum_{\tau} \alpha_{\tau}$, summing over labellings $\ell : \{1..n\} \rightarrow T$ with $X_i \in \text{inst}(\ell_i)$, and $n_{\tau}(\ell)$ the number of data labelled τ ,*

$$P_T^{\text{Dir}}(D) = \sum_{\ell} \frac{\Gamma(A)}{\Gamma(A+n)} \prod_{\tau \in T} \frac{\Gamma(\alpha_{\tau} + n_{\tau}(\ell))}{\Gamma(\alpha_{\tau})} \prod_{i=1}^n \mathcal{Q}_{\ell_i}(X_i).$$

P_T^{Dir} is exchangeable and consistent in n , and its normaliser does not depend on D . For disjoint instance sets the sum has one term, and $P(X_{n+1} = s \mid D) = \sum_{\tau} \frac{\alpha_{\tau} + n_{\tau}}{A+n} \mathcal{Q}_{\tau}(s)$; at $\alpha_{\tau} = \frac{1}{2}$ this is, per template, the Krichevsky–Trofimov (KT) estimator, the Dirichlet($\frac{1}{2}$) predictive rule.

Definition 2.9 (L1: derivation grammar). ^{model §1.5} Fix rule probabilities $\alpha_{\text{ax}}, \alpha_{\text{lg}}, \alpha_{\text{MP}}, \alpha_{\text{Gen}}, \alpha_{\text{VE}}$ summing to 1, with $\alpha_{\text{ax}} + \alpha_{\text{lg}} > 0$; put $\alpha_r := \alpha_{\text{MP}} + \alpha_{\text{Gen}} + \alpha_{\text{VE}}$ and $m := 2\alpha_{\text{MP}} + \alpha_{\text{Gen}} + \alpha_{\text{VE}}$. A tree is grown from the root; each node independently cites a theory axiom (α_{ax} : τ with probability w_{τ} , then $\theta \sim \mathcal{Q}$; conclusion $\tau\theta$), cites a logical axiom (α_{lg} : a schema of K uniformly, instantiated by $\mathcal{Q}$), or applies MP (two children), Gen (one child; a parameter from a geometric law is abstracted) or VE (one child; a term $t \sim \mathcal{Q}$). A tree is *valid* if at each MP node the right child concludes (left conclusion $\rightarrow B$), the node concluding B , and each VE node's child has root $\forall$. Then $\mu_T(s) := \Pr[\text{valid, conclusion } s]$, $Z_T := \sum_s \mu_T(s)$ and $P_T(s) := \mu_T(s)/Z_T$.

Subscripted $\alpha_{\bullet}$ are rule probabilities; the Dirichlet parameter is α or α_{τ} . The code length $-\log_2 \Pr(\text{tree})$ counts *grammar choices*, not the symbols of the formulas; the two can differ exponentially (Remark 6.11).

Lemma 2.10 (L1 is proper; support). *[proved; extinction criterion known; computed]^{model Lemma 1.7} (a) Let $\mathcal{Q}$ be uniformly subcritical. The tree is finite a.s. iff $m \leq 1$ (Athreya and Ney, 1972, Ch. I; numbering not checked); for $m < 1$ it has $1/(1-m)$ derivation nodes in expectation; for $m > 1$ it is infinite with probability $1 - q > 0$, q the least root of $q = (\alpha_{\text{ax}} + \alpha_{\text{lg}}) + (\alpha_{\text{Gen}} + \alpha_{\text{VE}})q + \alpha_{\text{MP}}q^2$. (b) $Z_T \geq 1 - \alpha_r > 0$ for $T \neq \emptyset$, and $Z_{\emptyset} \geq \alpha_{\text{lg}}$; so, when $Z_T > 0$, P_T is a probability whose normaliser does not depend on the data. (c) If all rule probabilities and weights are positive, $\mathcal{Q}$ has full support and parameters are admissible (in logical-axiom instances, $\forall E$ terms and Gen), then $\text{supp } P_T = \text{Th}(T)$. Without parameters Gen is vacuous, and for example $\forall x(R(x) \rightarrow R(x))$ has no tree.*

Definition 2.11 (Variants). ^{model §1.5, Ex 3.6; universal §1.5; pa Def 0.3; experiments §1.7} **L2:** L1 with citation forced at depth d ; support $\text{Th}_{\leq d}(T)$, the conclusions of valid trees of depth $\leq d$. **L1 $^{\sigma}$:** $\mu_T^{\sigma}(s) := \sum_{\pi \text{ valid, concluding } s} \Pr(\pi) e^{-\kappa|\pi|}$, $\kappa > 0$, $|\pi|$ the total symbol size of the tree's conclusions; $P_T^{\sigma} := \mu_T^{\sigma}/Z_T^{\sigma}$, a probability with a data-independent normaliser and the support of L1 (Lemma A.7). **L1 $^{\text{max}}$** (two-part, Viterbi): $V_T(s) :=$ the largest probability of a single valid tree concluding s (under L0: of a single citation). **L1 $^{\text{sch}}$, L1 $^{\text{naive}}$** (Definition 7.1): two-part codes over ND derivations with β bits per *written symbol*, a two-part form of L1 $^{\sigma}$, not of plain L1. **L0 $^{\text{cl}}$:** as L0, but a universal sentence $\forall \bar{x}\chi$ is cited through its instances $\chi(\bar{t})$. **L1 $^{\text{sel}}$** (stream filter): $P_T(s \mid S) = P_T(s)/P_T(S)$ for $s \in S$, a set known to contain every datum. **L1 $^{\text{sel}}_{\text{cit}}$** (per-citation rejection): $\sum_i w_i a_i(s)/c_i$, with $a_i(s)$ the probability that a chain from component i ends in s and c_i that it ends in S ; it equals L1 $^{\text{sel}}$ for one-component theories only (Remark A.6). Noise mixtures: $P_T^{\eta} := (1 - \eta)P_T + \eta N$, N a fixed full-support law, $\eta \in (0, 1)$. **L $_{\varepsilon}$:** $(1 - \varepsilon)P_T(s) + \varepsilon\mu_0(s)[T \not\vdash_k \neg s]$, μ_0 fixed, $\not\vdash_k$ a bounded refutation search; a sub-probability whose mass depends on T , not on the data. **L1 $^{\text{eq}}$:** μ_T/Z_T for an equational grammar (Appendix A.3), terms of size $\leq N$.

2.6 Hänni's scores

Hänni's working notes (Hänni, 2026)² keep the axiom systems that prove the givens “and not their negations”, or those “that do not contradict the given statements”, and suggest that the weight “depend [...] on how many of the given statements can be proven. (one could also penalize longer proofs i guess?)”.

Definition 2.12 (Hänni's scores). model §1.5; universal §1.5; pa Def 0.3 For labelled data $D = D_+ \cup D_-$ (true, false): $S_{\text{prove}}(T; D) := 1[T \text{ consistent}] \prod_{s \in D_+} 1[T \vdash s] \prod_{s \in D_-} 1[T \vdash \neg s]$; $S_{\text{nc}}(T; D) := 1[T \cup D_+ \cup \neg D_- \text{ consistent}]$; $S_g(T; D) := S_{\text{nc}}(T; D) \prod_{s \in D_+} g(\ell_T(s))$, with $\ell_T(s)$ the size of a shortest derivation (∞ if none), g non-increasing and $g_\infty := g(\infty) \in [0, 1]$. The posterior is $\pi(T)S(T; D)$. These are scores, not likelihoods: $\sum_D S(T; D) \neq 1$ and depends on T . For $\beta > 1$, the score $S_{\text{nc}, \beta}$ of Section 3 equals $\beta^n S_g$ with $g \equiv 1$ on provable data and $g_\infty = 1/\beta$; $S_{\text{nc}, 1} = S_{\text{nc}}$. The noisy likelihood L_ε (Definition 2.11) is related but not equal: at $\varepsilon = 1$ it is $\mu_0(D) \prod_{s \in D} [T \not\vdash_k \neg s]$, a per-datum, bounded-search relaxation of S_{nc} on positive data, which equals S_{nc} up to the common factor $\mu_0(D)$ only when joint consistency of $T \cup D$ reduces to non-refutation of each datum within k steps; as $\varepsilon \rightarrow 0$ it tends to the generative P_T , whose support ($\mathbf{L1}$, parameters admissible) is $\text{Th}(T)$.

2.7 Hänni's trichotomy

His notes also give each sentence “a p(true/false/independent)” and ask three questions about it.

Definition 2.13 (Belief, disbelief, independence). model §7; universal §1.6 For a posterior π_D on a countable class, a sentence s and $\vdash$ derivability in $\mathbf{K}$ without a bound: $\text{Bel}(s) := \pi_D(T \vdash s)$, $\text{Dis}(s) := \pi_D(T \vdash \neg s)$, $\text{Indep}(s) := \pi_D(T \not\vdash s \text{ and } T \not\vdash \neg s)$, $\text{Inc} := \pi_D(T \text{ inconsistent})$, $\text{Pl}(s) := 1 - \text{Dis}(s)$ and $\text{Bel}_{\text{cons}}(s) := \pi_D(T \text{ consistent and } T \vdash s)$.

Bounded $\text{Th}_d(T)$ is not closed under consequence, so Proposition 2.14(a), (c), (d) and Proposition 2.18 need $d = \infty$; Proposition 2.19 holds for every d (proofs in Appendix A.7).

Proposition 2.14 (Basic laws). [proved]^{model Prop 7.1} (a) $\text{Bel}(s) + \text{Dis}(s) + \text{Indep}(s) = 1 + \text{Inc}$; from now on the posterior is on consistent theories, so the three sum to 1. (b) $\text{Dis}(s) = \text{Bel}(\neg s)$; $\text{Bel}(\top) = 1$, $\text{Bel}(\perp) = 0$; $\text{Pl}(s) = \text{Bel}(s) + \text{Indep}(s)$. (c) If $s \vdash s'$ then $\text{Bel}(s) \leq \text{Bel}(s')$. (d) If $\vdash \neg(s \wedge s')$ then $\text{Bel}(s \vee s') \geq \text{Bel}(s) + \text{Bel}(s')$.

Example 2.15 (Renormalising is incoherent). [proved; computed]^{model Ex 7.4} Hänni's first question proposes to “get true/false by either just removing the independent fraction and renormalizing”; this is incoherent in general, as he later suspects. Let $R(s) := \text{Bel}(s)/(\text{Bel}(s) + \text{Dis}(s))$. With atoms a, b and theories $\{a\}$, $\{b\}$, $\{\neg(a \wedge b)\}$ of posterior $\frac{1}{3}$ each, $R(a) = R(b) = 1$ but $R(a \wedge b) = 0$; no probability does this.

Example 2.16 (The 50/50 rule is incoherent). [proved; computed]^{model Ex 7.5} His second proposes to “turn the independent fraction into 50/50” and expects “the downside of not giving coherent probabilities usually”; confirmed, with an exact criterion in Proposition 2.17. Let $F_{1/2}(s) := \text{Bel}(s) + \frac{1}{2}\text{Indep}(s)$. Already for the empty theory over atoms b, c : $F_{1/2}(b) = F_{1/2}(b \wedge c) = F_{1/2}(b \wedge \neg c) = \frac{1}{2}$, while coherence needs $P(b \wedge c) + P(b \wedge \neg c) = P(b)$.

²File `research/prior/hanni-solomonoff-axiom-induction.md` of the accompanying repository.

Proposition 2.17 (When the 50/50 rule is coherent). *[proved; computed]^{model Prop 7.6} Over finitely many propositional atoms, $F_{1/2}$ is coherent (equal to some probability on valuations, on every sentence) iff every theory of positive posterior mass has at most two models.*

Proposition 2.18 (Bel is a belief function). *[proved; framework known]^{model Prop 7.2} For all sentences $s_1, \dots, s_k$, $\text{Bel}(s_1 \vee \dots \vee s_k) \geq \sum_{\emptyset \neq I \subseteq [k]} (-1)^{|I|+1} \text{Bel}(\bigwedge_{i \in I} s_i)$. So Bel is a totally monotone capacity on the Lindenbaum algebra, with focal elements $\text{Mod}(T)$, and Pl is its plausibility (Shafer, 1976).*

Proposition 2.19 (Coherent bracketing). *[proved]^{model Prop 7.3} There is a probability P on complete consistent theories with $\text{Bel}(s) \leq P(s) \leq \text{Pl}(s)$ for every s .*

So his third, “hmm maybe this motivates not removing the independent fraction?”, gets a yes: (Bel, Pl) is a coherent lower and upper probability. Like the semimeasure he mentions, it keeps the mass on “undecided” (the convention under which Theorem 6.3 holds with a constant); a coherent point value would need a distribution over the completions of each theory.

Remark 2.20 (How the likelihood shapes Indep). *[proved for ground ψ ; proof sketch when ψ is generated by a schema]^{model §7} Let ψ be independent of the data. Under Hänni’s 0/1 scores the posterior odds of $T^* \cup \{\psi\}$ against T^* stay at their prior value (Proposition 2.22(b)). Under L0 and L1 a theory that proves *and generates* ψ loses mass while ψ never appears: polynomially with Dirichlet weights (Proposition 4.14), exponentially with fixed weights (Lemma 2.21). So that part of $\text{Bel}(\psi)$ falls (all of it under L1, parameters admissible), and $\text{Indep}(\psi)$ grows if the mass moves to theories that neither prove nor refute ψ .*

2.8 The size principle

Lemma 2.21 (Size principle, exact form). *[proved; computed]^{model Lemma 1.8 (c1); universal Thm U4(1)} Let P^*, P be probabilities on a countable $\mathcal{S}$, $\varepsilon := P(\mathcal{S} \setminus \text{supp } P^*) < 1$ and $\tilde{P} := P(\cdot \mid \text{supp } P^*)$. (a) $\text{KL}(P^* \| P) = \text{KL}(P^* \| \tilde{P}) + \ln \frac{1}{1-\varepsilon} \geq \ln \frac{1}{1-\varepsilon}$. (b) On data in $\text{supp } P^*$, $P(D) = (1 - \varepsilon)^n \tilde{P}(D)$, and $\mathbb{E}_{P^*}[\tilde{P}(D)/P^*(D)] = 1$. Computed checks: Appendix A.5.*

So a hypothesis that puts mass ε outside the data’s support loses at least $\ln \frac{1}{1-\varepsilon}$ nats per datum: the size principle (Tenenbaum and Griffiths, 2001) (for sub-probabilities: Theorem B.4).

Proposition 2.22 (Which likelihoods have the size principle). *[proved]^{model Prop 1.9} (a) L0, L1, L2 and L1 $^\sigma$ have it: each P_T is a probability on $\mathcal{S}$ with a data-independent normaliser, so Lemma 2.21 applies to every T that puts mass outside the data law’s support (under L0 with full-support $\mathcal{Q}$: every T with $\bigcup \text{inst}(T) \not\subseteq \text{supp } P^*$). (b) Hänni’s scores do not. Let $\text{Th}(T) \subseteq \text{Th}(T')$. Then $S_{\text{nc}}(T') \leq S_{\text{nc}}(T)$. If T' is consistent with the labelled data, $S_{\text{prove}}(T') \geq S_{\text{prove}}(T)$; if moreover $\bigcup \text{inst}(T) \subseteq \bigcup \text{inst}(T')$, $S_g(T') \geq S_g(T)$. So if $\psi \notin \text{Th}(T^*)$ and $T^* \cup \{\psi\} \cup D_+ \cup \neg D_-$ is consistent, the posterior odds of $T^* \cup \{\psi\}$ against T^* never fall below the prior odds under S_{prove} or S_g , and equal them under S_{nc} . The consistency hypothesis is needed ($\psi = \forall x R(x)$, $T^* = \emptyset$, $D_+ = \{\neg R(0)\}$). (c) Generative likelihoods penalise that strengthening (ψ ground): exactly by $(1 - w_\psi)^n$ under L0 with fixed weights (T^* ’s weights scaled by $1 - w_\psi$), on data not containing ψ ; at rate at least $\ln \frac{1}{1 - \alpha_{\text{ax}} w_\psi}$ nats per datum under L1; like $n^{-\alpha}$ with Dirichlet weights (Proposition 4.14).*

Remark 2.23 (Hänni’s graded score, normalised). *[proved; refuted (old sentence); computed]^{model Rem 1.10; referee m3 (r6)}*

Let $\ell_T(s)$ be the least length in bits of an explicit derivation of s from T in a prefix-free code that writes every formula out, $g(\ell) := 2^{-\kappa \ell}$, $\kappa \geq 1$, $g_\infty = 0$. Then $Z_T := \sum_s 2^{-\kappa \ell_T(s)} \leq 1$ and $P_T^g := 2^{-\kappa \ell_T} / Z_T$ is a probability; with $g_\infty > 0$ the normaliser is infinite. Like L1 $^\sigma$, P_T^g charges

the *written* length of the derivation (Proposition E.5). If $\bigcup \text{inst}(T) \subseteq \bigcup \text{inst}(T')$, then $Z_{T'} \geq Z_T$, so $P_{T'}^g(s) \leq P_T^g(s)$ whenever $\ell_{T'}(s) = \ell_T(s)$. Hänni’s remark that one can “think of this as there being some model for how the statements get generated” holds once the score is divided by Z_T and $g_\infty = 0$. *Refuted* (Appendix H.2): the pre-referee sentence “a stronger theory has more short theorems, hence a larger Z_T , hence a smaller $P_T(s)$ on each datum”. For $T = \{a, a \rightarrow b\}$ and $T' = T \cup \{b\}$ (same theorems, nested instances) b has a one-citation derivation from T' and needs MP from T , so $P_{T'}^g(b)/P_T^g(b) = 2^{\kappa(\ell_T(b) - \ell_{T'}(b))} Z_T/Z_{T'} \geq 2^{\kappa(\ell_T(b) - \ell_{T'}(b))} Z_T$, which exceeds 1 once $\kappa(\ell_T(b) - \ell_{T'}(b)) > \log_2(1/Z_T)$: a datum whose shortest derivation gets shorter can gain.

2.9 Computing the likelihood

Proposition 2.24 (What can be computed). *[proved; (b), (c) given Church’s theorem (known)]^{model Prop 2.7}*
Assume L1 with $m < 1$, $\mathcal{Q}$ uniformly subcritical, computable grammar probabilities and weights, and the hypotheses of Lemma 2.10(c). (a) $\mu_T(s)$, Z_T and $P_T(s)$ are computable reals, uniformly in (T, s) . (b) $\{(T, s) : P_T(s) > 0\} = \{(T, s) : s \in \text{Th}(T)\}$ is recursively enumerable and, for a language with a binary relation symbol, undecidable already for $T = \emptyset$ (Church, 1936; Turing, 1936). (c) No algorithm decides whether a theory has likelihood 0 on a datum. For a finite class, the posterior $\pi_n(T)$ is a computable real whenever it is defined, but which theories it has excluded is undecidable.

The proof enumerates trees, exponentially many in $1/\epsilon$ for precision ϵ , and nothing better is known (Appendix A.6); later sections restrict the calculus (Ch_J , $J \leq 2$) or score explicit derivations (L1^{sch}).

The two-part L1^{max} is not the sum: the maximum can break a tie inside the generator class, with two-part odds $(\pi(T_2)/\pi(T_1)) 2^{-n} \rightarrow 0$ between theories of equal L0 law and equal theorems (Proposition A.8, in Appendix A.6); whether it can change a *deductive* limit is open. The codes of Section 7 are also two-part.

3 The case $\forall x\varphi$: what instances support

Hänni’s question starts from statements $\varphi(t)$ and the law $\forall x\varphi$, which “implies all the other statements” (Section 1). In the model of Section 2 the data favour the generator that emits exactly the instances. In every comparison analysed (Theorem 3.4) they never favour $\forall x\varphi$ over its instance schema σ_φ , except through non-logical weight effects under a selection-aware likelihood (Proposition 3.6); with $\wedge$ -rules on quantified formulas the question is open. “All future data are instances” is confirmed; “the axioms prove $\forall x\varphi$ ” is not, a Bayesian form of the ω -gap of AS §4.5: all closed instances $\varphi(0), \varphi(S0), \dots$ together still do not entail $\forall x\varphi$.

So the intuition is right against hypotheses that fit worse (over-general templates, memorisers), for prediction, and for truth under a Gaifman-type credence, safely in $\mathbb{N}$. It needs three refinements: σ_φ is a second simple law that implies exactly the data, brought into the class by templates; “the axioms prove $\forall x\varphi$ ” gains from closed instances only through non-logical differences between the laws, as under L1^{sel} with a background (Proposition 3.6), and with data from a model generator it tends to the prior share of provers in the generator class (Theorem 3.9); and a derivation likelihood rewards the most specific generator that fits, and frequently used lemmas, not the logically strongest law. $\forall x\varphi$ is licensed by an instance at a parameter, a quantified consequence, or a selection-aware likelihood without a guard (Proposition 3.10(a), Proposition B.14(a), Proposition 3.12(c)), soundly in $\mathbb{N}$ and unsoundly in $\mathbb{R}$; data at parameters are common in proof-assistant practice. Proofs and further tables: Appendix B; logarithms are natural.

3.1 Setup

Definition 3.1 (Instances, schema, term laws). ^{universal §1.1–1.3} $\varphi(x)$ is an L_A -formula with exactly one free variable x , which occurs in it; the examples $(0 + x = x, x + 0 = x, x \cdot 0 = 0, \neg Sx = 0, x < Sx)$ are quantifier-free unless stated. $I_c(\varphi) := \{\varphi(t) : t \text{ closed}\}$; $I_o(\varphi)$ also allows parameters in t . The instance schema is $\sigma_\varphi := \varphi[z/x]$, z a 0-ary term metavariable, closed guard unless stated. Term laws: $\mathcal{Q}_{\text{num}}(S^j 0) = (1 - q)q^j$, $q = \frac{1}{2}$ by default; $\mathcal{Q}_{\text{GW}}$, each node picking its root $0, S, +, \cdot$ with probabilities $.4, .3, .2, .1$; $\mathcal{Q}_{\text{open}} := \rho[p] + (1 - \rho)\mathcal{Q}_{\text{num}}$ for a parameter p . The first two are root-generated: $\mathcal{Q}(f(t_1, \dots, t_r)) = p_f \prod_i \mathcal{Q}(t_i)$. One law supplies all bodies and $\forall E$ terms.

The hypotheses ^{universal §1.3} (Table 13 in Appendix B) are $H_\forall = \{\forall x\varphi\}$; $H_{\text{sch}} = \{\sigma_\varphi\}$; H_{open} , σ_φ with an open guard ($z \sim \mathcal{Q}_{\text{open}}$); $H_{\text{both}} = \{\forall x\varphi, \sigma_\varphi\}$ with weight $w_\forall$ on $\forall x\varphi$; memorisers, finite sets of closed instances; over-specific templates such as $\sigma_\varphi[z := Sz]$; over-general ones, with a subterm or subformula of σ_φ made a fresh metavariable (down to the inconsistent bare F_0); and root splits $\{\sigma_\varphi[z := f(z_1, \dots, z_r)] : f \text{ a root of } \mathcal{Q}\}$. For $\varphi = 0 + x = x$, $H_\forall$, H_{open} and H_{both} prove $\forall x\varphi$; H_{sch} , memorisers, over-specific templates and root splits do not, even over $\mathcal{Q}$ (AS Ex 4.9). $B \oplus_w A$ adds A with weight w to B , scaling B 's weights by $1 - w$. Calculi and likelihoods are as in Definitions 2.3 and 2.11. In $\mathbf{C}_{\text{min}}$ each chain node cites (probability $1 - c$) or applies $\forall E$ with $t \sim \mathcal{Q}$ (probability c); $\mathbf{C}_{\text{open}}(c, g)$ adds Gen on the parameter (g ; cite $p_{\text{cite}} = 1 - c - g$); μ_T is the unnormalised derivation mass and $\text{L1} = \mu_T / Z_T$. Hänni's scores are S_{prove} , S_{nc} and S_g with $g \equiv 1$ on provable data (Definition 2.12); for $\beta > 1$, $S_{\text{nc}, \beta} := \beta^n S_g$ with $g_\infty = 1/\beta$ rewards each proved datum by a factor β , and $S_{\text{nc}, 1} := S_{\text{nc}}$. Theorems about countable classes assume a proper prior (Definition 2.4); the computations use finite or summable families with the prior $2^{-(\sum_A |A| + \#\text{axioms})}$. Defaults: $c = 0.3$; $(c, g, \rho, q) = (0.3, 0.2, 0.1, \frac{1}{2})$ in $\mathbf{C}_{\text{open}}$; tables state α .

3.2 Instance data do not favour $\forall x\varphi$ over its schema

Proposition 3.2 (Factorisation). *[proved; computed]^{universal Prop U1} In $\mathbf{C}_{\text{min}}$, let φ have one free variable and m leading universal quantifiers. Then $\mu_{H_\forall}(s) = c\mu_{H_{\text{sch}}}(s)$ for every sentence $s \neq \forall x\varphi$; $\mu_{H_\forall}(\forall x\varphi) = 1 - c$, $\mu_{H_{\text{sch}}}(\forall x\varphi) = 0$; $Z_{H_{\text{sch}}} = (1 - c) \sum_{k \leq m} c^k$, $Z_{H_\forall} = (1 - c) + cZ_{H_{\text{sch}}}$. With a background, $\mu_{B \oplus_w \forall x\varphi} = (1 - w)\mu_B + wc\mu_{H_{\text{sch}}}$ and $\mu_{B \oplus_w \sigma_\varphi} = (1 - w)\mu_B + w\mu_{H_{\text{sch}}}$ on $s \neq \forall x\varphi$. For k variables the factor is c^k on each full instance.*

Theorem 3.3 (Odds on instance data). *[proved; computed]^{universal Thm U2} In $\mathbf{C}_{\text{min}}$, after n closed instances the odds $O_n := \pi_n(H_\forall)/\pi_n(H_{\text{sch}})$ satisfy: O_n/O_0 is 1 under L0^{cl} , under $\text{L1}^{\text{sel}}(S)$ with $\forall x\varphi \notin S$, and under the scores ($\forall x\varphi$ consistent); c^n under μ_T and L1^{max} ; $(cZ_{H_{\text{sch}}}/((1 - c) + cZ_{H_{\text{sch}}}))^n$ under L1 , which is $(c/(1 + c))^n$ for quantifier-free φ ; 0 for $n \geq 1$ under strict L0 ; and, with a background B under μ_T , $\prod_i r(X_i)$ with $r(s) \in [c, 1]$.*

So at $c = 0.3$, under L1 with quantifier-free φ , a generator whose axiom is $\forall x\varphi$ outputs $\forall x\varphi$ itself with probability $1/(1 + c) = 0.77$, and each observed instance multiplies its odds against σ_φ by $c/(1 + c) = 0.23$ (Table 2; checks in Table 14).

Theorem 3.4 (Instance data do not favour $\forall x\varphi$). *[proved]^{universal Thm B} On data not containing $\forall x\varphi$, the posterior odds of the $\forall$ -version against the schema version are at most the prior odds: (B1) in $\mathbf{C}_{\text{min}}$, single axioms, closed instances, under L0^{cl} , μ_T , L1 , L1^{sel} , L1^{max} and the scores (equality under L0^{cl} , under L1^{sel} when $\forall x\varphi \notin S$, and under the scores when $\forall x\varphi$ is consistent); (B2) in $\mathbf{C}_{\text{min}}$, $B \oplus_w \forall x\varphi$ against $B \oplus_w \sigma_\varphi$ for any background B and any data, under μ_T and L1 , with a fixed w or a common weight prior (equality under L0^{cl}); (B3) in $\mathbf{C}_{\text{U3}}$ calculi, quantifier-free*

likelihood	hypothesis	$n = 0$	1	5	20	200
$L0^{cl}$ (= $L1^{sel}$ here)	$H_{\forall}; H_{sch}$	.021; .042	.194; .387	.303; .605	.332; .665	.332; .665
	over-general; over-specific	.926; .011	.307; .110	.022; .067	.000; .000	.000; .000
	root split, weights p_f	.000	.002	.002	.003	.003
μ_T	$H_{\forall}; H_{sch}$	.021; .042	.067; .448	.001; .891	.000; .996	.000; .996
	root split, weights p_f	.000	.002	.003	.004	.004
L1	$H_{\forall}; H_{sch}$	.021; .042	.053; .455	.000; .892	.000; .996	.000; .996

Table 2: Running example (c2): mean posterior over 50 seeds, $\varphi = 0 + x = x$, $\mathcal{Q}_{num}$, $\mathcal{C}_{min}$, prior $2^{-(symbols+1 \text{ per axiom})}$. Class: $H_{\forall}$, H_{sch} , four over-general templates (among them the inconsistent bare formula metavariable, 0.674 at $n = 0$), $\varphi(Sz)$, and the root split with fixed and with learned ($\alpha = 1$) weights; the fixed-weight split has exactly H_{sch} ’s law (Proposition 4.6). Four other formulas and $\mathcal{Q}_{GW}$ behave alike.

φ , any background (Proposition B.2), under μ_T and $L1^{max}$; and, for single axioms and quantifier-free data, under the scores, with equality (by Herbrand’s theorem, $\{\forall x\varphi\} \vdash s$ iff $I_c(\varphi) \vdash s$ for quantifier-free s).

Proof idea. The mechanism is the size principle, through the extra $\forall E$ step that $\forall x\varphi$ needs to reach an instance (Lemma 2.21). (B2) holds at every w , and $Z_{H_{\forall}} \geq Z_{H_{sch}}$; (B3) rests on a simulation inequality (Proposition B.2 in Appendix B.1). $\square$

Not covered: $L1^{sel}$ with a background. There $B \oplus_w \forall x\varphi$ has the selected law of $B \oplus_{w_e} \sigma_{\varphi}$ with the effective weight $w_e = wc/(1 - w + wc)$ (proof of Proposition 3.6): with a common fixed w the two theories have different laws and whichever matches the data’s mixture proportion wins at a linear rate, and with a learned weight the $\forall$ -version can gain a bounded factor (Proposition 3.6). Neither effect is logical, and under the per-citation $L1^{sel}_{cit}$ both vanish (Remark 3.7). Also not covered: $\wedge$ -detours (Remark 3.5), normalised likelihoods in $\mathcal{C}_{U3}$, and the scores with a background or with quantified data, where the $\forall$ -version can be favoured (the datum $\neg\exists x\neg\varphi$ is proved by $H_{\forall}$ and not by H_{sch}).

The direction is robust (Proposition B.1): in $\mathcal{C}_{min}$, at every $s \neq \forall x\varphi$, a full-support noise component leaves the ratio of the laws of $H_{\forall}$ and H_{sch} in $[c, 1]$ under μ_T , ≤ 1 under **L1** and $= 1$ under $L1^{sel}$ with noise after selection; chain weights non-increasing in the number of eliminations keep $P_{H_{\forall}}^f(s) \leq P_{H_{sch}}^f(s)$; and under **L1**, quantifier-free φ and any prior on c , $O_n/O_0 \leq 2^{-n}$.

Remark 3.5 ($\wedge$ -detours). [refuted (conjecture); counterexample: proof sketch; computed]^{universal §2.3} The factor c of Proposition B.2 depends on the calculus (an earlier version conjectured otherwise; Appendix H.2): in $\mathcal{C}_{\wedge}$ at $(p_{cite}, c, a, e) = (0.35, 0.15, 0.25, 0.25)$, $\varphi = 0 + x = x$, the ratio $\mu_{H_{\forall}}(\varphi(t))/\mu_{H_{sch}}(\varphi(t))$ is $0.15553 > c$ (Appendix B.1); on a 9139-point grid it stays ≤ 0.925 (normalised ≤ 0.48). Whether detours can reverse the direction is open.

Proposition 3.6 (Learned weights under $L1^{sel}$). [proved; computed]^{universal Prop B2} In $\mathcal{C}_{min}$ let $B = \{\sigma_{\psi}\}$, ψ quantifier-free, $\text{inst}(\sigma_{\psi}) \cap \text{inst}(\sigma_{\varphi}) = \emptyset$, and compare $B \oplus_w \forall x\varphi$ with $B \oplus_{w_e} \sigma_{\varphi}$ (uniform prior on w , equal prior masses) under $L1^{sel}$, S the closed quantifier-free sentences. On data i.i.d. from $(1 - w^*)\mathcal{Q}_{\sigma_{\psi}} + w^*\mathcal{Q}_{\sigma_{\varphi}}$ the Bayes factor tends a.s. to $h(w^*)$, $h(v) := c/(c + v(1 - c))^2 \in [c, 1/c]$, which exceeds 1 iff $w^* < \sqrt{c}/(1 + \sqrt{c})$.

The cause is the prior induced on the effective weight w_e (computed to $4 \cdot 10^{-4}$ at $n = 10^5$). With learned weights the effect is bounded; it is not evidence that the axioms prove $\forall x\varphi$.

Remark 3.7 (Per-citation selection). [proved]^{editor}, from experiments §1.7, Prop X6(b) and universal Prop B2 Under the per-citation $L1_{\text{cit}}^{\text{sel}}$ the selected law of the component $\forall x\varphi$ is $\mathcal{Q}_{\sigma_\varphi}$, so both theories of Proposition 3.6 have law $(1-w)\mathcal{Q}_{\sigma_\psi} + w\mathcal{Q}_{\sigma_\varphi}$ at every w : the Bayes factor is 1 (Remark A.6).

A spare-slot prover dies only polynomially: with a uniform prior on $w_\forall$, under $L1$ and quantifier-free φ , H_{both} 's mass falls like $\Theta(1/n)$, as $\forall x\varphi$ is a spare slot, a template beside the generator's own that the data never or rarely use (Proposition B.3).

The experiments reproduce Theorem 3.3 in their chain Ch_1 (factor $\frac{1}{2}$: one bit per datum, to $1.1 \cdot 10^{-11}$) and, under $L1_{\text{cit}}^{\text{sel}}$, an exact tie with $P(T \vdash \forall x\varphi \mid D_n) \rightarrow 0.484$ (0.269 for $\neg Sx = 0$), the prior share in their code (E1, Table 7).

3.3 Memorisers, over-general and over-specific templates

Over-general templates lose exponentially (Theorem B.4 in Appendix B.2): a law with mass $p < 1$ on $I_c(\varphi)$ loses at least $\ln \frac{1}{p}$ nats per datum, a.s. and in expectation ($\mathbb{E} P'(D_n)/P^*(D_n) \leq p^n$), and under the citation law ($L0$, $L0^{\text{cl}}$) a first-order template strictly more general than σ_φ , with metavariables i.i.d. from $\mathcal{Q}$ and a formula law F , has $p \leq \max(\max_f p_f, \max_g F(\text{root} = g))$. Over-specific templates $\sigma_\varphi[z := f(z_1, \dots, z_r)]$ fail too: under a root-generated $\mathcal{Q}$ their odds against σ_φ are a martingale that grows like p_f^{-n} while all data have root f (probability p_f^n) and then vanishes (Proposition B.5). The memoriser class, however, is not exponentially small in general (Proposition B.6): with uniform weights and $\mathcal{Q}$ of infinite support and finite entropy its log-odds against H_{sch} fall faster than linearly, but with Laplace weights under $\mathcal{Q}_{\text{num}}$ they stay $\geq -C \ln^2 n$ a.s. for all large n (conjecturally $-\Theta(\ln^2 n)$); under $\mathcal{Q}_{\text{GW}}$ the computed rate is exponential, as distinct data grow linearly. E1 tests only the single memoriser $\text{Mem}(D_n)$, whose axioms are the distinct data seen so far; every fixed memoriser dies under well-specification (Remark C.4).

3.4 Predictive confirmation

Theorem 3.8 (Predictive confirmation). [proved; computed]^{universal Thm U8} Assume setting W (Definition 4.1). Let $M(\cdot \mid D_n) = \sum_T \pi_n(T) P_T$, $G := \{T : P_T(I) = 1\}$, and $T^* \in G$ the generator. (1) $\sum_{n \geq 0} \mathbb{E}[1 - M(I \mid D_n)] \leq \ln \frac{1}{\pi(T^*)}$. (2) The probability that all future data lie in I is $\pi_n(G)$, and $\pi_n(G) \rightarrow 1$ a.s. (3) $\mathbb{E}[1 - \pi_n(G)] \leq \min(1, \sum_{T \notin G} \frac{\pi(T)}{\pi(T^*)} P_T(\text{supp } P_{T^*})^n)$. (4) If $T \in G$ have law $\hat{P}$ and $T \notin G$ have law $(1 - \varepsilon_T)\hat{P} + \varepsilon_T N$, $N(I) = 0$, then on instance sequences $1 - \pi_n(G) = \sum_{T \notin G} \pi(T)(1 - \varepsilon_T)^n / (\pi(G) + \sum_{T \notin G} \pi(T)(1 - \varepsilon_T)^n)$.

The rate depends on near-universal competitors: $\approx (1 - \pi_0)/(\pi_0 n)$ for Laplace's rule with a point mass π_0 , but $1 - \pi_n(G) = 0.37$ at $n = 10^{50}$ for a dyadic family ($\varepsilon_j = 2^{-j}$ with prior $(1 - \pi_0)/(j(j+1))$ on $j \geq 1$, $\pi_0 = 0.01$). Part (2) is the countable-class analogue of Hutter's result for the universal prior, but G contains H_{sch} , which does not prove $\forall x\varphi$; and confirmation is not stepwise monotone (Remark B.7).

3.5 The Bayesian ω -gap

The next theorem answers “do the axioms prove $\forall x\varphi$?”. Its headline is (a) with (b): the posterior probability of a prover tends to the prior share of provers in the generator class $C^* := \{T : P_T = P_{T^*}\}$, which no data move. A likelihood is *faithful* for a calculus $\mathcal{C}$ if $\text{supp } P_T = \text{Th}_{\mathcal{C}}(T) \cap X$ for all T , with X the data space.

Theorem 3.9 (Bayesian ω -gap). *[(a)–(c), (d1), (e) proved; computed]^{universal Thm U10} Let $\mathcal{H}$ be countable with a proper prior $\pi > 0$, $\text{Prov}_\forall := \{T \in \mathcal{H} : T \vdash \forall x\varphi\}$ (first-order; inconsistent theories included), and the data i.i.d. from P_{T^*} , $T^* \in \mathcal{H}$, unless stated.*

- (a) $P(T \vdash \forall x\varphi \mid D_n) \rightarrow \pi(\text{Prov}_\forall \cap C^*)/\pi(C^*)$ a.s.; also for $\vdash_C$, Bel, Dis, Indep, Inc (Definition 2.13).
- (b) Prior share. If C^* has a prover and a non-prover, the limit is in $(0, 1)$: $H_\forall, H_{\text{sch}}$ under L0^{cl} or L1^{sel} ; $B \oplus \forall x\varphi$, $B \oplus \sigma_\varphi$ under L0^{cl} with a common weight or weight prior. For the two theories of Proposition 3.6 under L1^{sel} the laws differ, but that proposition gives such a limit.
- (c) Faithful regime. (c1) If $\forall x\varphi \in X$, $P(T \vdash_C \forall x\varphi \mid D_n) \rightarrow 1[T^* \vdash_C \forall x\varphi]$. (c2) If also C is sound and all axiom instances lie in X , members of C^* have T^* 's first-order theorems and $P(T \vdash \forall x\varphi \mid D_n) \rightarrow 1[T^* \vdash \forall x\varphi]$. (c3) In $C_{\min}$, and C_{open} with a guard, L1 with full-support $\mathcal{Q}$ is faithful and $\text{supp } P_{H_{\text{sch}}} = I_c(\varphi)$ for quantifier-free φ , so for $T^* = H_{\text{sch}}$ the limit in (c1) is 0. If C has a rule that maps an instance to a non-instance with positive probability ($\wedge I$, $\vee I$, symmetry of $=$, $\rightarrow I$, MP with cited logical axioms), every T with $P_T(I_c) > 0$ emits non-instances: instance-only data are then misspecified for every hypothesis, and the limit is set by the best-fitting generators of the class (Theorem 4.16).
- (d) Filtered data from $H_\forall$, seen only in $S = I_c(\varphi)$. (d1) For quantifier-free φ in $C_{\min}$ the filtered law is H_{sch} 's: limit 0 under L1 , the prior share under L1^{sel} . (d2) [proved for the pair; proof sketch in general] For quantified φ , H_{sch} is the KL-minimiser in $\{H_\forall, H_{\text{sch}}\}$ and the limit is 0; for larger classes the minimiser was not determined.
- (e) Unknown filter. With a countable family $\mathcal{F}$ of filters S with $P_T(S) > 0$ for every $T \in \mathcal{H}$, a prior ν on $\mathcal{F}$ independent of π , and pair laws $P_T(s)1[s \in S]/P_T(S)$, (a) applies to pairs. For $\{H_\forall, H_{\text{sch}}\}$, $C_{\min}$, quantifier-free φ and data law $\mathcal{Q}_{\sigma_\varphi}$ the limit is $\frac{\pi(H_\forall)\nu(\mathcal{F}_1)}{\pi(H_\forall)\nu(\mathcal{F}_1) + \pi(H_{\text{sch}})\nu(\mathcal{F}_0)}$, $\mathcal{F}_0 = \{S \supseteq I_c\}$, $\mathcal{F}_1 = \{S \in \mathcal{F}_0 : \forall x\varphi \notin S\}$.

In the prior-share regime the limit lies in $(0, 1)$, robustly, but its value depends on the code: for $\varphi = 0 + x = x$ the share of $H_\forall$ in $\{H_\forall, H_{\text{sch}}\}$ is 0.333 in the symbol-count prior above and in π_1 of Definition 2.4, 0.200 in π_2 , 0.484 in the experiments' code and 0.030–0.042 in the per-symbol codes of Section 7 (Table 16). The data never move it.

Hänni's own scores leave the question open (Proposition B.8 in Appendix B.4): for $\varphi = 0 + x = x$ under S_{prove} the posterior tends a.s. in total variation to the prior on the consistent provers of $I_c(\varphi)$, which include independent, true and false theories (Q , $Q + \forall x\varphi$, $Q + \neg \forall x\varphi$), so all three trichotomy values keep positive mass and over-generalisation is never penalised; under S_{nc} and S_g ($g \equiv 1$ on provable data) the provers of all the data share one factor, with the same conclusion, and under S_{nc} the limit is the prior on the larger set $\{T : T \cup I_c(\varphi) \text{ consistent}\}$, which contains non-provers such as $\emptyset$. Inconsistent theories inflate Bel and Dis alike, so the verifier should use Bel_{cons} (Remark 5.9). Closed instances cannot tell Q4 from its closed schema $z + 0 = z$, as $(Q - Q4)$ plus all closed instances of $x + 0 = x$ does not prove $\forall x(x + 0 = x)$ (Proposition B.9). For truth, a credence on structures believes $\forall x\varphi$ in the limit of its true closed instances if $P(\forall x\varphi) > 0$ and almost every structure is an elementary extension of its closed-term substructure (Gaifman's condition, cited from memory): safe for $\mathbb{N}$, unsafe for credences that weigh $\mathbb{R}$, V , models of $\text{PA} + \neg \text{Con}(\text{PA})$ or some models of Q (Proposition B.10).

3.6 Open instances and the closedness guard

Proposition 3.10 (Parameter data and the guard). *[proved; computed]^{universal Prop U11(a)–(d)} Assume the closure reading, Gen over parameters, theories without parameters, and a likelihood supported on theorems. (a) Once D contains some $\varphi(p)$, every T with $\pi(T \mid D) > 0$ proves $\forall x\varphi$. (b) If $P_{T^*}(\varphi(p)) = \beta > 0$, then $P(P(T \vdash \forall x\varphi \mid D_n) < 1) \leq (1 - \beta)^n$. (c) In $\mathbf{C}_{\text{open}}(c, g)$ with $\mathbf{Q}_{\text{open}}$, per closed datum under L1: $H_{\text{open}} : H_{\text{sch}} = \frac{1-\rho}{1+g\rho}$, $H_{\forall} : H_{\text{sch}} = \frac{c(1-\rho)}{1+c}$, $H_{\forall} : H_{\text{open}} = \frac{c(1+g\rho)}{1+c}$. (d) If the class is finite and contains the guarded H_{sch} , closed data give $P(T \vdash \forall x\varphi \mid D_n) \rightarrow 0$: the guard is learned at rate $\mathcal{W}_o := \ln \frac{1+g\rho}{1-\rho}$ (0.1252 nats per datum at the defaults); the spare-slot prover H_{both} decays only polynomially (like $1/n$ in $\mathbf{C}_{\text{min}}$, Proposition B.3).*

For countable classes with learned-weight provers the limit 0 is not proved. Here ρ is fixed; in the two-part code of Section 7, where it is learned, a head-symbol split of $x + 0 = x$ closes in on the schema at only about 5 bits per doubling of n (Remark 7.34).

Without a guard (DT° , λ -convention, parameters admissible) the outcome depends on which competitors the class contains; with a selection-aware likelihood the ω -step is taken. The competitors are the unguarded splits $R_k := \{\varphi(0), \dots, \varphi(S^{k-1}0), \varphi(S^k z)\}$ with weights $(v_0, \dots, v_{k-1}, u)$, provers of $\forall x\varphi$ over $\mathbf{Q}$ but not in pure logic; on closed numeral data under L1 they lose only $q^k \mathcal{W}_o$ nats per datum at their best weights, against $\mathcal{W}_o$ (H_{open}) and $\ln \frac{1+c}{c(1-\rho)}$ ($H_{\forall}$; Proposition B.13).

Remark 3.11 (Without a guard). *[refuted; computed]^{universal §7.2} Without a guard the posterior need not take the ω -step on closed data (contrary to an earlier version; Appendix H.2): in $\{H_{\forall}, H_{\text{open}}, R_1\}$, where $R_1 = \{\varphi(0), \varphi(Sz)\}$ does not prove $\forall x\varphi$ in pure logic, $P(T \vdash \forall x\varphi \mid D_n)$ is 0.697 at $n = 100$ and 0.000 from $n = 1000$. The claim survives for $\{H_{\forall}, H_{\text{open}}\}$ and under L1^{sel} (cf. AS Prop 4.5(b), (c)).*

Proposition 3.12 (Who wins without a guard). *[proved; computed]^{universal Prop N3} Closed data i.i.d. from $\mathbf{Q}_{\text{num}}$; the class contains $H_{\forall}, H_{\text{open}}, R_k$ for k in a nonempty $\mathcal{K}$ (learned or optimal weights), possibly memorisers. In pure logic $H_{\forall}, H_{\text{open}}$ prove $\forall x\varphi$, and R_k and memorisers do not; over $\mathbf{Q}$ all but the memorisers do. (a) L1, pure logic, provers only $H_{\forall}, H_{\text{open}}$: $P(T \vdash \forall x\varphi \mid D_n) \rightarrow 0$ a.s., exponentially, at rate $\geq (1 - q^k) \mathcal{W}_o$ for each $k \in \mathcal{K}$. (b) L1, over $\mathbf{Q}$, $\mathcal{K}$ finite: $\rightarrow 0$ with Laplace-weighted memorisers, = 1 at every n without them. (c) [proof sketch] L1^{sel} , finite $\mathcal{K}$: $\rightarrow 1$ in pure logic. (d) [computed; limit conjecture] L1, $\mathcal{K} = \{1, \dots, 40\}$, memorisers: 1.000 over $\mathbf{Q}$ from $n = 10$ to 10^{12} , with ever deeper splits.*

(b) is proved for Laplace-weighted memorisers; Table 18 uses Dirichlet($\frac{1}{2}$) weights. There the over- $\mathbf{Q}$ value falls to 0 with $R_{1..4}$ but stays 1.000 to $n = 10^{12}$ with $R_{1..40}$, the MAP moving to ever deeper splits. A likelihood that ignores the selection thus approximates the missing guard by ever deeper splits, whose waste $q^k \mathcal{W}_o$ tends to 0; over $\mathbf{Q}$ the answer is a race between the split family (provers) and the memoriser family (non-provers), a property of the class. A misspecified term law causes a related drift (Remark 4.20).

3.7 Quantified data, lemmas and the equational fragment

Quantified data decide (Proposition B.14 in Appendix B.6). A datum s with $B \cup I_c(\varphi) \not\models s$ gives $B \oplus \sigma_\varphi$ posterior 0 under any likelihood supported on theorems; over $\mathbf{Q}$, one datum $\forall y(0 + Sy = Sy)$ leaves only provers of $\forall x(0 + x = x)$. In $\mathbf{C}_{\text{min}}$, between provers that differ in cost ($\forall x\varphi$ behind L vacuous quantifiers against $\forall x\varphi$, both beside σ_φ , uniform weight priors, equal prior masses) the log Bayes factor tends to $\ln(1 + c + \dots + c^L)$ on instance data and grows linearly, a little above $f_q L \ln \frac{1}{c}$, when a fraction $f_q > 0$ of the data is $\forall x\varphi$ (proof sketch). In L1^{max} form a derived

sentence becomes an axiom when the derivation steps it saves, at $\geq \ln \frac{1}{c}$ each, exceed its prior cost and an $O(\ln n)$ dilution (proof sketch): frequently used lemmas become axioms. So the posterior identifies the data producer’s working set of primitives, not a canonical axiomatisation (compare Proposition 7.5). On the equational fragment $E = \{Q4, Q5\}$ every derivation of $0 + S^k 0 = S^k 0$ has at least $k + 1$ axiom leaves, so the two-part ($L1^{\max}$) comparison favours the schema $0 + z = z$ linearly in k at $q = \frac{1}{2}$, and the full sum does so at least when $q > 0.61$ (Proposition B.15); full Q, and the full sum at $q = \frac{1}{2}$, are open.

3.8 Without templates

Without templates σ_φ is unavailable and hypotheses are finite sets of sentences. Setting: $C_{\min}$, $c = 0.3$, numeral data, $q = \frac{1}{2}$, $\varphi = 0 + x = x$, learned weights; besides $H_\forall$ and memorisers the class has $\text{Split}_k := \{\varphi(0), \dots, \varphi(S^{k-1}0), \forall y \varphi(S^k y)\}$ and $\text{Both0} := \{\varphi(0), \forall x \varphi\}$.

Under $L0^{\text{cl}}$, $H_\forall$ has the true law and the probability of a prover tends to 1 (proof sketch). Under L1, Split_k loses only $q^k \ln \frac{1+c}{c}$ nats per datum at its best weights, against $\ln \frac{1+c}{c} = 1.466$ for $H_\forall$; it proves $\forall x \varphi$ over Q but not in pure logic, and memorisers prove it in neither. So in pure logic $P(T \vdash \forall x \varphi \mid D_n) \rightarrow 0$ in every class computed, and over Q the limit is a race between families: with depth ≤ 4 memorisers win (limit 0), with depth ≤ 40 splits win to $n = 10^{12}$ under four prior variants, and with all depths the limit is conjecturally 1 (Proposition B.16 in Appendix B.7). Every finite sentence set that emits infinitely many numeral instances (numeral $\forall E$ terms) proves $\forall x \varphi$ over Q (Lemma B.17). Split_1 shows that “all instances have short derivations, so $\forall x \varphi$ is derivable”, a naive form of Kreisel’s conjecture, fails without axioms such as Q3 (Remark B.18).

3.9 A good version for this case, and its soundness

For $\forall x \varphi$ the good version of Section 9.2 models the selection of reported data by a known filter ($L1^{\text{sel}}$, or the per-citation $L1_{\text{cit}}^{\text{sel}}$, under which the background effect of Proposition 3.6 vanishes, Remark 3.7) or by a prior over filters (Theorem 3.9(e)), and its verifier accepts s iff $\text{Bel}_{\text{cons}}(s) \geq 1 - \delta$. On closed instances it accepts $\forall x \varphi$ only if, eventually, the prior share of provers over theories and filters is $\geq 1 - \delta$ (for finite n see Corollary 3.13); one parameter instance, or one quantified datum entailing it over the background, suffices.

Corollary 3.13 (Soundness for $\forall x \varphi$). *[proved]^{universal §11} Let the data be i.i.d. from P_{T^*} with fixed weights, $T^* \in \mathcal{H}$, $T^* \not\vdash \forall x \varphi$, and $C_\forall^* := \{T \in C^* : T \vdash \forall x \varphi \Leftrightarrow T^* \vdash \forall x \varphi\}$. The probability that the verifier (on Bel or Bel_{cons}) ever accepts $\forall x \varphi$ is at most $\delta/\pi(C_\forall^*) \leq \delta/\pi(T^*)$, uniformly in time: Theorem 5.2 read at the sentence $\forall x \varphi$ (proof in Appendix B.4).*

Without well-specification there is no guarantee. In the misspecified settings of Theorem 3.9(d) and Propositions 3.12 and B.16 the limit theories (σ_φ , R_k , Split_k , memorisers) are true in $\mathbb{N}$, so in pure logic the verifier errs towards incompleteness; misspecification can also make it accept a falsehood with probability 1 (Example 4.19 and Proposition 5.4). The learned-weight computations here are outside the corollary (see Theorems 5.6 and 5.7).

Remark 3.14 (Splits of $\forall x \varphi$ and the MDL finding). *[proved (from the cited results)]^{universal §10} Well specified, a root split with weights p_f ties with the schema under L0 and L1 (Proposition 4.6) and with learned weights pays a polynomial Occam factor (Proposition 4.10). An unmodelled selection creates linear split wins, at rates $(1 - q^k)\mathcal{W}_o$ for R_k and $(1 - q^k) \ln \frac{1+c}{c}$ for Split_k (AS Prop E.16, split by numeral depth); modelling the selection removes them (Proposition 3.12(c)). This is Remark 4.13 for $\forall x \varphi$.*

4 What the posterior identifies

Well specified, with fixed weights, the posterior concentrates on the theories with the data's law and is the prior among them: it identifies the theorem set (under L1, parameters admissible) or the instance union (under L0), not the axioms. With learned (Dirichlet) weights the exact generator class has mass 0, and under L0 the instance union is identified for almost every weight vector. Splits of a schema are decided by Occam terms or by usage statistics. With Dirichlet weights, *spare templates* (templates beside the generator's own that the data never or rarely use) are never refuted and decay at best polynomially (exponentially with fixed weights). Misspecified, the posterior goes to the best-fitting generator, which can be weaker than the axioms behind the data, or unsound. Logarithms are natural, KL in nats; the experiments report bits.

4.1 Well-specified consistency

Definition 4.1 (Setting W). ^{model §2} $\mathcal{H}$ is a countable class with a prior π , $\sum_T \pi(T) = 1$; each T carries a *fixed* law P_T on the countable set $\mathcal{S}$ (fixed weights). The data $D_n = (X_1, \dots, X_n)$ are i.i.d. from P_{T^*} , $\pi(T^*) > 0$. The posterior is $\pi_n(T) := \pi(T)P_T(D_n)/M(D_n)$, $M(D_n) := \sum_T \pi(T)P_T(D_n)$; the predictive law is $M_n := \sum_T \pi_n(T)P_T$; the *generator class* is $C^* := \{T : P_T = P_{T^*}\}$.

Theorem 4.2 (Concentration on the generator class). ^{[proved; known (Doob 1949)]^{model Thm 2.1; universal Lemma U7} Under W, $\pi_n(C^*) \rightarrow 1$ a.s.; moreover $\pi_n(T) \rightarrow \pi(T)1[T \in C^*]/\pi(C^*)$ for every T , and in total variation.}

Corollary 4.3 (Inside C^* the posterior is the prior). ^{[proved; computed]^{model Cor 2.2} For $T, T' \in C^*$ and every n , $\pi_n(T)/\pi_n(T') = \pi(T)/\pi(T')$.}

Corollary 4.4 (Limits of deductive beliefs). ^{[proved]^{model Cor 2.3} Under W, a.s., for all s and $d \leq \infty$ at once, $\pi_n(\{T : s \in \text{Th}_d(T)\}) \rightarrow \pi(\{T \in C^* : s \in \text{Th}_d(T)\})/\pi(C^*)$.}

The proof is Doob's argument (Doob, 1949) (Appendix C.1). The data decide a deductive question only through C^* ; inside it the prior decides. The limit is $1[s \in \text{Th}_d(T^*)]$ under L0 (full-support $\mathcal{Q}$, positive weights) for every d ($\text{supp } P_T = \bigcup \text{inst}(T)$) and under L1 with parameters admissible for $d = \infty$ ($\text{supp } P_T = \text{Th}(T)$; Lemmas 2.10 and A.2); under L2 only $\text{Th}_{\leq d}$ is fixed, and for Hänni's scores nothing of this kind holds.

The rates (Proposition C.1, in Appendix C.1) are the usual ones: $\text{KL}(P_{T^*} \| P_T)$ per datum against each $T \notin C^*$, cumulative expected predictive KL at most $\ln \frac{1}{\pi(C^*)}$, and decay through the Hellinger affinity, which degenerates for spare templates (Proposition 4.14).

4.2 Identification is of the generator

Proposition 4.5 (L1 separates deductively equivalent theories). ^{[proved (open parameter region); computed]^{model Prop 2.5} Let $a := \neg(S0 = 0)$, $b := \neg(SS0 = 0)$, $T_1 := \{a, b\}$, $T_2 := \{a, a \rightarrow b\}$ (ground, weights $\frac{1}{2}$); $\text{Th}(T_1) = \text{Th}(T_2)$. Under L1 or L2, $P_{T_1}(b) \geq \alpha_{\text{ax}}/2$ and $P_{T_2}(b) \leq \alpha_r/(1 - \alpha_r)$; so $P_{T_1} \neq P_{T_2}$ if $\alpha_{\text{ax}}/2 > \alpha_r/(1 - \alpha_r)$.}

The laws also differ in every computed setting outside the proved region (Appendix C.2); for all parameters it is open. With data from P_{T_1} or P_{T_2} the posterior picks the generating axiomatisation, not a logically privileged one; from a third generator the drift is a difference of two KL divergences and can vanish.

Proposition 4.6 (Equal laws; splits are exact ties). *[proved; computed]^{model Prop 2.6 and remark; pa Prop 2.1} Let $\mathcal{Q}$ have full support and all weights be positive. (a) Under $\mathbf{L0}$, $P_{T,w} = P_{T',w'}$ implies $\bigcup \text{inst}(T) = \bigcup \text{inst}(T')$, hence $\text{Th}_d(T) = \text{Th}_d(T')$ for all d . (b) Let $\tau \in T$ have a metavariable M whose law factors at the root over finitely many productions, $\mathcal{Q}_M(\lambda \bar{z}. r(\beta_1, \dots, \beta_a)) = p_r \prod_j \mathcal{Q}(\beta_j)$, and let τ_r be τ with M 's body fixed to root r and fresh metavariables for its children. The root split T' , with τ replaced by the τ_r at weights $w'_{\tau_r} := w_\tau p_r$, consists of DT° templates, and $P_{T',w'} = P_{T,w}$ under $\mathbf{L0}$; $\mu_{T'} = \mu_T$, $Z_{T'} = Z_T$ under $\mathbf{L1}$, $\mathbf{L2}$. (c) A renamed or argument-permuted copy (AS Lemma A.2) has the same law if $\mathcal{Q}$ is symmetric in holes. Equal laws give equal instance sets, hence equivalence for first-order templates with only 0-ary term metavariables (AS Prop C.1, under its richness condition); in general (e.g. A1, or DT°) this converse is open.*

So with weights matched to $\mathcal{Q}$ no data tell a schema (e.g. T_{Ind}) from its root split; only the prior does. The tie needs $\mathcal{Q}$ to factor at the root, which the learned grammar of Section 7 does not (Proposition 7.12). In the experiments' chain Ch_J the tie holds for a 0-ary closed-guard term metavariable of a component that is not an MP major premise, also with latent citations (Remark C.2, in Appendix C.2; E6: Remark 7.10).

4.3 Positive data and Gold

Theorem 4.7 (Identification in the limit with probability one). *[(a), (b) proved; (b') known, proof sketch; (c) conjecture]^{model Thm 5.1} Let $\mathcal{H}$ be all theories, with a prior positive on T^* . (a) For fixed generators ($\mathbf{L0}$ with fixed positive weights; or $\mathbf{L1}$ with fixed positive parameters, $m \leq 1$, parameters admissible; full-support $\mathcal{Q}$) and data i.i.d. from P_{T^*} , the learner $g_n := L$ if $\pi_n(\{T : \text{supp } P_T = L\}) > \frac{1}{2}$ equals $\text{supp } P_{T^*}$ for all large n , a.s.: $\bigcup \text{inst}(T^*)$ under $\mathbf{L0}$, $\text{Th}(T^*)$ under $\mathbf{L1}$. No bound on the number of templates is needed. (b) With Dirichlet weights (prior $\pi(T) \text{Dir}_T(dw; \alpha)$, $\mathbf{L0}$, full-support $\mathcal{Q}$, data i.i.d. from P_{T^*,w^*}), for Lebesgue-a.e. w^* , $\pi_n(\{T : \bigcup \text{inst}(T) = \bigcup \text{inst}(T^*)\}) \rightarrow 1$ a.s. (b') Under (b), $\pi_n(\{(T, w) : \text{TV}(P_{T,w}, P_{T^*,w^*}) < \varepsilon\}) \rightarrow 1$ a.s. for all $\varepsilon > 0$ (Ghosal and van der Vaart, 2017, Thm 6.9; numbering checked only via citing papers). (c) Conjecture: (b) holds for every w^* in the open simplex.*

(b) runs Doob's argument with the support functional $\bigcup \text{inst}(T)$, which takes countably many values and a.s. equals the set of sentences seen (Appendix C.3). Of the computed posteriors, these theorems cover one-component generators in fixed hand pools; data-dependent pool members are covered only by the soundness bounds, and samples at one fixed weight vector are a yardstick, not an instance of a theorem (Remark C.3, in Appendix C.3).

Remark 4.8 (The exact class has posterior mass zero). *[refuted; counter-statement proved]^{model Thm 5.1(b'')} Under the hypotheses of (b), if $|T^*| \geq 2$ with linearly independent component laws, the exact class $\{(T, w) : P_{T,w} = P_{T^*,w^*}\}$ has posterior mass 0 at every n for all w^* off a countable set, since each $\{w : P_{T,w} = P_{T^*,w^*}\}$ is Lebesgue-null. So only the instance union is identified (an earlier version claimed more; Appendix H.2).*

Proposition 4.9 (Gold's limit point $\mathcal{L}_\infty$ against $\mathcal{L}_5$). *[proved]^{model Prop 5.2} Let $\mathcal{L}_1 \subsetneq \mathcal{L}_2 \subsetneq \dots$, $\mathcal{L}_\infty := \bigcup_i \mathcal{L}_i$, each $\mathcal{L}_i$ ($i \leq \infty$) with a generator P_i , $\text{supp } P_i = \mathcal{L}_i$, whose class has prior mass $\pi_i > 0$ (setting W). (a) On data i.i.d. from P_i , $g_n = \mathcal{L}_i$ eventually, a.s. (b) Some text for $\mathcal{L}_\infty$ has $g_n \neq \mathcal{L}_\infty$ infinitely often (Gold, 1967). (c) Such texts have P_∞^∞ -probability 0.*

The learner has no fixed preference between the ends of the limit point; the size principle decides (on P_5 -data the ratio of $\mathcal{L}_\infty$'s generator to $\mathcal{L}_5$'s is $P_\infty(\mathcal{L}_5)^n$ times a factor of mean at most 1; Lemma 2.21). E5 shows both ends identified and the MAP changing at each of 14 stages of Gold's

text (Table 36). For the $I\Sigma_n$ chain both directions separate, after presumably astronomical waiting times (a heuristic; Appendix C.3, Proposition 7.24). Fixed memorisers die at the first datum outside their set, so the Bayesian needs no bound on the number of templates, unlike the cautious learner; the memoriser class can keep mass $\exp(-O(\ln^2 n))$ (Laplace-weighted memorisers, geometric numerals; Remark C.4, in Appendix C.3).

4.4 Splits and the MDL finding

AS §6.10 found that MDL “tracks the statistics of usage, not the logical boundaries of schemas” (a naive code splits T_{Ind} by a linear margin, a well-specified one gains at most $O(\ln n)$; AS Props E.16, E.17). Does a derivation likelihood change this? With learned weights a root split must learn the root law that the schema takes from $\mathcal{Q}$.

Proposition 4.10 (Schema against root split, L0). *[proved; known (the χ^2 limit); computed]^{model Prop 5.4}*

Let $T = \{\tau\}$, T' its root split over K productions with $\text{Dir}(\alpha)$ weights, the data n instances of τ with root law r , root counts n_f , $\hat{r} := n_f/n$, and p the root law of $\mathcal{Q}$. (a) $\Delta_n := \ln P_{T'}^{\text{Dir}}(D_n) - \ln P_T(D_n) = \ln \text{DirMult}(n_f; \alpha) - \sum_f n_f \ln p_f$, $\text{DirMult}(n_f; \alpha) := \frac{\Gamma(K\alpha)}{\Gamma(K\alpha+n)} \prod_f \frac{\Gamma(\alpha+n_f)}{\Gamma(\alpha)}$. (b) If all $n_f \rightarrow \infty$: $\Delta_n = n\text{KL}(\hat{r}||p) - \frac{K-1}{2} \ln n + c_{K,\alpha} + (\alpha - \frac{1}{2}) \sum_f \ln \hat{r}_f + o(1)$, $c_{K,\alpha} := \ln \frac{\Gamma(K\alpha)}{\Gamma(\alpha)^K} + \frac{K-1}{2} \ln 2\pi$. (c) If $r = p$: $\Delta_n = -\frac{K-1}{2} \ln n + O_P(1)$ ($2n\text{KL}(\hat{r}||p) \rightarrow \chi_{K-1}^2$); the schema wins at rate $n^{-(K-1)/2}$. (d) If $r \neq p$: $\Delta_n/n \rightarrow \text{KL}(r||p) > 0$ a.s.; the split wins linearly. (e) The prior adds $\lambda(\ell(T') - \ell(T)) > 0$ bits for the schema.

Proposition 4.11 (Schema against root split, derivation likelihoods). *[(a)–(c) proved; (d) conjecture; computed]^{model Prop 5.6}* Let $T = \{\tau\}$ root-factor as in Proposition 4.6(b), $|R| = K$, root law p interior, T' the split with $\text{Dir}(\alpha)$ weights, under L0, L1 ($m < 1$) or L2 with other parameters fixed and shared; $P^{(p')}$ is the law of $\{\tau\}$ with root law p' , and $\text{BF}_n := P_{T'}^{\text{Dir}}(D_n)/P_T(D_n)$. (a) $P_{T',w'} = P^{(w')}$, so $P_{T'}^{\text{Dir}}(D) = \int \prod_i P^{(p')}(X_i) \text{Dir}(dp')$: the split with learned weights is the schema with a learned root law. (b) If $\mathcal{Q}$ fits the usage (data i.i.d. from P_T), BF_n is a nonnegative supermartingale from 1: $P(\sup_n \text{BF}_n \geq 1/\delta') \leq \delta'$. (c) If not (data i.i.d. from $P^{(r)} \neq P^{(p)}$, r interior), $\frac{1}{n} \ln \text{BF}_n \rightarrow \text{KL}(P^{(r)}||P^{(p)}) \in (0, \infty]$ a.s. (d) Conjecture: under (b), with $p' \mapsto P^{(p')}$ identifiable and nonsingular Fisher information, $\ln \text{BF}_n = -\frac{K-1}{2} \ln n + O_P(1)$.

(b) is Ville’s inequality (IL Lemma C.4); (d) would need a Laplace theorem for a model with latent citations. Computed: Tables 20 and 21 (slope -0.507 for (d)).

Remark 4.12 (E8: latent citations). *[computed]^{experiments E8}* $\{\varphi(z), \psi(z), \varphi(z) \rightarrow \psi(z)\}$ against the split of $\varphi(z)$ by its term’s root ($K = 4$; chain L1 data, 10 seeds; prior excluded): well-specified usage makes the split lose 1.43 bits per doubling of n under L1 and 1.51 under L0 ($n = 256 \dots 4096$; conjectured 1.5); skewed usage makes it win 0.202 against 0.074 bits per datum ($n = 1024 \dots 4096$), since a ψ -datum may be φ followed by MP, so under L1 its term informs the split too (Table 38). E8 does not test Proposition 4.11(b), whose setting differs.

Remark 4.13 (What this says about the MDL finding). *[proved; computed (as cited)]^{model §5.3; pa §2; universal §10; experiment}*

When the instantiation model is misspecified, the posterior tracks usage, not the logical boundary of a schema; as far as proved, a derivation likelihood does not change the MDL finding. A schema and its root split tie at matched weights. With learned weights and usage that $\mathcal{Q}$ fits, the split loses under L0 and gains at most $\ln(1/\delta')$ nats with probability $1 - \delta'$ under L1 (that it loses there is Proposition 4.11(d), a conjecture); with usage that $\mathcal{Q}$ misfits, it wins linearly under both. Three qualifications. (i) Partial splits that only L1 keeps alive are not covered: a

split of T_{Ind} with a non-atomic connective derives every induction instance, an unseen-connective datum costing 369–489 bits in the two-part code instead of being impossible (Proposition 7.14); a general comparison is open. (ii) A learned grammar can reverse the Occam drift: under the shared positional grammar of Section 7 the split into roots F costs $\Delta_{\text{prior}} + \frac{|F|-9}{2} \log_2 n + O(1)$ bits more than the schema (Proposition 7.12); with seven roots it gains 1 bit per doubling yet is 762 bits behind at $n = 256000$. (iii) Unmodelled selection of the reported data creates linear split wins; for $\forall x\varphi$ at rates $(1 - q^k)\mathcal{W}_o$ for R_k and $(1 - q^k) \ln \frac{1+c}{c}$ for Split_k , which modelling the selection removes (Proposition 3.12(c), Remark 3.14).

4.5 Spare templates

Proposition 4.14 (Cost of a spare template). *[(a), (d1) proved; (b), (c), (d2) proof sketch; computed]^{model Prop 5.5; experim}*
Let the truth be (T^*, w^*) under L0 with $\text{Dir}(\alpha_\tau)$ priors, $A := \sum_{\tau \in T^*} \alpha_\tau$, σ a spare with parameter α_σ , $P^* := P_{T^*, w^*}$, and $\text{BF}_n^\sigma := P_{T^* \cup \{\sigma\}}^{\text{Dir}}(D_n) / P_{T^*}^{\text{Dir}}(D_n)$. (a) If no datum is in $\text{inst}(\sigma)$, exactly $\text{BF}_n^\sigma = \frac{\Gamma(A+\alpha_\sigma)\Gamma(A+n)}{\Gamma(A)\Gamma(A+\alpha_\sigma+n)} \sim \frac{\Gamma(A+\alpha_\sigma)}{\Gamma(A)} n^{-\alpha_\sigma}$; at $\alpha_\sigma = \frac{1}{2}$, $\frac{1}{2} \log_2 n$ bits plus a constant and the prior bits. (b) If σ covers data but $\mathcal{Q}_\sigma(\text{supp } P^*) = 1 - c < 1$: $\ln \text{BF}_n^\sigma = -\alpha_\sigma \ln n + O_P(1)$. (c) If $\text{inst}(\sigma) \subseteq \text{supp } P^*$, $\mathcal{Q}_\sigma$ is outside the span of T^* 's components, and $\sum_s \mathcal{Q}_\sigma(s)^2 / P^*(s) < \infty$: $\ln \text{BF}_n^\sigma = -\frac{\alpha_\sigma}{2} \ln n + O_P(1)$. (d1) If $\mathcal{Q}_\sigma = \mathcal{Q}_{\tau_1}$, $\tau_1 \in T^*$ (e.g. an argument-permuted copy), and T^* 's instance sets are disjoint, $\text{BF}_n^\sigma \rightarrow \frac{\Gamma(A+\alpha_\sigma)\Gamma(\alpha_1)}{\Gamma(A)\Gamma(\alpha_1+\alpha_\sigma)} (w_1^*)^{\alpha_\sigma} > 0$ a.s.: the spare never vanishes. (d2) If $\mathcal{Q}_\sigma$ is in the convex hull of T^* 's components but equals none, and the induced prior on $P_{T^* \cup \{\sigma\}, w}$ has a continuous positive density at P^* , BF_n^σ tends to a positive constant. The case of a $\mathcal{Q}_\sigma$ in the span of T^* 's components but outside their convex hull is not covered.

(b) and (c) need a boundary Laplace approximation (not written out) of the kind studied for overfitted mixtures by Rousseau and Mengersen (2011). In E5 unused spares follow (a) exactly and a nested spare, case (c), has fitted exponent -0.251 ± 0.002 on $n = 10^2 \dots 10^8$; without the density hypothesis of (d2) a spare can gain slowly (Table 22).

Remark 4.15 (All spares; what spares cost). *[proved (the sum); waiting time in case (b) from a sketch; computed]^{model §5.4; experiments E2, E5}* With a common $\alpha_\sigma = \alpha$, BF_n^σ in (a) does not depend on σ ; so the posterior mass of all $T^* \cup \{\sigma\}$ with σ covering no datum is at most $\text{BF}_n^\sigma / \pi(T^*) = O(n^{-\alpha})$. With fixed weights the factor is $(1 - w_\sigma)^n$. Spares inside the support do not threaten identification of the instance union. What spares cost is slow rejection: about $(\text{prior ratio} / \delta_r)^{1/\alpha}$ data to fall below δ_r in case (b) (from the sketch of Proposition 4.14(b)). A spare sentence that T^* refutes makes $T^* \cup \{\sigma\}$ inconsistent, and positive data remove it only like $n^{-1/2}$ (E2, 0 = S0 beside Q1: posterior 0.004 at $n = 512$); the size principle does not see inconsistency (Proposition 7.36).

4.6 Misspecification

Human data are theorems of the actual axioms, but are not drawn from any P_T of the model.

Theorem 4.16 (Finite class: concentration on the KL-minimisers). *[proved; known (general form)]^{model Thm 3.1}*
Let $\mathcal{H}$ be finite, $\pi > 0$, the data i.i.d. from any P_H , $K_T := \text{KL}(P_H \| P_T)$, $\min_T K_T < \infty$, $\mathbb{M} := \arg \min_T K_T$. Then $\pi_n(\mathbb{M}) \rightarrow 1$ a.s., with $\frac{1}{n} \ln \frac{\pi_n(T)}{\pi_n(T_0)} \rightarrow K_{T_0} - K_T < 0$ for $T \notin \mathbb{M} \ni T_0$; inside $\mathbb{M}$ the posterior need not converge.

The general form is known (Berk, 1966; Kleijn and van der Vaart, 2006) (abstracts read, conditions not). In the full class with Dirichlet weights even this picture can fail: hybrids of memorisation and an unsound escape approach KL 0, and unsound mass $\rightarrow 1$ is conjectured, with a computed illustration (Remark C.5, in Appendix C.6).

Example 4.17 (The axiom’s own theorems lead to a weaker theory). *[proved]^{model Ex 3.2}* In L_A with parameters not admissible, under $L0$ with fixed full-support $\mathcal{Q}$ and $\pi(\{0+z=z\}) > 0$, let the human’s axiom be $\forall x(0+x=x)$ and the data its theorems $0+t=t, t \sim \mathcal{Q}$. They are well specified for $\{0+z=z\}$, whose generator class has instance union $I_c(0+x=x)$ (Proposition 4.6), and no such theory proves $\forall x(0+x=x)$ (take $\mathbb{N} \cup \{a\}$ with $0+a:=0$). So $P(T \vdash \forall x(0+x=x) \mid D_n) \rightarrow 0$ a.s.: the ω -gap of AS §4.5 (all closed instances together still do not entail $\forall x\varphi$), forced by the likelihood.

Under $L1$ without parameters $\text{supp } P_T = \text{Th}(T)$ fails; Theorem 3.9 treats that case.

Lemma 4.18 (Sound templates cover at most one zero-sum equation). *[proved]^{model Lemma 3.3}* Let $Z := \{t = 0 : t \text{ a closed term over } 0, +\}$. A DT° template all of whose instances are true in $\mathbb{N}$ (parameters, if admissible, read universally) covers at most one member of Z .

Example 4.19 (An unsound limit). *[proved; computed]^{model Ex 3.4}* Let the human’s axioms be $\mathcal{Q}$ and the data true equations $t = 0, t \in Z$, from a law of infinite support; the class is finite, made of sound theories $T_1, \dots, T_m$ and $T_{\text{esc}} := \{z = 0\}$, all with positive prior; $L0$ with full-support $\mathcal{Q}$. By Lemma 4.18 the T_j cover finitely many members of Z , so a.s. $\pi_n(T_{\text{esc}}) = 1$ eventually. But $T_{\text{esc}} \vdash S0 = 0$ and $\mathcal{Q} \vdash \neg S0 = 0$; $\mathcal{Q}$ itself, if in the class, dies at the first datum, as its axioms are not in Z . Computed: against twenty best-possible sound theories at prior 0.99, $S0 = 0$ reached posterior ≥ 0.99 in 2000/2000 runs (median 17 data).

In Example 4.19 $\mathcal{Q}$ fails because the data are not axiom instances; a derivation likelihood removes that failure, but not reliably its outcome: in a truncated equational grammar the best fit is unsound in all 16 settings with $p_+ = 0.25$ (Example C.6, in Appendix C.6; Conjecture C.7 for the untruncated grammar).

Remark 4.20 (E3(a): misspecified usage of instances). *[computed]^{experiments E3(a)}* On instances of $x + 0 = x$ with a term law not the model’s, the posterior moves at a linear rate to a best-fitting family member: under heavy tails a depth-2 nesting with the same instances ($\{\varphi(z), \varphi(Sz), \varphi(SSz)\}$ in 3 seeds, $\{\varphi(z), \varphi(SSz)\}$ in 2); under numerals closed-guard splits R_m , m growing in n (9–10 at $n = 512$, 11–12 at 1024; $\varphi(2+3)$ underived); on a finite selected support $\text{Mem}(D_n)$. No unsound mass at large n ; completeness is lost (Table 9).

Table 23 (in Appendix C.6) collects the cases: misspecified usage of axiom instances gives the same or weaker theorems; derived theorems, absent counterexamples, mistakes and small samples can give unsound ones.

4.7 Predictive versus deductive

Theorem 4.21 (Predictive versus deductive). *[proved]^{model Thm 8.1}* Assume W . (a) $\sum_{n \geq 0} \mathbb{E} \text{KL}(P_{T^*} \| M_n) \leq \ln \frac{1}{\pi(C^*)}$. (b) $\pi_n(\{T : T \vdash s\}) \rightarrow \pi(\{T \in C^* : T \vdash s\})/\pi(C^*)$ a.s., which is $1[T^* \vdash s]$ iff the members of C^* agree on s up to prior-null sets; they do when the generator determines the theorem set ($L0$ with full support; $L1$ with parameters admissible), and need not under $L2$ or when $\text{supp } P_T \subsetneq \text{Th}(T)$. (c) If $T_1 \vdash s, T_2 \not\vdash s$ and the data come from P_{T_1} , the expected log-odds of T_1 against T_2 grow by exactly $\text{KL}(P_{T_1} \| P_{T_2})$ per datum, while the per-datum predictions differ in total variation by at most $(\text{KL}(P_{T_1} \| P_{T_2})/2)^{1/2}$.

So when two theories differ in what they prove but hardly in what they generate, prediction is settled early and the deductive question stays near its prior odds for about $1/\text{KL}$ data. Misspecified, predictions go to the KL-projection and beliefs to its theorems, so predictive success certifies

nothing deductive. For $\forall x\varphi$ this is the gap between Theorem 3.8 and Theorem 3.9 (compare Hutter, 2007). Data that come with their proofs identify the instance union of the generator (Proposition C.8, in Appendix C.7). Hänni’s “grade near misses”, read as a corruption channel that is part of the well-specified model, keeps Theorems 4.2 and 5.2 and Corollary 4.4; with a full-support channel the size principle is vacuous, and how much the generator class grows is open (Remark C.9, in Appendix C.7).

5 Soundness of the thresholded verifier

A posterior over theories can serve as a proof checker: it accepts a sentence when theories deriving it carry at least $1 - \delta$ of the mass. With fixed weights and data drawn from the model, no adaptive prover ever gets a non-theorem accepted, with probability at least $1 - \delta/\pi(C_d^*)$, uniformly in time (Theorem 5.2). With learned (Dirichlet) weights this holds on average over the weight prior (Theorem 5.6), and, for likelihoods linear in the weights, at each weight vector with a threshold that shrinks with n (Theorem 5.7); a constant threshold can fail (Example 5.8). For a general derivation-grammar likelihood with learned weights only the averaged guarantee is proved. Misspecified, there is no guarantee (Proposition 5.4).

5.1 Protocol

Definition 5.1 (Protocol; thresholded verifier). ^{model §4} Fix a countable class $\mathcal{H}$ with prior π , a truth $T^* \in \mathcal{H}$ and a derivability bound $d \leq \infty$. Data $X_1, X_2, \dots$ form an infinite sequence. In each round $t = 1, 2, \dots$ either the next datum arrives or a *prover* submits a query q_t ; the prover chooses which. Let $n(t)$ be the number of data that have arrived by round t . The verifier $V_{\delta,d}$ answers **ACCEPT** iff $\pi_t(\{T : q_t \notin \text{Th}_d(T)\}) \leq \delta$. Otherwise it may **ESCALATE** to a deterministic oracle that says whether $q_t \in \text{Th}_d(T^*)$; the j -th answer becomes the constraint $c_j(T) := 1[q \in \text{Th}_d(T) \Leftrightarrow q \in \text{Th}_d(T^*)]$. The posterior at round t is $\pi_t(T) \propto \pi(T) \prod_{i \leq n(t)} P_T(X_i) \prod_j c_j(T)$ with fixed weights; with Dirichlet weights the marginal $P_T^{\text{Dir}}(X_1, \dots, X_{n(t)})$ replaces the product over data. The verifier is *sound* on a run if it never accepts a sentence outside $\text{Th}_d(T^*)$. Finally $C_d^* := \{T : P_T = P_{T^*}, \text{Th}_d(T) = \text{Th}_d(T^*)\}$ and $W_d^* := \pi(C_d^*) \geq \pi(T^*)$.

The prover may be randomised, adaptive and computationally unbounded, may know T^* , the verifier and the whole future data sequence, and chooses when data arrive. Soundness is relative to $\text{Th}_d(T^*)$, not to truth; d is symbol size (Definition 2.2), chain steps ($\vdash_J$) in the experiments, and 0 (being an instance) in Remark 7.32. This is the protocol of AS Def 2.6 with $V_{\delta,d}$ in place of the version-space verifier.

5.2 Fixed weights

Theorem 5.2 (Time-uniform soundness, fixed weights). *[proved; computed]^{model Thm 4.1} Assume setting W (Definition 4.1): every $T \in \mathcal{H}$ carries a fixed law P_T (fixed mixture weights), the data are i.i.d. from P_{T^*} with $\pi(T^*) > 0$, the prior does not depend on n , and the oracle is truthful. If $\delta \leq W_d^* \delta'$, there is an event G with $P(G) \geq 1 - \delta'$, depending only on the data sequence, on which no prover ever gets a sentence outside $\text{Th}_d(T^*)$ accepted, in any round.*

Proof idea. As in IL Thms 4.15(b) and 10.9: members of C_d^* share T^* ’s likelihood and pass every constraint, so $\pi_t(C_d^*) = W_d^*/Z_t$ with $Z_t \leq M(X^{n(t)})/P_{T^*}(X^{n(t)})$ (M the Bayes mixture), a nonnegative supermartingale from 1 that by Ville’s inequality (IL Lemma C.4; Ville, 1939) reaches $1/\delta'$ with probability at most δ' . $\square$

One event covers all queries and rounds, so no union bound over the prover's attempts is needed. The constant improves from $\pi(T^*)$ to W_d^* , and it cannot be improved further: a pair of ground theories with fixed weights attains IL's acceptance probability, which tends to δ' (Remark D.1, in Appendix D.1). The guarantee is per target: under π_λ it covers the T^* with $\lambda\ell(T^*) \leq \log_2(\delta'/\delta)$, so for long axiom systems it is weak. IL's escalation bound (IL Thm 4.16) transfers to escalations of valid queries; those of invalid queries are not bounded, as the protocol has no reject threshold (Appendix D.1). Read at the single sentence $\forall x\varphi$, with the members of C^* that agree with T^* on it, the theorem gives Corollary 3.13.

Remark 5.3 (Hypotheses). [proved]^{model §4.1 (referee m8)} What matters: (i) well-specification of the *data law*, not only of the theorem set (Proposition 5.4; selected data in Remark 5.10(A)); (ii) fixed weights, or Section 5.4; (iii) a proper prior that does not depend on n (a penalty λ_n growing with n breaks the supermartingale step; no replacement is known); (iv) truthful constraints. A finite truncation $\mathcal{H}_N \ni T^*$ keeps the theorem with W_d^* replaced by $\pi(C_d^* \cap \mathcal{H}_N)/\pi(\mathcal{H}_N) \geq \pi(T^*)$, and so does a computable verifier for a finite class and $d < \infty$ (Appendix D.1; an earlier version claimed more, Appendix H.2).

Proposition 5.4 (Misspecification: a waiting prover wins). [proved; computed]^{model Prop 4.2} *In the setting of Example 4.19 (human axioms $\mathbf{Q}$; data $t = 0$ for closed $\{0, +\}$ -terms t from a law with infinite support; a finite class of sound theories, possibly with $\mathbf{Q}$, plus $T_{\text{esc}} = \{z = 0\}$; $\mathbf{L0}$), let the prover wait until $\pi_t(\{T : T \not\models S0 = 0\}) \leq \delta$ and then submit $S0 = 0$. For every $\delta > 0$, with probability 1 the verifier $V_{\delta, \infty}$ accepts $S0 = 0$ at some finite time, whether or not it escalates; yet $\mathbf{Q} \vdash \neg S0 = 0$.*

By Example 4.19, almost surely $\pi_n(T_{\text{esc}}) = 1$ for all large n , and $T_{\text{esc}} \vdash S0 = 0$; the prover's only query is made when it is accepted. Against an escalating verifier the waiting prover won 400/400 runs, a prover asking in every round 11/400 (Appendix D.1). Misspecification can also make the verifier incomplete: in Example 4.17, where the best-fitting generator is weaker than T^* , it accepts $\forall x(0 + x = x)$ at most finitely often.

5.3 The cautious limit and completeness

In the limit $\delta \rightarrow 0$ the verifier is the cautious verifier of AS, read on theorems: $V_{0,d}$ accepts exactly the common Th_d of the theories of positive prior mass whose support contains the data, and is sound deterministically and without well-specification (Proposition D.2, in Appendix D.2).

Proposition 5.5 (Completeness without anchors). [proved]^{model Prop 4.4} *Assume setting W with $\mathbf{L0}$ (full-support $\mathcal{Q}$, any d) or with $\mathbf{L1}$ ($d = \infty$, parameters admissible). For every $s \in \text{Th}_d(T^*)$ and every $\delta > 0$, almost surely $V_{\delta,d}$ accepts s at all sufficiently large n . With Dirichlet weights under $\mathbf{L0}$ the same holds for Lebesgue-almost every w^* .*

By Corollary 4.4 the non-deriving mass of s tends to $\pi(C^* \cap \{T : s \notin \text{Th}_d(T)\})/\pi(C^*)$, which is 0 because under these likelihoods every member of C^* has the theorems of T^* ; with Dirichlet weights use Theorem 4.7(b). Without a bound on the number of templates the cautious verifier accepts only $\text{Th}_d(D_n)$ (under $\mathbf{L0}$ with a prior that charges every finite theory; AS Prop 6.4), and with a bound it becomes exact only once the data contain an *anchor*, a set of data that only the target template covers among its competitors (AS Def 2.6, Lemma 2.8). The Bayesian verifier needs neither, because the likelihood moves mass away from memorisers and splits; it gives up deterministic soundness (Table 26).

5.4 Dirichlet weights

Read with parameter (T, w) and prior $\pi(T)\text{Dir}_T(dw)$, Theorem 5.2 is vacuous for Dirichlet weights under L0 and likelihoods affine in w : if $|T^*| \geq 2$ with linearly independent component laws, its class has prior mass 0 for all w^* off a countable set, which forces $\delta = 0$; under L1 this is plausible but not proved (Remark D.3, in Appendix D.3).

At a fixed weight vector and for likelihoods linear in w , the Dirichlet marginal loses at most a regret factor: $P_T^{\text{Dir}} \geq R_\alpha(n, K) P_{T,w}$ with $K = |T|$, where $-\ln R_\alpha(n, K) = \frac{K-1}{2} \ln n + O(1)$ for $\alpha_\tau \equiv \frac{1}{2}$, and $R_\alpha(n, K) \geq \frac{\Gamma(A)}{e^{(K-1)!} \prod_\tau \Gamma(\alpha_\tau)} (n+1)^{-(K-1)}$ if all $\alpha_\tau \leq 1$, $A := \sum_\tau \alpha_\tau$ (Lemma D.4, in Appendix D.3).

Theorem 5.6 (Soundness on average over w^*). *[proved]^{model Thm 4.7; experiments Prop X8(a), (b)} Let $\mathcal{H}$ be countable with prior π and Dirichlet weights, each T with its marginal law P_T^{Dir} on data sequences (under L0, L1 or L2), and truthful constraints. Let the data come from $P_{T^*}^{\text{Dir}}$: first $w^* \sim \text{Dir}_{T^*}$, then i.i.d. from P_{T^*,w^*} . Put $C_d^{\text{Dir}} := \{T : P_T^{\text{Dir}} = P_{T^*}^{\text{Dir}} \text{ on } \mathcal{S}^n \text{ for all } n, \text{Th}_d(T) = \text{Th}_d(T^*)\} \ni T^*$. If $\delta \leq \pi(C_d^{\text{Dir}}) \delta'$, there is a data event G with $P_{T^*}^{\text{Dir}}(G) \geq 1 - \delta'$ on which no prover ever gets a sentence outside $\text{Th}_d(T^*)$ accepted. Hence $\text{Dir}_{T^*}\{w^* : P_{T^*,w^*}^\infty(G^c) > \delta'/\gamma\} \leq \gamma$ for every $\gamma \in (0, 1]$. Pool version (experiments): with the experiments' prior weights $2^{-\text{bits}(T)}$ and $\vdash_J$ for Th_d , the verifier may use the posterior restricted to any pool $R_n \ni T^*$, even one chosen from the data, and the conclusion holds for $\delta \leq 2^{-\text{bits}(T^*)} \delta'$. If the family is a fixed pool H plus all memorisers and the memorisers' codes lie outside the template code, the prior weights may sum to more than 1, and the threshold must be $\delta \leq 2^{-\text{bits}(T^*)} \delta' / (\sum_{T \in H} 2^{-\text{bits}(T)} + 1)$.*

Theorem 5.7 (Soundness at a fixed w^*). *[proved]^{model Thm 4.8; experiments Prop X8(c)} As in Theorem 5.6, but with data i.i.d. from P_{T^*,w^*} for a fixed w^* in the closed simplex, a likelihood linear in w (L0; also the experiments' chain Ch_J), $K := |T^*|$, and a verifier that accepts at round t iff the non-deriving mass is at most $\delta_{n(t)}$. If $\delta_n \leq \pi(C_d^{\text{Dir}}) \delta' R_\alpha(n, K)$ for every n , then with probability at least $1 - \delta'$ no prover ever gets a sentence outside $\text{Th}_d(T^*)$ accepted. For $\alpha \leq 1$, Lemma D.4(b) gives an explicit $\delta_n \propto (n+1)^{-(K-1)}$; with $\alpha = \frac{1}{2}$ and the exact R_α , $\delta_n \asymp n^{-(K-1)/2}$. To cover every target with at most k templates, use $\min_{K \leq k} R_\alpha(n, K)$. Pool version (experiments): with the experiments' prior weights $2^{-\text{bits}(T)}$ and $\vdash_J$ for Th_d , the conclusion holds for the posterior restricted to any pool $R_n \ni T^*$, even one chosen from the data, if $\delta_n \leq 2^{-\text{bits}(T^*)} R_\alpha(n, K) \delta'$ for every n .*

The proofs (Appendix D.3) run that of Theorem 5.2 under $P_{T^*}^{\text{Dir}}$, or under P_{T^*,w^*} with the regret bound. Theorem 5.7 is time-uniform, but its threshold slows acceptance; whether completeness (Proposition 5.5) survives is open. Theorem 5.6 covers a fixed w^* only outside a set of small prior mass. The shrinking threshold is needed:

Example 5.8 (Constant threshold). *[computed; mechanism proof sketch]^{model Ex 4.9; experiments check_fixed_weight}*

L0; root law of $\mathcal{Q}(p_0, p_S, p_+, p_-) = (0.5, 0.5 - \varepsilon, \varepsilon/2, \varepsilon/2)$; truth $T^* = \{0+0=0, (Sz)+0=Sz\}$ with $\text{Dir}(\frac{1}{2}, \frac{1}{2})$ weights, data from P_{T^*,w^*} with the interior $w^* = (p_0, p_S)/(1 - \varepsilon)$. The competitor $T' = \{z+0=z\}$ proves $q := (0 \cdot 0) + 0 = 0 \cdot 0$, which T^* does not. Prior $\frac{1}{2}$ each; the verifier accepts q iff $\pi_n(T^*) \leq \delta = 0.01$, so Theorem 5.6's bound is 0.02. In three implementations (Table 27) q was accepted with probability 0.977–0.997 ($\varepsilon = 10^{-5}$) and 1.000 ($\varepsilon = 10^{-6}$) at the fixed w^* , 0.003–0.010 with w^* drawn from the prior, and in 0/300 runs at the fixed w^* with the threshold of Theorem 5.7; in the referee's runs at $\varepsilon = 10^{-3}$ and 10^{-4} , never. The mechanism (Appendix D.3): on the data's support $P_{T'} = (1 - \varepsilon)P_{T^*,w^*}$, so T' gains the regret $\frac{1}{2} \ln n + O(1)$ against $P_{T^*}^{\text{Dir}}$ and

loses only $n\varepsilon$; the constant threshold fails when T' is so close to T^* 's law that $1 \ll n \ll 1/\varepsilon$ is a long window. With fixed weights the ratio is $(1 - \varepsilon)^n \leq 1$ and q is never accepted.

Remark 5.9 (The Bel_{cons} verifier). [computed; transfer proved]^{universal §6.2, §11; editor, from model §4} Inconsistent theories inflate Bel and Dis alike (Definition 2.13): in the class of Table 2 at $n = 0$, $\text{Bel}(\forall x\varphi) \approx 0.69$ is almost all Inc, and generative likelihoods remove such theories within a few data (Table 17). Section 3.9 therefore recommends “accept s iff $\text{Bel}_{\text{cons}}(s) \geq 1 - \delta$ ”. When this verifier accepts, so would $V_{\delta,d}$ at the same posterior. The proofs of Theorems 5.2, 5.6 and 5.7 bound $\pi_t(C_d^*)$ (or $\pi_t(C_d^{\text{Dir}})$) from below for all truthful constraints, so all three transfer to it (Bel_{cons} read at level d).

5.5 What the experiments show

Remark 5.10 (E4: adaptive provers in the code). [computed]^{experiments E4} Verifier on $\vdash_1$, $\text{Dir}(\frac{1}{2})$ weights, one-component T^* , so Theorem 5.6 applies (Table 35). (A) The tight construction of IL Prop C.5 with a Dirichlet decoy $T' = \{\varphi(0), 0 = S0\}$: win probability exactly $(1 - u)^{t^*}$, 9–46 times below δ' in five settings. With *selected* data (only $\varphi(0)$, each a theorem of T^*) the prover wins 20/20 in every setting. (B) Well specified ($\varphi = x + 0 = x$, $T^* = H_{\text{sch}}$, bound 0.05, every $n \leq 256$): 0/100 wins under L0 and L1; the mass on theories deriving an invalid query peaks at 0.45 and 0.53 at $n \leq 6$, always on the true but underivable $\forall x(x + 0 = x)$. (C1) 10% false near misses $t + 0 = St$, with the mistake schema in the pool: 100/100 wins, first acceptance at the first mistaken datum in 90/100 streams. (C2) Heavy-tailed terms: 0/100. (C3) Selection with the decoy and a clean memoriser in the pool: 0/100.

In (C1) the data's actual generator is a two-schema theory with fixed weights (0.9, 0.1), and the queries first accepted, $0 + 0 = 1$ and $2 + 0 = 3$, are theorems of it: the verifier is unsound only relative to the intended T^* , since nothing forces the best-fitting generator to be the actual axioms. Theorem 5.7 would cover a verifier with its shrinking threshold on that generator, not the constant-threshold verifier used in E4. Comparing (C3) with (A): under misspecification soundness depends on what else is in the pool, and no theorem covers this.

Remark 5.11 (Early unsound lumps). [computed]^{experiments E2; pa §5.4} E2 (L0, $T^* = \text{Q1..Q7} + T_{\text{Ind}}$ at fixed weights, 25 seeds, verifier on citation; causal pools with $\text{SeenQ}(D_n)$, the theory of T_{Ind} and the Q axioms cited so far, Section 8.1): a $\delta = 0.05$ verifier accepts the false $\forall x(x + 0 = 0)$, $\forall x(Sx = x)$, $\forall x\forall y(y = x)$ in 5 of 25 seeds, at $n = 8$ or 16, never from $n = 32$ (Table 8). The two-part code of Remark 7.32 behaves alike. No contradiction with the pool version of Theorem 5.7: for E2 it needs $\delta_n \leq 2^{-226} R_\alpha(n, 8) \delta' \asymp 2^{-226} n^{-3.5}$ (T^* costs 226.2 bits in the experiments' code).

6 Time penalties and Hänni's collapse

Hänni's working notes (Hänni, 2026)³ compare axiom induction, which weighs axiom systems that prove the labelled data, with Solomonoff function induction, which weighs programs that label statements directly. He argues that the two are equivalent for consistent assigners: an assigner T becomes the schema “T(quoted-phi) = accept => phi”, where “being an axiom of the right form is in fact decidable”, and the result is consistent because “a model of the phis alone would be a model of the added sentences also”. So unrestricted axiom induction collapses into function induction, and he considers restricting axioms to “second-order substitution schemas of a specific

³File `research/prior/hanni-solomonoff-axiom-induction.md`; working notes, quoted verbatim.

kind”. The question adds a time penalty: “it should not take too long to generate axioms / to check whether a statement is an axiom”. The result (proofs in Appendix E):

Remark 6.1 (What the time penalty does). [(1), (2), (4) proved; (3) lower bound proved, upper bound proof sketch]^{model §6.5} (1) Penalties on the time to check or generate axioms (Kt, speed prior) do not block the collapse: Craig’s sets A_f^C have membership decidable in polynomial time (Proposition 6.8), like finite DT° theories, so such a penalty charges them as it charges template theories; no time-penalised prior is defined here, so no bound on how much it changes Theorem 6.3 is claimed. (2) Templates block Hänni’s schema (its acceptance part, Proposition 6.6) but, over PA, not the collapse for Σ_n -sound assigners on Σ_n sentences: one ground reflection sentence reproduces them at a cost linear in $|f|$ (Proposition 6.7); for merely consistent assigners that sentence can be inconsistent with PA, and whether templates block the collapse for them is open. (3) Derivation size in written symbols prices the assigner: one decidable X for all finite DT° theories, and one for each e for all axiom sets with membership in time $O(n^e)$, forces every such theory consistent with Γ_{f_X} to use derivations of symbol size above $s(m)$ for infinitely many lengths m , i.e. at least a fixed polynomial root of the assigner’s nondeterministic time (Theorem 6.9 and Corollary 6.10); $L1^\sigma$ then charges at least $\kappa s(m)/4 - \ln(Z_T/Z_T^\sigma)$ nats and the graded score at least $\kappa s(m)$ bits ($\kappa s(m) + \log_2 Z_T$ after normalisation) on those data (Proposition 6.13). The collapse constructions through A_f^C and the two-sorted A_f pay at most a polynomial of the deterministic running time (proof sketch, Remark E.3); for $\rho_{f,n}$ no upper bound is established. (4) Plain $L1$ is proved to charge at least $\zeta s(m) - \ln(C/Z_T)$ nats when the assigner’s language lies outside $NTIME(2^{(d+1)s})$, about the logarithm of its nondeterministic time (Corollary 6.12); a polynomial charge is Conjecture 6.14. Function induction’s total log loss on f_X ’s labels is at most $|f_X|$ bits (when $B \cup \Gamma_{f_X}$ is consistent).

6.1 The two inductors

Fix a countable first-order language L with sentences Sent_L , a universal prefix machine U (program p has weight $2^{-|p|}$), and a decidable background $B \subseteq \text{Sent}_L$, possibly empty, that is part of every axiom hypothesis. Data are labelled sentences $D = ((\varphi_1, b_1), \dots, (\varphi_n, b_n))$, $b_i \in \{1, 0\}$ (true, false).

Definition 6.2 (Axiom and function induction). ^{model §6.1}

- **AI:** a hypothesis is a program p that halts on every input and decides a set $A_p \subseteq \text{Sent}_L$. It is *compatible* with D if $B \cup A_p$ is consistent and proves φ_i when $b_i = 1$ and $\neg\varphi_i$ when $b_i = 0$. $W_{\text{AI}}(D) := \sum_{p \text{ compatible}} 2^{-|p|}$.
- **FI_{cons}:** a hypothesis is a program f computing a partial map $\text{Sent}_L \rightarrow \{\text{acc}, \text{rej}\}$ with $B \cup \Gamma_f$ consistent, where $\Gamma_f := \{\varphi : f(\varphi) = \text{acc}\} \cup \{\neg\varphi : f(\varphi) = \text{rej}\}$; it is compatible if it gives each φ_i the label b_i ; $W_{\text{FI}}(D) := \sum_{f \text{ compatible}} 2^{-|f|}$. FI_{all} is the same over all assigners.
- **Semimeasure convention:** $P_X(b_1 \dots b_n \mid \varphi_1 \dots \varphi_n) := W_X(D_n)/W_X(\emptyset)$, and log loss is $-\log_2$ of it. Hypotheses that leave φ undecided drop out, so $W(D + (\varphi, 1)) + W(D + (\varphi, 0)) \leq W(D)$ (Hänni: “maybe we should speak of a semimeasure here instead of renormalizing”). $W_X(D)$ depends only on the set D .

Function induction over all assigners is not the fair comparison: on a consistent labelling of fair coin flips FI_{all} has bounded log loss while AI and FI_{cons} lose at least n bits in expectation (Proposition E.1), as Hänni argued informally.

Theorem 6.3 (Hänni’s equivalence, repaired). [*proved; Craig’s theorem known*]^{model Thm 6.1} *There is a constant c , depending only on U , the coding of syntax and the decider for B , such that*

$2^{-c}W_{\text{AI}}(D) \leq W_{\text{FI}}(D) \leq 2^cW_{\text{AI}}(D)$ for every finite labelled D . Hence for every reveal order and labelling the semimeasure log losses of AI and FI_{cons} differ by at most $2c$ bits.

Proof idea. $\text{AI} \rightarrow \text{FI}$ is Hänni's argument: map p to “search for proofs of φ and of $\neg\varphi$ from $B \cup A_p$ ”. For $\text{FI} \rightarrow \text{AI}$ his schema is replaced by Craig's trick (Craig, 1953): f becomes a decider of $A_f^C := \{\varphi^{\wedge(k+1)} : f(\varphi) = \text{acc in exactly } k \text{ steps}\} \cup \{(\neg\varphi)^{\wedge(k+1)} : f(\varphi) = \text{rej in exactly } k \text{ steps}\}$, $\psi^{\wedge m}$ the right-nested m -fold conjunction. A_f^C is decidable and has the consequences of Γ_f ; its decider is f plus a fixed wrapper (Appendix E.1). $\square$

The constant equivalence is proved only in the semimeasure convention; renormalising at each step, the argument gives only $2c$ bits per step (Remark E.2).

6.2 Hänni's schema

Let $\text{Accept}_f(x)$ be a Σ_1 formula of arithmetic for “ f halts on input x with output acc” (via Kleene's T predicate), and $\text{Reject}_f(x)$ likewise. Both are written without $<$ (bounded quantifiers as $\exists z (z+x = y)$), so that Σ_1 -completeness of $\mathbf{Q}$, which in L_A holds for $<$ -free Σ_1 sentences, applies. Hänni's schema is $A_f := \{\text{Accept}_f(\ulcorner\varphi\urcorner) \rightarrow \varphi : \varphi \in \text{Sent}_L\} \cup \{\text{Reject}_f(\ulcorner\varphi\urcorner) \rightarrow \neg\varphi : \varphi \in \text{Sent}_L\}$.

Proposition 6.4 (Two sorts: his argument works). *[proved; Σ_1 -completeness of $\mathbf{Q}$ for $<$ -free sentences known]^{model Prop 6.2} Let L -sentences live in one sort and arithmetic in another, with $B := \mathbf{Q}$ on the second. If Γ_f has a model, then $B \cup A_f$ is consistent and $\text{Cn}(B \cup A_f) \cap \text{Sent}_L = \text{Cn}(\Gamma_f)$.*

Proposition 6.5 (One sort over PA). *[proved; Hänni's argument refuted in this reading]^{model Prop 6.3} Assume $\text{Con}(\text{PA})$ and the standard formalisation of computations. Let $L = L_A$, $B = \text{PA}$, and let f accept $\ulcorner\neg\text{Con}(\text{PA})\urcorner$, search on input $\ulcorner\text{Con}(\text{PA})\urcorner$ for a PA-proof of $\perp$ and accept if it finds one, and loop otherwise. Then $\Gamma_f = \{\neg\text{Con}(\text{PA})\}$ is consistent with PA; $\text{PA} \cup A_f$ is not.*

His consistency argument assumes that the antecedents are evaluated correctly; in a model of $\text{PA} + \neg\text{Con}(\text{PA})$, $\text{Accept}_f(\ulcorner\text{Con}(\text{PA})\urcorner)$ holds via a nonstandard proof of $\perp$. So his *conclusion*, the equivalence, stands (Theorem 6.3); his *argument* needs two sorts or a sound f .

6.3 What templates do

Proposition 6.6 (No non-ground template in the collapse schema). *[proved]^{model Prop 6.4; pa Prop 4.3} Write Gödel numbers as unary numerals S^k0 (or any numerals in which some term symbol, here $+$, never occurs), and let $C_f := \{\text{Accept}_f(\ulcorner\varphi\urcorner) \rightarrow \varphi : \varphi \in \text{Sent}_L\}$, the acceptance part of A_f , single- or two-sorted. Every DT° template τ with $\text{inst}(\tau) \subseteq C_f$ is ground. So C_f is not $\bigcup \text{inst}(T)$ for any finite set T of DT° templates, and such a set with instances in C_f covers finitely many of its members. Likewise for the reflection schema $\text{Refl}_T = \{\text{Prv}_T(\ulcorner\varphi\urcorner) \rightarrow \varphi\}$ (Proposition 7.22).*

The rejection part, and templates whose instances mix the two parts, are not treated here (Appendix H.5). For the acceptance part, Hänni's restriction blocks his construction: inside DT° it costs one ground template per sentence, which is memorisation. But one ground sentence can carry the collapse over PA.

Proposition 6.7 (One reflection sentence carries the collapse). *[proved, given partial truth predicates (known)]^{model Prop 6.5} For $n \geq 1$ let $\text{Tr}_n(x)$ be a Σ_n partial truth predicate, $\text{PA} \vdash \varphi \leftrightarrow \text{Tr}_n(\ulcorner\varphi\urcorner)$ for every Σ_n sentence φ (known; Hájek and Pudlák, 1993, Ch. I; Kaye, 1991, Ch. 9; chapter numbers*

not checked), and $\text{Sent}_{\Sigma_n}(x)$ a Δ_1 formula for “ x codes a Σ_n sentence”. The single sentence (ground DT° template)

$$\rho_{f,n} := \forall x [(\text{Sent}_{\Sigma_n}(x) \wedge \text{Accept}_f(x)) \rightarrow \text{Tr}_n(x)] \wedge \forall x [(\text{Sent}_{\Sigma_n}(x) \wedge \text{Reject}_f(x)) \rightarrow \neg \text{Tr}_n(x)]$$

has $\ell(\rho_{f,n}) \leq a|f| + b_n$ for constants a, b_n (binary numeral for f ’s index), and: (a) $\text{PA} + \rho_{f,n}$ proves every Σ_n sentence that f accepts and the negation of every one it rejects; (b) if f is Σ_n -sound (it accepts only true and rejects only false Σ_n sentences), $\text{PA} + \rho_{f,n}$ is true in $\mathbb{N}$.

So PA ’s templates plus one ground sentence reproduce any Σ_n -sound assigner on Σ_n sentences, at a cost linear in $|f|$ (not $|f| + c$); for merely consistent assigners $\rho_{f,n}$ can be inconsistent with PA (as in Proposition 6.5).

6.4 Time penalties

Proposition 6.8 (Membership is cheap in both collapse sets). *[proved]^{model Prop 6.6} (a) With binary numerals, membership of χ in A_f is decidable in time polynomial in $|\chi|$ without running f : parse χ as $\text{Accept}_f(t) \rightarrow \varphi$ or $\text{Reject}_f(t) \rightarrow \neg\varphi$, compute $\ulcorner\varphi\urcorner$, compare with t . With unary numerals likewise, since then $|\chi| \geq \ulcorner\varphi\urcorner$. (b) Membership in A_f^C is decidable in time polynomial in $|\chi|$: at most $|\chi|$ decompositions $\chi = \psi^{\wedge m}$, each running f for $m - 1 < |\chi|$ steps.*

So a penalty in the style of Levin’s Kt or the speed prior on the time to *check* axiomhood puts these sets in the same polynomial class as DT° templates (Remark A.1(iii)): it does not price the collapse. Penalising the time to *generate* axioms fails too: $\varphi^{\wedge(k+1)} \in A_f^C$ comes from k steps of f and is longer than k .

The computation of f moves into the proofs (A_f : a derivation must prove $\text{Accept}_f(\ulcorner\varphi\urcorner)$) or into the axioms’ length (A_f^C), both counted by derivation size in symbols; no theory with polynomial-time membership escapes this. Let $L \supseteq \{0, S, +, \cdot, R\}$, R unary. For $X \subseteq \{0, 1\}^*$ put $\varphi_w := R(\underline{w})$, $\underline{w}$ a binary numeral term, and let f_X accept φ_w iff $w \in X$ and reject it otherwise; Γ_{f_X} is consistent (interpret R as X).

Theorem 6.9 (Derivation size bounds the assigner’s nondeterministic time). *[proved]^{model Thm 6.7} Let T be any axiom set with membership decidable in time $O(n^e)$ and $T \cup \Gamma_{f_X}$ consistent. If every φ_w , $w \in X$, has a $\mathbf{K}$ -derivation from T of symbol size at most $\ell(|w|)$, with $\ell(m) \geq m$ time-constructible, then $X \in \text{NTIME}(\ell(m)^{c_e})$, c_e depending only on e and $\mathbf{K}$ ’s checking overhead.*

Proof idea. Guess a derivation of size at most $\ell(|w|)$ and check it. If $w \notin X$, then $\neg\varphi_w \in \Gamma_{f_X}$, so consistency rules out a derivation (Appendix E.4). $\square$

Corollary 6.10 (Hard assigners force long derivations, for every theory). *[proved, given the non-deterministic time hierarchy theorem (known)]^{model Cor 6.8 (referee m1)} For finite DT° theories membership takes time $O(|T| \cdot |\chi|)$ (AS Thm 2.5), so $e = 1$ and c_1 is one constant for all of them. (a) Let $s(m) \geq m$ be time-constructible with $s(m+1)^{c_1} = o(s(m)^{c_1+1})$ (every polynomial). There is a decidable X such that every finite DT° theory T with $T \cup \Gamma_{f_X}$ consistent has, for infinitely many m , some $w \in X$ of length m with no $\mathbf{K}$ -derivation of φ_w from T of symbol size $\leq s(m)$. (b) For each e the same holds, with X depending on e , for every axiom set with membership in time $O(n^e)$; a single X for all e exists under a growth condition [proof sketch] (Appendix E.4).*

The hierarchy theorem is cited from Cook (1972); Seiferas et al. (1978); Žák (1983), unchecked.

6.5 What the derivation likelihoods charge

Theorem 6.9 bounds the *symbol* size of derivations. The code length $-\ln \Pr(\pi)$ of an L1 tree counts its grammar choices instead (Definition 2.9), and the two can differ exponentially.

Remark 6.11 (Refuted: plain L1 charges symbol size). [refuted; counterexample proved and computed]^{model Prop 6.9} Under L1 with subcritical parameters, $-\ln P_T(\varphi) \geq \kappa \ell_{\min}(\varphi) - \ln(C/Z_T)$, $\ell_{\min}$ the least symbol size of a derivation, fails (an earlier sketch claimed it; Appendix H.2). From $\forall x (x = x)$, repeat k times “ $\forall E$ with $t = p + p$, then Gen on p ”, and close with one $\forall E$: the conclusion has size $2^{k+3} - 1$, while $-\ln \Pr(\text{tree}) = 12.77 + 12.21k$ nats (12.206 per round) and $-\ln P_T \leq -\ln \Pr(\text{tree})$ (Appendix E.5, Table 28).

$\forall E$ and Gen substitute into every occurrence of a variable at the price of one choice, so Theorem 6.9 and Corollary 6.10 transfer to $L1^\sigma$ and to Hänni’s graded score, not to plain L1. Let $\nu(\pi)$ count the grammar nodes of a tree π (derivation, $\mathcal{Q}$ and parameter-chain nodes) and $\nu_{\min}(s)$ be the least ν of a valid tree concluding s . L1’s code length counts grammar choices: a tree with ν grammar nodes from a finite T has conclusions of symbol size at most exponential in ν , and L1 with a uniformly subcritical grammar and a geometric parameter law charges at least $\zeta \nu_{\min}(\varphi) - \ln(C/Z_T)$ nats for φ ; so if each φ_w , $w \in X$, has a tree with at most $\ell(|w|)$ nodes from a finite DT° theory consistent with Γ_{f_X} , then $X \in \text{NTIME}(2^{(d+1)\ell(m)})$, d independent of T (Lemma E.4, Proposition E.5, and Theorem E.6 in Appendix E.5).

Corollary 6.12 (Hard assigners make plain L1 pay, logarithmically). [proved, given the hierarchy theorem (known)]^{model Cor 6.13} Let $s(m) \geq m$ be time-constructible with $s(m)^2 - (d+1)s(m+1) \rightarrow \infty$ (every polynomial; 2^m). There is a decidable X such that for every finite DT° theory T with $T \cup \Gamma_{f_X}$ consistent, for infinitely many m some $w \in X$ of length m has $\nu_{\min}(\varphi_w) > s(m)$; under the hypotheses of Proposition E.5, $-\ln P_T(\varphi_w) \geq \zeta s(m) - \ln(C/Z_T)$ on those data.

Symbol-size penalties, by contrast, charge a polynomial root of the nondeterministic time:

Proposition 6.13 (Symbol-size penalties charge polynomially). [(a) proved; (b), (c) proved, given the hierarchy theorem (via Corollary 6.10)]^{model Prop 6.14} (a) Under $L1^\sigma$ (Definition 2.11), $P_T^\sigma(\varphi) \leq e^{-\kappa \ell_{\min}^\sigma(\varphi)} Z_T/Z_T^\sigma$, with $\ell_{\min}^\sigma(\varphi)$ the least total symbol size of a valid L1 tree concluding φ . (b) A valid L1 tree of total symbol size ℓ yields a K-derivation of symbol size at most 4ℓ . So with X from Corollary 6.10(a), for every finite DT° theory T consistent with Γ_{f_X} , infinitely often $-\ln P_T^\sigma(\varphi_w) \geq \kappa s(m)/4 - \ln(Z_T/Z_T^\sigma)$. (c) Hänni’s graded score with $g(\ell) = 2^{-\kappa \ell}$ on explicit K-derivations, in a code with at least one bit per written symbol (Remark 2.23), charges at least $\kappa s(m)$ bits on the same data ($\kappa s(m) + \log_2 Z_T$ normalised).

Conjecture 6.14 (Plain L1 charges polynomially too). [conjecture]^{model Conj 6.15} Theorem E.6 holds with $\text{NTIME}(\ell(m)^c)$, for a constant c , in place of $\text{NTIME}(2^{(d+1)\ell(m)})$.

The missing step is checking trees on shared representations, with de Bruijn indices and Gen under sharing (Appendix E.5). Unless Conjecture 6.14 is proved, the question’s derivation-length prior should therefore measure length in written symbols ($L1^\sigma$, or the graded score normalised as in Remark 2.23). Whether axiom induction with a symbol-size penalty is equivalent, up to polynomial factors, to function induction over consistent assigners with short certificates is open.

6.6 Links to the other sections

Links to Sections 3 and 7 are in Remark E.7: no penalty monotone in the number of derivation steps reverses Theorem 3.4; a Kt-style penalty charges only $O(\ln k)$ for $0 + S^k 0 = S^k 0$ on the

equational fragment; and a time-bounded likelihood favours making theorems with long proofs axioms.

Remark 6.15 (Prior variants on the E2 data). `[computed]`^{experiments E7} On the E2 data (25 seeds), a membership-time factor $(1 + |T|)^{-\tau}$ shifts the log-odds of T^* against a hand competitor by at most $\log_2(78/2) \approx 5.3$ bits at $\tau = 1$; it changes nothing visible from $n = 64$ on and at $n \leq 32$ favours small unsound lumps, slightly at $\tau = 1$ (unsound MAP at $n = 8$ in 9 instead of 7 seeds) and markedly at $\tau = 4$ (mean unsound mass at $n = 8$ 0.372 against 0.256; a false probe accepted in 5 instead of 2 seeds). A steeper simplicity penalty $\lambda = 2$ makes the MAP at $n = 8$ unsound in 22 seeds ($\lambda = 1$: 7) but removes the false spare slot faster ($2 \cdot 10^{-4}$ against 0.004 unsound mass at $n = 512$); $\lambda = \frac{1}{2}$ (proper only on finite pools) the reverse: neither removes both failures (Table 39).

Conjecture 6.16 (A time-bounded analogue). *[conjecture; not written out]*^{model §9.3} *Hänni’s note on time-bounded Solomonoff induction⁴ gives, for each polynomial p , a predictor whose log loss exceeds that of any “prediction algorithm P with runtime $O(p(n))$ ” by “at most a constant c_P ”. By his argument, the derivation-bounded L2 inducer with a growing depth budget plausibly has constant regret against theories whose data have derivations within the budget.*

7 The axioms we actually have: PA and ZF

Which axioms does the posterior pick up for arithmetic and set theory? Sources: track pa; track experiments for E2, E3(b), E6 and Props X9–X11; AS (Claude (Anthropic), 2026a) is cited by its own numbering. Units are bits; proofs and full tables are in Appendix F. Three caveats hold throughout. (i) Track pa computes with the two-part code $L1^{\text{sch}}$ or with L0 under a learned grammar, never with the generative L1 of Section 2; its numbers are not L1 results. (ii) Candidate sets are hand-picked and finite. (iii) Derivation costs are those of explicit derivations, hence upper bounds. “Proved; computed” means a proof plus a derivation checked line by line by track pa’s checker `nd.py` (Appendix F.1).

7.1 Setting

Definition 7.1 (The two-part derivation code $L1^{\text{sch}}$).

pa Defs 0.2, 0.3 Derivations are in a sound and complete natural-deduction calculus ND (Table 11). An instance $f(\varphi)$ of a schema $f(P)$ is derived schematically in P and costs $\text{bits}(\pi) + L_Q(\varphi)$, the motive being coded once by the grammar. Per line, $\text{bits}(\pi)$ charges $\log_2 10$ for the rule, $\log_2 |T|$ per axiom citation, $\log_2(\text{line index})$ per premise reference and β per *written symbol*. The *decodable* variant also codes the number of lines, the premise counts of tc lines and the variables of subst and $\exists$ I lines. $L1^{\text{naive}}$ writes every formula out at the instance. On a finite candidate set the prior is $\pi(T) \propto 2^{-\beta \sum_{\tau \in T} |\tau|}$: $\beta = \log_2 23 \approx 4.52$ bits per symbol of `nd.size` (a quantifier and its variable count 2) in the derivation computations, $\beta = 5$ per symbol of the `dtlib` size (a de Bruijn $\forall$ counts 1) in those built on AS’s code u7. $\text{CL}(T; D)$ is the total code length, ℓ_{cite} the cost of a one-line citation.

The decodable variant adds 4.8–7.3% to derivation costs and is a prefix code, so Kraft holds (proved): $L1^{\text{sch}}$ is the two-part form of a sub-probability likelihood, of the $L1^\sigma$ kind, not of plain

⁴File `research/prior/hanni-polytime-solomonoff.md`; working notes, in which he says he has not “checked all the technical claims completely carefully”.

L1 (Proposition E.5). The grammar $\mathcal{Q}$ is *learned*: each metavariable body is coded at its read-off position by sequential KT in the context (sort, depth, parent, child index), over 9 formula symbols ($=, \neg, \wedge, \vee, \rightarrow, \forall, \exists, \top, \perp$) and 14 term symbols. Sequential KT is a Dirichlet($\frac{1}{2}$) marginal (Krichevsky and Trofimov, 1981), as are the weights, so CL differences in the L0 computations are exact $\log_2$ posterior odds, except where templates overlap (Remark F.7).

Definition 7.2 (Refutation rule R_d).

pa Def 0.4 With data D and negative data N , R_d sets the likelihood of T to 0 if $T \vdash_{\leq d} \neg s'$ for some $s' \in D$ or $T \vdash_{\leq d} s$ for some $s \in N$ (Th_d in derivation size; under L0, $d = 0$ and “derives” means “has as an instance”). R_∞ is not computable.

7.2 Equivalent axiomatisations: usage decides

Let $\text{Dlt} := \forall u \forall v (u < v \leftrightarrow \exists z (u + Sz = v))$. For a motive φ in x (parameters allowed), $\text{Ind}(\varphi) := \varphi(0) \wedge \forall x (\varphi \rightarrow \varphi(Sx)) \rightarrow \forall x \varphi$; $\text{CVI}(\varphi) := \forall x (\theta_\varphi \rightarrow \varphi) \rightarrow \forall x \varphi$ with $\theta_\varphi := \forall y (y < x \rightarrow \varphi(y))$; $\text{LNP}(\varphi) := \exists x \varphi \rightarrow \exists x (\varphi \wedge \forall y (y < x \rightarrow \neg \varphi(y)))$. For nonempty $S \subseteq \{\text{Ind}, \text{CVI}, \text{LNP}\}$ let $\text{PA}_S := \mathbf{Q} + \text{Dlt} + \{f(P) : f \in S\}$, $\text{PA}_f := \text{PA}_{\{f\}}$, and PA_{all} with all three.

Proposition 7.3 (Three forms of induction). *[proved; computed]*

pa Prop 1.1 With $B := \{\text{Q1}, \dots, \text{Q5}, \text{Dlt}\}$, for every motive φ : (A) $B \vdash \text{CVI}(\varphi) \rightarrow \text{Ind}(\varphi)$; (B) $B \vdash \text{Ind}(\varphi) \rightarrow \text{CVI}(\varphi)$; (C1) $\vdash \text{CVI}(\neg \varphi) \rightarrow \text{LNP}(\varphi)$; (C2) $\vdash \text{LNP}(\neg \varphi) \rightarrow \text{CVI}(\varphi)$. So PA_{Ind} , PA_{CVI} , PA_{LNP} have the same theorems.

The derivations were checked schematically in P , replayed on 200 random motives, and re-checked by the referee’s independent checker (Appendix F.2).

Remark 7.4 (Patterns). *[proved]*

pa Rem 1.2 $\text{CVI}(P), \text{LNP}(P) \in \text{PAT}$, while $T_{\text{Ind}} \in \text{DT}^\circ \setminus \text{PAT}$ ($P(0), P(Sx)$). So PA is $\mathbf{Q}$, Dlt and one higher-order pattern (compare AS Prop 5.18: $\mathbf{Q}$ plus the first-order pattern IndEq).

Usage. Let the data be i.i.d. uses: a form f with probability p_f and a motive $\varphi \sim \mathcal{Q}$, the datum being $f(\varphi)$. In PA_S a use costs $c_S(f) + L_{\mathcal{Q}}(\varphi)$, $c_S(f)$ the cost of the cheapest derivation of $f(P)$ in a fixed library of checked derivations (a citation if $f \in S$; Table 3).

Proposition 7.5 (Usage limit). *[proved, given the library constants]*

pa Prop 1.3 $\frac{1}{n}(\text{CL}(\text{PA}_S; D_n) - \text{CL}(\text{PA}_{S'}; D_n)) \rightarrow \sum_f p_f (c_S(f) - c_{S'}(f))$ a.s.; the posterior concentrates on the S minimising $\mathbb{E}_p[c_S(f)]$ if it is unique.

Proposition 7.6 (Lower bound). *[proved]*pa Prop 1.4 Every L1^{sch} -derivation of $\text{CVI}(P)$ from PA_{Ind} costs at least $\log_2 10 + 11\beta \approx 53$ bits more than citing it in PA_{CVI} ; every derivation of $\text{Ind}(P)$ from PA_{CVI} at least $\log_2 10 + 9\beta \approx 44$ bits more than citing it in PA_{Ind} .

The first version’s argument (“at least two lines”) was wrong, since a citation also writes $P(x)$; the proof in Appendix F.2 follows the premises back to a line that writes the form’s skeleton or antecedent.

Remark 7.7 (The usage-closed axiomatisation). *[computed]*pa §1.2 For L1^{sch} with this library: (i) PA_{Ind} beats PA_{CVI} iff $p_{\text{CVI}}/p_{\text{Ind}} < 0.600$, and PA_{LNP} iff $p_{\text{LNP}}/p_{\text{Ind}} < 0.757$ (0.609, 0.765 under the decodable code; under L1^{naive} the first ratio is 0.671–1.052, so the ratios are code-specific). (ii) PA_{all} pays $81 + 86$ prior bits and 0.29 bits per citation more than PA_{Ind} , so it is ahead after about one use of a non-primitive form (in expectation after 1–12 schema uses for the mixes tried).

theory	forms derived	lines	symbols	overhead	decodable
PA _{Ind}	CVI, LNP	63, 82	204, 332	1468.0, 2234.1	1551.2, 2341.8
PA _{CVI}	Ind, LNP	39, 20	127, 117	880.4, 646.4	945.1, 676.9
PA _{LNP}	Ind, CVI	65, 27	253, 128	1692.1, 757.5	1792.3, 799.1

Table 3: Overhead in bits per use of deriving a non-primitive form $f(P)$ over citing it ($L1^{\text{sch}}$, $\beta = \log_2 23$); a citation costs 15.5–15.8 bits; template priors Ind 72, CVI 81, LNP 86 bits. Computed; upper bounds (pa §1.2).

(iii) So the posterior concentrates on the *usage-closed* axiomatisation: every form used at a rate above about $5 \cdot 10^{-4}$ per use is primitive. All candidates are PA-equivalent, so the theorems are right whichever wins (compare Proposition 4.5).

Proposition 7.8 (Recursion conventions). *[proved; computed]*

pa Prop 1.5 $Q + T_{\text{Ind}}$ and the theory with $Q4$, $Q5$ replaced by $0 + x = x$, $Sx + y = S(x + y)$ prove each other’s recursion laws with one induction each, so they are equivalent; the per-use overheads are 213.8 and 593.4 bits one way, 213.8 and 545.1 the other. The posterior adopts the convention the data use.

Remark 7.9 (ZF). *[proved; computed; known; conjecture (reverse overheads)]*^{pa §1.4} Separation from Replacement, Replacement from Collection and Foundation from $\in$ -induction are checked, at 574–1432 bits per use against template priors of 72–185 bits (Table 29). Collection from Replacement holds over ZF, fails without Power Set (Zarach, 1996; Gitman et al., 2016) (known) and is open to us without Foundation; EInd from Found needs transitive closures (known; Kaye and Wong, 2007). The reverse overheads were not formalised (conjecture: much larger). So the textbook ZF axioms win exactly when they are used directly.

Remark 7.10 (E6: commutativity). *[computed]*^{experiments E6} Among eight forms of commutativity (Proposition 8.2), the chain L1 identifies the generating form by $n = 32$ in every run, both among logically equivalent forms and among instance-equivalent but weaker ones; under $L1_{\text{cit}}^{\text{sel}}$ five closed-guard forms tie, and the mass on provers of $\forall x \forall y x + y = y + x$ tends to their prior share 0.088 (Table 37).

7.3 The MDL finding in PA

AS found that “MDL tracks the statistics of usage, not the logical boundaries of schemas” (AS §6.10, Props E.16, E.17, Table 12); its codes PC and DPC are sequential KT, so its table already gives $\log_2$ posterior odds. Track pa re-ran AS’s code u7 (reproducing AS Table 12 at $n = 1000$) on $T^* := Q + T_{\text{Ind}}$ against the root split H_{root} (7 templates T_f) and the split H_{used} into the roots F that occur, adding the grammars SDPC (one learned positional grammar shared by all templates), RDPC (T^* ’s grammar may condition on the motive’s root) and CF (one nonterminal per sort), under usage laws G1 (u7’s natural law, 7 roots), G2 and G3 (4 and 3 roots).

Proposition 7.11 (Chain rule). *[proved]*^{pa Prop 2.1} *With fixed parameters, if T^* has weight w on T_{Ind} and body law $Q(\varphi) = Q(f)Q(\varphi \mid f)$, f the root, then H_F ($F \supseteq$ the roots in the support) with weights $wQ(f)$ on T_f and body laws $Q(\cdot \mid f)$ has $P_{H_F} = P_{T^*}$.*

Proposition 7.12 (Occam identity). *[(a), (c) proved; (b) proved under its growth condition; computed]*^{pa Prop 2.3} *Use L0 with a Dirichlet($\frac{1}{2}$) citation index and a shared learned positional grammar (KT*

per context c , alphabet A_c); $\text{KT}(k; A)$ is $-\log_2$ of the Dirichlet($\frac{1}{2}$) marginal of counts k over A symbols. Suppose every datum is covered by exactly one template of H and one of H' , and, for a set C of contexts, H' codes datum by datum exactly the (context, symbol) pairs that H codes outside C . (a) $\text{CL}(H') - \text{CL}(H) = \Delta_{\text{prior}} + \text{KT}(n'; |H'|) - \text{KT}(n; |H|) - \sum_{c \in C} \text{KT}(m_c; |A_c|)$ (n, n' citation counts, m_c symbol counts in c). (b) If all nonzero counts grow linearly, $\text{KT}(k; A) = N \hat{H}(k) + \frac{A-1}{2} \log_2 N + O(1)$, $N = \sum k$. (c) If $C = \{c\}$, the data visiting c are those of one template of H , and H' 's template and the pair (H 's template, symbol in c) determine each other, then $\text{CL}(H') - \text{CL}(H) = \Delta_{\text{prior}} + \frac{(|H'| - |H|) - (|A_c| - 1)}{2} \log_2 n + O(1)$; for a split of T_{Ind} into roots F under SDPC, $\Delta_{\text{prior}} + \frac{|F|-9}{2} \log_2 n + O(1)$.

The identity equals the computed SDPC code length to 0.1 bit at every n and law (slopes $-1, -2.5, -3$ bits per doubling for $|F| = 7, 4, 3$). It reconciles the signs of the Occam drift: with a fixed $\mathcal{Q}$ only the split pays Occam terms (Proposition 4.10; E2: about 4 bits per doubling); with a learned grammar the schema also pays $\frac{1}{2} \log_2 n$ per formula symbol never used at the root.

Remark 7.13 (Refuted: a constant margin). [refuted]

pa §2.2; referee m2 The first version called the SDPC margin of H_{root} constant ($+770 \rightarrow +762$ bits for $n = 10^3 \rightarrow 2.56 \cdot 10^5$). It falls by 1 bit per doubling (Proposition 7.12(c), $|F| = 7$); extrapolated crossover $n \approx 2^{780}$.

On G1 at $n = 256000$ the split leads by 274656, 94089, 163028 and 17242 bits under NAIVE, PC, DPC and CF, and trails by 762 (SDPC) and 5986 (RDPC) bits (Table 30, Appendix F.3). So the finding is about the grammar, not the likelihood family: a split pays off linearly whenever $\mathcal{Q}$ ignores a feature that the split conditions on and that usage depends on; SDPC and RDPC hold it at the prior margin up to Occam terms. A derivation likelihood changes the consequences:

Proposition 7.14 (Detours). [(a) proved; computed; (b) known]

pa Prop 2.2 (a) A split H_F derives every induction instance if F contains a non-atomic connective, through the wrappers $\neg\neg P, P \wedge P, P \vee P, (0 = 0) \rightarrow P, \forall zP, \exists zP$ (six checked derivations; under L1^{sch} they cost 369.2–488.9 bits more than a citation). (b) If $F = \{=\}$, $\mathcal{Q} + H_F \subseteq \text{IOpen}$, which does not prove the irrationality of $\sqrt{2}$ (Shepherdson, 1964) (secondary sources only), while PA does.

So under L1 MDL's incompleteness for schemas (AS App. E.7) disappears once a split has a non-atomic root; the usage effect and the unsoundness error (Remark 7.32) remain (qualification (i) of Remark 4.13).

7.4 Theorem data: statements given without proof

Hänni proposed that “statements given without proof in our data set should be fairly easily derived from axioms”. Then the data are theorems. A theory T can derive a datum s , at cost $D_T(s)$ (cheapest library derivation under L1^{sch}), or adopt it as a ground axiom.

Proposition 7.15 (Memorisation threshold). [proved]

pa Prop 3.1 If s occurs r times and the other data have the same best derivations under T and $T \cup \{s\}$, then $\text{CL}(T \cup \{s\}) - \text{CL}(T) = \beta|s| + \Delta I - r(D_T(s) - \ell_{\text{cite}})$, with ΔI the index cost change ($\log_2 \frac{k+1}{k}$ per citation over k axioms). Up to ΔI , memorising at the first occurrence is cheaper iff $\beta < \beta^*(s) := (D_T(s) - \ell_{\text{cite}})/|s|$.

theorem s	$ s $	lines	$D_T(s)$	$\beta s + \ell_{\text{cite}}$	ratio	$\beta^*(s)$
$\forall x \neg Sx = x$	7	11	224.6	38.2	5.9	31.2
$\forall x 0 \cdot x = 0$	7	14	281.3	38.2	7.4	39.3
$\forall x 0 + x = x$	7	11	220.1	38.2	5.8	30.5
$\forall a \forall v Sa + v = S(a + v)$	13	23	576.0	65.3	8.8	43.8
$\forall q \forall x x + q = q + x$	11	55	1358.6	56.3	24.2	122.9
$\forall a \forall b \forall x (a + b) + x = a + (b + x)$	17	27	793.4	83.4	9.5	46.3

Table 4: Deriving versus memorising in $\mathbf{Q} + T_{\text{Ind}}$ ($\mathbf{L1}^{\text{sch}}$, $\beta = \log_2 23 \approx 4.52$; bits): memorising wins at the first occurrence for every theorem. Computed; checked derivations; upper bounds (pa §3.2; more in Appendix F.4).

Remark 7.16 (Lemmas, compressible data, theorem streams). [computed; known]

pa §§3.3–3.5 (i) Commutativity costs 1358.6 bits with its two lemmas derived in line and 504.7 with them adopted (prior 90.5): lemmas used often become axioms (compare Proposition B.14(c)). (ii) A schema compresses its direct instances about fourfold; for a non-primitive form, adopting $f(P)$ beats deriving and memorising after 1–2 uses. (iii) On streams of fresh induction uses (probability u) and library theorems, for $u = 0$ the MAP is $\mathbf{Q}$ plus the library from the first datum, a finite subtheory of PA and so strictly weaker (known; Ryll-Nardzewski, 1952); for $u = 0.1, 0.5$ it is $\mathbf{Q} + T_{\text{Ind}}$ plus the library, with PA’s theorems.

Proposition 7.17 (Gibbs). [proved]^{pa Prop 3.2} If $\sum_s 2^{-\ell(s)} \leq 1$ (e.g. $\ell(s) = \beta|s|$) and P has finite entropy, then $\mathbb{E}_P[-\log_2 P] \leq \mathbb{E}_P[\ell]$. So on data from P_{T^*} the generator’s code length for a new datum is in expectation at most the prior cost of memorising it.

Proposition 7.18 (Well specified, the generator wins). [proved, given Theorem 4.2 and known facts; proof sketch (last part)]

pa Prop 3.3 Let the data be i.i.d. from PA’s L1 generator in setting W (Definition 4.1), weights positive and parameters admissible ($\text{supp } P_{\text{PA}} = \text{Th}(\text{PA})$, Lemma 2.10), in a countable class with $\pi(\text{PA}) > 0$. Then a.s. every $\mathbf{Q} \cup S$ (S a finite set of PA-theorems) has posterior 0 eventually, and $\pi_n(C^*) \rightarrow 1$. Theories $\text{PA} \cup S$ (generic weights) decay polynomially, like spare slots.

Proposition 7.19 (“Must prove” with refutation). [proved]

pa Prop 3.4 Under S_{prove} with bounded derivability and $\mathbf{R}_d$: (a) the posterior is the prior restricted to the score-1 theories; (b) $\mathbf{Q} + T_{\text{Ind}}$ beats $\mathbf{Q} \cup \{s_1, \dots, s_m\}$ once $\beta \sum |s_i| > 72$ bits (two library theorems suffice), if d covers its derivations; (c) any unrefuted theory with smaller prior deriving all data within d beats both, e.g. the bare formula metavariable F_0 , or an inconsistent theory whose shortest refutation exceeds d .

Answer. A derivation-length likelihood finds $\mathbf{Q} + T_{\text{Ind}}$ from theorem data (1) when the data come from its own derivation process (Propositions 7.17 and 7.18), which human theorems, selected for short statements and interest, do not; (2) when they contain direct uses of the schema; (3) when the axiom prior is much steeper than the proof code ($\beta \geq \beta^*$); (4) under the 0/1 constraint with refutation, at the price of the size principle. Otherwise $\mathbf{L1}^{\text{sch}}$ memorises short theorems, and the MAP is strictly weaker than PA. $\mathbf{L1}^{\text{eq}}$ (Example C.6) and the fragment $\{Q4, Q5\}$ (Proposition B.15) show the same pattern: a derivation likelihood favours the theory that makes the data cheap to state. Not computed: theorem data under plain L1 or the full sum. A time bound on computing the likelihood sums over fewer derivations and so pushes further towards memorisation (Remark E.7).

7.5 Data from $\text{Th}(\mathbb{N})$, the $\text{I}\Sigma_n$ chain and narrow practice

Proposition 7.20 (No consistent r.e. theory survives). *[proved]*

pa Prop 4.1 If $P_T(s) > 0$ iff $T \vdash s$ (e.g. L1, weights positive, parameters admissible) and the data are i.i.d. from a law with full support on $\text{Th}(\mathbb{N})$, every consistent r.e. theory has posterior 0 eventually, a.s.

Proposition 7.21 (Refutation). *[proved; known]*

pa Prop 4.2 Without a refutation rule inconsistent theories are never refuted; R_∞ refutes them at once but is not computable; R_{d_n} with $d_n \rightarrow \infty$ refutes each eventually; no computable rule does better uniformly, since consistency of r.e. theories is Π_1 -complete.

Along PA, $\text{PA} + \text{Con}(\text{PA})$, ... the posterior drifts; a toy model with logarithmic drift is in Remark F.2. In DT° each step must add a ground sentence:

Proposition 7.22 (Reflection is not a template). *[proved]^{pa Prop 4.3}* Every DT° template whose instances lie in $\text{Ref}_T := \{\text{Prv}_T(\ulcorner \varphi \urcorner) \rightarrow \varphi\}$ (unary numerals) is ground. The proof is that of Proposition 6.6 with Prv_T for Accept_f .

Proposition 7.23 (Regret against a theory plus exceptions). *[proved]^{pa Prop 4.4}* Under L_ε , for every T and D_n , with E the data T neither derives nor refutes: $-\log_2 P_{\text{mix}}(D_n) \leq -\log_2 \pi(T) + \sum_{s \notin E} -\log_2((1 - \varepsilon)P_T(s)) + \sum_{s \in E} -\log_2(\varepsilon\mu_0(s))$. With 0/1 support the same holds for $T \cup E$, the exceptions memorised at $\beta|e|$ prior bits each.

Pointwise limits are trivial: under 0/1 support memorisation alone eventually decides every sentence correctly (Remark F.1). So for $\text{Th}(\mathbb{N})$ what can be promised is regret against every theory plus its exceptions and depth-relative soundness, not identification. In Gold's model the chain $\text{I}\Sigma_1 \subset \text{I}\Sigma_2 \subset \dots$ with union PA is a limit point.

Proposition 7.24 (The $\text{I}\Sigma_n$ chain). *[(a) proved, given known facts; (b) proof sketch; (c) refuted]*

pa Prop 4.6; referee M4, m7 (a) For i.i.d. theorems from PA's L1 generator (setting W , parameters admissible), each $\text{I}\Sigma_n = \text{Q} + \sigma_n$ has posterior 0 eventually and $\pi_n(C_{\text{PA}}^*) \rightarrow 1$ a.s. in any countable class with $\pi(\text{PA}) > 0$; the waiting times are presumably astronomical (a heuristic). (b) Before that, σ_n pays at each induction use for a Tarski biconditional that grows with the motive, so the schema wins linearly on induction-use data. (c) Refuted: "if the community uses only Σ_n motives, every DT° theory covering its practice contains PA-strength induction, so the posterior lands on the L_∞ side".

(a) uses $\text{I}\Sigma_{n+1} \vdash \text{Con}(\text{I}\Sigma_n)$ (known; Hájek and Pudlák, 1993, location not verified), Gödel's second theorem and Theorem 4.2. (c) fails because AS Thm 5.16 concerns a single covering template and the wrappers of Proposition 7.14 need a formula metavariable (AS Thm 5.16 does hold with parameters, contrary to a remark of the referee). What is true, using that equivalent motives give equivalent induction instances (Lemma F.3):

Proposition 7.25 (Read-once induction templates). *[proved]^{pa Prop 4.8}* Let $\tau = T_{\text{Ind}}[P := \lambda x.C]$ with C read-once: built by connectives and quantifiers from atoms $A(x, \bar{z}) = B(x, \bar{z})$ (A, B term metavariables) and $F(x, \bar{z})$ (F a formula metavariable), each metavariable occurring once. (a) If C has a formula metavariable, $\text{Q1} + \text{inst}(\tau) \vdash \text{Ind}(\psi)$ for every ψ . (b) Otherwise every instance has C 's formula skeleton, so $\text{Q} + \text{inst}(\tau) \subseteq \text{I}\Sigma_k$ for some k , and $\text{I}\Sigma_k \subsetneq \text{PA}$.

The read-once condition is sufficient, not necessary (Remark F.4); which other templates give full induction is open.

Example 7.26 (Narrow practice ends on the weaker side). [computed]^{pa §4.3 (1); referee r8} Practice: $\mathbf{Q}$ plus induction on motives $\exists z(s = t)$ and $\neg(s = t)$. $H_{\text{narrow}} := \mathbf{Q} + \text{Ind}(\lambda x. \exists z A(x, z) = B(x, z)) + \text{Ind}(\lambda x. \neg C(x) = D(x))$ (term metavariables) covers it and lies in $\mathbf{I}\Sigma_1 \subsetneq \mathbf{PA}$. $\text{CL}(H_{\text{narrow}}) - \text{CL}(T^*) = +230.8, +212.5, +192.5, +174.6$ bits at $n = 100, 300, 1000, 3000$, matching Proposition 7.12(a) to 0.1 bit; slope $\frac{9-8}{2} - 3 \cdot \frac{9-1}{2} = -11.5$ bits per doubling (three fixed formula contexts), crossover $n \approx 2^{26.7}$.

Example 7.27 (G1 keeps PA). [computed; proof sketch (slope)]^{pa §4.3 (2)} u7’s law G1 has 157 motive skeletons, each equivalent to a Σ_2 formula, so a term-only skeleton theory covers G1 inside $\mathbf{I}\Sigma_2$; coverage does not force PA. But it is 42838.5 bits behind at $n = 3000$ (prior part 43040), with asymptotic slope $\frac{164-8}{2} - 19 \cdot \frac{9-1}{2} = +2.0$ bits per doubling (19 contexts; Wilks’s theorem, from memory).

So deductive strength beyond usage is set by an Occam balance: a term-only theory gains $\frac{|A_c|-1}{2}$ bits per doubling for each context it fixes and pays $\frac{1}{2}$ per extra template. Narrow practice lands on the weaker side, broad practice on PA’s; this is the Bayesian L_∞ -versus- L_5 choice (Proposition 4.9).

Conjecture 7.28 (Broad practice). [conjecture]^{pa Conj 4.10} *If the motives have unbounded depth and use every connective at every depth with positive probability, the mass on theories deriving all of PA tends to 1. (Open because covering theories may use non-read-once templates.)*

Track experiments (fixed PCFG, L0, causal pools; language with $<$ and nine motive roots) finds the narrow effect with atomic motives. Let T_f be T_{Ind} with motive root f ; for a connective root, $\mathbf{Q} + T_f$ has the same consequences as $\mathbf{Q} + T_{\text{Ind}}$ (Proposition F.5), and Q3 is redundant given T_{Ind} (Proposition F.6).

Proposition 7.29 (Atomic fragments are weaker). [proved, given two known facts]^{experiments Prop X10} $T^* := \mathbf{Q}1..Q7 + T_{\text{Ind}}$ contains PA and proves $\sigma := \neg \exists x \exists y (\neg y = 0 \wedge x \cdot x = (SS0 \cdot y) \cdot y)$; $\mathbf{Q} + \{T_-, T_<\} \not\vdash \sigma$ (known facts: $\mathbf{PA} \vdash \sigma$; Shepherdson’s model, Shepherdson, 1964, secondary sources only).

Remark 7.30 (E3(b)). [computed]^{experiments E3(b)} L0 data from T^* with misspecified motive laws (5 seeds, $n \leq 2048$). Three connective-rich laws keep the theorems (mass 0.998–1.000 on T^* -equivalents at $n = 2048$); under two of them the MAP leaves T^* for $T^* + T_\exists$ or an observed-root split. Atomic motives: T^* has mass 0.988 at $n = 64$ in 4 of 5 seeds, then $\mathbf{Q} + \{T_-, T_<\}$ has mass 1.000 in 5 of 5 seeds from $n = 1024$ (T^* -equivalents: $2 \cdot 10^{-178}$ at $n = 2048$): the posterior moves to a strictly weaker theory (Table 10).

7.6 Unlabelled mixtures: a Bayesian DTRC

Hypotheses: finite sets of DT° templates, prior 5 bits per symbol, Dirichlet($\frac{1}{2}$) weights, one shared positional grammar, negative data under $\mathbf{R}_0$. Data: G1 practice (Q1–Q7 as ground sentences, induction with probability $\frac{1}{2}$), one seed; hand-picked candidates (Table 5).

Proposition 7.31 (Spare slot). [proved; computed]^{pa Prop 5.1} A never-used template τ added to K templates costs $\beta|\tau| + \frac{1}{2} \log_2 n + \log_2 \frac{\Gamma(K/2)}{\Gamma((K+1)/2)} + o(1)$ bits (Proposition 4.14(a)); 19.03 and 19.82 bits at $n = 1000, 3000$ for $K = 8$, $\tau = F \rightarrow F$, against the computed 19.0, 19.8.

Without negatives $\mathbf{Q} + T_{\text{Ind}}$ has mass $2.6 \cdot 10^{-53}$ at $n = 10$, $2.2 \cdot 10^{-13}$ at 30, and $\geq 1 - 6 \cdot 10^{-6}$ from $n = 100$; with two negatives ($0 = S0$ and a false instance of T_{L_∞}) $\geq 1 - 1.6 \cdot 10^{-5}$ from $n = 10$.

candidate	$n = 10$	30	100	3000
root split ($7 T_f$)	+776.4	+773.6	+772.7	+768.3
spare + ($F \rightarrow F$)	+15.9	+16.6	+17.4	+19.8
memorise instances	+1792.3	+5392.7	+15901.1	+399693.6
over-general $Q + T_{L\infty}$ (AS Prop 5.12)	+549.5	+1500.6	+3832.5	+82717.0
lump $F_0 + T_{\text{Ind}}$	-174.7	-42.0	+319.3	+8929.8
bare F_0	+584.0	+2148.6	+6067.9	+137387.5

Table 5: Bayesian DTRC: $\text{CL}(\text{candidate}) - \text{CL}(Q + T_{\text{Ind}})$ in bits on G1 practice; one seed; hand-picked candidates. Computed (pa §5.2; all n in Appendix F.7).

The root split and the spare are PA-equivalent, so the mass on PA-equivalents is $\geq 1 - 10^{-96}$ from $n = 100$, *within this candidate set*. The split gains 1 bit per doubling (Proposition 7.12), the spare loses $\frac{1}{2}$; the rest lose linearly.

Remark 7.32 (Unsound lumps). [computed]

pa §5.4; experiments E2 At $n = 10, 30$ the unsound lump $F_0 + T_{\text{Ind}}$ leads by 174.7 and 42.0 bits (F_0 codes the rarely seen Q axioms through the grammar for less than seven ground templates cost), and $V_{\delta,0}$ accepts the false $0 = S0$ for every $\delta \geq 3 \cdot 10^{-13}$; the negative $0 = S0$ refutes the lump. This is AS’s “errs towards unsoundness for rarely used ground axioms” (AS App. E.7) in Bayesian form, and no contradiction with Theorems 5.6 and 5.7 ($\pi(Q + T_{\text{Ind}}) = 2^{-330}$ here; u7’s data are not from a model generator). One run in a hand-picked set; for the frequency see E2.

Remark 7.33 (E2). [computed]

experiments E2 L0 citations from T^* at fixed weights, 25 seeds, $n \leq 512$, causal pools. T^* takes the mass once Q1, Q2, Q4–Q7 have each been cited (median datum 55); before that the sound but weaker SeenQ(D_n) leads, a competitor pa did not score (data laws and candidate sets differ). Mean mass of T^* : 0.751 at $n = 64$, 0.996 at 512. A $\delta = 0.05$ verifier accepts a false probe in 5 of 25 seeds, at $n \leq 16$ (Remark 5.11). The complete root split stays about 700 bits behind; the false spare $T^* + \{0 = S0\}$ keeps 0.004 at $n = 512$ (Table 8).

Remark 7.34 ($\forall x\varphi$ as a sound merge). [computed; identity proved]pa §5.5; referee M5 On data $t + 0 = t$ (t random closed terms) the schema $z + 0 = z$ wins (AS Prop 6.13). The sound split by the head of t covers every closed instance but does not entail $\forall x(x + 0 = x)$; it closes in at 5 bits per doubling (Proposition 7.12(c), $\frac{3-13}{2}$), crossover near $n = 10^8$: a Bayesian ω -gap (Theorem 3.9), slower than in Proposition 3.10 because the parameter probability is learned. $\forall x(x + 0 = x)$ itself loses at a rate set by the rule code (Table 32): $\log_2 10$ bits per datum under the fixed code (the first version’s “ $2^{-3.3n}$, as H4 predicts” holds only for it), $\frac{R-1}{2} \log_2 n + O(1)$ under a learned depth-indexed code, $h(f)$ bits per datum in a practice with a fraction f of φ -instances (h the binary entropy). The direction holds under every code (Theorem 3.4; universal’s learned shared c : Proposition B.1).

Proposition 7.35 (“Does not contradict” learns nothing). [proved]

pa Prop 5.2 If the likelihood is $\mu_0(s)[T \not\models \neg s]$ with μ_0 independent of T , the posterior is the prior restricted to compatible theories.

L_ε is the minimal model that credits derivations without killing gaps, as Hänni asks (Proposition 2.22(b)).

Proposition 7.36 (Inconsistency is invisible to the size principle). [proved]pa Prop 5.3 Under $L1^{\max}$ with citation weights, adding σ with fixed weight fraction w gives $P_{T \cup \{\sigma\}}(s) \geq (1 - w)^{c(s)} P_T(s)$, $c(s)$ the citations in s ’s best T -derivation; with Dirichlet weights an unused σ costs its prior plus about $\frac{1}{2} \log_2 n$ bits. Consistency must be enforced by R_d .

	what fails	evidence	fixable?
F1	usage over logic	Proposition 7.5 and Remark 7.7	change the target
F2	misspecified grammar: linear fragmentation	Table 30	shared positional grammar
F3	never-used parts split off (ω -gap)	Proposition 7.12 and Remark 7.34	harmless for theorems (Proposition 7.14)
F4	false generalisation, no counterexamples	Example 7.37	only by refutation
F5	unsound lumps at small n	Remarks 7.32 and 7.33	a negative datum, more data
F6	inconsistency unseen by the size principle	Proposition 7.36	only R_{d_n} , not certifiable
F7	no well-specified theory for $\text{Th}(\mathbb{N})$	Proposition 7.20 and Remark F.2	as regret (Proposition 7.23)
F8	strength beyond usage set by Occam terms	Example 7.26 and Remark 7.30	broad practice (Conjecture 7.28)
F9	bounded search memorises long proofs	Section 7.4	not in general (Proposition 2.24)
F10	theorem data memorised, $\text{MAP} \subsetneq \text{PA}$	Proposition 7.15 and Remark 7.16	schema uses, steep prior

Table 6: Robust failures (pa §6); statuses as in the cited items.

7.7 Robust failures and the overall answer

Example 7.37 (Euler’s polynomial (F4)). [computed; proof sketch (covering, eventual behaviour)]
pa §6 F4; referee m9 Data: $\text{Prime}(k \cdot k + k + 41)$, k geometric(0.9) conditioned on primality (false k : 40, 41, 44, ...). The false schema $\text{Prime}(z \cdot z + z + 41)$ beats memorisation by 51445 bits at $n = 100$ and 314225 at $n = 10^6$ (flat grammar); no finite unguarded covering union is true without memorising. Under a depth-indexed grammar the predictive probability of $k = 40$ falls like $1/n$, yet the schema entails a falsehood; under the flat grammar memorisation should win beyond $n = 10^6$.

7.7.1 The overall answer

The inducer does *not* robustly pick up the textbook axioms; it picks up the axioms that are *used*. Within the hand-picked candidate sets, and for data that are direct uses of axioms, it puts nearly all mass on theories deductively equivalent to PA ; in three of the four cases this is automatic, since every candidate is PA -equivalent (usage mixes, recursion conventions, root splits), and in the DTRC run PA -equivalents get mass $\geq 1 - 10^{-96}$ from $n = 100$. It fails robustly on theorem data, narrow practice, absent counterexamples, small n and inconsistency (Table 6). Not scored: all finite template unions, non-read-once templates, stronger true theories such as $\text{PA} + \text{Con}(\text{PA})$ (spare slots), further ZF candidates. The first version’s “it robustly gets a theory deductively equivalent to PA ” is refuted as stated (referee M2–M4).

8 Experiments: a small exact laboratory

Track experiments wrote a Python package, `bai` (`code/bai`), that computes exact posteriors over finite pools of theories made of DT° templates, reusing the code of AS (Claude (Anthropic), 2026a) for syntax, matching, Min and DTRC, and ran eight experiments, E1–E8. It is a laboratory for checking the claims of §§3–7 on small cases, not a prover. Three facts frame every number.

Each posterior is exact *over its pool*, i.e. conditioned on “the theory is in the pool” (Remark 8.3). The derivation likelihood is a chain of at most two steps, so induction followed by MP and $\forall E$ is outside it, and the PA runs use L0. And E2, E3(b), E5(b), E8 sample multi-component theories at one fixed weight vector, where Theorem 4.2 does not cover the Dirichlet posterior; there the runs are a yardstick, not an instance of a theorem (Remark C.3). Code lengths are in bits, $\alpha = \frac{1}{2}$, every run is seeded (Appendix G.5), and all findings below are computed unless a proposition is cited.

8.1 What is implemented

A *component* is a DT° template with a closed or open *guard* on each metavariable (may its body contain the parameter?); a theory is a finite set of components. The language is L_A ; no axiom of Q mentions $<$. There is one parameter p : a datum $\varphi(Sp)$ means $\forall y \varphi(Sy)$, and data with two parameters have probability 0. The instantiation grammar $\mathcal{Q}$ is a fixed PCFG over bodies (Appendix G.1) that satisfies Definition 2.6 and counts holes and bound variables alike, so splitting a schema by the root of a body is an exact reparametrisation (Proposition 4.6(b), Remark C.2). The prior is $\pi(T) \propto 2^{-\lambda \text{bits}(T)} (1 + |T|)^{-\tau}$, where $\text{bits}(T)$ is the length of T in a prefix code that satisfies Kraft’s inequality, $|T|$ is the template size and τ is the time exponent of Remark A.1; by default $\lambda = 1$ and $\tau = 0$. For instance $\{z + 0 = z\}$ costs 17.1 bits, $\{\forall x (x + 0 = x)\}$ 17.2 bits and $T^* := Q1..Q7 + T_{\text{Ind}}$ 226.2 bits.

The mixture weights get a $\text{Dir}(\frac{1}{2})$ prior and are integrated out. L0 is citation: $P(s \mid T, w) = \sum_i w_i a_i(s)$ with $a_i(s) = \prod_M \mathcal{Q}(\theta_s(M))$ for the unique matcher θ_s . L1 is the forward chain Ch_J (Definition 2.3; rules in Appendix A.3): cite component i with probability w_i , then take at most $J \in \{1, 2\}$ steps of $\forall E$ (term from $\mathcal{Q}_e$), Gen on p , or MP with a *cited* major premise, stopping with probability $c_{\text{stop}} = \frac{1}{2}$ while a rule applies; $c_{\text{ch}} := 1 - c_{\text{stop}}$. The chain never fails, so it needs no normaliser, and its law is linear in w . L1^{sel}_{cit} observes the chain through “closed quantifier-free sentences only” by per-citation rejection, a_i/c_i with c_i the probability that a chain from component i ends there (Remark A.6). The marginal $P_T^{\text{Dir}}(D_n)$ is computed exactly by a dynamic programme over count vectors (Lemma 2.8). Above 20000 states rigorous bounds replace it; the largest posterior mass a theory so bounded could have had never exceeded $8 \cdot 10^{-14}$ (Appendix G.4).

The pool at sample size n holds: hand-picked theories; theories built *causally* from D_b at build points $b \leq n$, namely Min of random subsets of D_b (AS Thm 3.3), skeleton clusters (DTRC without refutation, stopped at k clusters; AS §7.1) and, in PA runs, DTRC with the PA refuter and the observed-root split; in PA runs, $\text{Trim}(T, D_n)$ for every pool theory T , the components that could have produced a seen datum, so that $\text{Trim}(T^*, D_n) = \text{SeenQ}(D_n)$ is T_{Ind} plus the Q axioms cited so far; and $\text{Mem}(D_n)$. So every member uses only data already seen, which the track’s first version did not ensure (Remark 8.4). $T \vdash_J s$ iff some chain of at most J steps from T , with all rules enabled, ends in s : a sound under-approximation of derivability, Th_d of Definition 2.2 with d in chain steps. $V_{\delta, J}$, the verifier $V_{\delta, d}$ of Definition 5.1 on the pool, accepts s iff the pool posterior of $\{T : T \vdash_J s\}$ is at least $1 - \delta$; $J = 1$ in E1, E3(a), E4 and $J = 0$ in E2, E3(b), E7.

Proposition 8.1 (The computed likelihoods are exact). [*proved; computed*]

experiments Props X1–X4 (a) For a component τ and a sentence s with at most the parameter p , citing τ with independent bodies from $\mathcal{Q}$ yields s with probability $\prod_M \mathcal{Q}(\theta_s(M))$ if τ matches s by a matcher θ_s that respects the guards, and 0 otherwise. (b) For $a_{ij} \geq 0$, $\mathbb{E}_{w \sim \text{Dir}(\alpha)} \prod_{j=1}^n \sum_{i=1}^K w_i a_{ij} = \sum_z \prod_j a_{zjj} \cdot \frac{\Gamma(K\alpha)}{\Gamma(K\alpha+n)} \prod_i \frac{\Gamma(\alpha+n_i(z))}{\Gamma(\alpha)}$, summed over assignments z of the data to the K components, with counts $n_i(z)$. The dynamic programme computes this sum, and the bounds bracket it. (c)

data	lik.	mass of the generating theory at $n = 4, 8, 32, 256$	MAP at 256	$P(\vdash_1 \forall x\varphi)$	first n
<i>sch</i>	L0	0.87, 0.97, 1.000, 1.000	H_{sch}	$2 \cdot 10^{-7}$	16
<i>sch</i>	L1	0.84, 0.96, 1.000, 1.000	H_{sch}	$3 \cdot 10^{-7}$	16
<i>sch</i>	L1 ^{sel} _{cit}	0.43, 0.42, 0.53, 0.59	H_{sch}	0.484; 0.269	never
<i>all</i>	L1	0.87, 0.99, 1.000, 1.000	$H_{\forall}$	1.000	16
<i>allq</i>	L1	0.73, 0.90, 1.000, 1.000	$H_{\forall}$	1.000	16
<i>open</i>	L1	0.89, 1.000, 1.000, 1.000	H_{open}	1.000	8
<i>allq2</i>	L1	0.88, 0.997, 1.000, 1.000	$H_{\forall}$	1.000	16

Table 7: E1. Means over 5 seeds and, where the three formulas agree, over φ . $P(\vdash_1 \forall x\varphi)$ at $n = 256$; under L1^{sel}_{cit} it is 0.484 for $x + 0 = x$ and $0 + x = x$ and 0.269 for $\neg Sx = 0$. Last column: first n from which the generator keeps mass ≥ 0.99 , worst run. Under L0 (rows omitted) $H_{\forall}$ dies at its first instance; on *all* data H_{both} has mass ≥ 0.995 from $n = 32$, on *allq* data $H_{\forall} \cup H_{\text{open}}$ has 1.000 from $n = 16$, and on *open* and *allq2* data $\text{Mem}(D_n)$ has 1.000 from $n = 16$ and 64; $P(\vdash_1 \forall x\varphi) = 1.000$ at $n = 256$ in all four.

For every theory, weight vector and J , the law of the chain's output is a probability on sentences.
(d) *The backward recursion over predecessors computes the L1 coefficients $a_i(s)$ exactly whenever the predecessor sets it enumerates are complete. They are complete for the component shapes of Appendix G.2, and the code admits into L1 pools only theories of the shapes it checks (this left out 65 and 43 data-derived theories in two E1 configurations, none elsewhere).*

Proof idea. (a) is the matching theorem, AS Thm 2.5. (b) expands the product and takes Dirichlet moments (Lemma 2.8); the upper bound uses concavity of the log-likelihood on the simplex. (c) Every step chooses among finitely many outcomes of total probability 1, and no run fails. (d) Each predecessor set is in bijection with the matchers of a derived DT^o template, which are unique. Details in Appendix G.2. $\square$

The code agrees with brute force on 500 random Dirichlet sums ($1.4 \cdot 10^{-14}$) and with the referee's independent reimplementations, written from the notes, on 400 marginals ($1.4 \cdot 10^{-14}$) and 904 L1 coefficients at $J = 1, 2$ ($2.1 \cdot 10^{-14}$); a forward sampler passes χ^2 tests; 21 unit tests pass.

8.2 Results

E1: $\forall x\varphi$ from data about φ . For $\varphi \in \{x + 0 = x, 0 + x = x, \neg Sx = 0\}$, seeds 0–4 and $n = 1, 2, 4, \dots, 256$, five generators are used: *sch* (L0 citations of $\sigma_{\varphi} = \varphi(z)$, closed terms); *all* (Ch₁ from $H_{\forall}$, closed $\forall E$ terms, so about half of the data are $\forall x\varphi$ itself); *allq* (the same with open terms, e.g. $\varphi(Sp)$); *open* (Ch₁ from H_{open} , whose Gen steps emit sentences such as $\forall y \varphi(Sy)$); and *allq2* (Ch₂ from $H_{\forall}$: $\forall E$ with an open term, then Gen). The pool (15–28 theories) holds $H_{\forall}$, H_{sch} , H_{open} , H_{both} , $H_{\forall} \cup H_{\text{open}}$, the over-specific $\{\varphi(0), \varphi(Sz)\}$, two root splits, three spare-slot theories, over-general generalisations of $\varphi(z)$ and F_0 , causal theories and $\text{Mem}(D_n)$. The likelihoods are L0, L1 with the generator's own chain, and L1^{sel}_{cit} on *sch* data. Every generator has one component, so Theorem 4.2 applies (Table 7).

(1) *The factor of Theorem 3.3, exactly.* In Ch _{J} with closed $\forall E$ terms, $H_{\forall}$ emits $\forall x\varphi$ with probability c_{stop} and otherwise $\varphi(t)$ with probability $c_{\text{ch}} Q(t)$, after which no rule applies; H_{sch} emits $\varphi(t)$ with probability $Q(t)$. So on closed instances the log₂ odds $H_{\text{sch}} : H_{\forall}$ gain $\log_2(1/c_{\text{ch}}) = 1$ bit per datum under L1 and stay at the prior odds under L1^{sel}_{cit} ($c_{\forall} = c_{\text{ch}}$, $c_{\text{sch}} = 1$); this is experiments Prop X6 (proved), the Ch _{J} case of the *unnormalised* row of Theorem 3.3. Computed: $n + 0.090$ bits ($x + 0 = x, 0 + x = x$) and $n + 1.440$ ($\neg Sx = 0$) at every n and seed, within $1.1 \cdot 10^{-11}$

n	T^*	equiv.	weaker	unsound	acc.	MAP [tag] (count of 25)
8	$2 \cdot 10^{-21}$	$4 \cdot 10^{-13}$	0.744	0.256	2	skel4@8 [w] 12, DTRC@8 [w] 6, Trim(Q-lumped) [u] 4, ...
16	$9 \cdot 10^{-7}$	$5 \cdot 10^{-6}$	0.803	0.197	3	skel6@16 [w] 9, skel4@16 [w] 5, Q-lumped [u] 3, ...
32	0.197	0.197	0.727	0.077	0	$T^* - Q2$ [w] 7, skel6@32 [w] 6, T^* 5, ...
64	0.751	0.791	0.200	0.009	0	T^* 19, $T^* - Q3$ [e] 1, ...
128	0.991	0.991	0	0.009	0	T^* 25
512	0.996	0.996	0	0.004	0	T^* 25

Table 8: E2: posterior mass by tag (means over 25 seeds). “Acc.”: seeds in which $V_{0.05,0}$ accepts a false probe. Tags: [e] equivalent to T^* , [w] sound and strictly weaker, [u] unsound. At $n = 512$ the unsound mass is the false spare $T^* + \{0 = S0\}$.

bits of n plus the prior difference from the referee’s code; the $L1_{\text{cit}}^{\text{sel}}$ tie holds to 10^{-13} . (2) *The ω -gap*. Under $L1_{\text{cit}}^{\text{sel}} P(T \vdash \forall x \varphi \mid D_n)$ tends to the prior share, in this code, of the provers among the theories tied with H_{sch} (Theorem 3.9(b), Table 16), while held-out closed instances get mass ≥ 0.999 from $n = 8$ in every run except *open* data under L0 (Theorem 3.8). (3) *The size principle*. At $n = 8$ on *sch* data, $\text{Mem}(D_n)$ has mass $\leq 2.4 \cdot 10^{-12}$ and the over-general theories $\leq 7 \cdot 10^{-5}$, both falling exponentially (Theorem B.4). This is one memoriser; the memoriser class can decay more slowly (Proposition B.6). Under L1 $\text{Mem}(D_n)$ is not provably the best memoriser, since a ground $\forall$ -sentence can raise its likelihood (the referee found $\text{Mem}(D_n) \cup \{\forall x (x + 0 = x)\}$ 11–16 bits worse, the margin shrinking). (4) *Quantified data*. From a prover of $\forall x \varphi$, L1 finds the generator by $n = 16$ and $P(T \vdash \forall x \varphi \mid D_n) \rightarrow 1$ (Proposition 3.10(a), Proposition B.14(a)). Under L0 derived sentences can only be axioms, so theories that list them win, and with Gen outputs of unbounded variety L0 ends at $\text{Mem}(D_n)$.

E2: PA without labels. Data are L0 citations from $T^* = Q1..Q7 + T_{\text{Ind}}$ (numbering of §2) at fixed weights 0.05 per Q axiom and 0.65 for T_{Ind} , with one-hole motives from $\mathcal{Q}$; seeds 0–24; $n = 8, \dots, 512$; the likelihood is L0 with the same $\mathcal{Q}$. Besides the causal theories, $\text{Trim}(T, D_n)$ and $\text{Mem}(D_n)$, the pool holds T^* , $T^* - Q_i$, *frag-complete* (Q plus the nine root fragments T_f), *frag-atoms* ($Q + \{T_=&, T_<\}$), the spares $T^* + T_\wedge$, $T^* + \{\forall x (0 + x = x)\}$ and $T^* + \{0 = S0\}$, *IndSwap* (Q plus a swapped, equivalent induction template), two over-general induction templates, *Q-lumped* = $\{\forall x \forall y P(x, y), \forall x P(x), T_{\text{Ind}}\}$ and F_0 . Tags come from Propositions 7.29, F.5 and F.6 and a budgeted refuter (“unsound” is certain). The weights are fixed, so the applicable soundness statement is Theorem 5.7, not Theorem 5.6 (Table 8; SeenQ and code lengths in Table 33).

(1) T^* wins once Q1, Q2, Q4–Q7 have each been cited (the last at datum 20–97, median 55; Q3 is redundant by Proposition F.6); the mass on its equivalents is ≥ 0.95 from $n = 32, 64$ or 128 (5, 15, 5 seeds). Before that the sound but weaker $\text{SeenQ}(D_n)$ leads, since an uncited axiom costs its prior bits and the factor of Proposition 4.14(a). (2) At $n \leq 16$ *Q-lumped* or its trim beats SeenQ in some seeds (by up to 12.7 bits), and $V_{0.05,0}$ then accepts $\forall x (x + 0 = 0)$, $\forall x (Sx = x)$, $\forall x \forall y (y = x)$: in 5 of 25 seeds, never from $n = 32$ (19 of 25 with the first version’s pool, Table 33). This is consistent with Theorem 5.7, which needs $\delta_n \leq 2^{-226} R_\alpha(n, 8) \delta' \asymp 2^{-226} n^{-3.5}$ (Remark 5.11). (3) *frag-complete* has T^* ’s instance set and its laws include T^* ’s (Proposition 4.6(b)), so it loses the prior difference (698.9 bits) plus 4.2 bits per doubling of n ($n = 64 \dots 512$; asymptotically $(16 - 8)/2 = 4$, the Occam term of Proposition 4.10(c)). (4) Unused spare sentences lose 0.49 bits per doubling and the nested $T_\wedge$ 0.25 (Proposition 4.14(a), (c)); the inconsistent $T^* + \{0 = S0\}$ keeps 0.004 at $n = 512$. (5) The equivalent but uncited *IndSwap* has likelihood 0 from the first induction datum: citation identifies the cited axioms. Over-general induction templates lose

data	MAP at the largest n (5 seeds; L0 and L1 alike)	inst. set	gain	$P(\vdash \text{ held-out})$	superset
<i>heavy</i>	C_2 (3 seeds), $\{\varphi(z), \varphi(SSz)\}$ (2)	same	$212 \rightarrow 455$	1.000	$3 \cdot 10^{-63}$
<i>numerals</i>	R_9 or R_{10} at $n = 512$; R_{11} or R_{12} at 1024	subset	$1271 \rightarrow 2897$	0.333	0
<i>small</i>	H_{sch} up to $n = 512$; $\text{Mem}(D_n)$ at 1024 (L0)	same $\rightarrow$ finite	$0 \rightarrow 156$	$1.000 \rightarrow 0.000$	$7 \cdot 10^{-5}$
<i>skewQ</i>	frag1 , a root split (from $n = 512$)	same	$70 \rightarrow 212$	1.000	$2 \cdot 10^{-20}$

Table 9: E3(a). Gain: code length of H_{sch} minus that of the MAP under L0, bits, $n = 512 \rightarrow 1024$ (means). $P(\vdash \text{ held-out})$: mean mass deriving $\varphi(9)$, $\varphi(2 + 3)$, $\varphi(1 \cdot 4)$. Superset: largest mass on over-general theories at the largest n over seeds and likelihoods. Families: Table 34.

motives	MAP at $n = 256 / 1024 / 2048$ (5 seeds)	equiv.	weaker	T^*	unsound at 16
<i>dtrc-motive</i>	$T^* / T^* + T_{\exists} / T^* + T_{\exists}$	1.000	0	$5 \cdot 10^{-70}$	0.25; 1.0
<i>root-skew</i>	$T^* + T_{\rightarrow} / \text{frag-observed@16, @32} / \text{same}$	1.000	0	0	0.52; 1.0
<i>deep</i>	$T^* / T^* / T^*$	0.998	0	0.998	$2 \cdot 10^{-11}$; $7 \cdot 10^{-11}$
<i>atomic</i>	frag-atoms (4), T^* (1) / frag-atoms / frag-atoms	$2 \cdot 10^{-178}$	1.000	$4 \cdot 10^{-298}$	0.21; 0.999

Table 10: E3(b). Masses at $n = 2048$; unsound mass at $n = 16$ as mean; max over seeds. Code lengths relative to T^* at $n = 2048$: *root-skew* **frag-observed@b** -1387 to -1509 bits; *dtrc-motive* $T^* + T_{\exists}$ -228 to -318 ; *deep* **frag-complete** $+665$ to $+674$; *atomic* **frag-atoms** -986 to -1028 .

linearly.

E3: misspecified usage. All data are theorems of the target, but their law is no pool theory’s (Theorem 4.16). (a) Instances of $\varphi = (x + 0 = x)$ under other term laws: *heavy* (a zeta(1.5) numeral with probability 0.7, else $S^k(t_1 \circ t_2)$), *numerals* ($S^k 0$, $k \sim \text{Geom}(0.2)$), *small* (Q -terms with at most 4 symbols: 12 distinct sentences) and *skewQ* (another PCFG); L0 ($n \leq 1024$) and L1 (Ch_1 , $n \leq 512$); seeds 0–4. The pool adds to E1’s hand pool the closed-guard numeral splits $R_m = \{\varphi(0), \dots, \varphi(S^{m-1}0), \varphi(S^m z)\}$, $m \leq 16$, and the nested $C_1 = \{\varphi(z), \varphi(Sz)\}$ and $C_2 = C_1 \cup \{\varphi(SSz)\}$. (b) E2’s T^* and weights with other motive laws: *dtrc-motive* (AS’s motive generator), *root-skew* (roots $\rightarrow, \wedge, =, \exists$ with .5, .2, .2, .1), *deep* (more connectives) and *atomic* ($a(x) = b(x)$ with probability 0.6, $a(x) < b(x)$ with 0.4); L0; $n \leq 2048$; E2’s pool plus $T^* + T_f$ for every root f (Tables 9 and 10).

(a) The posterior moves at a linear rate to the best-fitting member of a family; the winner keeps changing as n grows. On numerals the best split deepens (R_4/R_5 at $n = 128$, R_9/R_{10} at 512, R_{11}/R_{12} at 1024, interior to the family), like the drift of Proposition 3.12(d) in track universal’s calculus; $n = 1024$ is too small to test its part (b). On heavy tails the deepest nesting available wins, and a deeper one would probably win if the exact programme could hold it (conjecture). Nothing unsound survives at large n , but completeness is lost: with numerals only the held-out $\varphi(2 + 3)$, $\varphi(1 \cdot 4)$ are not derived, and with a finite selected support the axioms become the 12 observed sentences. As in Example 4.17, all data are theorems of the target and the posterior goes to a strictly weaker theory. L1 gives the same MAPs as L0 for $n \leq 512$. (b) Under connective-rich motive laws the winners contain T_{Ind} or a connective fragment and Q1, Q2, Q4–Q7, so they are equivalent to T^* (Proposition F.5); yet under *dtrc-motive* and *root-skew* the MAP leaves T^* : the theorems survive, the axioms do not. Under *atomic* motives T^* has mass 0.988 at $n = 64$ in 4 of 5 seeds, and then $Q + \{T_=: T_<\}$ takes mass 1.000 in every seed from $n = 1024$. That theory does

not prove the irrationality of $\sqrt{2}$, which T^* proves (Proposition 7.29): the posterior moves *away* from T^* to a strictly weaker theory (Remark 7.30). Unsound lumps recur at $n = 16$.

E4: an adaptive prover. The prover wins if $V_{\delta,1}$ ever accepts some s with $T^* \not\vdash_1 s$. T^* always has one component, so Theorem 5.6 bounds the win probability by δ' when $\delta = w^*\delta'$ (Table 35). (A) The tight construction of IL Prop C.5 (Claude (Anthropic), 2026b) with a Dirichlet decoy: $T^* = \{\varphi(z)\}$ on numerals with $P(t = 0) = 1 - u$, decoy $T' = \{\varphi(0), 0 = S0\}$, prior w^* on T^* ; the prover asks $0 = S0$ every round. It wins iff the first t^* data are $\varphi(0)$, with t^* the first n at which $\pi_n(T^*) \leq \delta$ given $\varphi(0)^n$, so $P(\text{win}) = (1 - u)^{t^*}$ exactly. (B, C) $\varphi = (x + 0 = x)$, $T^* = H_{\text{sch}}$, E1's hand pool plus $\text{Mem}(D_n)$; $w^* = 7 \cdot 10^{-6}$ (the pool version of Theorem 5.6), bound 0.05; each round the prover asks eight sentences that T^* does not derive; 100 streams, every $n \leq 256$, L0 and L1. Well specified, wins stay below δ' : 9–46 times below in (A), whose Dirichlet decoy is slower than IL's fixed-weight one (Remark D.1), and 0/100 in (B), where the largest invalid mass (0.45, 0.53) sits on the true but underivable $\forall x (x + 0 = x)$ at $n \leq 6$. Misspecified, the prover wins every time when 10% of the data are false near misses $t + 0 = St$ and the mistake schema is in the pool (C1), and when only a decoy fits selected data better ((A) with only $\varphi(0)$ shown); heavy tails (C2) and selection with a clean memoriser in the pool (C3) did no harm. No theorem covers the misspecified case (Remark 5.10).

E5: spare slots, L_∞ against L_5 , and Gold's text (Table 36). (a) Exact Bayes factors of $\{\varphi(z)\}$ plus a spare against $\{\varphi(z)\}$ on its own L0 data: the unused spares ($0 = S0$; $z \cdot 0 = 0$) match Proposition 4.14(a) to all printed digits (slope -0.500 per doubling); the nested $\varphi(Sz)$ has slope -0.281 on $n = 256 \dots 4096$ and, by quadrature, -0.251 ± 0.002 on $10^2 \dots 10^8$, supporting the $n^{-1/4}$ of Proposition 4.14(c), itself a proof sketch. A false spare sentence is thus removed by positive data only like $n^{-1/2}$. (b) With L_k the k ground sentences $\varphi(S^j 0)$, $j < k$, and $L_\infty = \{\varphi(z)\}$ under Q_{num} : on data from L_5 the posterior switches to L_5 between $n = 64$ and 128 (mass 1.000 from 256, 10/10 seeds), and on data from L_∞ it keeps L_∞ (1.000 from $n = 16$). Proposition 4.9(a) proves this for fixed generators; here L_5 has Dirichlet weights and is sampled at one weight vector, so the run is a yardstick (Remark C.3). (c) Against the MAP learner over $\{L_1, \dots, L_{60}, L_\infty\}$ the first 14 stages of Gold's construction all end (740 data): the start of a text on which, if every stage ends, the MAP changes infinitely often (Proposition 4.9(b), (c)).

E6: generating axiomatisations. Eight forms of commutativity of $+$ are compared.

Proposition 8.2 (Forms of commutativity). *[proved; computed]*

experiments Prop X12 Let $A_{xy} := \{\forall x \forall y (x + y = y + x)\}$, $A_{yx} := \{\forall y \forall x (x + y = y + x)\}$, $S_{ab} := \{z + z' = z' + z\}$, $M_x := \{\forall x (x + z = z + x)\}$ and $M_y := \{\forall y (z + y = y + z)\}$, with closed guards, and let S_{ab}^{open} , M_x^{open} , M_y^{open} be the same templates with open guards. (a) A_{xy} , A_{yx} , M_x^{open} and M_y^{open} are logically equivalent. (b) $S_{ab} \cup M_x \cup M_y \cup S_{ab}^{\text{open}} \not\vdash \forall x \forall y (x + y = y + x)$. So S_{ab} , M_x , M_y and S_{ab}^{open} are strictly weaker than A_{xy} , although S_{ab} , M_x and M_y yield exactly its closed instances $t + t' = t' + t$.

The proof (Appendix G.3) uses the structure $\mathbb{N} \sqcup \{\alpha, \beta\}$ with $\alpha + \beta = \alpha \neq \beta = \beta + \alpha$. Part (a) corrects a remark of the track's referee that open guards cannot reach $\forall x \forall y$ under the single-parameter convention: one bound variable and the parameter already give two variables. The fixed pool is the eight forms, $A_{xy} \cup A_{yx}$, $A_{xy} \cup S_{ab}$ and $\text{Mem}(D_n)$. Data come from Ch₂ (closed $\forall E$ terms) on A_{xy} and on M_x , and from L0 citations of S_{ab} ; seeds 0–4; $n \leq 256$ (Table 37).

L1 identifies the generating form by $n = 32$ in every run, among logically equivalent and among instance-equivalent weaker forms (compare Proposition 4.5); A_{xy} is told from A_{yx} and the open forms by partial instances such as $\forall y (2 + y = y + 2)$. Under $L1_{\text{cit}}^{\text{sel}}$ the five closed-guard forms have equal likelihoods on every datum (as in E1(1)), so the mass on provers of commutativity tends to their prior share 0.088, not to 1: the ω -gap of Theorem 3.9 again, not a tie among equivalents. A redundant second form is a spare slot ($< 10^{-9}$; Remark 7.10).

E7: time factor and steeper penalty (E2’s data and pools, 25 seeds; Table 39). The time factor costs $\tau \log_2(1 + |T|)$ bits (T^* has 77 symbols), so at $\tau = 1$ it moves the log-odds between T^* and any hand competitor by at most $\log_2(78/2) \approx 5.3$ bits. It changes nothing visible from $n = 64$ on; at $n \leq 32$ it slightly favours small unsound lumps (unsound MAP at $n = 8$ in 9 instead of 7 seeds at $\tau = 1$; largest change of a reported mass 0.181 at $\tau = 1$, 0.563 at $\tau = 4$). This follows from the factor’s definition and does not test the brief’s H6 (Proposition 6.8 and Theorem 6.9). The steepness $\lambda = 2$ makes early unsound lumping far more frequent (unsound MAP at $n = 8$ in 22 seeds) but removes the false spare faster; $\lambda = \frac{1}{2}$ does the reverse (Remark 6.15).

E8: splits under a derivation likelihood. Let $\varphi(t) := (t+0 = t)$, $\psi(t) := (0+t = t)$, $\text{whole} := \{\varphi(z), \psi(z), \varphi(z) \rightarrow \psi(z)\}$, and $\text{split} := \text{whole}$ with $\varphi(z)$ replaced by $\varphi(0), \varphi(Sz), \varphi(z+z'), \varphi(z \cdot z')$ ($K = 4$ root productions). Under Ch_1 a citation of $\varphi(t)$ stops or applies MP with the cited conditional, giving $\psi(t)$, which can also be cited: the component behind a ψ -datum is latent. Data come from Ch_1 on whole at fixed weights (0.5, 0.25, 0.25), with bodies from $\mathcal{Q}$ (*wellspec*) or from another PCFG (*skew*: roots 0, S , $+$, $\cdot$ with .5, .1, .2, .2); seeds 0–9; $n \leq 4096$. Table 38 gives Bayes factors without the prior, which adds 76.4 bits for whole.

At matched weights ($w_{\varphi p_f}$) the two theories give every datum the same L1 probability, to $1.1 \cdot 10^{-13}$ (experiments Prop X13, Remark C.2). With learned weights and well-specified usage the split loses 1.43 bits per doubling under L1 and 1.51 under L0 ($n = 256 \dots 4096$; 1.60, 1.76 on $1024 \dots 4096$), near the $(K - 1)/2 = 1.5$ of conjecture Proposition 4.11(d). With misspecified usage it wins linearly, more under L1 (0.202 against 0.074 bits per datum), because there the term of a ψ -datum also informs the split (Proposition 4.11(c)). E8 does not test Proposition 4.11(b), which concerns a one-component schema with fixed weights. In this model a derivation likelihood does not change the MDL finding (Remark 4.13); what it changes in PA, the consequences of a split (Proposition 7.14), the chain cannot test.

8.3 What the experiments do not show

Remark 8.3 (Limitations). [(i) computed; (ii)–(v) proved (properties of the code)]^{experiments §§11, 12, 16.5}
(i) *Pool, not class.* The winners, and even the frequency of unsound acceptance, changed with the pool: E2’s acceptances fell from 19 to 5 of 25 seeds with causal pools and trimmed theories, and E3(a)’s winners moved when the families R_m, C_k were extended. The pool version of Theorem 5.6 keeps soundness under pool restriction; nothing comparable is shown for identification. Search beyond a pool is open. (ii) *Chains, $J \leq 2$.* MP needs a cited major premise, so no run computes PA under a derivation likelihood. (iii) *One parameter; fixed $\mathcal{Q}$.* A misfit of usage statistics is absorbed by extra components (E3, E8); a learned grammar can change the sign of the Occam drift (Proposition 7.12). (iv) *Bounded derivability.* $\vdash_J$ under-approximates provability, so the $P(T \vdash s)$ columns are lower bounds. (v) *Small runs.* 25 seeds in E2 and E7, 10 in E5 and E8, 5 otherwise; $n \leq 4096$ (quadrature to 10^8). The exact Dirichlet sum is exponential in the number of overlapping components, which also stops E3(a)’s nesting at C_2 . No run uses human-

written theorem lists: every misspecified generator perturbs or selects from the model’s own term or motive law. Neither Hänni’s collapse (Section 6) nor his “do not contradict” variant is implemented.

Remark 8.4 (Claims of the first version). [refuted]

experiments §16.4; referee M1–M4 Each claim is kept with its counterexample. (1) E2: “in 3 of 5 seeds an unsound lump has mass ≥ 0.997 at some $n \leq 32$ and a $\delta = 0.05$ verifier accepts a false sentence”. True of a pool that used unseen data and lacked $\text{SeenQ}(D_n)$; with causal pools, 2 of seeds 0–4 and 5 of 25. Counterexample to its representativeness: seed 2 at $n = 32$, where the old `skel4` (built from 40 data) had mass 1.0 and the causal `skel4@32` has 0.85. (2) E2: “MAP at $n = 8$: $\text{DTRC}(n = 16)$ three times”: built from data 9–16, unseen at $n = 8$. (3) E6: “the five closed-guard forms are logically equivalent”: Proposition 8.2(b). (4) “Soundness at a constant threshold for any well-specified data”: fails for multi-component T^* at fixed weights (Example 5.8). (5) PA: “robust at the level of theorems, not of axioms”: E3(b) atomic motives (Proposition 7.29). (6) E1: “ $P(T \vdash \varphi(t^*) \mid D_n) = 1.000$ from $n = 8$ in every configuration”: *open* data under L0 give 0 at $n = 4, 8$. (7) E2: “no false probe is accepted”: the mean over seeds hid seeds 1, 2 and 4.

9 Discussion

The answers of Section 1.3 are conditional: the positive ones need data drawn from a law of the model, and the data the question has in mind, the theorems humans state, are not. This section says what the answers mean for the three refinements of the question, what a good version looks like and what it cannot promise, how the results relate to Hänni’s note and to the literature, and what remains open.

9.1 The three refinements

Templates. Hänni proposed templates to block his collapse. They block his schema: no non-ground DT° template has all its instances in the acceptance part of the collapse schema (Proposition 6.6), and the reflection schema contains none either (Proposition 7.22). They do not block the collapse at the level of theorems: in arithmetic one ground reflection sentence reproduces any Σ_n -sound assigner, at a cost linear in its length (Proposition 6.7). Inside the class membership is matching, so likelihoods can be computed exactly over a pool (Proposition 8.1). Templates have a second effect that the intuition behind the question does not anticipate. They bring the instance schema $\varphi(z)$ into the class, and $\varphi(z)$ is a simple law that implies exactly the observed instances; on closed instances $\forall x\varphi$ never overtakes it (Theorem 3.4). In a class of sentences only, under citation with the closure reading, $\forall x\varphi$ does take the posterior (Proposition B.16(a), proof sketch). Nor do templates make the posterior see the logical boundaries of a schema: a schema and its root split tie exactly at matched weights, and with learned weights they are told apart by Occam terms or by usage statistics (Propositions 4.6, 4.10 and 4.11).

A derivation-length prior. The phrase has two readings. As a generative derivation grammar with a data-independent normaliser (L1), it supplies the size principle that Hänni’s 0/1 scores lack (Proposition 2.22); this is what makes memorisers and over-general templates lose. But L1’s code length counts grammar choices, not written symbols, and can be exponentially shorter than the written derivation (Remark 6.11). If the length of a derivation is meant to price computation, it should be measured in written symbols (L1^σ , or Hänni’s graded score divided by its normaliser; Remark 2.23 and Proposition 6.13). The second reading, “statements given without proof in our

data set should be fairly easily derived from axioms”, makes the data theorems. Then a derivation likelihood favours the theory that makes the data cheap to state, not the logically minimal one. Under a two-part code this is memorisation for short theorems with long proofs (Proposition 7.15 and Remark 7.16); the same pattern appears in the fragment $\{Q4, Q5\}$ (Proposition B.15, two-part comparison) and in the equational grammar $L1^{eq}$ (Example C.6). The generator wins when the data come from its own derivation process (Propositions 7.17 and 7.18) or contain direct uses of its schemas. So a derivation-length prior identifies the working set of primitives of whoever produced the data, including frequently used lemmas (Proposition B.14(c), Proposition 7.5), not a canonical axiomatisation.

A time penalty. The penalty the question names, on the time “to generate axioms / to check whether a statement is an axiom”, does not bite. Inside DT° membership is matching (Remark A.1); and the collapse sets themselves have polynomial membership, so the brief’s conjecture that such a penalty prices the collapse is refuted (Proposition 6.8). What prices the collapse is derivation size in written symbols: every theory consistent with a hard assigner needs long derivations (Theorem 6.9 and Corollary 6.10), and $L1^\sigma$ and the graded score charge this polynomially (Proposition 6.13). The charge lies between the assigner’s nondeterministic time, a proved lower bound, and its deterministic time, an upper bound with a proof sketch (Remark 6.1); for plain $L1$ only a logarithmic charge is proved (Corollary 6.12 and Conjecture 6.14). Two other limits act differently. A bound on the time spent evaluating the likelihood sums over fewer derivations and so favours adopting theorems with long proofs as axioms (Remark E.7(iv)). A steeper simplicity penalty trades early unsound lumping for faster removal of a false spare slot, and neither steepness removes both (Remark 6.15).

9.2 A good version, and what it cannot promise

Collecting the pieces, a good version of the inducer has:

- *Hypotheses and prior:* finite sets of DT° templates with a proper prefix-code prior and Dirichlet weights (Definition 2.4 and Lemma 2.5); inconsistent theories removed by refutation at growing depth (Proposition 7.21) or reported as a fourth value Inc (Definition 2.13).
- *Likelihood:* a derivation grammar with a data-independent normaliser and a uniformly sub-critical instantiation grammar (Definitions 2.6 and 2.9), measured in written symbols ($L1^\sigma$) if it is to price computation; a model of which theorems get reported, by a known filter ($L1^{sel}$) or a prior over filters (Theorem 3.9(e)); and a noise component or corruption channel, so that one false datum does not refute the true theory (Proposition B.1, Remark C.9).
- *Output:* the triple (Bel, Dis, Indep), a belief function with its plausibility, not a point estimate (Section 2.7); and a verifier that accepts s when $Bel_{cons}(s) \geq 1 - \delta$ (Remark 5.9), with a threshold that shrinks with n when the weights are learned (Theorem 5.7).

Its likelihood is a computable real whose positivity is undecidable (Proposition 2.24); exact computation is feasible over finite pools with short derivations (Section 8).

What it can promise, when the data are drawn from a law of the model: concentration on the generator class and convergence of predictions (Theorems 4.2 and 4.21); identification of the theorem set under $L1$ with parameters admissible, or of the instance union under $L0$, with no bound on the number of templates and around Gold’s theorem (Corollary 4.4, Theorem 4.7, and Proposition 4.9); soundness of the verifier against adaptive provers with probability $1 -$

$\delta/\pi(C_d^*)$ for fixed weights, and the averaged or shrinking-threshold versions for learned weights (Theorems 5.2, 5.6 and 5.7).

What it cannot promise. It cannot single out the actual axioms: inside the generator class the posterior is the prior (Corollary 4.3). It cannot certify $\forall x\varphi$ from closed instances (Theorem 3.9). It removes a false spare sentence only polynomially (Proposition 4.14) and an inconsistent theory only by refutation (Propositions 7.21 and 7.36). Above all, it promises nothing when the data are not drawn from a law of the model, which is the case the question cares about: the posterior then goes to the best-fitting generator, weaker or unsound by turns (Table 23), and a verifier on it can be made to accept a falsehood with probability 1 (Proposition 5.4). Under misspecification the soundness of the verifier depended, in the experiments, on what else was in the pool (Remark 5.10). The main lever is to model how the data were produced: which theorems are reported, how often each axiom form is used, and which mistakes occur. Each of these is a modelling choice that the data cannot check from positive instances alone.

9.3 Hänni’s note, point by point

His two variants are scores without the size principle, and “does not contradict” alone learns nothing beyond compatibility (Proposition 2.22(b), Proposition 7.35). His weighting by provable givens and proof length, divided by its normaliser, is a two-part code that charges written derivation length; this makes good his remark that one can “think of this as there being some model for how the statements get generated” (Remark 2.23). The trichotomy is best kept as a triple (Section 2.7). The equivalence holds with the constant he expects, via Craig’s trick, in a semimeasure convention (his note also considers speaking “of a semimeasure here instead of renormalizing”); both weights depend on the data only as a set, so it holds for every reveal order (Theorem 6.3 and Definition 6.2). His “easy” separation from unrestricted function induction is Proposition E.1. Of his other ideas, “only generating a very small subset of observations” is what a likelihood with a data-independent normaliser does, with the selection modelled by $L1^{\text{sel}}$ or a prior over filters; “making some mistakes” is a noise mixture, harmless for soundness only if the noise model is part of the well-specified generator (E4(C1) in Remark 5.10); grading near misses is a corruption channel (Remark C.9). His time-bounded note has an analogue in a derivation-bounded inducer (Conjecture 6.16).

9.4 Relation to the literature

We cite and do not re-prove; Appendix H.6 lists the references that were not checked against their sources. Theorem 3.8 is the countable-class analogue of the confirmation of universal hypotheses by the universal prior (Hutter, 2007); Theorem 3.9 shows that what gets confirmed is “all future data are instances”, not derivability. Confirmation is not stepwise monotone (Leike and Hutter, 2015), and for truth rather than derivability Gaifman’s condition applies (Proposition B.10; Gaifman, 1964). Theorem 4.2 is the argument of Doob (1949) for a countable class; consistency in larger models goes back to Schwartz (1965) (see Ghosal and van der Vaart, 2017), and misspecified posteriors concentrate on KL-minimisers in general (Berk, 1966; Kleijn and van der Vaart, 2006), under conditions we did not check for template classes. Learning from text is limited by Gold’s theorem (Gold, 1967; Angluin, 1980); learning from stochastic positive data goes back to Horning (1969) and Angluin (1988). The size principle is that of Tenenbaum and Griffiths (2001), the code-length reading is MDL (Rissanen, 1978) with the KT estimator (Krichevsky and Trofimov, 1981), and the spare-slot rates are of the kind studied for overfitted mixtures (Rousseau and Mengersen, 2011). Bel and Pl are a belief function and its plausibility in the sense of Shafer

(1976). The collapse results use Craig’s trick (Craig, 1953) and the nondeterministic time hierarchy (Cook, 1972; Seiferas et al., 1978; Žák, 1983); Levin’s Kt and the speed prior enter only as examples of membership- or generation-time penalties; the upper bounds recall Pudlák (1998); Kreisel’s conjecture is related (Remark B.18). Inductive logic programming (Muggleton, 1991) learns clauses from examples and background knowledge; AS is the non-Bayesian counterpart of this paper, and IL supplies its Ville argument (Claude (Anthropic), 2026b; Ville, 1939).

9.5 Open problems

Collected from the four tracks and from the writing of this paper, and deduplicated; the cited statements give the details.

1. *Rates*: identification for every weight vector (Theorem 4.7(c)); rigorous spare-slot rates (Proposition 4.14); the Occam rate under L1 (Proposition 4.11(d)); the memoriser rate (Proposition B.6(iii)).
2. *Competitors only L1 keeps alive*: partial splits that derive the rest (Remark 4.13); whether the two-part maximum can change a deductive limit (Proposition A.8); whether L1 separates equivalent theories for all grammar parameters (Proposition 4.5).
3. *Misspecification*: unsound mass in the full class (Remark C.5); the KL-minimising families for instance-only data; Conjecture C.7; soundness bounds in terms of the distance from the data law to the pool, and robust posteriors.
4. *The verifier*: completeness under the shrinking threshold; strictness of the inclusion in Proposition D.2(d).
5. $\forall x\varphi$: reversal through $\wedge$ -detours (Remark 3.5); the race between split and memoriser families over Q (Propositions 3.12 and B.16); an unknown selection learned jointly with the templates (Theorem 3.9(e)); the full-sum comparison at the default law (Proposition B.15); formulas with several variables.
6. *The collapse*: Conjecture 6.14; whether axiom induction with a symbol-size penalty equals function induction over assigners with short certificates up to polynomial factors; a single-sorted repair of Hänni’s schema; the $I\Sigma_n$ comparison under a derivation-time penalty; Conjecture 6.16.
7. *PA and ZF*: the posterior over all template unions; non-read-once induction templates (Conjecture 7.28); theorem data under plain L1, the full sum or Dirichlet weights; Collection from Replacement without Foundation.
8. *Computation*: search beyond a pool; tree derivations in code, so that PA can be run under L1; a learned instantiation grammar; cheap remedies for early unsound lumps (Remark 8.3).
9. *Near misses*: how much natural corruption channels enlarge the generator class (Remark C.9).
10. *Write-ups*: full proofs of the proof sketches listed in Appendix H.5.

Closing. This was a check of an idea, not the design of a prover. The idea survives in a precise form: a template prior with a generative derivation likelihood is a coherent inducer, it confirms predictions, and with well-specified data it finds a system with the right theorems (or instances) and supports a verifier that adaptive provers fool only with small probability. What it does not do is turn human-stated theorems into the axioms behind them. The gap lies between how the model generates data and how mathematicians select what they write down.

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A The model: proofs and details

Scripts cited here are in `research/tracks/model/checks/` (the track’s) and `.../referee_code/` (its referee’s); each is seeded and writes its output next to itself.

A.1 Prior

The template code (Definition 2.4). Templates are coded in preorder, one token per node from a fixed alphabet (the symbols of the language, connectives, quantifiers, and escapes IDX, PAR, MV) at $\lceil \log_2(\text{alphabet size}) \rceil$ bits. The escapes are followed by Elias- γ codes: $\gamma(k+1)$ for a de Bruijn index k , $\gamma(j+1)$ for p_j , and for a metavariable a new/old bit, then its arity ($\gamma(n+1)$) and sort bit if new or its index if old, then its arguments. A theory $\{\tau_1, \dots, \tau_k\}$ is coded as $\gamma(k+1)$ and the k template codes in lexicographic order.

Proof of Lemma 2.5. The template code is prefix-free: each token fixes the number of children that follow, the escapes are followed by self-delimiting Elias- γ codes and fixed bits, and an old metavariable's arity is already known. $\gamma(k+1)$ fixes how many template codes follow, so theory codes are prefix-free. The code determines a template up to renaming of metavariables, which is how Definition 2.1 identifies templates, and sorting makes the code of a set canonical; so distinct theories have distinct codes, and Kraft's inequality gives $\sum_T 2^{-\ell(T)} \leq 1$. Hence $0 < Z_\lambda \leq 1$ for $\lambda \geq 1$. For $\lambda \geq 2$, $\sum_T \pi_\lambda(T)^{1/2} = Z_\lambda^{-1/2} \sum_T 2^{-\lambda \ell(T)/2} \leq Z_\lambda^{-1/2} \sum_T 2^{-\ell(T)} < \infty$. $\square$

Implicit in the notes. (1) Theories must be taken up to renaming of metavariables: renamings share a code, so otherwise infinitely many theories would share one code and the sum would diverge. (2) The notes code the number of templates as $\gamma(k)$; Elias- γ codes positive integers, and the empty theory (used in Proposition 2.24(b)) needs a code, so we write $\gamma(k+1)$. Neither changes a statement.

Remark A.1 (Other priors; time factors). [proved]^{experiments §1.3; universal §1.6; pa Def 0.2; model §1.3} (i) The experiments' prior $\propto 2^{-\lambda \text{bits}(T)}(1 + |T|)^{-\tau}$ uses a stochastic template code that satisfies Kraft, so it is proper for $\lambda \geq 1$ ($\lambda = \frac{1}{2}$ only on finite pools); $\mathbf{Q} \cup \{T_{\text{Ind}}\}$ costs 226.2 bits. Its factor $(1 + |T|)^{-\tau}$ (time exponent τ , default 0) prices membership time. (ii) The priors $2^{-\lambda(\sum_A |A| + \#\text{axioms})}$ of Section 3 and $2^{-\beta_{\text{ax}} \sum_\tau |\tau|}$ of Section 7 ($\beta_{\text{ax}} = 5$ or $\log_2 23$ bits per axiom symbol) are improper over all theories and are used only on finite or summable families. Prior shares depend on the code (Table 16). (iii) A Levin-style time factor for a decider of the axiom set is polynomial in $\ell(T)$ inside DT° , since membership is matching; it changes $\ln \pi$ by $O(\ln \ell(T))$ and barely bites. Time matters in Section 6.

Proof of Remark A.1. (i) The experiments' template code is $-\log_2$ of a subcritical stochastic grammar over templates whose arities fix the tree shape, so $\sum_\tau 2^{-\text{bits}(\tau)} \leq 1$. A set of m distinct components costs $|\gamma(m)| + \sum_i \text{bits}(\tau_i) - \log_2 m!$, so the sum over such sets is the sum over ordered tuples of distinct components times $2^{-|\gamma(m)|}$, at most $2^{-|\gamma(m)|} (\sum_\tau 2^{-\text{bits}(\tau)})^m \leq 2^{-|\gamma(m)|}$; summing over m gives at most 1, and $(1 + |T|)^{-\tau} \leq 1$. An independent implementation of the code gives the same lengths to four decimals (experiments §1.3). (iii) Membership in a finite DT° theory is decided by matching against each template, in time $O(\sum_{\tau \in T} (|\tau| + |s|))$ (AS Thm 2.5); so the time constant is polynomial in $\ell(T)$ and its logarithm is $O(\ln \ell(T))$. $\square$

A.2 Instantiation grammar

Lemma A.2 ($\mathcal{Q}_\tau$ is a law on $\text{inst}(\tau)$). [proved]^{model Lemma 1.6} Let $\mathcal{Q}$ be uniformly subcritical, $\tau \in \text{DT}^\circ$, and N the size of a body drawn from type i . (a) $\mathbb{E}_i N \leq h(i)/(h_{\min}(1 - \rho))$; bodies are finite a.s. (b) With $\Lambda := 2/((1 - \rho)h_{\min})$ and $0 < \zeta \leq (\Lambda b h_{\max})^{-2}$: $\mathbb{E}_i[e^{\zeta N}] \leq e^{\zeta \Lambda h(i)}$, so $P_i(N \geq \ell) \leq e^{\zeta(\Lambda h_{\max} - \ell)}$. (c) $\mathcal{Q}_\tau(s) := \mathcal{Q}(\theta_s)$ if $\tau\theta_s = s$, else 0, is a probability on $\mathcal{S}$; with full

support its support is $\text{inst}(\tau)$. (d) (a) and (b) hold for every multi-type branching process with independent children given the type, at most b children and the uniform condition.

Proof of Lemma A.2. (a) Let G_j be the sum of h over generation j . Given the first j generations, $\mathbb{E}[G_{j+1}] \leq \rho G_j$ by the uniform condition and linearity, so $\mathbb{E}G_j \leq \rho^j h(i)$. Generation j has at most $G_j/h_{\min}$ nodes; sum over j .

(b) Let N_D count the nodes of depth at most D . We show $\mathbb{E}_j[e^{\zeta N_D}] \leq e^{\zeta \Lambda h(j)}$ for all types j by induction on D . For $D = 0$, $N_0 = 1 \leq \Lambda h_{\min} = 2/(1-\rho)$. For the step, let the root's children have types $c_1, \dots, c_r$ ($r \leq b$) and $H := \sum_l h(c_l) \in [0, b h_{\max}]$. The subtrees are independent given the types, so $\mathbb{E}_j[e^{\zeta N_{D+1}}] \leq e^{\zeta \Lambda H} \mathbb{E}_j[e^{\zeta \Lambda H}]$. Put $x := \zeta \Lambda H \in [0, 1]$ (as $\zeta \leq (\Lambda b h_{\max})^{-2}$ and $\Lambda b h_{\max} \geq 2$). For $x \in [0, 1]$, $e^x \leq 1 + x + x^2$, since $e^x - 1 - x \leq x^2(e-2)$. So $\mathbb{E}_j[e^{\zeta \Lambda H}] \leq 1 + \zeta \Lambda \rho h(j) + \zeta^2 \Lambda^2 b^2 h_{\max}^2 \leq \exp(\zeta \Lambda \rho h(j) + \zeta^2 \Lambda^2 b^2 h_{\max}^2)$, and $\mathbb{E}_j[e^{\zeta N_{D+1}}] \leq e^{\zeta \Lambda h(j)}$ iff $1 + \zeta \Lambda^2 b^2 h_{\max}^2 \leq \Lambda(1-\rho)h(j)$. The right side is at least 2, the left at most 2. Monotone convergence gives the bound for N ; the tail is Markov's inequality.

(c) By AS Thm 2.5(a), $\theta \mapsto \tau\theta$ is injective, so $\sum_s \mathcal{Q}_\tau(s) = \sum_\theta \mathcal{Q}(\theta) = \prod_M 1 = 1$, each body law being a probability on finite bodies by (a). With full support $\mathcal{Q}(\theta) > 0$ for every θ .

(d) The proofs of (a) and (b) use only independence given the type, the bound b and the uniform condition. $\square$

Remark A.3 (The earlier condition). [refuted (old condition); exclusion proved; computed]^{model §1.4; referee M5 (r1); c10}

The pre-referee condition “every type has mean offspring below 1” is refuted (referee M5) as a usable hypothesis. It dominated the process by one Galton–Watson process with the largest mean. (1) No formula type of L_A or L_∞ satisfies it: every formula production has a child. (2) With infinitely many types it does not give finiteness: with a nullary predicate $\top$, let a formula node with k variables choose $\forall$ (child with $k+1$ variables) with probability $1 - 1/(k+2)^2$ and $\top$ otherwise; the chain never stops with probability $\prod_{m \geq 2} (1 - 1/m^2) = \frac{1}{2}$ (r1_subcritical.out: product 0.500000, simulated 0.4989). (3) Means do not give domination: $\{0:0.55, 2:0.45\}$ and $\{0:0.1, 1:0.9\}$ both have mean 0.9, and a law dominating both has mean ≥ 1.35 . The uniform condition excludes the chain of (2) [proved]: it requires $h(k+1)(1 - 1/(k+2)^2) \leq \rho h(k)$, so $h(k) \leq h(0)\rho^k \prod_{j < k} \frac{(j+2)^2}{(j+1)(j+3)} = h(0)\rho^k \frac{2(k+1)}{k+2} \rightarrow 0$, against $h \geq h_{\min} > 0$ (c10_subcritical.out, Part C: e.g. $6.1 \cdot 10^{-92}$ at $k = 2000$ for $\rho = 0.9$). The uniform condition holds for a grammar for L_A with $\rho = 0.75$ (Example A.4).

Example A.4 (Grammars that satisfy Definition 2.6). [computed]^{model §1.4 (c10); experiments §1.2} $\mathcal{Q}_A$ for L_A : term node 0.4, S .2, $+$.1, $\cdot$.1, a variable .2 (uniform over the k variables; renormalised away when $k = 0$); formula node $=$.3, $<$.2, $\neg, \wedge, \rightarrow, \forall, \exists$.1 each. Term nodes have mean 0.75 ($k = 0$) or 0.6 term children; formula nodes 0.7 formula and 1.0 term children. With $h(o) = 1$, $h(\iota) = 0.05$ the weighted ratio is at most 0.75 at every type, so $\rho = 0.75$. From $2 \cdot 10^5$ bodies: mean size 14.62 (s.e. 0.04) from a formula root, against the bound 80; 4.016 (s.e. 0.015) from a term root at $k = 0$, against the bound 4, which is the exact mean $1/(1 - 0.75)$. Both tails decay exponentially. Track experiments' grammar has mean 0.72 same-sort children at formula nodes and at most 0.75 at term nodes, and satisfies the condition with $h(o) = 1$ and $h(\iota)$ small.

A.3 Calculi

One law $\mathcal{Q}$ instantiates metavariables and, unless said otherwise, supplies the $\forall E$ terms (universal §1.2).

$\mathcal{C}_{\min}$. Chains. A node cites (probability $1 - c$: an axiom by weight, metavariables from $\mathcal{Q}$) or applies $\forall E$ (probability c) to a recursively generated premise with $t \sim \mathcal{Q}$, failing if the

premise has no leading $\forall$. For $\{\sigma_\varphi\}$, $Z = (1 - c) \sum_{k \leq m'} c^k$ with m' the number of leading $\forall$ of φ (Proposition 3.2). $C_{\text{open}}(c, g)$. Adds Gen (probability g), which abstracts the parameter of the premise and fails if there is none; $p_{\text{cite}} = 1 - c - g$; terms from $\mathcal{Q}_{\text{open}} = \rho[p] + (1 - \rho)\mathcal{Q}_{\text{num}}$ (ρ here is the parameter probability). Per closed datum, normalised: $H_{\forall}/H_{\text{sch}} = c(1 - \rho)/(1 + c)$, $H_{\text{open}}/H_{\text{sch}} = (1 - \rho)/(1 + g\rho)$ (Proposition 3.10). C_{U3} . Each node picks cite, $\forall E$, or a rule whose premises and conclusion are quantifier-free (equality rules; propositional rules and MP on quantifier-free formulas), with fixed probabilities; trees. $C_{\wedge}$. Cite (p_{cite}), $\forall E$ (c), $\wedge I$ (a), $\wedge E$ (e ; side left or right with probability $\frac{1}{2}$), the $\wedge$ -rules on all formulas; trees. Used for Remark 3.5.

Ch_J (experiments §1.6). A chain $s_0, \dots, s_k$, $k \leq J$. s_0 cites component i with probability w_i (bodies from $\mathcal{Q}$, guards respected). At step $j < J$, if some rule applies, the chain stops with probability c_{stop} , otherwise it applies an applicable rule r with probability $\propto c_r$: *elim* ($s_j = \forall x\psi \mapsto \psi[t/x]$, t from $\mathcal{Q}_e$, closed terms, with p iff the run is open); *gen* ($s_j \mapsto \forall x s_j[x/p]$ if s_j contains p); *mp* (a uniformly chosen component $A \rightarrow B$, $A, B \in \text{DT}^\circ$ with the metavariables of A in B , whose A matches s_j ; $s_{j+1} = B\theta$, θ read off s_j and drawn from $\mathcal{Q}$ on the rest). If no rule applies, or $j = J$, it stops. Defaults $c_{\text{stop}} = \frac{1}{2}$, all $c_r = 1$, $J = 1$ ($J = 2$ in parts of E1 and in E6); one parameter p . The chain never fails, so its law is a probability (experiments Prop X3), and $P(s \mid T, w) = \sum_i w_i a_i(s)$. It is the special case of L1 with cited major premises and depth $\leq J$, so induction followed by MP and $\forall E$ is outside it (Remark 8.3).

ND and L1^{sch} (pa Def 0.3). Sequents $\Gamma \vdash A$ with rules hyp, ax (an axiom, or a schema instance at a motive), tc (from $\Gamma_j \vdash A_j$ infer $\bigcup_j \Gamma_j \vdash B$ if $\bigwedge_j A_j \rightarrow B$ is a tautology), $\rightarrow I$, $\forall E$, $\forall I$, $\exists I$, $\exists E$, refl, subst, with the usual side conditions; a standard sound and complete calculus, checked by `nd.py`. The derivation is written schematically in a predicate metavariable P and the motive is coded once by the grammar. Per line it costs $\log_2 10$ bits for the rule, $\log_2 |T|$ per axiom citation, $\log_2(\text{line index})$ per premise reference and $\beta = \log_2 23 \approx 4.52$ bits per written symbol. The decodable variant also codes the number of lines, the premise count of each tc line and the variables named by subst and $\exists I$; it adds 4.8–7.3% to the costs of pa §1’s derivations. With at least $\log_2$ of the alphabet size bits per written symbol (about $\log_2 25$ with the variable names used) it is a prefix code, so Kraft holds and L1^{sch} is the two-part form of a sub-probability likelihood [proved for $\beta \geq \log_2(\text{alphabet size})$]. The reported $\beta = \log_2 23$ under-charges symbols by about 3%, which changes no conclusion (pa §0).

The grammar of L1^{eq} (model Ex 3.6). A node cites a template (α_{ax} ; metavariables from $\mathcal{Q}$) or applies refl ($t \sim \mathcal{Q}$), sym, trans (valid iff the middle terms agree), congruence for S , or congruence for $+$ (side term $u \sim \mathcal{Q}$, left or right). A tree is valid iff its steps are and all its terms are closed of size $\leq N$. μ_T is the exact least fixpoint; Monte Carlo agrees ($Z = 0.54909$ against 0.54965 ± 0.00091). The rules are complete for closed equational consequence (known; Birkhoff, 1935, not checked).

Table 11 compares the calculi.

Lemma A.5 (A4 is not a template). *[(a) proved; (b) proof sketch]^{model Lemma 1.2} (a) A4, written as $\forall x P(x) \rightarrow P(f)$ with f a 0-ary term metavariable, is not in SO° , so not in DT° . (b) No finite set of DT° templates has the instances of A4 as its instance union.*

(a) AS Def 2.1 requires metavariable-free arguments. The other logical axioms are templates: A1–A3 in FO, A5 in PAT as $\forall x(P \rightarrow F(x)) \rightarrow (P \rightarrow \forall x F(x))$, substitutivity in DT° . (b) is sketched next: a template inside A4 covers $\forall x B_m \rightarrow B_m[t/x]$, $B_m := (x + \dots + x = x)$, for one m or one t .

calculus	rules	normalisation	factor $P_{H_V}/P_{H_{sch}}$ per closed instance	used in
K	cite theory or logical axiom, MP, Gen, $\forall E$	μ_T/Z_T ; finite a.s. iff $m \leq 1$	not computed; with parameters admissible no hypothesis emits only instances (Theorem 3.9(c3))	§§4–6
C_{min}	cite $(1 - c)$, $\forall E (c)$	μ_T or μ_T/Z_T	c unnormalised; $c/(1 + c)$ normalised (φ quantifier-free)	§3
C_{open}	$C_{min} + \text{Gen on } p (g)$	same	$c(1 - \rho)/(1 + c)$ normalised, against guarded H_{sch} (ρ : parameter probability of Q_{open})	§3
C_{U3}	cite, $\forall E$, quantifier-free rules	trees, μ_T	$\leq c$ on the mass that cites the axiom (Proposition B.2)	§3
$C_{\wedge}$	cite, $\forall E$, $\wedge I$, $\wedge E$	trees, μ_T	$> c$ (refuted conjecture, Remark 3.5); ≤ 0.925 on the computed grid	§3
Ch_J	cite, then $\leq J$ steps ($\forall E$, Gen, MP)	proper, no normaliser	$c_{ch} = 1 - c_{stop}$	§8
ND	hyp, ax, tc, $\rightarrow I$, $\forall E$, $\forall I$, $\exists I$, $\exists E$, refl, subst	two-part code; sub-probability	1/10 per $\forall E$ step (fixed rule code; learned codes: Table 32)	§7
$L1^{eq}$	cite, refl, sym, trans, $cong_S$, $cong_+$	μ_T/Z_T , least fixpoint	— (closed equations only)	§4

Table 11: The calculi of Definitions 2.2 and 2.3. “Factor” is the likelihood ratio of $H_V = \{\forall x\varphi\}$ to its instance schema H_{sch} (Table 13) on one closed instance $\varphi(t)$; m is the mean offspring of Definition 2.9. Rates are comparable only within one calculus and one code: Ch_J is proper, yet its factor equals the *unnormalised* C_{min} factor with $c = c_{ch}$. The direction (factor at most 1) is proved for C_{min} (all listed likelihoods) and, unnormalised, for C_{U3} (Theorem 3.4); it follows from the exact laws for C_{open} (Proposition 3.10), and holds in Ch_J (experiments Prop X6) and under pa’s codes (pa §5.5); in $C_{\wedge}$ it holds on the computed grid and is open in general.

Proof sketch of Lemma A.5(b) Let $s_{m,t} := \forall x B_m \rightarrow B_m[t/x]$, $B_m := (x + \dots + x = x)$ with m summands. Suppose $\tau \in DT^\circ$ with $\text{inst}(\tau) \subseteq \text{inst}(A4)$ covers $s_{m,t}$ and $s_{m',t'}$, $m \neq m'$. The antecedents differ under $\forall x$ at positions whose subterms contain the bound x , so τ has an occurrence there with x among its arguments, say $f(x)$. The matching position of the consequent carries $B_m[t/x]$; in τ it is either $f(c)$ with a fixed metavariable-free c , forcing $t = t' = c$ there, or a metavariable independent of f , which gives instances outside A4. So each template covers $s_{m,t}$ for one m or one t , and a finite set misses some $s_{m,t}$. *Not written out:* turning “the matching position in the consequent” into a statement about every DT° shape (model §10, open problem 9).

A.4 Likelihoods

Table 12 lists the likelihoods of Definitions 2.7, 2.9 and 2.11 and the scores of Definition 2.12.

Proof of Lemma 2.8. $\prod_i P_{T,w}(X_i) = \sum_\ell \prod_\tau w_\tau^{n_\tau(\ell)} \prod_i Q_{\ell_i}(X_i)$, a finite sum over the labellings with $X_i \in \text{inst}(\ell_i)$; integrate term by term with the Dirichlet moments (known)

$$\mathbb{E} \prod_\tau w_\tau^{n_\tau} = \frac{\Gamma(A)}{\Gamma(A+n)} \prod_\tau \frac{\Gamma(\alpha_\tau + n_\tau)}{\Gamma(\alpha_\tau)}.$$

The result is symmetric in the data. P_T^{Dir} is a mixture of the i.i.d. laws $P_{T,w}^{\otimes n}$, each a probability on S^n consistent in n (Lemma A.2(c)), so its normaliser is 1 for every D . With disjoint instance sets each datum has one label, and $P_T^{\text{Dir}}(D, s)/P_T^{\text{Dir}}(D) = \frac{\Gamma(A+n)}{\Gamma(A+n+1)} \frac{\Gamma(\alpha_\tau + n_\tau + 1)}{\Gamma(\alpha_\tau + n_\tau)} Q_\tau(s) = \frac{\alpha_\tau + n_\tau}{A+n} Q_\tau(s)$ for $s \in \text{inst}(\tau)$. $\square$

name	definition	normaliser	size pr.	model / universal / pa / experiments
L0	$\sum_{\tau} w_{\tau} Q_{\tau}(s)$, or P_T^{Dir}	probability	yes	L0 / L0-strict / L0 / L0
L0 ^{cl}	L0; $\forall \bar{x} \chi$ cited via instances	probability	yes [†]	— / L0-closure / — / —
μ_T	$\Pr[\text{valid tree concluding } s]$	mass Z_T	yes [‡]	μ_T / L1(c) / — / —
L1	μ_T / Z_T	$Z_T \geq 1 - \alpha_r$	yes	L1 / L1-norm / L1 (not computed) / L1 (Ch _J)
L2	citation forced at depth d	$Z_T^{(d)}$	yes	L2 / — / “bounded” / —
L1 ^σ	$L1 \cdot e^{-\kappa \pi }$, renormalised	Z_T^{σ}	yes	L1 ^σ / — / — / —
L1 ^{max}	best single tree	sub-prob.	yes [‡]	two-part / L1-max / L1-max / —
L1 ^{sch}	two-part ND code	sub-prob.	yes [‡]	— / — / L1-sch / —
L1 ^{sel}	$P_T(s)/P_T(S)$ on S	$P_T(S)$	yes [†]	— / L1-sel(S) / — / —
L1 ^{sel} _{cit}	$\sum_i w_i a_i(s)/c_i$	c_i	yes [†]	— / — / — / L1sel
P_T^{η}	$(1 - \eta)P_T + \eta N$	as P_T	vacuous [¶]	(channel P_T^K) / P_T^{η} / — / —
L_{ε}	$(1 - \varepsilon)P_T + \varepsilon \mu_0[T \not\vdash_k \neg s]$	mass dep. on T	partly [§]	— / — / L_{ε} / —
L1 ^{eq}	equational grammar	Z_T	yes [†]	L1-eq / — / — / —
P_T^g	$2^{-\kappa \ell_T(s)} / Z_T$	$Z_T \leq 1$	restricted	P_T^g / — / — / —
scores	Definition 2.12	none	no	$S_{\text{prove}}, S_{\text{nc}}, S_g$ / $S_{\text{nc}, \beta}$ / (L_{ε} , a relaxation) / —

Table 12: Likelihoods and scores. Normalisers are data-independent. “yes”: Proposition 2.22(a); “restricted”: Remark 2.23; “no”: Proposition 2.22(b). [†]A fixed probability on $\mathcal{S}$, so Lemma 2.21 applies as in the proof of Proposition 2.22(a). [‡]A fixed sub-probability, so Theorem B.4 applies. [§]Only $(1 - \varepsilon)P_T$ separates theories that refute no datum; at $\varepsilon = 1$ nothing is learned (Proposition 7.35). [¶]With full-support noise (or a full-support channel, Remark C.9) $\text{supp } P_T^{\eta} = \mathcal{S}$, so no competitor puts mass outside the data’s support and Lemma 2.21 says nothing; theories are separated only through KL. The marks say that mass outside the data’s support is lost, not how strongly two theories are separated.

Computed. `c8_formulas.out` (1): closed form $9.908294678 \cdot 10^{-6}$ against quadrature over $w_1 \sim \text{Beta}(\frac{1}{2}, \frac{1}{2})$ $9.908294680 \cdot 10^{-6}$ (asserted to 10^{-12}). Experiments Prop X2: the exact dynamic programme agrees with brute force on 500 random instances and with the referee’s independent count-vector sum on 400 cases to $1.4 \cdot 10^{-14}$; above 20000 states it returns rigorous bounds (Appendix G).

Proof of Lemma 2.10. (a) A derivation node has 0, 1 or 2 premises with probabilities $\alpha_{\text{ax}} + \alpha_{\text{lg}} > 0$, $\alpha_{\text{Gen}} + \alpha_{\text{VE}}$, α_{MP} , independently: a Galton–Watson tree with mean m . The extinction theorem (known) gives finiteness iff $m \leq 1$, given positive mass at 0 (without it the tree is infinite even at $m = 1$; referee m11), and extinction probability q for $m > 1$. $\mathbb{E}N = 1 + m\mathbb{E}N$, with $\mathbb{E}N < \infty$ for $m < 1$ since generation k has mean m^k . Bodies and terms are finite a.s. by Lemma A.2(a). (b) A one-node citation tree is valid, so $Z_T \geq \alpha_{\text{ax}} + \alpha_{\text{lg}} = 1 - \alpha_r$ for $T \neq \emptyset$; for $T = \emptyset$ only logical-axiom citations are valid, so $Z_{\emptyset} \geq \alpha_{\text{lg}}$. (c) $\subseteq$: a valid tree is a K-derivation from $\bigcup \text{inst}(T)$, each $\forall E$ node replaced by an A4 instance and MP. $\supseteq$: a K-derivation from $\bigcup \text{inst}(T)$ and logical axioms is a grammar tree of positive probability: citations have $w_{\tau} Q(\theta) > 0$, logical-axiom instances (A4’s term included) have positive Q -probability with parameters where used, and each Gen has positive parameter probability. Without parameters all formulas in a tree are closed and Gen adds only vacuous quantifiers (referee m12). $\square$

Implicit in the notes. For $T = \emptyset$ no theory axiom can be cited; reading such a node as invalid gives the bound $Z_\emptyset \geq \alpha_{\text{lg}}$ of (b), positive under the hypotheses of (c), which is what Proposition 2.24(b) uses.

Computed (c8_formulas.out (2), derivation nodes): $(\alpha_{\text{MP}}, \alpha_{\text{Gen}} + \alpha_{\text{VE}}) = (0.15, 0.2)$, $m = 0.5$: mean 2.010 against 2; $(0.3, 0.3)$, $m = 0.9$: 9.965 against 10; $m = 1$: heavy tail; $(0.4, 0.3)$, $m = 1.1$: $P(\text{finite}, < 20000 \text{ nodes}) = 0.7489$ against $q = 0.75$.

Remark A.6 (Two selection-aware likelihoods). [proved]^{experiments §1.7 (referee m8)} L1^{sel} and $\text{L1}_{\text{cit}}^{\text{sel}}$ coincide for one-component theories and differ for mixtures $(\sum_i w_i a_i / \sum_i w_i c_i \text{ against } \sum_i w_i a_i / c_i)$. Proposition 3.6 uses L1^{sel} ; experiments E1 and E6 use $\text{L1}_{\text{cit}}^{\text{sel}}$, for which multi-component pool members are a modelling choice (Remark 3.7).

Lemma A.7 (L1^σ is proper). [proved]^{model §1.5} $0 < Z_T^\sigma \leq Z_T \leq 1$; so L1^σ is a probability with a data-independent normaliser and the support of L1 .

Proof of Lemma A.7 and Remark A.6. A one-node citation tree π has $\Pr(\pi) e^{-\kappa|\pi|} > 0$, and $e^{-\kappa|\pi|} \leq 1$ gives $\mu_T^\sigma \leq \mu_T$; $\mu_T^\sigma(s) > 0$ iff some valid tree concludes s with positive probability, iff $\mu_T(s) > 0$. For Remark A.6: with one component $P_T = a_1$ and $P_T(S) = c_1$; with several, the stream filter divides by $\sum_i w_i c_i$ and the per-citation form each term by its own c_i . $\square$

A.5 The size principle

Proof of Lemma 2.21. On $\text{supp } P^*$, $P = (1 - \varepsilon)\tilde{P}$; substitute into

$$\text{KL}(P^* \| P) = \sum_{s \in \text{supp } P^*} P^*(s) \ln(P^*(s)/P(s)).$$

For (b), $\mathbb{E}_{P^*}[\tilde{P}(X)/P^*(X)] = \sum_{s \in \text{supp } P^*} \tilde{P}(s) = 1$; extend to n data by independence. $\square$

Computed (c1_size_principle.out): root law $(0, S, +, \cdot) = (.5, .3, .1, .1)$; P^* the law of $\{z + 0 = z\}$, P that of $\{z_1 + 0 = z_2\}$ on the 30901 terms of size ≤ 11 . (a) holds to 10^{-9} on this truncated grammar (asserted; there $\text{KL} = 2.618$ nats). Separately, the simulated per-datum log-ratio of the untruncated grammar is 3.81, 4.19, 3.94 nats at $n = 10^2, 10^3, 10^4$ (no standard error reported), against its $\text{KL} = H(\mathcal{Q}) = 3.894$ nats.

Proof of Proposition 2.22. (a) Each P_T is a fixed probability on $\mathcal{S}$ (Lemmas 2.10, A.2 and A.7; under L2 all trees are finite and one-node citations valid); apply Lemma 2.21. Under L0 with full support and positive weights, $\text{supp } P_{T,w} = \bigcup \text{inst}(T)$. (b) A model of $T' \cup D_+ \cup \neg D_-$ is a model of $\text{Th}(T') \supseteq \bigcup \text{inst}(T)$, so $S_{\text{nc}}(T') \leq S_{\text{nc}}(T)$. If $S_{\text{prove}}(T) = 1$, T' proves what T proves and is consistent by hypothesis, so $S_{\text{prove}}(T') = 1$. For S_g : $S_{\text{nc}}(T') = 1$ by hypothesis, and instance inclusion makes every derivation from T one from T' , so $\ell_{T'} \leq \ell_T$ and $g(\ell_{T'}) \geq g(\ell_T)$. The consequence is the case $T = T^*$, $T' = T^* \cup \{\psi\}$, where the instances are nested; if T^* proves the labelled data, the consistency hypothesis reduces to that of $T^* \cup \{\psi\}$. (c) Under L0 , $P_{T'}(s) = (1 - w_\psi)P_{T^*}(s) + w_\psi[s = \psi]$. Under L1 , the one-node tree citing ψ gives $P_{T'}(\psi) \geq \alpha_{\text{ax}} w_\psi$ (as $Z_{T'} \leq 1$), while $\psi \notin \text{Th}(T^*) \supseteq \text{supp } P_{T^*}$ by the inclusion $\subseteq$ of Lemma 2.10(c), which needs no hypothesis. So $\varepsilon \geq \alpha_{\text{ax}} w_\psi$ in Lemma 2.21, and by the strong law $n^{-1} \ln \frac{P_{T'}(D)}{P_{T^*}(D)} \rightarrow -\text{KL}(P_{T^*} \| P_{T'}) \leq -\ln \frac{1}{1 - \alpha_{\text{ax}} w_\psi}$ a.s. The Dirichlet case is Proposition 4.14(a). $\square$

Proof of Remark 2.23. A derivation determines its conclusion, so distinct sentences have distinct shortest derivations with codes in one prefix-free set; with $\kappa \geq 1$, Kraft gives $Z_T \leq 1$. With $g_\infty > 0$ the infinitely many sentences without derivation make the sum infinite. The code writes

axioms as formulas, so instance inclusion makes every derivation from T one from T' with the same code: $\ell_{T'} \leq \ell_T$, $Z_{T'} \geq Z_T$. The refuted sentence fails where the numerator grows too: $P_{T'}^g(b)/P_T^g(b) = 2^{\kappa(\ell_T(b) - \ell_{T'}(b))} Z_T/Z_{T'}$, and $Z_{T'} \leq 1$.

Computed by the referee (`r6_stronger_theory.out`), in a different setting: ℓ is the symbol size of the least MP tree, normalised over a finite universe of 570 formulas of size ≤ 7 , where the normaliser need not be ≤ 1 . There $\ell_T(b) = 5$ and $\ell_{T'}(b) = 1$; $\kappa = 1$ gives $P_T(b) = 0.0321$, $P_{T'}(b) = 0.3456$ ($\kappa = \frac{1}{2}$: 0.0432, 0.1491), 10 sentences gaining and 36 losing. Under L2 (depth 2) the probability of b is 0.0134, 0.0156, 0.0168 under T and 0.263, 0.240, 0.196 under T' , in three rule settings. $\square$

A.6 Computing the likelihood

Proof of Proposition 2.24. (a) Let $\nu(\pi)$ count the grammar nodes of a tree: derivation nodes, $\mathcal{Q}$ nodes of bodies and terms, and the nodes of the geometric chains that pick parameters. Weight them 1, $\eta h_{\mathcal{Q}}$ and η . A derivation node then has expected weighted offspring at most $m + \eta(v_T h_{\mathcal{Q}, \max} + 1)$, with v_T the largest number of metavariable slots of a template of T or of a logical schema; a $\mathcal{Q}$ node has ratio at most $\rho_{\mathcal{Q}}$; a chain node continues with a fixed probability below 1. As $m < 1$, a small η makes all ratios less than 1, so the whole tree satisfies Lemma A.2(d) (as in the proof of Proposition E.5) and $\mathbb{E}\nu \leq C_\nu$, computable from the parameters and T . Finitely many trees have $\nu \leq N$ (each node chooses from a finite list: rule, template, logical schema, production, one of at most the arity plus N variables, chain step); validity is decidable and each probability is a computable real. So $\mu_T^{\leq N}(s) \leq \mu_T(s) \leq \mu_T^{\leq N}(s) + C_\nu/N$ by Markov's inequality, likewise for Z_T , and $Z_T \geq 1 - \alpha_r > 0$ ($Z_\emptyset \geq \alpha_{lg} > 0$; Lemma 2.10(b)). (b) The identity is Lemma 2.10(c). $\bigcup \text{inst}(T)$ is decidable by matching, so $\text{Th}(T)$ is recursively enumerable. $\text{Th}(\emptyset)$ is the set of valid formulas, undecidable for a language with a binary relation symbol (Church, Turing; known). (c) Deciding $P_T(s) = 0$ would decide $s \notin \text{Th}(T)$. For a finite class, $\pi_n(T)$ is a quotient of computable reals whose denominator is positive when defined. $\square$

Cost. Precision ϵ needs all trees with up to $N \approx C_\nu/\epsilon$ grammar nodes (C_ν bounds their mean number), exponentially many in N ; a likelihood ratio needs ϵ below likelihoods that are exponentially small in the datum's least tree size (Proposition E.5). No better general algorithm and no lower bound are known.

Proposition A.8 (The maximum is not the sum). *[proved]^{model Prop 2.8} Let $\tau := \forall x \forall y F(x, y)$ ($F : \iota^2 \rightarrow o$), $\tau' := \forall x \forall y F(y, x)$, $T_1 := \{\tau\}$, $T_2 := \{\tau, \tau'\}$ with weights $\frac{1}{2}, \frac{1}{2}$, and $\mathcal{Q}$ symmetric in the holes. Under L0, $P_{T_2} = P_{T_1}$, so the posterior odds equal the prior odds at every n . Under L1^{max}, $V_{T_2}(s) = \frac{1}{2}V_{T_1}(s)$ on every instance, so the two-part odds are $(\pi(T_2)/\pi(T_1))2^{-n} \rightarrow 0$: the maximum breaks a tie inside the generator class.*

Proof of Proposition A.8. For a body β of type $\iota^2 \rightarrow o$ let β^{sw} swap its holes. Then $\tau'[F := \beta] = \tau[F := \beta^{\text{sw}}]$, so $\text{inst}(\tau) = \text{inst}(\tau')$, and hole symmetry gives $\mathcal{Q}_\tau = \mathcal{Q}_{\tau'}$. Under L0, $P_{T_2} = \frac{1}{2}\mathcal{Q}_\tau + \frac{1}{2}\mathcal{Q}_{\tau'} = P_{T_1}$, and the posterior odds are the prior odds (Corollary 4.3). The codes of τ and τ' differ, so $T_1 \neq T_2$. Under the maximum each instance has exactly one matcher per template (AS Thm 2.5), so two citations of probability $\frac{1}{2}\mathcal{Q}_\tau(s)$ each, and $V_{T_2}(s) = \frac{1}{2}V_{T_1}(s)$. $\square$

A.7 Hänni's trichotomy

Derivability is unbounded here (Definition 2.13), except in Proposition 2.19.

Proof of Proposition 2.14. (a) A consistent theory proves at most one of s , $\neg s$, so it is counted once in $\text{Bel}(s) + \text{Dis}(s) + \text{Indep}(s)$; an inconsistent one proves both and is counted twice. The sum is $(1 - \text{Inc}) + 2\text{Inc}$. (b) $\text{Dis}(s) = \text{Bel}(\neg s)$ by definition; classically $T \vdash \neg\neg s$ iff $T \vdash s$. On consistent theories $\text{Bel}(\top) = 1$, $\text{Bel}(\perp) = 0$, and $1 - \text{Dis} = \text{Bel} + \text{Indep}$ by (a). (c) $\text{Th}(T)$ is closed under consequence. (d) A consistent T proves at most one of two incompatible sentences, and proves $s \vee s'$ if it proves either. $\square$

Proof of Proposition 2.18. For one consistent T put $u(s) := 1[T \vdash s]$. Closure under $\wedge$ -introduction and elimination gives $u(\bigwedge_{i \in I} s_i) = \prod_{i \in I} u(s_i)$. So the right side equals $\sum_{\emptyset \neq I} (-1)^{|I|+1} \prod_{i \in I} u(s_i) = 1 - \prod_i (1 - u(s_i)) = 1[T \text{ proves some } s_i] \leq u(s_1 \vee \dots \vee s_k)$. The inequality is linear in u , so it survives the mixture $\text{Bel} = \sum_T \pi_D(T) u_T$. By completeness, $u_T(s) = 1[\text{Mod}(T) \subseteq \text{Mod}(s)]$, which identifies the focal elements. $\square$

Proof of Proposition 2.19. For each T in the posterior's (countable) support choose a complete consistent extension $\bar{T}$ (Lindenbaum), and put $P(s) := \sum_T \pi_D(T) 1[s \in \bar{T}]$, a mixture of point masses on complete theories. Then $1[T \vdash s] \leq 1[s \in \bar{T}] \leq 1 - 1[T \vdash \neg s]$, the second because $\bar{T}$ is consistent and contains $\neg s$ when $T \vdash \neg s$. Average over T . The argument uses only $\text{Th}_d(T) \subseteq \text{Th}(T) \subseteq \bar{T}$, so it holds for bounded derivability as well. $\square$

Proof of Examples 2.15 and 2.16. Example 2.15: $\text{Bel}(a) = \frac{1}{3}$ (from $\{a\}$) and $\text{Dis}(a) = 0$, since no theory proves $\neg a$; so $R(a) = 1$, and likewise $R(b) = 1$. No theory proves $a \wedge b$, and $\{\neg(a \wedge b)\}$ refutes it, so $R(a \wedge b) = 0$. A probability with $P(a) = P(b) = 1$ has $P(a \wedge b) = 1$. Example 2.16: the empty theory proves none of b , $b \wedge c$, $b \wedge \neg c$ or their negations, so each gets $F_{1/2} = \frac{1}{2}$, and $\frac{1}{2} + \frac{1}{2} \neq \frac{1}{2}$. $\square$

Proof of Proposition 2.17. Identify a consistent theory with its nonempty set of models $\text{Mod}(T)$ among the valuations, and a sentence with its set of models; over finitely many atoms every set of valuations is a sentence. For one theory

$$F_T(s) = 1[\text{Mod}(T) \subseteq s] + \frac{1}{2} 1[\text{Mod}(T) \text{ meets both } s \text{ and its complement}],$$

and $F_{1/2} = \sum_T \pi_D(T) F_T$. *If.* With $|\text{Mod}(T)| = 1$, F_T is the point mass on the model. With $\text{Mod}(T) = \{v, v'\}$, $F_T(s)$ is 1, $\frac{1}{2}$ or 0 as s contains both, one or none of v, v' , i.e. $F_T = \frac{1}{2}(\delta_v + \delta_{v'})$. A mixture of probabilities is a probability. *Only if.* For valuations $v \neq v'$ put $D_T := F_T(\{v, v'\}) - F_T(\{v\}) - F_T(\{v'\})$. If $v, v' \in \text{Mod}(T)$ and $|\text{Mod}(T)| \geq 3$, all three sets meet $\text{Mod}(T)$ and its complement, so $D_T = \frac{1}{2} - \frac{1}{2} - \frac{1}{2} = -\frac{1}{2}$. In every other case $D_T = 0$: if $\text{Mod}(T) = \{v, v'\}$, $D_T = 1 - \frac{1}{2} - \frac{1}{2}$; if exactly one of them, say v , is a model, then $F_T(\{v'\}) = 0$ and $F_T(\{v, v'\}) = F_T(\{v\})$, because $\{v, v'\}$ and $\{v\}$ meet $\text{Mod}(T)$ and its complement in the same way; if neither is a model, all three terms are 0. So if some T_0 with $\pi_D(T_0) > 0$ has three or more models and v, v' are two of them, $\sum_T \pi_D(T) D_T \leq -\frac{1}{2} \pi_D(T_0) < 0$, whereas a probability gives 0. $\square$

Linear-programming checks. c5: Examples 2.15 and 2.16 are infeasible as probabilities; 300 random posteriors on three atoms gave no violation of 2- or 3-monotonicity and no bracketing failure; renormalisation was incoherent in 90/300, the 50/50 rule in 290/300. r5 (independent generator): no violation of 2-, 3- or 4-monotonicity on two atoms, no bracketing failure with other completions; the 50/50 rule was coherent in 12/300 trials, exactly those in which every theory had at most two models. c14: the criterion of Proposition 2.17 agrees with LP coherence in 3225/3225 posteriors on two atoms (all one- and two-theory supports plus 3000 random) and 600/600 on

three. (Model §7 also credits c5 with 4-monotonicity; c5’s output records only 2 and 3, and 4 was checked by the referee on two atoms.)

Remark A.9 (Renormalisation and complete theories). [refuted; computed]^{model §7 (referee m4)} The model track’s first version said: “the coherent cases [of renormalisation] are essentially those where every theory is complete”. **Refuted**: in the referee’s trials renormalisation was coherent in 199 of 300 posteriors, only 2 of them with all theories complete; in c14 it was coherent in 1761 cases on two atoms (218 with all theories complete) and 306 on three atoms (42). For the 50/50 rule the exact condition is Proposition 2.17; for renormalisation no characterisation is given.

B Proofs and computations for Section 3

Proofs follow the universal track’s final notes; steps they leave out are named. Checks are the seeded scripts c1–c10 in `research/tracks/universal/checks/`; the referee’s independent scripts r1–r7 (which do not import the track’s code) are in `referee_code/`. A second run of all scripts reproduced every output byte for byte. Dirichlet $\alpha = 1$ in c2–c7, $\alpha = \frac{1}{2}$ in c8–c10.

B.1 Factorisation, odds and robustness

name	axioms	$\vdash \forall x\varphi$: pure / over Q
$H_\forall$; H_{sch}	$\{\forall x\varphi\}$; $\{\sigma_\varphi\}$, closed guard	yes/yes; no/no
H_{open} ; H_{both}	$\{\sigma_\varphi\}$, open guard ($z \sim \mathcal{Q}_{\text{open}}$); $\{\forall x\varphi, \sigma_\varphi\}$, weight $w_\forall$ on $\forall x\varphi$	yes/yes; yes/yes
memorisers; over-specific	finite sets of closed instances; e.g. $\sigma_\varphi[z := Sz]$, closed guard	no/no; no/no
over-general	a subterm or subformula of σ_φ made a fresh metavariable; the bare F_0 (inconsistent)	varies
root split	$\{\sigma_\varphi[z := f(z_1, \dots, z_r)] : f \text{ a root of } \mathcal{Q}\}$, closed guards	no/no

Table 13: Hypotheses of Section 3; provability for $\varphi = 0 + x = x$ (closed instances do not prove $\forall x\varphi$ over Q: AS Ex 4.9). $B \oplus_w A$ adds A with weight w to B , scaling B ’s weights by $1 - w$. R_k , Split_k , Both_0 : Sections 3.6 and 3.8.

Proof of Proposition 3.2. A $\mathcal{C}_{\text{min}}$ -derivation from $\{\forall x\varphi\}$ is a chain: cite $\forall x\varphi$, then $k \geq 0$ eliminations with terms $t_1, \dots, t_k$, with probability $c^k(1 - c) \prod_j \mathcal{Q}(t_j)$. With $k = 0$ it yields $\forall x\varphi$. For $k \geq 1$ map it to the σ_φ -chain “cite σ_φ with $z := t_1$, then eliminate with $t_2, \dots, t_k$ ”, of probability $c^{k-1}(1 - c) \prod_j \mathcal{Q}(t_j)$. The conclusions coincide, since $\forall E$ with t_1 gives $\varphi(t_1) = \sigma_\varphi[z := t_1]$ (t_1 has no bound variable) and the remaining steps agree; failing chains map to failing chains. The map is a bijection onto all σ_φ -chains, and no σ_φ -chain yields $\forall x\varphi$ (its conclusions have fewer quantifiers). Summing over chains with conclusion s gives the factor c . Citation factorises over axioms, $\mu_T = \sum_A w_A \mu_A$, which gives the background form. The leading universal quantifiers of φ are rigid, so exactly $m + 1$ chain lengths succeed from σ_φ and $m + 2$ from $\forall x\varphi$, which gives the normalisers. For k variables the first k eliminations are absorbed into one citation. $\square$

c1 checks the identity on every output of $2 \cdot 10^5$ sampled derivations ($0 + x = x$, $\neg Sx = 0$, $\forall y(x + y = y + x)$; $\mathcal{Q}_{\text{num}}$, $\mathcal{Q}_{\text{GW}}$) to $5 \cdot 10^{-14}$; referee r1 to $1.6 \cdot 10^{-15}$.

Proof of Theorem 3.3. Each case follows from Proposition 3.2. **L0**: $H_\forall$ cites only $\forall x\varphi$. **L0^{cl}**: identical laws by definition. μ_T : factor c per datum. **L1**: divide by the normalisers; for quantifier-free φ , $Z_{H_{\text{sch}}} = 1 - c$ and the factor is $c(1 - c)/((1 - c) + c(1 - c)) = c/(1 + c)$. **L1^{sel}**:

$\mu_{H_\forall}(s)/\mu_{H_\forall}(S) = c\mu_{H_{\text{sch}}}(s)/(c\mu_{H_{\text{sch}}}(S))$ as $\forall x\varphi \notin S$. **L1^{max}**: best chains map to best chains, each scaled by c . Scores: both theories prove every instance and, when $\forall x\varphi$ is consistent, are consistent with the data (if it is not, $S_{\text{prove}}(H_\forall) = 0$, which still agrees with Theorem 3.4). Background: $r(s) = \frac{(1-w)\mu_B(s)+wc\mu_{H_{\text{sch}}}(s)}{(1-w)\mu_B(s)+w\mu_{H_{\text{sch}}}(s)} \in [c, 1]$. $\square$

likelihood	O_n/O_0
L0 (strict citation)	0 for $n \geq 1$
L0 ^{cl} ; L1 ^{sel} (S) with $\forall x\varphi \notin S$; S_{prove} , S_{nc} , S_g ($\forall x\varphi$ consistent)	1
μ_T (unnormalised); L1 ^{max}	c^n
L1	$(cZ_{H_{\text{sch}}} / ((1-c) + cZ_{H_{\text{sch}}}))^n$; $(c/(1+c))^n$ for quantifier-free φ
background B , under μ_T	$\prod_i r(X_i)$, $r(s) \in [c, 1]$; $r(s) = c$ if $\mu_B(s) = 0$

Table 14: Odds of $H_\forall$ against H_{sch} after n closed instances, relative to the prior odds (Theorem 3.3); asserted on every sampled path (5 formulas, 2 term laws, 50 seeds, $n \leq 200$), referee's code to $7 \cdot 10^{-13}$. The consistency condition in the score row is ours; the notes leave it implicit.

Proof of Theorem 3.4. (B1) is Theorem 3.3. (B3) under μ_T and L1^{max} is Proposition B.2, whose map acts derivation by derivation. Under the scores, for single axioms and quantifier-free data: by Herbrand's theorem a quantifier-free sentence s follows from $\{\forall x\varphi\}$ iff it follows from finitely many closed instances of φ , i.e. from $I_c(\varphi) = \bigcup \text{inst}(H_{\text{sch}})$; in particular $\{\forall x\varphi\}$ is consistent iff $I_c(\varphi)$ is. So $H_\forall$ and H_{sch} prove, and are consistent with, the same quantifier-free data, and S_{prove} , S_{nc} , S_g give them equal scores. (B2): fix w and $s \neq \forall x\varphi$. Under μ_T , $\mu_{B \oplus w \forall}(s) = (1-w)\mu_B(s) + wc\mu_{H_{\text{sch}}}(s) \leq (1-w)\mu_B(s) + w\mu_{H_{\text{sch}}}(s) = \mu_{B \oplus w \sigma}(s)$. Under L1, $Z_{B \oplus w \forall} = (1-w)Z_B + w((1-c) + cZ_{H_{\text{sch}}}) \geq (1-w)Z_B + wZ_{H_{\text{sch}}} = Z_{B \oplus w \sigma}$, since $(1-c) + cZ_{H_{\text{sch}}} - Z_{H_{\text{sch}}} = (1-c)(1 - Z_{H_{\text{sch}}}) \geq 0$; so the normalised likelihoods obey the same inequality datum by datum. A pointwise inequality at every w survives integration against a common weight prior. Under L0^{cl} the laws coincide at every w . $\square$

Proposition B.1 (Robustness of the direction). *[proved; computed]^{universal Props U2n, U2t, U2c} In $\mathbf{C}_{\min}$, for every $s \neq \forall x\varphi$: (U2n) with a full-support noise law N and $\eta \in (0, 1)$, the ratio of the noisy laws of $H_\forall$ and H_{sch} at s is in $[c, 1]$ under μ_T , ≤ 1 under L1, and 1 under L1^{sel} with noise after selection; a false datum no longer refutes. (U2t) If chains are weighted by $f(\ell)w_A \prod_j \mathcal{Q}(t_j)$, f non-increasing in the number ℓ of eliminations (e.g. Levin-style $f(\ell) \propto (\ell+1)^{-2}$), then $P_{H_\forall}^f(s) \leq \sup_\ell \frac{f(\ell+1)}{f(\ell)} P_{H_{\text{sch}}}^f(s) \leq P_{H_{\text{sch}}}^f(s)$. (U2c) Under L1, quantifier-free φ and any prior on c , $O_n/O_0 \leq 2^{-n}$.*

Proof of Proposition B.1. (U2n) The unnormalised claim is the background case of Theorem 3.3 with $\mu_B := N$. For L1, $c\mu_{H_{\text{sch}}}(s)/Z_{H_\forall} \leq \mu_{H_{\text{sch}}}(s)/Z_{H_{\text{sch}}}$ because $Z_{H_\forall} = (1-c) + cZ_{H_{\text{sch}}} \geq Z_{H_{\text{sch}}}$ (as $Z_{H_{\text{sch}}} \leq 1$); mixing both with the same ηN keeps the inequality. Under L1^{sel} the selected laws coincide. A false datum has likelihood $\geq \eta N(s) > 0$. (U2t) The bijection maps a $\forall x\varphi$ -chain with $\ell+1$ eliminations to a σ_φ -chain with ℓ , with the same term product; $f(\ell+1)$ becomes $f(\ell)$. (U2c) H_{sch} 's normalised law is $\mathcal{Q}_{\sigma_\varphi}$ for every c , and $c/(1+c)$ increases in c to $\frac{1}{2}$ at $c=1$; a bound valid for every c survives integration. $\square$ c10(h) checked (U2n) at $2 \cdot 10^5$ random parameter points.

Proposition B.2 (Simulation inequality). *[proved]^{universal Prop U3} In a $\mathbf{C}_{U3}$ calculus, for quantifier-free φ and s , $\mu_{B \oplus \forall x \varphi}(s) - P^0(s) \leq c(\mu_{B \oplus \sigma_\varphi}(s) - P^0(s))$, with $P^0(s)$ the mass of derivations of s that never cite the added axiom.*

Proof of Proposition B.2. In a derivation of a quantifier-free s from $B \oplus \forall x \varphi$, a leaf citing $\forall x \varphi$ is not the root, and its parent takes a premise that is not quantifier-free; in a $\mathbf{C}_{U3}$ calculus only $\forall E$ does. So each such leaf sits in a pattern “ $\forall E(\text{cite } \forall x \varphi, t)$ ” concluding $\varphi(t)$. Replace each pattern by “cite $\sigma_\varphi, z := t$ ”: probability $c \mathcal{Q}(t) p_{\text{cite}} w$ becomes $p_{\text{cite}} w \mathcal{Q}(t)$, and conclusions are kept. The map is injective, because cited axioms are labelled and every σ_φ -leaf in the image came from a pattern. Derivations with $m \geq 1$ patterns contribute $\sum c^m \Pr(f(\pi)) \leq c \sum_{\pi' \text{ citing } \sigma_\varphi} \Pr(\pi')$; those with $m = 0$ never cite the added axiom and have the same probability in both theories. $\square$

The $\wedge$ -detour counterexample (Remark 3.5; proof sketch). $\mathbf{C}_\wedge$ with probabilities p_c (cite), c ($\forall E$), a ($\wedge I$), e ($\wedge E$, side chosen with probability $\frac{1}{2}$); $B = \emptyset$; $\varphi = 0 + x = x$. A formula G made at a base node (cite or $\forall E$) reaches the root through a balanced word of $\wedge I$ (into a conjunction with a free partner) and $\wedge E$ (take G ’s side), each pair of weight eaZ_T with Z_T the success probability. So $P_T(G) = \text{base}_T(G)D(eaZ_T)$ with $D(x) = \sum_j \text{Cat}_j x^j = (1 - \sqrt{1 - 4x})/(2x)$; the ansatz solves $P(G) = \text{base}(G) + e \sum_W P(G \wedge W)$ exactly when $D = 1 + xD^2$. For $H_\forall$, $\text{base}(\varphi(t)) = cP(\forall x \varphi) \mathcal{Q}(t)$ and $P(\forall x \varphi) = p_c D_\forall$, so $P_{H_\forall}(\varphi(t)) = cp_c D_\forall^2 \mathcal{Q}(t)$; for H_{sch} , $P(\varphi(t)) = p_c D_\sigma \mathcal{Q}(t)$. Z_T is the least fixed point of $Z = D(eaZ)(p_c + cP_U + aZ^2)$, $P_U = p_c D$ for $H_\forall$, 0 for H_{sch} ; the first map dominates, so $D_\forall \geq D_\sigma \geq 1$, with $D_\forall > 1$ if $a, e > 0$, and $R = cD_\forall^2/D_\sigma \geq cD_\forall > c$. Not written out: that the least fixed point is the success probability. At (0.35, 0.15, 0.25, 0.25), $D_\forall = 1.03160$, $D_\sigma = 1.02636$, $R = 0.15553$ (c10(a), referee r2); two samplers give 0.15528 ± 0.00067 and 0.1562 ± 0.0005 .

Proof of Proposition 3.6. Under $\mathbf{L1}^{\text{sel}}$, T_σ gives $\varphi(t)$ probability $w \mathcal{Q}(t)$ and $\psi(t)$ probability $(1 - w) \mathcal{Q}(t)$. $T_\forall$ reaches S through $\forall x \varphi$ only after one elimination, so it gives $w_e \mathcal{Q}(t)$ and $(1 - w_e) \mathcal{Q}(t)$ with $w_e = wc/(1 - w + wc)$. The likelihoods are $K_D w_e^{n_\varphi} (1 - w_e)^{n_\psi}$ and $K_D w^{n_\varphi} (1 - w)^{n_\psi}$ with the same K_D . $w \mapsto w_e$ is a C^1 bijection of $[0, 1]$ with inverse $w = v/(c + v(1 - c))$ and $dw/dv = h(v)$, so

$$\text{BF}_n = \int_0^1 v^{n_\varphi} (1 - v)^{n_\psi} h(v) dv / \int_0^1 v^{n_\varphi} (1 - v)^{n_\psi} dv = \mathbb{E}[h(V)], \quad V \sim \text{Beta}(n_\varphi + 1, n_\psi + 1).$$

$n_\varphi/n \rightarrow w^*$ a.s. and the Beta variance is $\leq 1/(4(n + 3))$, so $V \rightarrow w^*$ in law; h is bounded and continuous. $h(v) > 1$ iff $\sqrt{c} > c + v(1 - c)$ iff $v < \sqrt{c}/(1 + \sqrt{c})$. $\square$

c10(b) ($c = 0.3$, 10 seeds, $n = 10^5$): 2.1918, 1.1537, 0.7102, 0.4056 at $w^* = 0.1, 0.3, 0.5, 0.8$, against $h(w^*) = 2.1914, 1.1534, 0.7101, 0.4056$.

Proof of Remark 3.7. Under $\mathbf{L1}_{\text{cit}}^{\text{sel}}$ each citation is repeated until its chain lands in S . The component $\forall x \varphi$ lands only after one elimination, so its selected law is $\mathcal{Q}_{\sigma_\varphi}$, as is that of σ_φ ; σ_ψ has $\mathcal{Q}_{\sigma_\psi}$. Both theories therefore have law $(1 - w) \mathcal{Q}_{\sigma_\psi} + w \mathcal{Q}_{\sigma_\varphi}$ at every w . $\square$

Proposition B.3 (Spare-slot prover). *[proved; computed]^{universal §4} In $\{H_{\text{sch}}, H_\forall, H_{\text{both}}\}$ with a uniform prior on $w_\forall$, under $\mathbf{L1}$ with quantifier-free φ , the likelihood ratio $H_{\text{both}} : H_{\text{sch}}$ is $I_n = \int_0^1 \left(\frac{1 - w(1 - c)}{1 + wc}\right)^n dw \in [\frac{1}{n+1}, \frac{1}{n(1 - c)}]$, so $P(T \vdash \forall x \varphi \mid D_n) \rightarrow 0$ like $\Theta(1/n)$: $\forall x \varphi$ is a spare slot of H_{both} , a template beside the generator’s own that the data never or rarely use (Proposition 4.14).*

Proof of Proposition B.3. H_{both} with weight w on $\forall x\varphi$ has $\mu(\varphi(t)) = (1 - w + wc)(1 - c)\mathcal{Q}(t)$ and $Z = (1 - c)(1 + wc)$, so its ratio to H_{sch} is $(1 - w(1 - c))/(1 + wc)$. Since $1 - w + wc - (1 - w)(1 + wc) = w^2c \geq 0$, $I_n \geq \int_0^1 (1 - w)^n dw = \frac{1}{n+1}$; and the integrand is $\leq e^{-nw(1-c)}$, so $I_n \leq \frac{1}{n(1-c)}$. $\square$ Computed (c5 Part C): $nI_n = 0.958, 0.996, 1.000$ at $n = 10, 10^2, 10^5$.

B.2 Size principle, over-specific templates and memorisers

Theorem B.4 (Size principle on instances). *[proved; computed]^{universal Thm U4} Let the data be i.i.d. from P^* , $P^*(I) = 1$ for $I = I_c(\varphi)$, and P' a sub-probability with $P'(I) = p < 1$. Then $\text{KL}(P^* \| P') \geq \ln \frac{1}{p}$, $\frac{1}{n} \ln \frac{P'(D_n)}{P^*(D_n)} \rightarrow -\text{KL}(P^* \| P')$ a.s., and $\mathbb{E} \frac{P'(D_n)}{P^*(D_n)} \leq p^n$. Under the citation law $(\mathbf{L0}, \mathbf{L0}^{\text{cl}})$, a first-order template strictly more general than σ_φ , with metavariables i.i.d. from $\mathcal{Q}$ and a formula law F , has $P_\tau(I) \leq \max(\max_f p_f, \max_g F(\text{root} = g))$.*

Proof of Theorem B.4. (1) Log-sum inequality: $\sum_{s \in I} P^*(s) \ln \frac{P^*(s)}{P'(s)} \geq P^*(I) \ln \frac{P^*(I)}{P'(I)} = \ln \frac{1}{p}$. (2) With $Y_i = \ln \frac{P'(X_i)}{P^*(X_i)}$, $y^+ \leq e^y$ gives $\mathbb{E} Y^+ \leq \mathbb{E} e^Y = P'(\text{supp } P^*) \leq 1$, so $\mathbb{E} Y = -\text{KL} \in [-\infty, 0)$ and the strong law applies (it allows $\mathbb{E} Y = -\infty$ when $\mathbb{E} Y^+ < \infty$). (3) Tonelli: $\mathbb{E} \prod_i P'(X_i)/P^*(X_i) = P'(\text{supp } P^*)^n$. (4) Write $\sigma_\varphi = \tau\rho$, ρ not a renaming. Either (a) some metavariable u of τ has $\rho(u)$ with a rigid root f , or (b) ρ identifies two metavariables $u \neq u'$; otherwise ρ would be a renaming. If $\tau\theta \in \text{inst}(\sigma_\varphi)$, then $\tau\theta = \tau\rho\theta'$ and unique matching gives $\theta = \rho\theta'$ on τ 's metavariables. In case (a) $\theta(u)$ has root f (probability $\mathcal{Q}(\text{root} = f)$, or $F(\text{root} = f)$); in case (b) $\theta(u) = \theta(u')$, of probability $\sum_t \mathcal{Q}(t)^2 \leq \max_t \mathcal{Q}(t) \leq \max_f \mathcal{Q}(\text{root} = f)$. $\square$

c2: under $\mathcal{Q}_{\text{num}}$, $?y + ?z = ?z$ has $P_\tau(I) = \frac{1}{2}$ and KL 0.693, $0 + ?y = ?z$ has $\frac{1}{3}$ and 1.386; (4) held in 38 of 38 cases and in the referee's 130 random generalisations (r7).

Proposition B.5 (Over-specific templates). *[proved; computed]^{universal Prop U5} For $\tau = \sigma_\varphi[z := f(z_1, \dots, z_r)]$ and root-generated $\mathcal{Q}$, $P_\tau(\varphi(t))/P_{H_{\text{sch}}}(\varphi(t))$ is $1/p_f$ if $\text{root}(t) = f$ and 0 otherwise. So the odds $\tau : \sigma_\varphi$ grow like p_f^{-n} while all data have root f (probability p_f^n), then vanish; they form a martingale. Some single-root template survives with probability $\sum_f p_f^n$, the failure probability of AS Thm 4.4(a).*

Proof of Proposition B.5. $\mathcal{Q}(f(\bar{t})) = p_f \prod_i \mathcal{Q}(t_i)$ and τ matches $\varphi(t)$ iff $\text{root}(t) = f$, with $z_i := t_i$; so $P_\tau(\varphi(t)) = \mathcal{Q}(t)/p_f$ there and 0 elsewhere. Under H_{sch} 's law the ratio has mean $\sum_{\text{root}(t)=f} \mathcal{Q}(t)/p_f = 1$. $\square$ c2: gain per datum $\ln 2$ under $\mathcal{Q}_{\text{num}}$ and $\ln(1/0.3) = 1.204$ under $\mathcal{Q}_{\text{GW}}$; over-specific mass 0.110 at $n = 1$, 0 at $n = 10$ (Table 2).

Proposition B.6 (Memorisers). *[(i), (ii), (ii')] proved; (iii) conjecture; computed]^{universal Prop U6} Give the memorisers a product-Bernoulli prior over a named universe of sentences s , with $r_s = 2^{-(|s|+1)}$ and $\sum r_s < \infty$. Let $Z_0 := \prod_s (1 - r_s) > 0$, S_n the distinct data, $m_n = |S_n|$, μ the prior of the memoriser class. (i) The class marginal lies in $[Z_0, 1] \cdot \prod_{s \in S_n} r_s m_n^{-n}$ for uniform weights, with the Dirichlet(α)-multinomial probability $\text{DirMult}_\alpha(D_n)$ of the counts (formula in Proposition 4.10) in place of m_n^{-n} for Dirichlet weights. (ii) Uniform weights, $\mathcal{Q}$ of infinite support and finite entropy: the log-odds against H_{sch} divided by n tend to $-\infty$ a.s. (ii') Laplace weights: deterministically the log-odds are $\geq \ln \frac{\mu Z_0}{\pi(H_{\text{sch}})} + \sum_{s \in S_n} \ln r_s - (m_n - 1) \ln(n + m_n - 1)$, which under $\mathcal{Q}_{\text{num}}$ (numeral universe) is $\geq -C \ln^2 n$ a.s. for all large n : the class is not exponentially small. (iii) Conjecture: $-\Theta(\ln^2 n)$ under $\mathcal{Q}_{\text{num}}$.*

Proof of Proposition B.6. (i) $\pi(F) = Z_0 \prod_{s \in F} r_s / (1 - r_s)$. $F = S_n$ gives the lower bound $(r_s / (1 - r_s) \geq r_s)$. For the upper, $\sum_{F \supseteq S_n} \pi(F) = \prod_{s \in S_n} r_s$, and each likelihood is at most that of S_n : $|F|^{-n} \leq m_n^{-n}$, and j unused components multiply DirMult_α by $\frac{\Gamma((m+j)\alpha)\Gamma(n+m\alpha)}{\Gamma(m\alpha)\Gamma(n+(m+j)\alpha)} \leq 1$. (ii) The log-odds are at most a constant $-n \ln m_n + \sum_i \ln \frac{1}{Q(t_i)}$, the sum is $n(H(Q) + o(1))$ a.s., and $m_n \rightarrow \infty$ a.s. (ii') $\text{DirMult}_1(D_n) = 1 / [\binom{n}{(n_s)} \binom{n+m-1}{m-1}]$; the multinomial coefficient is $\leq 1/\text{ML}(D_n)$ (a type class has probability ≤ 1 under the maximum-likelihood law), $\binom{n+m-1}{m-1} \leq (n+m-1)^{m-1}$, and $\text{ML}(D_n) \geq P_{H_{\text{sch}}}(D_n)$; with the lower bracket this is the deterministic bound. For numerals $|\varphi(S^k 0)| = |\sigma_\varphi| + \text{occ} \cdot k$. With K_n the largest k seen, $P(K_n \geq 3 \log_{1/q} n) \leq n \cdot n^{-3} = n^{-2}$, summable, so by Borel–Cantelli a.s. $K_n < 3 \log_{1/q} n$ eventually; then $m_n \leq K_n + 1$ and $\sum_{s \in S_n} (|s| + 1) = O(\ln^2 n)$. $\square$

(The first version had probability $1 - 1/n$ at threshold $2 \log_{1/q} n$; c10(g) checks the n^{-2} tail exactly.) The universe must be named: over all closed sentences $\sum r_s = \infty$ at $\lambda = 1$ (referee r6). c3 ($\alpha = 1$, 10 seeds, log-odds up to a constant in $[\ln Z_0, 0]$): under $\mathcal{Q}_{\text{num}}$, $q = \frac{1}{2}$, $m_n = 10.4, 16.4$ at $n = 10^3, 10^5$ and the Laplace log-odds are $-133.8, -327.7$ ($-2.80 \ln^2 n, -2.47 \ln^2 n$; uniform weights $-1062, -1.4 \cdot 10^5$); at $q = 0.9$, $D_l / \ln^2 n$ moves from -35 to -53 (pre-asymptotic); under $\mathcal{Q}_{\text{GW}}$, $m_n = 2.2 \cdot 10^4$ at $n = 10^5$ and $D_l / n \approx -4.0$: exponential.

B.3 Predictive confirmation

Proof of Theorem 3.8. (1) By the chain rule, $\sum_{n < N} \mathbb{E} \text{KL}(P^* \| M(\cdot \mid D_n)) = \mathbb{E} \ln \frac{P^*(D_N)}{M(D_N)} \leq \ln \frac{1}{\pi(T^*)}$, as $M \geq \pi(T^*)P^*$. Pushing forward to the indicator of I , $\text{KL}(P^* \| M(\cdot \mid D_n)) \geq \ln \frac{1}{M(I \mid D_n)} \geq 1 - M(I \mid D_n)$; a nonnegative series with finite mean is finite a.s. (2) $P_T(I)^\infty = 1[T \in G]$, and $C^* \subseteq G$, so Theorem 4.2 applies. (3) $1 - \pi_n(G) \leq \min(1, \sum_{T \notin G} \pi_n(T) / \pi_n(T^*))$ and Theorem B.4(3). (4) $P_T(D_n) = (1 - \varepsilon_T)^n \hat{P}(D_n)$ on instance sequences. $\square$

Remark B.7 (Relation to Hutter). [known; computed]^{universal §5} In the Bayes–Laplace model the universal hypothesis, even its observable form, has posterior 0, while the universal prior confirms it (Hutter, 2007) (abstract checked; full text not). Theorem 3.8(2) is the analogue for countable classes, by the same mechanism: prior mass on instance-only hypotheses. But G contains H_{sch} , which does not prove $\forall x \varphi$. Nor is confirmation stepwise monotone: with prior $\frac{1}{2}$ on each of σ_φ , $\varphi(Sz)$, the datum $\varphi(S0)$ lowers σ_φ to $\frac{1}{3}$, a Nicod-type effect (Leike and Hutter, 2015).

c4: with a point mass $\pi_0 = 0.01$ and $\text{Uniform}(0, 1)$ on ε , $P(\text{all future} \mid n) = \pi_0 / (\pi_0 + \frac{1-\pi_0}{n+1})$; with $\pi_0 = 0$ it is 0 while $P(\text{next } m) = \frac{n+1}{n+m+1}$. Summed misses 4.22 (Laplace) and 2.70 (dyadic) $\leq \ln 100 = 4.61$.

B.4 The Bayesian ω -gap

Proof of Theorem 3.9. (a) Sum the limit of Theorem 4.2 over $\text{Prov}_\forall$, or over the theories counted by Bel, Dis, Indep or Inc. (b) Both theories lie in C^* , one in $\text{Prov}_\forall$. For the learned-weight pair the marginal laws are exchangeable, not equal, so (a) does not apply; Proposition 3.6 gives the limit $\pi(T_\forall)h / (\pi(T_\forall)h + \pi(T_\sigma)) \in (0, 1)$ directly. (Under L1^{sel} , $\{\sigma_\varphi, \exists x \neg(0 + x = x)\}$ with S excluding the second sentence is a consistent refuter in C^* , so all three trichotomy values can keep mass.) (c1) $T \in C^*$ implies $\text{Th}_C(T) \cap X = \text{Th}_C(T^*) \cap X$. (c2) Each axiom instance of $T \in C^*$ lies in $\text{Th}_C(T^*) \cap X$, hence follows from T^* by soundness; symmetrically. (In $\mathbf{C}_{\min}$, $T^* = \{\forall x(\varphi \wedge \psi)\}$ has limit 0 in (c1) and 1 in (c2): different questions, conflated in the first version.) (c3) In $\mathbf{C}_{\min}$ every chain has positive probability under full-support $\mathcal{Q}$, so L1 is faithful, and a σ_φ -chain with quantifier-free φ can only cite. For the second part apply the rule to a derivation

of an instance. (d1) $\mu_{H_\forall}(\cdot \mid S) = c\mu_{H_{\text{sch}}}(\cdot \cap S)/(c\mu_{H_{\text{sch}}}(S)) = P_{H_{\text{sch}}}$ since $\text{supp } \mu_{H_{\text{sch}}} = I_c \subseteq S$; then (c3) with $T^* := H_{\text{sch}}$, or (b) under L1^{sel} . (d2) For quantified φ , $\tilde{P} := \mu_{H_{\text{sch}}}(\cdot \mid S)$ is the law of neither hypothesis. On S , $\mu_{H_\forall} = c\mu_{H_{\text{sch}}}$, so both log-ratios are constant there and $\text{KL}(\tilde{P} \parallel P_{H_{\text{sch}}}) = \ln \frac{Z_{H_{\text{sch}}}}{\mu_{H_{\text{sch}}}(S)} < \ln \frac{Z_{H_\forall}}{c\mu_{H_{\text{sch}}}(S)} = \text{KL}(\tilde{P} \parallel P_{H_\forall})$, because $cZ_{H_{\text{sch}}} < Z_{H_\forall}$; the posterior concentrates on H_{sch} (Theorem 4.16). For a general class this is a sketch. (e) Pairs form a countable class of i.i.d. laws. (H_{sch}, S) has the data law iff $S \supseteq I_c$; $(H_\forall, S)$ has it iff $S \supseteq I_c$ and $\forall x \varphi \notin S$ (Theorem 3.3); otherwise some instance gets probability 0 or $\forall x \varphi$ positive probability. Summing prior masses over C^* gives the limit. $\square$

c10(e): filters {all sentences, closed quantifier-free, closed instances, closed instances with S -rooted argument}, $\pi = (\frac{1}{3}, \frac{2}{3})$, ν uniform: 0.333, 0.284, 0.252, 0.250 at $n = 0, 1, 5, 10$, predicted 0.2500.

Proof of Corollary 3.13. Members of $C_\forall^*$ do not prove $\forall x \varphi$ and share T^* 's likelihood, so

$$\pi_n(C_\forall^*) = \pi(C_\forall^*) P_{T^*}(D_n)/M(D_n).$$

Acceptance needs $\pi_n(C_\forall^*) \leq \delta$, that is, $M(D_n)/P_{T^*}(D_n) \geq \pi(C_\forall^*)/\delta$. This ratio is a nonnegative supermartingale started at 1; by Ville's inequality it ever reaches that level with probability at most $\delta/\pi(C_\forall^*)$. $\square$

stream	likelihood	class	$n = 0$	10	50	100
closed instances	L1	full	0.750	0.364	0.004	0.000
closed instances	L1	$\{H_\forall, H_{\text{open}}\}$	1.000	1.000	1.000	1.000
closed instances	L1^{sel}	full	0.750	0.750	0.750	0.750
output of H_{open}	L1	full	0.750	0.860	1.000	1.000
output of $H_\forall$	L1	full	0.750	1.000	1.000	1.000

Table 15: $P(T \vdash \forall x \varphi \mid D_n)$ in C_{open} (c5 Part A); closed instances are also $H_\forall$'s output filtered to closed instances; $\varphi = 0 + x = x$, defaults, prior $2^{-(\text{symbols}+1 \text{ per axiom})}$; full class $\{H_\forall, H_{\text{open}}, H_{\text{sch}}$ (guarded), $H_{\text{both}}\}$, with a uniform prior on H_{both} 's weight, prior share of provers 0.750. On the full output of $H_\forall$ (itself reported), $H_\forall$ has 0.993 at $n = 10$.

prior code	bits($H_\forall$)	bits(H_{sch})	share of $H_\forall$
universal: dtrc symbols +1 per axiom, $\lambda = 1$	7	6	0.333
model π_1 (Definition 2.4; 4 bits per token)	27	26	0.333
model π_2			0.200
experiments' template code, closed guard	17.18	17.09	0.484
pa, $\beta = 5$ bits per symbol	30	25	0.030
pa, $\beta = \log_2 23$	27.14	22.62	0.042

Table 16: Prior share of $H_\forall$ in $\{H_\forall, H_{\text{sch}}\}$ for $\varphi = 0 + x = x$ under the four tracks' codes (c10(f); the model rows by hand from Definition 2.4). The experiments' value equals their E1 limit under $\text{L1}_{\text{cit}}^{\text{sel universal §6.6}}$.

Proposition B.8 (Hänni's variants on $\forall x \varphi$). *[proved; computed]^{universal §6.2} Let $\varphi = 0 + x = x$, $\mathcal{Q}$ of full support. Under S_{prove} the posterior converges a.s. in total variation to the prior restricted to $\{T \text{ consistent} : T \vdash I_c(\varphi)\}$, which contains $\mathcal{Q}$ (independent), $\mathcal{Q} + \forall x \varphi$ (true), $\mathcal{Q} + \exists x \neg \varphi$ (false),*

H_{sch} , $H_{\forall}$ and consistent over-general templates such as $\{0 + z_1 = z_2\}$: all three trichotomy values keep positive mass, and over-generalisation is never penalised. Under S_{nc} and S_g ($g \equiv 1$ on provable data) the theories that prove all the data share one factor, so their mutual odds keep the prior odds and again all three trichotomy values keep positive mass; under S_{nc} the limit is the prior restricted to the larger set $\{T : T \cup I_c(\varphi) \text{ consistent}\}$, which also contains non-provers of the instances such as $\emptyset$. Under generative likelihoods inconsistent theories die within a few data (Table 17).

Proof of Proposition B.8. Under S_{prove} the posterior is the prior restricted to $\text{Adm}_n = \{T \text{ consistent} : T \vdash D_n\}$; every closed instance eventually appears a.s., so $\text{Adm}_n \downarrow \{T \text{ consistent} : T \vdash I_c(\varphi)\}$, and continuity of measure gives convergence in total variation. The model of AS Ex 4.9 shows that Q , $Q + \forall x\varphi$ and $Q + \exists x\neg\varphi$ are consistent with all closed instances, each of which Q proves; $\{0 + z_1 = z_2\}$ has a one-element model. Under S_{nc} and S_g ($g \equiv 1$ on provable data) all theories proving the data get the same factor, so their mutual odds stay at the prior odds. Under S_{nc} the score is $1[T \cup D_n \text{ consistent}]$; as every closed instance eventually appears a.s., compactness and continuity of measure give the limit: the prior restricted to $\{T : T \cup I_c(\varphi) \text{ consistent}\}$, which contains $\emptyset$. $\text{Inc} = 0$ under the scores by definition. $\square$

φ , law, likelihood	$n = 0$	1	2	5	10	20
$0 + x = x$, $\mathcal{Q}_{\text{num}}$; $L0^{\text{cl}}$, $L1$, $L1^{\text{sel}}$	0.674	0.000	0.000	0.000	0.000	0.000
$\neg Sx = 0$, $\mathcal{Q}_{\text{num}}$; $L0^{\text{cl}}$ / $L1$	0.931	.585/.635	.319/.390	.056/.081	.002/.003	0.000
$x < Sx$, $\mathcal{Q}_{\text{num}}$; $L0^{\text{cl}}$	0.627	0.080	0.004	0.000	0.000	0.000
$\neg Sx = 0$, $\mathcal{Q}_{\text{GW}}$; $L0^{\text{cl}}$	0.931	0.454	0.191	0.010	0.000	0.000

Table 17: Mass Inc of inconsistent theories in the class of Table 2 without its learned-weight root split, so all members have fixed weights and no Dirichlet α enters (c10(d), 20 seeds): the bare formula metavariable and, for $\neg Sx = 0$, three templates with instances such as $\neg(0 = 0)$. $\text{Bel}_{\text{cons}}(\forall x\varphi)$ tends to the prior share ≈ 0.332 under $L0^{\text{cl}}$, $L1^{\text{sel}}$ and to 0 under $L1$.

Proposition B.9 (An ω -gap for Q4). *[proved; computed]^{universal Prop U13} ($Q - Q4$) plus all closed instances of $x + 0 = x$ does not prove $\forall x(x + 0 = x)$ (a model on $\mathbb{N} \cup \{a, b\}$). So closed instances cannot tell the axiom $Q4$ from its closed schema $z + 0 = z$, which differ in their theorems over $Q - Q4$.*

Proof of Proposition B.9. On $\mathbb{N} \cup \{a, b\}$, standard on $\mathbb{N}$, let $Sa = a$, $Sb = b$; $a + m = m$, $b + m = b$ ($m \in \mathbb{N}$); $x + a = a$ for $x \in \mathbb{N} \cup \{a\}$, $b + a = b$, $x + b = b$ for all x ; $x \cdot 0 = 0$; $a \cdot m = a$, $b \cdot m = b$ ($m \geq 1$); $n \cdot a = n \cdot b = b$ ($n \geq 1$); $0 \cdot a = 0 \cdot b = 0$; $a \cdot a = a \cdot b = a$; $b \cdot a = b \cdot b = b$. Q1–Q3: S is the identity on $\{a, b\}$ and $n \mapsto n + 1$ on $\mathbb{N}$. Q5: for $y = m$, $a + (m + 1) = m + 1 = S(a + m)$, both sides are b for $x = b$, and $x \in \mathbb{N}$ is standard; for $y \in \{a, b\}$, $Sy = y$ and $x + y \in \{a, b\}$ is fixed by S . Q6 holds by definition. Q7: for $y = m$, $a \cdot (m + 1) = a = a \cdot m + a$ (as $0 + a = a + a = a$) and $b \cdot (m + 1) = b = b \cdot m + b$; for $y \in \{a, b\}$ we need $z + x = z$ with $z = x \cdot y$, which holds for $x = 0$ ($0 + 0 = 0$), $x = n \geq 1$ ($b + n = b$), $x = a$ ($a + a = a$) and $x = b$ ($b + b = b$). Closed terms denote standard numbers, so every closed instance holds, but $a + 0 = 0 \neq a$. $\square$

Each case was re-checked for this paper; c6 and referee r5 find no violation on $\{0..40\} \cup \{a, b\}$ and $\{0..80\} \cup \{a, b\}$.

Proposition B.10 (The Gaifman condition). *[proved; known]^{universal Prop U15} Let μ be a probability on L -structures, $P(\psi) := \mu\{M : M \models \psi\}$, and $t_1, t_2, \dots$ the closed terms. Then*

$$P(\forall x\varphi \mid \bigwedge_{i \leq n} \varphi(t_i)) \rightarrow P(\forall x\varphi) / \mu\{M : M \models \varphi(t) \text{ for all closed } t\}.$$

The Gaifman condition (Gaifman, 1964) $P(\forall x\varphi) = \inf_n P(\bigwedge_{i \leq n} \varphi(t_i))$ holds for all φ iff $M_0 \preceq M$ for μ -a.e. M , with M_0 the closed-term substructure (AS Prop 4.7); the limit is then 1 if $P(\forall x\varphi) > 0$ and 0 if $P(\forall x\varphi) = 0$.

So believing $\forall x\varphi$ from true closed instances is safe for $\mathbb{N}$ and unsafe for credences that weigh $\mathbb{R}$, V , models of $\text{PA} + \neg\text{Con}(\text{PA})$ (AS Ex 4.8) or some models of Q (AS Ex 4.9). The condition is cited from memory by the track; secondary sources state it with constants rather than closed terms.

Proof of Proposition B.10. $\forall x\varphi$ entails each instance, so $P(\forall x\varphi \wedge \bigwedge_{i \leq n} \varphi(t_i)) = P(\forall x\varphi)$, and the conjunctions decrease to “all closed instances hold”; continuity of μ gives the limit. The gap in the Gaifman condition is $\mu\{M : \text{all closed instances hold, } M \not\models \forall x\varphi\}$, which vanishes for every φ (countably many) iff μ -a.e. M satisfies (i) of AS Prop 4.7, i.e. $M_0 \preceq M$. $\square$

B.5 Open instances, guards and root splits

Proof of Proposition 3.10. (a) $P_T(D) > 0$ gives $T \vdash \varphi(p)$, and p occurs in no axiom, so Gen gives $T \vdash \forall x\varphi$. (b) follows from (a). (c) With weights $(w_\forall, w_{\text{open}}, w_{\text{sch}})$ on $\forall x\varphi$ and the open- and closed-guarded σ_φ , the only outputs are $\forall x\varphi$, $\varphi(p)$ and closed $\varphi(t)$ (Gen fails on closed formulas, $\forall E$ on quantifier-free ones). With $a := \mu(\forall x\varphi)$, $b := \mu(\varphi(p))$: $a = p_{\text{cite}}w_\forall + gb$ and $b = \rho(p_{\text{cite}}w_{\text{open}} + ca)$, so

$$a = \frac{p_{\text{cite}}(w_\forall + g\rho w_{\text{open}})}{1 - gc\rho}, \quad Z = a(1 + c) + p_{\text{cite}}(w_{\text{open}} + w_{\text{sch}}),$$

$$\mu(\varphi(t)) = \mathcal{Q}_{\text{num}}(t)[(1 - \rho)(p_{\text{cite}}w_{\text{open}} + ca) + p_{\text{cite}}w_{\text{sch}}] \quad (t \text{ closed}).$$

Hence $Z_{H_{\text{open}}} = p_{\text{cite}} \frac{1+g\rho}{1-gc\rho}$, $Z_{H_\forall} = p_{\text{cite}} \frac{1+c}{1-gc\rho}$, $Z_{H_{\text{sch}}} = p_{\text{cite}}$; unnormalised, $\mu_{H_\forall}/\mu_{H_{\text{open}}} = c$ at instances and $1/(g\rho)$ at $\forall x\varphi$, and $\mu_{H_{\text{open}}}/\mu_{H_{\text{sch}}} = \frac{1-\rho}{1-gc\rho}$ at closed instances; divide by the normalisers. (d) follows, for the finitely many hypotheses compared, from (c), Theorem B.4(2) and Proposition B.3; the rate $1/n$ for H_{both} is computed in $\mathbf{C}_{\text{min}}$, and in $\mathbf{C}_{\text{open}}$ its per-datum ratio to H_{sch} (the formulas above with $w_{\text{open}} = 0$) is still below 1 at every $w_\forall > 0$. $\square$

c1 (sampler, max $|z| = 1.60$), c5 and referee r7 (2.15) confirm (c).

Lemma B.11 (Waste). *[proved]^{universal Lemma N0} Let P^* be a law on a countable set, $H \cup T$ a partition of $\text{supp } P^*$ with $\tau := P^*(T) \in (0, 1]$, N_W a law on a set disjoint from $\text{supp } P^*$, and $r_w > 0$. Over the laws $p_h P^*(\cdot \mid H) + p_t P^*(\cdot \mid T) + r_w p_t N_W$ with $p_h + (1 + r_w)p_t = 1$, the minimum of $\text{KL}(P^* \parallel \cdot)$ is $\tau \ln(1 + r_w)$, attained at $p_t = \tau/(1 + r_w)$, $p_h = 1 - \tau$.*

Proof. The conditionals are exact and N_W lives off the support, so $\text{KL} = (1 - \tau) \ln \frac{1-\tau}{p_h} + \tau \ln \frac{\tau}{p_t}$, convex in p_t once $p_h = 1 - (1 + r_w)p_t$. The derivative $(1 - \tau)(1 + r_w)/p_h - \tau/p_t$ vanishes at $p_t = \tau/(1 + r_w)$, $p_h = 1 - \tau$, where the value is $\tau \ln(1 + r_w)$. $\square$

Proposition B.12 (Exact law of R_k). *[proved; computed]^{universal Prop N1} In $\mathbf{C}_{\text{open}}$, let $A_\rho := \frac{1-\rho}{1-gc\rho}$ and $B_\rho := \frac{1+g\rho}{1-gc\rho}$, and give R_k sentence weights $v_0, \dots, v_{k-1}$ and template weight u . Its unnormalised law is $p_{\text{cite}}v_j$ at $\varphi(S^j 0)$ ($j < k$); $p_{\text{cite}}uA_\rho \mathcal{Q}_{\text{num}}(S^m 0)$ at $\varphi(S^{k+m} 0)$; $p_{\text{cite}}u\rho/(1 - gc\rho)$ at $\varphi(S^k p)$; $p_{\text{cite}}ug\rho/(1 - gc\rho)$ at $\forall y\varphi(S^k y)$; and 0 elsewhere. $Z = p_{\text{cite}}(1 - u + uB_\rho)$. H_{open} is the case $k = 0$, $u = 1$.*

Proof. Citing a sentence yields it, and Gen and $\forall E$ then fail. Citing the template yields $\varphi(S^k t)$, $t \sim \mathcal{Q}_{\text{open}}$; Gen applies only to $\varphi(S^k p)$, giving $U := \forall y \varphi(S^k y)$, and $\forall E$ only to U . With $a := \mu(U)$, $b := \mu(\varphi(S^k p))$: $a = gb$, $b = \rho(p_{\text{cite}}u + ca)$, so $a = g\rho p_{\text{cite}}u/(1 - gc\rho)$, $b = \rho p_{\text{cite}}u/(1 - gc\rho)$, and a closed tail instance has mass $(1 - \rho)\mathcal{Q}_{\text{num}}(S^m 0)(p_{\text{cite}}u + ca) = p_{\text{cite}}u A_\rho \mathcal{Q}_{\text{num}}(S^m 0)$. Summing gives $Z = p_{\text{cite}}(1 - u) + p_{\text{cite}}u(1 + g\rho)/(1 - gc\rho)$. $\square$

c8 Part 1: sampler for $R_1..R_3$, $\max |z| = 2.07$ (referee r3: 2.19).

Proposition B.13 (Rates on closed data). *[proved; computed]^{universal Prop N2} On closed numeral data from $\mathcal{Q}_{\text{num}}$, under L1: $\text{KL}(P^* \| P_{H_\forall}) = \ln \frac{1+c}{c(1-\rho)}$, $\text{KL}(P^* \| P_{H_{\text{open}}}) = \mathcal{W}_0$, $\inf_w \text{KL}(P^* \| P_{R_k}) = q^k \mathcal{W}_0$. Each split level halves the loss at $q = \frac{1}{2}$ (0.0626, 0.0313, 0.0156 for R_1, R_2, R_3).*

The proof uses the exact law of R_k (Proposition B.12) and the waste lemma (Lemma B.11): a law that reproduces P^* on a tail of mass τ and wastes r_w times the tail's mass loses at best $\tau \ln(1 + r_w)$; for R_k , $\tau = q^k$ and $\ln(1 + r_w) = \mathcal{W}_0$.

Proof of Proposition B.13. Conditioned on closed outputs, $H_\forall$ and H_{open} have law P^* (Proposition 3.10(c)), so their KL is $-\ln$ of their closed mass, $c(1 - \rho)/(1 + c)$ and $A_\rho/B_\rho = (1 - \rho)/(1 + g\rho)$. For R_k (Proposition B.12) the normalised law has head mass $(1 - u)/Z_R$, $Z_R = 1 - u + uB_\rho$, tail mass $p_t = A_\rho u/Z_R$ with the exact tail shape (because $\mathcal{Q}_{\text{num}}(S^{k+m} 0) = q^k \mathcal{Q}_{\text{num}}(S^m 0)$), and waste $(B_\rho - A_\rho)u/Z_R = r_w p_t$ with $r_w = B_\rho/A_\rho - 1$. For fixed u the best head is $v_j \propto \mathcal{Q}_{\text{num}}(S^j 0)$ (Gibbs), and Lemma B.11 applies with $\tau = q^k$: p_t ranges over $(0, A_\rho/B_\rho] = (0, 1/(1 + r_w)]$, which contains the optimum, attained at $u = q^k/(B_\rho(1 - q^k) + q^k)$. The value is $q^k \ln(B_\rho/A_\rho) = q^k \mathcal{W}_0$. $\square$

c8 Part 2: numerical minimisation agrees to $1.8 \cdot 10^{-15}$ ($k \leq 6$, four settings). For Remark 3.11, the referee's independent run (r3) of $\{H_\forall, H_{\text{open}}, R_1\}$ gives 0.880 at $n = 100$, then 0.000 at $n = 300$.

Proof of Proposition 3.12. *Provability.* On $\mathbb{N} \cup \{e, e'\}$, standard on $\mathbb{N}$, with $Se = Se' = e'$ and $0 + e = 0 + e' = e'$, every $S^k x$ ($k \geq 1$) is a standard number $\geq k$ or e' , so every instance of $\varphi(S^k z)$, closed or under closure, and $\varphi(0), \dots, \varphi(S^{k-1} 0)$ hold, while $0 + e \neq e$. Over Q: Q proves each $0 + S^j 0 = S^j 0$ (Q4, Q5, induction on j outside the theory), and k uses of Q3 show that every x is one of $0, \dots, S^{k-1} 0$ or $S^k y$, where $\varphi(S^k p)$, read as $\forall y \varphi(S^k y)$, applies. Memorisers consist of closed instances, which Q proves, and $Q \not\vdash \forall x \varphi$ (AS Ex 4.9). (a) $H_\forall, H_{\text{open}}$ have fixed laws with $\text{KL} \geq \mathcal{W}_0$ (note $\ln \frac{1+c}{c(1-\rho)} \geq \mathcal{W}_0$), so their normalised log-likelihood ratio to P^* is $-n\text{KL} + o(n)$ a.s. For R_k with learned weights, take the interior optimum w_0 and an ε -neighbourhood N_ε on which the weights are bounded away from 0; there the per-datum log-likelihood ($\ln v_j - \ln Z_R$, or $\ln(A_\rho u) + \ln \mathcal{Q}_{\text{num}}(S^m 0) - \ln Z_R$, whose middle term does not depend on w) has w -gradient bounded by some M . So the marginal is $\geq \text{Dir}(N_\varepsilon) P_{w_0}(D_n) e^{-nM\varepsilon}$, and $\liminf \frac{1}{n} \ln \frac{\text{marginal}}{P^*(D_n)} \geq -q^k \mathcal{W}_0 - M\varepsilon$ for every ε ; the log-odds of R_k against each prover grow like $n(1 - q^k) \mathcal{W}_0 - o(n)$. Provers that add σ_φ as a spare component to some R_k decay only polynomially relative to R_k (proof sketch; Proposition 4.14(b)). (b) All members but the memorisers prove $\forall x \varphi$ over Q, which gives the second sentence. By Proposition B.6(ii') the memoriser class has log-marginal $\geq -C \ln^2 n$ relative to P^* . For each R_k the likelihood ratio to P^* depends only on the empirical frequencies $\hat{p}$ of the $k + 1$ categories (head values; tail), since the tail shape cancels; so $\frac{1}{n} \ln \sup_w \frac{P_{R_k, w}(D_n)}{P^*(D_n)} = \sup_w G(\hat{p}, w)$, convex and hence continuous in $\hat{p}$ near p^* , with limit $-q^k \mathcal{W}_0 < 0$, and the marginal is below the supremum. With $\mathcal{K}$ finite all non-memorisers lose linearly against the memorisers. (c) (Proof sketch.) Under L1^{sel} , R_k 's filtered law (head

v_j/N , tail $A_\rho u \mathcal{Q}_{\text{num}}(S^m 0)/N$, $N = 1 - u + A_\rho u$ is a smooth reparametrisation of a $(k+1)$ -category multinomial equal to P^* at one interior point. The Laplace (BIC) expansion (Schwarz, 1978; Clarke and Barron, 1990) (recalled, details not checked) gives log-marginal $-\frac{k}{2} \ln n + O_P(1)$ relative to P^* ; memorisers lose $O(\ln^2 n)$; $H_\forall$, H_{open} have log-ratio 0. Not written out: almost-sure control, uniform in k for infinite $\mathcal{K}$. $\square$

likelihood, class	quantity	$n = 0$	10	10^2	10^3	10^4	10^5	10^6
L1, $\{H_\forall, H_{\text{open}}, R_1\}$	$P(\vdash)$, pure	.997	.996	.697	.000	.000	.000	.000
	MAP	H_{open}	H_{open}	H_{open}	R_1	R_1	R_1	R_1
L1, $+R_{2..4}$, mem.	$P(Q \cup T \vdash)$	.601	1.000	1.000	1.000	1.000	.000	.000
	MAP	H_{open}	H_{open}	H_{open}	R_3	R_4	mem	mem
L1, $+R_{2..40}$, mem.	$P(\vdash)$, pure	.599	.996	.697	.000	.000	.000	.000
	$P(Q \cup T \vdash)$	.601	1.000	1.000	1.000	1.000	1.000	1.000
	MAP	H_{open}	H_{open}	H_{open}	R_3	R_6	R_9	R_{12}
L1 ^{sel} , each class	$P(\vdash)$, pure	.997/.599	.998	.987	1.000	1.000	1.000	1.000

Table 18: Without a guard (c8 Part 3): $P(T \vdash \forall x \varphi \mid D_n)$ in pure logic and over $\mathbf{Q}$, $\varphi = 0 + x = x$, closed numeral data ($q = \frac{1}{2}$), $\mathbf{C}_{\text{open}}$ at $(c, g, \rho) = (0.3, 0.2, 0.1)$, Dirichlet($\frac{1}{2}$) weights (Proposition 3.12(b) is proved for Laplace-weighted memorisers), prior $2^{-(\text{symbols}+1 \text{ per axiom})}$, memoriser class mass 2^{-6} (lower bracket), mean of 20 seeds; MAP = mode over seeds. Part 4 (multinomial counts, 10 seeds): with $R_{1..40}$ the MAP is $R_{15}, R_{18}, R_{21}, R_{24}, R_{30}$ at $n = 10^7, 10^8, 10^9, 10^{10}, 10^{12}$ and $P(Q \cup T \vdash) = 1.000$, $P(\vdash)$ pure = 0, also with the upper bracket and class mass $\frac{1}{2}$, with $\alpha = 1$, and with $\lambda = 2$ ($R_{14..R_{30}}$)^{universal Prop N3}.

The MAP depth is about $\log_2 n$ minus a slowly growing term (≈ 8 at $n = 10^6$, ≈ 10 at 10^{12}).

B.6 Quantified data and the equational fragment

Proposition B.14 (Quantified data). *[(a), (b) proved; computed]^{universal Prop U12} (a) If $s \in D$ and $B \cup I_c(\varphi) \not\models s$, then under any likelihood supported on theorems $B \oplus \sigma_\varphi$ has posterior 0; over $\mathbf{Q}$, one datum $\forall y(0 + Sy = Sy)$ leaves only provers of $\forall x(0 + x = x)$. (b) If all theories prove $\forall x \varphi$ at different costs ($\{\forall^L \forall x \varphi, \sigma_\varphi\}$ against $\{\forall x \varphi, \sigma_\varphi\}$ in $\mathbf{C}_{\text{min}}$, L vacuous quantifiers, uniform weight priors, equal prior masses, data $\forall x \varphi$ with probability f_q and instances otherwise), the log Bayes factor tends to $\ln(1 + c + \dots + c^L)$ when $f_q = 0$ (0.349 at $L = 3$) and grows linearly when $f_q > 0$, at a rate a little above $f_q L \ln \frac{1}{c}$ ([proof sketch] for $f_q > 0$). (c) [proof sketch] In L1^{max} form a derived sentence becomes an axiom when the derivation steps it saves, at $\geq \ln \frac{1}{c}$ each, exceed its prior cost and an $O(\ln n)$ dilution: frequently used lemmas become axioms.*

Proof of Proposition B.14. (a) If $B \cup I_c \not\models s$, then $B \oplus \sigma_\varphi \not\models s$. In AS Ex 4.9's model, $0 + Sa = 0 + a = b \neq a = Sa$, so $\mathbf{Q} + I_c \not\models \forall y(0 + Sy = Sy)$; and $\mathbf{Q} + \forall y(0 + Sy = Sy) \vdash \forall x(0 + x = x)$ by Q4 at 0 and Q3 elsewhere. (b) In units of $1 - c$, with $S_B := \sum_{k=0}^{L+1} c^k \cdot \{\forall x \varphi, \sigma_\varphi\}$ with weight w on $\forall x \varphi$ gives $P(\forall x \varphi) = \frac{w}{1+wc}$, $P(\varphi(t)) = \frac{1-w+wc}{1+wc} \mathcal{Q}(t)$; $\{\forall^L \forall x \varphi, \sigma_\varphi\}$ gives $P(\forall x \varphi) = wc^L/Z_B$, $P(\varphi(t)) = ((1-w) + wc^{L+1}) \mathcal{Q}(t)/Z_B$, $Z_B = wS_B + 1 - w$ (vacuous eliminations succeed with term mass 1). A quantified datum adds $L \ln \frac{1}{c} + \ln \frac{Z_B}{1+wc}$ to the log-ratio; an instance 0.021, 0.094, 0.307, 1.323 nats at $w = 0.05, 0.2, 0.5, 0.9$ ($L = 3$), so the two theories price instances alike only for small w (referee m5). For $f_q = 0$ both Bayes factors against H_{sch} are $\int_0^1 e^{nf_i(w)} dw$ with $f_i(0) = 0$, f_i decreasing, $f'_{\text{both}}(0) = -1$, $f'_B(0) = -(1 + c + \dots + c^L)$; Watson's lemma gives $(n|f'_i(0)|)^{-1}(1 + o(1))$ for each. For $f_q > 0$ (sketch) the rate is $\Delta(f_q) := \max_w \mathbb{E}_{\text{both}}(w) - \max_w \mathbb{E}_B(w)$, the Laplace

approximation at the interior maxima not written out. (c) (Sketch.) In $\mathbf{L1}^{\max}$ form a derivation's code length is a sum of rule and term costs; adding ψ costs its prior bits and an $O(\ln n)$ dilution, and saves $\ell_T(s) - \ell_{T+\psi}(s)$ steps of cost $\geq \ln \frac{1}{c}$ per datum s . $\square$

c10(c), equal priors, 20 seeds, $n \leq 10^5$: for $f_q = 0$ the log Bayes factor is 0.349 ($L = 3$) and 0.357 ($L = 10$) at every n ; for $L = 3$, $\Delta = 0.0396, 0.1980, 0.7921$ at $f_q = 0.01, 0.05, 0.2$ (measured 0.0395, 0.1977, 0.7921; $f_q L \ln \frac{1}{c} = 0.036, 0.181, 0.722$); for $L = 10$, 0.1240, 0.6198, 2.4793 (measured 0.1236, 0.6187, 2.4793). The first version's table (c5 Part B) gave $\{\forall^L \forall x \varphi, \sigma_\varphi\}$ a much smaller prior and so tracked only the refutation in (a): $P(T \vdash \forall x \varphi) = 1 - (1 - f_q)^n$ within Monte Carlo error.

Proposition B.15 (The equational fragment). *[(a)–(c) proved; (a) computed]^{universal Prop U14} (a) In equational logic over $E = \{Q4, Q5\}$ every derivation of $0 + S^k 0 = S^k 0$ has at least $k + 1$ axiom leaves. (b) If grammar nodes pick rules independently, with at most two premises and mean $\mu < 1$, then $P(\geq L \text{ nodes}) \leq e^{-(L-1)(1-\mu)^2/2}$. (c) The full sum favours the schema by $\Omega(k)$ on $0 + S^k 0 = S^k 0$ when $q > e^{-(1-\mu)^2/2}$; this needs $q > 0.61$, so it does not apply at $q = \frac{1}{2}$; the two-part ($\mathbf{L1}^{\max}$) comparison favours it linearly in k at $q = \frac{1}{2}$ when each leaf pays a citation and a term of point mass $\leq 1 - q$.*

Proof of Proposition B.15. (a) Let $w(t) := \sum_{+\text{-nodes } l+r} (1 + \#S(r))$, $\#S(r)$ the number of S symbols in r . Rewriting $u + 0$ to u , or $u + Sv$ to $S(u + v)$, anywhere in a term changes w by exactly -1 (the rewritten node contributes 1, resp. $2 + \#S(v)$ before and 0, resp. $1 + \#S(v)$ after; nodes inside are untouched; nodes above keep the S -count of their right argument), and the reverse by $+1$. $w(0 + S^k 0) = k + 1$, $w(S^k 0) = 0$. A derivation (reflexivity, symmetry, transitivity, congruence, substitution, axiom instances) converts by induction into a rewrite chain with at most as many steps as axiom leaves: congruence and transitivity concatenate chains, symmetry reverses one, substitution keeps lengths. (b) Couple the grammar with the Galton–Watson tree of rule labels and explore it depth-first with premise counts ξ_i ; $\geq L$ nodes requires $\sum_{i \leq L-1} (\xi_i - 1) \geq 0$; apply Hoeffding's inequality (Hoeffding, 1963) to $\xi_i - 1 \in [-1, 1]$ with mean $\mu - 1$ (typed grammars need the maximum of μ over contexts). (c) Full sum: by (a), (b), $P_E(0 + S^k 0 = S^k 0) \leq e^{-k(1-\mu)^2/2}$, while the schema gives $\geq p_{\text{cite}} w(1 - q) q^k$. Two-part: $\geq k + 1$ leaves of cost $\geq \ln \frac{1}{p_{\text{cite}}} + \ln \frac{1}{1-q} > \ln \frac{1}{q}$ (for $q \geq \frac{1}{2}$), against $O(1) + k \ln \frac{1}{q}$ for the schema. $\square$

c6: shortest chains of exactly $k + 1$ for $k \leq 5$ by breadth-first search; referee r5: the invariant on 339 250 random rewrites.

B.7 The sentence-only class

Proposition B.16 (Sentence-only class). *[(b), (c) proved; computed]^{universal Prop U16} (a) [proof sketch] Under $\mathbf{L0}^{\text{cl}}$, $H_\forall$ has the true law and Split_k , Both0 match it only at a point or at the boundary, so $P(T \vdash \forall x \varphi \mid D_n) \rightarrow 1$ (Laplace expansion uniform in k not written out). (b) Under $\mathbf{L1}$: $\text{KL}(P^* \| P_{H_\forall}) = \ln \frac{1+c}{c}$ (1.466), $\inf_w \text{KL}(P^* \| P_{\text{Split}_k}) = q^k \ln \frac{1+c}{c}$, $\inf_w \text{KL}(P^* \| P_{\text{Both0}}) = q \ln \frac{1+qc}{qc}$ (1.018). (c) Split_k proves $\forall x \varphi$ over $\mathbf{Q}$, not in pure logic; memorisers in neither. (d) [computed; limit conjecture] In pure logic $P(T \vdash \forall x \varphi \mid D_n) \rightarrow 0$ in every class computed. Over $\mathbf{Q}$, with depth ≤ 4 memorisers win (limit 0); with depth ≤ 40 splits win to $n = 10^{12}$ under four prior variants (Table 19). Conjecture: with all depths the limit is 1.*

Proof of Proposition B.16. (a) (Sketch.) Split_k equals the true law at one weight vector ($v_j = \mathcal{Q}_{\text{num}}(S^j 0)$, $u = q^k$) and pays $n^{-k/2}$ with learned weights; Both0 only at the boundary;

memorisers lose $O(\ln^2 n)$; uniformity in k not written out. (b) $H_\forall$ gives each instance $c/(1+c)$ times its true probability (Theorem 3.3). Split_k : the universal sentence yields itself with mass $(1-c)u$ or $\varphi(S^{k+m}0)$ with mass $(1-c)uc\mathcal{Q}_{\text{num}}(S^m0)$; normalised by $(1-c)(1+cu)$: head $v_j/(1+cu)$, tail $uc\mathcal{Q}_{\text{num}}(S^m0)/(1+cu)$, waste $u/(1+cu) = p_t/c$; Lemma B.11 with $\tau = q^k$, $r_w = 1/c$ gives $q^k \ln(1 + \frac{1}{c})$, attainable as p_t ranges over $(0, \frac{c}{1+c}]$. Both0 with weight w on $\varphi(0)$: with $M := (1-w)c/(1+c(1-w))$, the tail $\{j \geq 1\}$ has mass qM with the exact shape, waste M/c , and $\varphi(0)$ mass $1 - qM - M/c$; so $\text{KL} = (1-q) \ln \frac{1-q}{1-M(q+1/c)} + q \ln \frac{1}{M}$, convex, minimal at $M = \frac{qc}{1+qc} \leq \frac{c}{1+c}$ with value $q \ln \frac{1+qc}{qc}$. (c) The countermodel and derivation of Proposition 3.12. $\square$

c9 Parts 1–2 confirm the rates to $8.9 \cdot 10^{-16}$; referee r4 reproduces them.

Lemma B.17 (Survivors prove $\forall x\varphi$ over $\mathbf{Q}$). [*proved*]^{universal Lemma S1} *In $\mathbf{C}_{\min}$ with numeral elimination terms and $\varphi = 0 + x = x$, every finite set T of sentences with $P_T(\varphi(S^j0)) > 0$ for infinitely many j satisfies $\mathbf{Q} \cup T \vdash \forall x\varphi$.*

Proof of Lemma B.17. A chain cites some $A \in T$ and eliminates with numerals; T is finite, so one A yields $\varphi(S^j0)$ for infinitely many j . The conclusions are quantifier-free, so $A = \forall \bar{y}\chi$ with χ quantifier-free and $\chi[\bar{t}/\bar{y}] = (0 + S^j0 = S^j0)$ for numeral tuples with infinitely many j . Numerals contain only 0 and S , so χ is $\tau_0 + \tau_1 = \tau_2$ with τ_0 either 0 or a variable instantiated by 0 in every such output, and τ_1, τ_2 of the forms S^a0 or S^ay . As j varies, $\tau_1 = S^ay_m$ and $\tau_2 = S^by_{m'}$, and neither variable is that of τ_0 (which takes only the value 0). If $m = m'$, then $a = b$ and $A \vdash \forall y\varphi(S^ay)$ (instantiate τ_0 's variable, if any, by 0); over $\mathbf{Q}$, a uses of Q3 and the closed instances give $\forall x\varphi$. If $m \neq m'$, $A \vdash 0 + S^a0 = S^b0$ and $A \vdash 0 + S^a0 = S^{b+1}0$, so $S^b0 = S^{b+1}0$, which $\mathbf{Q}$ refutes; then $\mathbf{Q} \cup T$ is inconsistent. $\square$

The track's proof treats only $\tau_0 = 0$; the case of a variable always instantiated by 0 is added here and does not change the conclusion.

Remark B.18 (Kreisel's conjecture). [*known (secondary sources only)*]^{universal §9} “All instances have short derivations, so $\forall x\varphi$ is derivable” resembles Kreisel's conjecture, reported open for standard formalisations of PA, settled for some (Parikh, 1973; Baaz and Pudlák, 1993) and false for some theories close to PA (Hrubeš, 2007) (none of these was opened). Split_1 shows that the naive version fails without axioms such as Q3: every numeral instance has a one-step derivation, $\forall x\varphi$ none.

likelihood, class	quantity	$n = 0$	20	10^2	10^3	10^4	10^5	10^6
L1, $\{H_\forall, \text{Split}_1, \text{Both0}\}$	$P(\vdash)$, pure	0.996	0.038	0.000	0.000	0.000	0.000	0.000
L1, $+\text{Split}_{1..4}$, mem.	$P(\mathbf{Q} \cup T \vdash)$	0.338	0.926	0.947	0.475	0.000	0.000	0.000
	MAP	mem	Split ₁	Split ₃	mem	mem	mem	mem
L1, $+\text{Split}_{1..40}$, mem.	$P(\vdash)$, pure	0.336	0.024	0.000	0.000	0.000	0.000	0.000
	$P(\mathbf{Q} \cup T \vdash)$	0.338	0.926	0.947	1.000	1.000	1.000	1.000
	MAP	mem	Split ₁	Split ₃	Split ₆	Split ₉	Split ₁₂	Split ₁₅
L0 ^{cl} , $+\text{Split}_{1..40}$, mem.	$P(\vdash)$, pure	0.336	0.997	0.999	1.000	1.000	1.000	1.000

Table 19: Sentence-only class (c9 Part 3): $\varphi = 0 + x = x$, $\mathbf{C}_{\min}$, $c = 0.3$, numerals with $q = \frac{1}{2}$, Dirichlet($\frac{1}{2}$), prior $2^{-(\text{symbols}+1 \text{ per axiom})}$, memoriser class mass 2^{-6} (lower bracket), 20 seeds, MAP = mode; the larger classes also contain $H_\forall$ and Both0. With $\text{Split}_{1..40}$ under L1 the MAP is Split₁₈, Split₂₁, Split₂₄, Split₂₇, Split₃₄ at $n = 10^7, \dots, 10^{10}, 10^{12}$ with $P(\mathbf{Q} \cup T \vdash) = 1.000$, also with the upper bracket and class mass $\frac{1}{2}$, with $\alpha = 1$ and with $\lambda = 2$ (c9 Part 4).

The race (c9 Part 5). Log odds of the split family ($k \leq 40$) against the memoriser family, mean of 10 seeds: 16.5 nats at $n = 10^2$, 54.7 at 10^4 , 105.1 at 10^6 , 193.5 at 10^{10} and 286.4 at 10^{12} (minimum over seeds negative only at 10^2), of the order of $L \log_2 L$ with $L = \log_2 n$. *Heuristic (conjecture).* The best memoriser pays about L^2 bits of prior and $(L/2) \ln n$ nats of Dirichlet cost; the best split wastes about $nq^k \ln \frac{1+c}{c}$ nats, balancing the $\Theta(L)$ cost of one more level at $k \approx L - \log_2 L$, and so saves about $\log_2 L$ levels of $\Theta(L)$ nats. *History.* The first version's " $= 1$ over $\mathbf{Q}$ " came from $\{H_V, \text{Split}_1, \text{Both0}\}$, where all members prove $\forall x \varphi$ over $\mathbf{Q}$; the referee's " $\rightarrow 0$ over $\mathbf{Q}$ " holds with depth ≤ 4 , not with depth ≤ 40 to $n = 10^{12}$.

C Proofs and computations for Section 4

Proofs follow the model track's final notes; steps they leave out are named. The checks are the seeded scripts of `research/tracks/model/checks/` (c2–c16), each with its output file; the referee's independent scripts (r1–r7) are in `referee_code/`. The model track's referee re-did the proofs of Theorem 4.2, Corollaries 4.3 and 4.4, Proposition C.1, Proposition 4.5, Proposition 4.6(a), (b), Theorem 4.16, Examples 4.17 and 4.19 and Lemma 4.18, Theorem 4.7(a), Proposition 4.9, Proposition 4.10, Proposition 4.14(a) and Theorem 4.21 line by line and found no error beyond the corrections recorded in Remark 4.8 and Appendix H. Results added after the referee report were checked by the track and again while writing, not by a referee; the complete list is in Appendix H.1. Nothing is checked in a proof assistant.

C.1 Consistency and rates

Proof of Theorem 4.2. *Step 1 (joint law; Lévy).* Let Π be the law of $(T, X_1, X_2, \dots)$ on $\mathcal{H} \times \mathcal{S}^\infty$ with $\Pi(T = t, X \in E) = \pi(t)P_t^\infty(E)$, and $M := \sum_t \pi(t)P_t^\infty$ its data marginal. Since $\mathcal{H}$ is countable, Bayes' rule gives $\pi_n(A) = \Pi(T \in A \mid X_1, \dots, X_n)$ on the M -full event where $M(D_n) > 0$. By Lévy's upward theorem (known), for each fixed $A \subseteq \mathcal{H}$, $\pi_n(A) \rightarrow \Pi(T \in A \mid X_\infty)$ Π -a.s.; the convergence event is a data event, so it holds M -a.s. *Step 2 (the data determine the generator).* For $x \in \mathcal{S}^\infty$ let $F(x)(s) := \lim_n n^{-1} \# \{i \leq n : x_i = s\}$, and $F(x) := \perp$ if some limit fails to exist or the limits do not sum to 1. By the strong law applied to the countably many s , $P_t^\infty(F = P_t) = 1$ for each t (on that event the limits are the numbers $P_t(s)$, which sum to 1). Hence $\Pi(P_T = F(X)) = 1$. *Step 3 (the limit lives on $\{T : P_T = F(X)\}$).* Choose a version of $\{\Pi(T = t \mid X_\infty)\}_t$ that sums to 1 a.s. Each indicator $1\{P_t = F(X)\}$ is X_∞ -measurable, so $\mathbb{E}[1\{P_T = F(X)\} \mid X_\infty] = \sum_t 1\{P_t = F(X)\} \Pi(T = t \mid X_\infty)$ by monotone convergence. The left side is 1 a.s. by Step 2. $F(X)$ ranges Π -a.s. over the countable set $\{P_t : t \in \mathcal{H}\}$; apply Step 1 to the countably many sets $A_\nu := \{t : P_t = \nu\}$ and intersect the full-measure events. Then, M -a.s., $\lim_n \pi_n(A_{F(X)}) = \Pi(T \in A_{F(X)} \mid X_\infty) = 1$. *Step 4 (transfer to the truth).* $M \geq \pi(T^*)P_{T^*}^\infty$, so every M -null event is $P_{T^*}^\infty$ -null and Step 3 holds $P_{T^*}^\infty$ -a.s. Under $P_{T^*}^\infty$, $F(X) = P_{T^*}$ a.s. by Step 2, so $A_{F(X)} = C^*$, and $\pi_n(C^*) \rightarrow 1$. *Pointwise and total variation.* Write $q_T := \pi(T)1[T \in C^*]/\pi(C^*)$. By Corollary 4.3, $\pi_n(T) = \pi_n(C^*)q_T$ for $T \in C^*$, so

$$\sum_T |\pi_n(T) - q_T| = \sum_{T \in C^*} q_T (1 - \pi_n(C^*)) + \pi_n(\mathcal{H} \setminus C^*) = 2(1 - \pi_n(C^*)) \rightarrow 0.$$

□

The universal track proves the same statement independently (its Lemma U7, total variation via Scheffé's lemma); the exact identity above is ours and gives the same conclusion directly.

Proof of Corollary 4.3. Every $T \in C^*$ has $P_T = P_{T^*}$ as a law on $\mathcal{S}$, so $P_T(D_n) = P_{T^*}(D_n)$ for every sequence, and the posterior ratio is the prior ratio. $\square$

Computed (c9b; every likelihood by matching; three seeds, $n \leq 2000$): $\{z+0=z\}$ and its root split with weights $p_f = (0.5, 0.3, 0.1, 0.1)$ agree to $5.68 \cdot 10^{-14}$ in log-likelihood; an over-general template, a split missing a root and T^* plus the spare $0 = S0$ (weight 0.05, decaying as 0.95^n) go to 0. An earlier check (c9) was tautological for the split (referee m5).

Proof of Corollary 4.4. Put $A := \{T : s \in \text{Th}_d(T)\}$. Then $\pi_n(A) = \pi_n(A \cap C^*) + \pi_n(A \setminus C^*)$. The second term is at most $\pi_n(\mathcal{H} \setminus C^*) \rightarrow 0$. By Corollary 4.3 the first equals $\pi_n(C^*)\pi(A \cap C^*)/\pi(C^*)$. Both hold on the single event of Theorem 4.2, so simultaneously for all (s, d) . For the two cases named after the corollary: under L0 with full-support $\mathcal{Q}$ and positive weights, $\text{supp } P_T = \bigcup \text{inst}(T)$, so every $T \in C^*$ has T^* 's instance union and hence its Th_d ; under L1 with parameters admissible, $\text{supp } P_T = \text{Th}(T)$, so every $T \in C^*$ has $\text{Th}(T) = \text{Th}(T^*)$. $\square$

Proposition C.1 (Rates). *[proved]^{model Prop 2.4} Assume W. (a) For $T \notin C^*$, $\pi(T) > 0$: $\frac{1}{n} \ln \frac{\pi_n(T^*)}{\pi_n(T)} \rightarrow \text{KL}(P_{T^*} \| P_T) \in (0, \infty)$ a.s. (b) $\sum_{i=1}^n \mathbb{E} \text{KL}(P_{T^*} \| M_{i-1}) \leq \ln \frac{1}{\pi(C^*)}$. (c) For $A \subseteq \mathcal{H} \setminus C^*$ and $\rho_T := \sum_s (P_{T^*}(s)P_T(s))^{1/2}$, $\mathbb{E}[(\pi_n(A)/\pi_n(T^*))^{1/2}] \leq \sum_{T \in A} (\pi(T)/\pi(T^*))^{1/2} \rho_T^n$; so $P[\pi_n(A) \geq \varepsilon] \leq C \rho^n \varepsilon^{-1/2}$ if $\sup_A \rho_T \leq \rho < 1$ and $C := \sum_{T \in A} (\pi(T)/\pi(T^*))^{1/2} < \infty$ (true for π_λ , $\lambda \geq 2$).*

The rate (c) degenerates for spare templates ($\rho_T \geq (1-\varepsilon)^{1/2}$ for a spare of weight ε), whose decay the weight prior sets (Proposition 4.14).

Proof of Proposition C.1. (a) $\ln \frac{\pi_n(T^*)}{\pi_n(T)} = \ln \frac{\pi(T^*)}{\pi(T)} + \sum_{i \leq n} Y_i$ with $Y_i := \ln \frac{P_{T^*}(X_i)}{P_T(X_i)}$ i.i.d. Their negative part is integrable, since $(\ln x)^+ \leq x$ gives $\mathbb{E}Y^- = \mathbb{E}(\ln \frac{P_T}{P_{T^*}})^+ \leq \mathbb{E} \frac{P_T(X)}{P_{T^*}(X)} \leq 1$, and $\mathbb{E}Y = \text{KL}(P_{T^*} \| P_T) > 0$ by Gibbs' inequality, as $P_T \neq P_{T^*}$. The strong law holds with limit in $(0, \infty]$. If $P_T(X_i) = 0$ for some i , the ratio is $+\infty$ from then on. (b) $M(D_n) \geq \pi(C^*)P_{T^*}(D_n)$. By the chain rule $\ln \frac{P_{T^*}(D_n)}{M(D_n)} = \sum_{i \leq n} \ln \frac{P_{T^*}(X_i)}{M_{i-1}(X_i)}$, and the conditional expectation of the i -th term given D_{i-1} is $\text{KL}(P_{T^*} \| M_{i-1})$. Take expectations. (c) $\frac{\pi_n(A)}{\pi_n(T^*)} = \sum_{T \in A} \frac{\pi(T)}{\pi(T^*)} L_T$ with $L_T := \prod_{i \leq n} \frac{P_T(X_i)}{P_{T^*}(X_i)}$. Use $(\sum a_T)^{1/2} \leq \sum a_T^{1/2}$ and, by independence, $\mathbb{E}L_T^{1/2} = \rho_T^n$. For the probability bound, $\pi_n(A) \leq \pi_n(A)/\pi_n(T^*)$ and Markov's inequality applied to the square root. $\square$

For the degeneration: if $T = T^* \cup \{\tau\}$ with weight ε on τ and T^* 's weights scaled by $1 - \varepsilon$, then $P_T \geq (1 - \varepsilon)P_{T^*}$, so $\rho_T \geq (1 - \varepsilon)^{1/2} \sum_s P_{T^*}(s) = (1 - \varepsilon)^{1/2}$.

C.2 Identification of the generator

Proof of Proposition 4.5. $\text{Th}(T_1) = \text{Th}(T_2)$: b follows from a and $a \rightarrow b$ by MP, and $a \rightarrow b$ follows from b by an A1 instance and MP. Under T_2 : b has root $\neg$; every instance of a logical axiom has root $\rightarrow, \forall$ or $=$, and b is not an axiom of T_2 ; so no one-node tree concludes b , and $\mu_{T_2}(b) \leq \Pr[\text{the root is a rule node}] = \alpha_r$. With $Z_{T_2} \geq 1 - \alpha_r$ (Lemma 2.10(b)), $P_{T_2}(b) \leq \alpha_r/(1 - \alpha_r)$. Under T_1 the one-node tree citing b has probability $\alpha_{\text{ax}}/2$, and $Z_{T_1} \leq 1$. The same argument holds under L2. $\square$

Computed (c6; A1–A3 and MP, formulas of size ≤ 3 , depth ≤ 2 ; rule probabilities for (ax, lg, MP) (0.6, 0.25, 0.15) in the proved region, (0.5, 0.3, 0.2) on its boundary, (0.3, 0.3, 0.4) and (0.2, 0.4, 0.4) outside): $P_{T_1}(b) = 0.16\text{--}0.35$, $P_{T_2}(b) = 0.007\text{--}0.04$, $\text{TV} = 0.16\text{--}0.35$; the referee's implementation (r6) confirms both bounds. A route to all parameters, not carried out: $P_T(s)$ is a power series in the rule probabilities with nonnegative coefficients, so the difference is real-analytic on the subcritical region and nonzero near $\alpha_r = 0$.

Proof of Proposition 4.6. (a) $\text{supp } P_{T,w} = \bigcup \text{inst}(T)$ by Lemma A.2(c) with positive weights, and Th_d depends only on the instance union (Definition 2.2). (b) *Templates.* Substituting the body into a pattern occurrence $M(\bar{y})$ gives $r(N_1(\bar{y}, \dots), \dots)$; each N_j has a pattern occurrence, because $\bar{y}$ are distinct bound variables and a binder production adds one more. Other occurrences of M receive the same body with metavariable-free arguments, which DT° allows. So $\tau_r \in \text{DT}^\circ$. *Instances.* $\text{inst}(\tau) = \bigsqcup_r \text{inst}(\tau_r)$: at a pattern occurrence of M an instance shows the root production of $\theta(M)$, and distinct productions give distinct subtrees there (for a hole production z_i , the distinct bound variable y_i). *Probabilities.* For $s \in \text{inst}(\tau_r)$ with τ 's matcher θ , the matcher of τ_r gives the children's bodies of $\theta(M)$, and the root factorisation gives $\mathcal{Q}(\theta) = p_r \mathcal{Q}_{\tau_r}(s)$. (Holes and bound indices are drawn alike, so a child body under a binder has the law of a body with one more argument.) Hence $w_\tau \mathcal{Q}_\tau(s) = \sum_r w_\tau p_r \mathcal{Q}_{\tau_r}(s)$, and the other components are unchanged. *Under L1, L2.* A theory-citation node's conclusion has the same law under (T, w) and (T', w') , by the L0 statement; every other node has the same law, and a tree's validity and conclusion depend only on its nodes' conclusions. So $\mu_{T'} = \mu_T$ and $Z_{T'} = Z_T$. (c) A renaming permutes the factors of $\mathcal{Q}(\theta)$; an argument permutation permutes the holes, which a hole-symmetric $\mathcal{Q}$ ignores. The converse is (a) for single templates; for first-order templates with only 0-ary term metavariables, AS Prop C.1 (under its richness condition) turns equal instance sets into $\tau \equiv \tau'$. That proposition assumes one side has only 0-ary term metavariables (referee m10), so it says nothing about templates such as $A1 = B \rightarrow (C \rightarrow B)$. $\square$

With *learned* weights the likelihoods differ; that case is Propositions 4.10 and 4.11. Track pa's Prop 2.1 is (b) for T_{Ind} with a shared grammar $\mathcal{Q}(\varphi) = \mathcal{Q}(f)\mathcal{Q}(\cdot | f)$: an induction datum with motive root f has probability $w\mathcal{Q}(f)\mathcal{Q}(\varphi | f) = w\mathcal{Q}(\varphi)$ under both.

Computed (c2, Part A): $z+0 = z$ against its split over 0, S , $+$, $\cdot$ with root law (0.5, 0.3, 0.1, 0.1); maximal difference $1.08 \cdot 10^{-19}$ over all 2849 instances with $|t| \leq 9$, every instance matched against every template.

Remark C.2 (Chain likelihood, latent citations). [proved; computed]^{experiments Prop X13, E8} In the chain Ch_J the tie holds for a 0-ary closed-guard term metavariable of a component that is not an MP major premise; in E8, where a datum $\psi(t)$ may be cited or derived from $\varphi(t)$ by MP, the two log-probabilities agree to $1.1 \cdot 10^{-13}$. For E6 see Remark 7.10.

Proof of Remark C.2 (experiments Prop X13). Under $\mathcal{Q}$ a closed term has root f with probability q_f and, given the root, children drawn independently from the same closed-term law (every node of a closed term has the same context). So “cite τ with probability w_τ and draw the body of u from $\mathcal{Q}$ ” and “cite τ_f with probability $w_\tau q_f$ and draw the bodies of $u_1, \dots, u_r$ from $\mathcal{Q}$ ” give the same law on the cited sentence, jointly with the bodies of τ 's other metavariables. Every later chain step depends only on the current sentence and on the set of MP components, which is the same in T and T' because neither τ nor any τ_f is an MP component. $\square$

C.3 Identification in the limit and Gold

Proof of Theorem 4.7(a). By Theorem 4.2, $\pi_n(C^*) \rightarrow 1$, and every $T \in C^*$ has $\text{supp } P_T = \text{supp } P_{T^*}$. At most one L can have posterior mass above $\frac{1}{2}$; once $\pi_n(C^*) > \frac{1}{2}$ it is $\text{supp } P_{T^*}$, which is $\bigcup \text{inst}(T^*)$ under L0 (Lemma A.2(c)) and $\text{Th}(T^*)$ under L1 (Lemma 2.10(c)). $\square$

Proof of Theorem 4.7(b). The proof of Theorem 4.2, with the frequency functional replaced by a support functional. *A countable parameter that the data determine.* Put $U(T, w) := \bigcup \text{inst}(T)$; it takes countably many values. For a sequence x let $F'(x) := \{x_i : i \geq 1\}$. For

w in the open simplex, $\text{supp } P_{T,w} = \bigcup \text{inst}(T)$ is a countable set of points of positive probability, each of which occurs a.s., and nothing else occurs; so $P_{T,w}^\infty(F' = U(T,w)) = 1$. Dir_T charges only the open simplex. Hence $\Pi(U = F'(X)) = 1$ for the joint law Π of (T, w, X) . *Steps 1 and 3.* The posterior given D_n is $\Pi(\cdot \mid D_n)$ (Bayes' formula with the likelihood as a density with respect to the prior). For each of the countably many sets $A_\nu := \{(T, w) : \bigcup \text{inst}(T) = \nu\}$, Lévy's theorem gives $\pi_n(A_\nu) \rightarrow \Pi(A_\nu \mid X_\infty)$ M -a.s. Since $F'(X)$ is X_∞ -measurable, $1 = \mathbb{E}[1\{U = F'(X)\} \mid X_\infty] = \sum_\nu 1\{F'(X) = \nu\} \Pi(A_\nu \mid X_\infty)$ a.s. So, M -a.s., $\pi_n(A_{F'(X)}) \rightarrow 1$. *Step 4, changed.* Let E be the M -null event where this fails. Since $M \geq \pi(T^*) \int P_{T^*,w}^\infty(\cdot) \text{Dir}_{T^*}(dw)$, we get $\int P_{T^*,w}^\infty(E) \text{Dir}_{T^*}(dw) = 0$, so $P_{T^*,w}^\infty(E) = 0$ for Dir_{T^*} -a.e. w , which is Lebesgue-a.e. w because the Dirichlet density is positive on the open simplex. For such w^* , a.s. $F'(X) = \bigcup \text{inst}(T^*)$ and $\pi_n(A_{\bigcup \text{inst}(T^*)}) \rightarrow 1$. $\square$

The pre-referee proof said that the steps of Theorem 4.2 “go through with the countable sum replaced by an integral”. That sentence is withdrawn: Step 3 needs a countable family of sets, which the support functional provides and the law itself does not (referee M4, who suggested this route).

Sketch of Theorem 4.7(b'). $P_{T,w}$ lies in the Polish space of probability vectors on $\mathcal{S}$ (ℓ^1 metric), and the frequency functional F recovers it a.s. For a countable convergence-determining family of bounded continuous functions g , Lévy's theorem gives that the posterior mean of $g(P)$ converges to $g(F(X))$ a.s. So the posterior law of $P_{T,w}$ converges weakly to the point mass at P^* , which gives the statement for total-variation balls; transfer to Lebesgue-a.e. w^* as in (b). Not written out: the choice of the family and the passage from weak convergence to total-variation balls.

Proof of the counter-statement in Remark 4.8. The posterior on w given T is absolutely continuous with respect to Dir_T , so it suffices that each W_T below is Lebesgue-null. Fix T . The set $W_T := \{w : P_{T,w} = P^*\}$ is the intersection of the simplex with an affine subspace, since $w \mapsto P_{T,w}$ is affine. If it had positive Lebesgue measure in the simplex, the affine map would be constant on an open set, hence on the whole simplex, and every component of T would have law P^* (for $|T| = 1$ the condition is $\mathcal{Q}_\tau = P^*$). The posterior on w given T is absolutely continuous with respect to Dir_T . So the posterior mass of the exact class is 0 at every n unless some template σ has $\mathcal{Q}_\sigma = P^* = \sum_\tau w_\tau^* \mathcal{Q}_\tau$. For each σ , linear independence of $\{\mathcal{Q}_\tau : \tau \in T^*\}$ allows at most one such w^* , and there are countably many σ . $\square$

Remark C.3 (Which computed posteriors are covered). [known (Doob's theorem in the (T, w) form); not covered: data-dependent pools]^{experiments Prop X5} Theorem 4.2 is stated for fixed laws. Run on the parameter (T, w) with prior $\pi(T) \text{Dir}_T(dw)$, as in the proof of Theorem 4.7(b), Doob's argument gives concentration on $\{(T, w) : P_{T,w} = P_{T^*}\}$ for a one-component generator, which is an atom of the prior, in a fixed pool: this covers E1, E4 and E6 with their hand pools. The experiments' data-dependent pool members ($\text{Mem}(D_n)$, causal theories) are covered by no consistency theorem here, only by the pool versions of Theorems 5.6 and 5.7. E2, E3(b), E5(b) and E8 sample at one fixed weight vector, where only Theorem 4.7(b) and conjecture (c) speak: a yardstick, not an instance of a theorem.

Remark C.4 (Memorisation and the bound). [proved; known (prior work, flagged)]^{model §5.2} A fixed memoriser $\text{Mem}(E)$ dies once a datum outside E appears; before that it pays its prior and, with Dirichlet weights, about $\frac{|E|-1}{2} \ln n$ nats. The memoriser *class* can keep mass $\exp(-O(\ln^2 n))$ (Laplace-weighted memorisers, geometric numerals; Proposition B.6). So the Bayesian needs no

bound on the number of templates, unlike the cautious learner (AS Prop 6.4). Identification from stochastic positive data goes back to Horning (1969) and Angluin (1988) (existence checked, content not).

Proof of Proposition 4.9. (a) is Theorem 4.2. (b) Enumerate $\mathcal{L}_\infty = \{e_1, e_2, \dots\}$ and build the text in stages. At stage i , with σ the prefix built so far, pick $k_i > k_{i-1}$ with $\sigma \cup \{e_i\} \subseteq \mathcal{L}_{k_i}$ (possible since the chain increases to $\mathcal{L}_\infty$) and append e_i ; the new prefix σ' has positive P_{k_i} -probability. By (a), applied to $P_{k_i}^\infty$ conditioned on the positive-probability prefix σ' , almost every continuation reaches $g_n = \mathcal{L}_{k_i}$ at some finite n ; pick such a continuation, with elements in $\mathcal{L}_{k_i} \subseteq \mathcal{L}_\infty$, and cut it at a time where $g_n = \mathcal{L}_{k_i}$. The resulting sequence lists every e_i and only elements of $\mathcal{L}_\infty$, and $g_n = \mathcal{L}_{k_i} \neq \mathcal{L}_\infty$ at infinitely many times. (c) By (a) with $i = \infty$, P_∞ -a.s. $g_n = \mathcal{L}_\infty$ eventually. $\square$

Computed (E5; Table 36): on data uniform on $\mathcal{L}_5$ the mass of $\mathcal{L}_5$ is $2 \cdot 10^{-6}$ at $n = 64$, 0.661 at 128 and 1.000 from 256 in all 10 seeds ($\mathcal{L}_5$ gains $3 - \log_2 5 \approx 0.68$ bits per datum, repaying 72 prior bits and its Dirichlet cost); on Gold's text against the MAP learner over $\{\mathcal{L}_1, \dots, \mathcal{L}_{60}, \mathcal{L}_\infty\}$ the first 14 stages end (740 data), which shows only that these stages end (experiments' referee m10). *The $\text{I}\Sigma_n$ chain* (model §5.2, heuristic). Under L1 with theorem data, PA's generator charges $\text{Th}(\text{PA}) \setminus \text{Th}(\text{I}\Sigma_n)$, so $\text{I}\Sigma_n$ is eventually rejected on PA-data; and PA's generator charges $\text{Th}(\text{PA}) \setminus \text{Th}(\text{I}\Sigma_5)$, so PA is eventually rejected on $\text{I}\Sigma_5$ -data. Both masses presumably are tiny, as they need long derivations with complex induction; if so, the number of data needed is of order $1/P_{\text{PA}}(\text{Th}(\text{PA}) \setminus \text{Th}(\text{I}\Sigma_n))$. The masses were not estimated.

C.4 Splits

Proof of Proposition 4.10. (a) The split's instance sets are disjoint (Proposition 4.6(b)), so the labelling sum of Lemma 2.8 has one term, and $\mathcal{Q}_{\tau_f}(s) = \mathcal{Q}_\tau(s)/p_f$ for $s \in \text{inst}(\tau_f)$. So $P_T^{\text{Dir}}(D) = \text{DirMult}(n_f; \alpha) \prod_i \mathcal{Q}_\tau(X_i)/p_{f_i}$, while $P_T(D) = \prod_i \mathcal{Q}_\tau(X_i)$. (b) Stirling: $\ln \Gamma(a + m) = (a + m - \frac{1}{2}) \ln m - m + \frac{1}{2} \ln 2\pi + o(1)$ as $m \rightarrow \infty$. Applied to $\ln \Gamma(K\alpha + n)$ and the K terms $\ln \Gamma(\alpha + n_f)$,

$$\begin{aligned} \ln \text{DirMult}(n_f; \alpha) &= \sum_f n_f \ln n_f - n \ln n + (\alpha - \tfrac{1}{2}) \sum_f \ln n_f \\ &\quad - (K\alpha - \tfrac{1}{2}) \ln n + \frac{K-1}{2} \ln 2\pi + \ln \frac{\Gamma(K\alpha)}{\Gamma(\alpha)^K} + o(1). \end{aligned}$$

With $\ln n_f = \ln n + \ln \hat{r}_f$ the first two terms are $n \sum_f \hat{r}_f \ln \hat{r}_f$, and the $\ln n$ terms collect to $(K\alpha - \frac{K}{2} - K\alpha + \frac{1}{2}) \ln n = -\frac{K-1}{2} \ln n$. Subtracting $\sum_f n_f \ln p_f$ turns $n \sum_f \hat{r}_f \ln \hat{r}_f$ into $n \text{KL}(\hat{r} \| p)$. (c) Pearson's and Wilks's χ^2 limit (known), with all $n_f \rightarrow \infty$ a.s. since p is in the open simplex. (d) The strong law for $\hat{r}$; $\ln \text{DirMult}(n_f; \alpha) = n \sum_f \hat{r}_f \ln \hat{r}_f + O(\ln n)$ also when some $r_f = 0$. (e) The prior. $\square$

With $\alpha = \frac{1}{2}$ the term $(\alpha - \frac{1}{2}) \sum \ln \hat{r}_f$ vanishes, and in the well-specified case $\mathbb{E}\Delta_n \approx \frac{K-1}{2} - \frac{K-1}{2} \ln n + c_{K,\alpha}$; for $K = 4$, $c_{4,1/2} = \ln \pi^{-2} + \frac{3}{2} \ln 2\pi = 0.467$, which gives the predictions of Table 20.

Proof of Proposition 4.11. (a) Under (T', w') a theory-citation node chooses τ_r with probability w'_r and draws the children's bodies from $\mathcal{Q}$. By the root factorisation and $\text{inst}(\tau) = \bigsqcup_r \text{inst}(\tau_r)$, its conclusion has the law of a citation of τ whose M has root law w' . All other nodes have the same laws, and validity and conclusions depend only on the nodes' conclusions; so μ and Z coincide, under L0, L1 and L2. (b) By Tonelli,

$$\mathbb{E}[\text{BF}_{n+1} \mid D_n] = \int \prod_{i \leq n} \frac{P^{(p')}(X_i)}{P^{(p)}} \sum_{x: P^{(p)}(x) > 0} P^{(p')}(x) \text{Dir}(dp') \leq \text{BF}_n.$$

n	10^2	10^3	10^4	10^5	10^6
well specified, mean Δ_n	-4.93	-8.34	-11.77	-15.27	-18.85
prediction	-4.94	-8.39	-11.85	-15.30	-18.76
$P(\Delta_n > 0)$	0.007	0	0	0	0
misspecified, mean Δ_n	-0.06	39.7	482.2	4917.5	49356.3
prediction	-0.00	41.0	481.9	4921.9	49353.3

Table 20: Proposition 4.10, computed (model check c2, Part B): Δ_n in nats, $K = 4$, $\alpha = \frac{1}{2}$, $\mathcal{Q}$'s root law $p = (0.5, 0.3, 0.1, 0.1)$, 400 runs per n . Well specified: $r = p$. Misspecified: $r = (0.4, 0.3, 0.2, 0.1)$, $\text{KL}(r\|p) = 0.0494$ nats; predictions from (b) with $\hat{r}$ replaced by r , plus $\frac{K-1}{2}$ (the mean of $n\text{KL}(\hat{r}\|r)$) in both cases. The referee (r4) checked (b) for $\alpha = 0.3, 1, 2$ ($K = 3$): the residual tends to 0 like $1/n$.

Ville's inequality (known; IL Lemma C.4) and the supermartingale convergence theorem (known).
(c) Write

$$\ln \text{BF}_n = \ln \frac{P_{T'}^{\text{Dir}}(D_n)}{P^{(r)}(D_n)} + \sum_{i \leq n} \ln \frac{P^{(r)}(X_i)}{P^{(p)}(X_i)}.$$

The second sum divided by n tends to $\text{KL}(P^{(r)}\|P^{(p)})$ a.s., as in Proposition C.1(a). For the first term: *upper bound*: $P_{T'}^{\text{Dir}}(D_n)/P^{(r)}(D_n)$ is a nonnegative supermartingale under $P^{(r)}$, as in (b); it converges a.s. to a finite limit, so $\limsup_n n^{-1} \ln$ of it is ≤ 0 . *Lower bound*: for $\gamma > 0$ let $U_\gamma := \{p' : \text{KL}(P^{(r)}\|P^{(p')}) < \gamma\}$. By the continuity lemma below it contains a neighbourhood of the interior point r , so $\text{Dir}(U_\gamma) > 0$; let Dir_U be the normalised restriction. By Jensen's inequality,

$$\ln \int_{U_\gamma} \prod_i \frac{P^{(p')}(X_i)}{P^{(r)}(X_i)} \text{Dir}(dp') \geq \ln \text{Dir}(U_\gamma) + \sum_i Y_i, \quad Y_i := \int \ln \frac{P^{(p')}(X_i)}{P^{(r)}(X_i)} \text{Dir}_U(dp').$$

The Y_i are i.i.d., and by Tonelli on the positive and negative parts

$$\mathbb{E}Y^+ \leq \int \mathbb{E} \frac{P^{(p')}(X)}{P^{(r)}(X)} \text{Dir}_U(dp') \leq 1, \quad \mathbb{E}Y = - \int \text{KL}(P^{(r)}\|P^{(p')}) \text{Dir}_U(dp') \geq -\gamma.$$

By the strong law, $\liminf_n n^{-1} \ln[P_{T'}^{\text{Dir}}(D_n)/P^{(r)}(D_n)] \geq -\gamma$ a.s.; let $\gamma \downarrow 0$ along a sequence. *Continuity lemma*: $\text{KL}(P^{(r)}\|P^{(p')}) \rightarrow 0$ as $p' \rightarrow r$. For a tree π , $\Pr_{p'}(\pi)/\Pr_r(\pi) = \prod p'_f/r_f$ over π 's theory-citation leaves, f the root production chosen there. Let $L(\pi)$ be the number of these leaves, $\varrho := \min_f p'_f/r_f \leq 1$ and $\varrho' := \max_f p'_f/r_f \geq 1$. Then $\mu_{p'}(x) \geq \sum_{\pi \rightarrow x} \varrho^{L(\pi)} \Pr_r(\pi) \geq \mu_r(x) \varrho^{\mathbb{E}_r[L|x]}$ (Jensen, $\mathbb{E}_r[\cdot | x]$ over the valid trees concluding x), and $Z_{p'} \leq \mathbb{E}_r[\varrho'^L; \text{valid}]$. So

$$\text{KL}(P^{(r)}\|P^{(p')}) \leq \ln \frac{1}{\varrho} \cdot \mathbb{E}_r[L | \text{valid}] + \ln \frac{\mathbb{E}_r[\varrho'^L; \text{valid}]}{Z_r}.$$

Here $\mathbb{E}_r[L | \text{valid}] \leq \mathbb{E}_r[N]/Z_r < \infty$, with N the number of derivation nodes ($m < 1$). And $\mathbb{E}_r[\varrho'^L; \text{valid}] \rightarrow Z_r$ as $\varrho' \downarrow 1$ by dominated convergence, since $L \leq N$ and $\mathbb{E}_r[e^{\zeta N}] < \infty$ for small $\zeta > 0$ (Lemma A.2(d) for the derivation tree). Both terms tend to 0 as $p' \rightarrow r$. Under L2, $L \leq 2^d$ and the lemma is immediate. $\square$

C.5 Spare templates

Proof of Proposition 4.14(a). The labellings of the data by templates of $T^* \cup \{\sigma\}$ are those for T^* , with $n_\sigma = 0$. For each labelling, the Dirichlet moment of Lemma 2.8 for $T^* \cup \{\sigma\}$ divided

part	result
(a)	split with weights $(u, 1 - u)$ against the schema with root law u , $u \in \{0.1, 0.37, 0.8\}$: maximal difference $1.05 \cdot 10^{-14}$
latency	88 of 5885 conclusions depend on u through derived trees
(b)	$\delta' = 0.5, 0.2, 0.05$: $P(\max_{n \leq 3000} \text{BF}_n \geq 1/\delta') = 0.427, 0.133, 0.036$ (1000 runs, every n checked)
(c)	data root law 0.3, schema 0.6: KL = 0.11487 nats; mean $n^{-1} \ln \text{BF}_n = 0.11301, 0.11460, 0.11464$ at $n = 10^3, 10^4, 10^5$
(d)	mean $\ln \text{BF}_n = -1.765, -6.405$ at $n = 10^2, 10^6$ (400 runs); slope -0.507 (-0.498 on $n \geq 10^4$) against -0.5

Table 21: Proposition 4.11, computed (model check c13): a propositional L2 model, depth 2, atom a , logical axioms A1–A3 instantiated from a grammar truncated to size ≤ 5 , MP; $\tau = F \rightarrow F$ with F rooted at a or $\neg$, $K = 2$, root law $p_a = 0.6$; 5885 possible conclusions. Citations are latent only weakly here.

by that for T^* is $\frac{\Gamma(A+\alpha_\sigma)}{\Gamma(A+\alpha_\sigma+n)} \frac{\Gamma(\alpha_\sigma)}{\Gamma(\alpha_\sigma)} \Big/ \frac{\Gamma(A)}{\Gamma(A+n)}$, the same for every labelling; factor it out. Asymptotics: $\Gamma(a+n)/\Gamma(b+n) \sim n^{a-b}$. $\square$

For a symmetric prior ($\alpha_\tau = \alpha_\sigma = \alpha$, $A = K\alpha$) this is experiments' Prop X7(a) and, at $\alpha = \frac{1}{2}$, pa's Prop 5.1: $\log_2$ of the inverse is $\frac{1}{2} \log_2 n + \log_2 \frac{\Gamma(K/2)}{\Gamma((K+1)/2)} + o(1)$ bits. Its prediction for pa's code is compared with pa's computed spare costs in Proposition 7.31.

Proof of Proposition 4.14(d1). $P_{T^* \cup \{\sigma\}, w} = \sum_{\tau \neq \tau_1} w_\tau \mathcal{Q}_\tau + (w_{\tau_1} + w_\sigma) \mathcal{Q}_{\tau_1}$ for every w . By the aggregation property of Dirichlet laws, $(w_{\tau_1} + w_\sigma, (w_\tau)_{\tau \neq \tau_1}) \sim \text{Dir}(\alpha_1 + \alpha_\sigma, (\alpha_\tau)_{\tau \neq \tau_1})$. So $P_{T^* \cup \{\sigma\}}^{\text{Dir}}$ is the Dirichlet marginal of T^* with α_1 replaced by $\alpha_1 + \alpha_\sigma$, and with disjoint instance sets the one-labelling formula of Lemma 2.8 gives, for every D_n with n_1 data in $\text{inst}(\tau_1)$, exactly

$$\text{BF}_n^\sigma = \frac{\Gamma(A + \alpha_\sigma) \Gamma(A + n)}{\Gamma(A) \Gamma(A + \alpha_\sigma + n)} \cdot \frac{\Gamma(\alpha_1 + \alpha_\sigma + n_1) \Gamma(\alpha_1)}{\Gamma(\alpha_1 + \alpha_\sigma) \Gamma(\alpha_1 + n_1)}.$$

The limit follows from $\Gamma(a+n)/\Gamma(b+n) \sim n^{a-b}$ and $n_1/n \rightarrow w_1^*$ a.s. $\square$

We checked the limits in Table 22 against this formula: with $A = 1$, $\alpha_1 = \frac{1}{2}$, $w_1^* = \frac{1}{2}$ it gives $\ln \text{BF}_\infty^\sigma = \ln(\frac{\pi}{2} \cdot 2^{-1/2}) = 0.1050$ at $\alpha_\sigma = \frac{1}{2}$, 0 at $\alpha_\sigma = 1$ and $\ln \frac{2}{3} = -0.4055$ at $\alpha_\sigma = 2$, as c15 reports.

Sketch of Proposition 4.14(b), (c). Write $w = ((1-u)\tilde{w}, u)$ with $u \sim \text{Beta}(\alpha_\sigma, A)$ independent of $\tilde{w} \sim \text{Dir}(\alpha_\tau)$ (aggregation). The per-datum factor is $1 - u + u \mathcal{Q}_\sigma(X)/P_{T^*, \tilde{w}}(X)$. In (b), at $\tilde{w} = w^*$ its mean is $1 - uc$, so the likelihood decays like e^{-ncu} in u , and $\int u^{\alpha_\sigma-1} e^{-ncu} du \sim \Gamma(\alpha_\sigma)(cn)^{-\alpha_\sigma}$. In (c) the mean is 1 (zero drift) and the log-likelihood in u is locally $-nIu^2/2 + \sqrt{n}Zu$ with $I = \sum_s \mathcal{Q}_\sigma(s)^2/P^*(s) - 1$; this quadratic expansion is where the finite- χ^2 hypothesis of (c) is used. The integral against $u^{\alpha_\sigma-1}$ scales as $n^{-\alpha_\sigma/2}$. Not written out: uniformity in $\tilde{w}$ near w^* , and almost-sure or in-probability control of the fluctuations. This is a boundary Laplace approximation of the kind analysed for overfitted mixtures by Rousseau and Mengersen (2011), whose setting has continuous component parameters (checked by the model track's referee); their result is not applied here.

Sketch of Proposition 4.14(d2). T^* and $T^* \cup \{\sigma\}$ parametrise the same family of laws, so the two marginals integrate the same likelihood against two prior densities on that family, and Laplace’s method gives the ratio of the densities at P^* . Not written out: the Laplace step.

case	source	result
(a)	E5(a)	$0 = S0$ never cited, $u \cdot 0 = 0$ never matched: $\log_2$ BF $-1.00, -6.83$ at $n = 1, 4096$, equal to the formula on every seed; slope -0.500
(a), (b)	c3, r4	formula matched to 4 decimals; slopes $-0.499, -0.498$; referee: $-\alpha_\sigma$ to 3 decimals for $\alpha_\sigma = \frac{1}{2}, 1, 2$
(c)	E5(a2)	nested $\varphi(Sz)$ in $\{\varphi(z)\}$: mean $\log_2$ BF -2.196 at $n = 10^2$, -7.200 at 10^8 (1000 draws per n); slope -0.251 ± 0.002
(c)	c3b, r4	-0.265 on $10^3 \dots 10^7$ (s.e. ≈ 0.07 per point); referee $-0.230, -0.523, -0.961$ for $\alpha_\sigma = \frac{1}{2}, 1, 2$
(d1)	c15	formula equals the numerical integral to 5 decimals; at $n = 10^6$ within 0.003 of the limit
(d2)	c15	$w_1^* = 0.3$: $\ln \text{BF}_n^\sigma$ changes by $-0.031, -0.021, -0.004$ per decade; $w_1^* = \frac{1}{2}$ ($P^* = \mathcal{Q}_\sigma$, density infinite): $+0.22, +0.20, +0.13$

Table 22: Spare templates, computed. Slopes are of the mean log Bayes factor against the log of n in the same base, so they do not depend on the base. The primary estimate for (c) is E5(a2); its rate remains a proof sketch.

C.6 Misspecification

Proof of Theorem 4.16. On $\text{supp } P_H$, $\ln(P_T/P_{T_0}) = Y_0 - Y_T$ with $Y_T := \ln(P_H/P_T)$. Y_T has an integrable negative part, $\mathbb{E}(\ln \frac{P_T}{P_H})^+ \leq \mathbb{E} \frac{P_T}{P_H} \leq 1$, and mean K_T ; $Y_0 \in L^1$ because also $K_{T_0} < \infty$. By the strong law, $n^{-1} \sum_{i \leq n} (Y_{0,i} - Y_{T,i}) \rightarrow K_{T_0} - K_T$ a.s., the value $-\infty$ allowed when $K_T = \infty$. $\mathcal{H}$ is finite. $\square$

Inside $\mathbb{M}$, members with equal KL but different laws trade places like a zero-drift random walk.

Proof of Example 4.17. $P_H = P_\sigma$, so Theorem 4.2 applies with truth σ . Every $T \in C_\sigma$ has $\bigcup \text{inst}(T) = \text{inst}(\sigma) = I_c(0 + x = x)$ by Proposition 4.6(a) (parameters not admissible), hence $\text{Th}(T) = \text{Th}(I_c(0 + x = x))$. Take the structure $\mathbb{N} \cup \{a\}$, standard on $\mathbb{N}$, with $0 + a := 0$ and the other operations extended arbitrarily, for instance $Sa := a$ and $a \circ y := y \circ a := a$ for $\circ \in \{+, \cdot\}$ except $0 + a = 0$. Closed terms denote standard numbers, so every closed instance holds; $\forall x(0 + x = x)$ fails at a . By Corollary 4.4 the posterior probability that $T \vdash \forall x(0 + x = x)$ tends to 0. $\square$

Proof of Lemma 4.18. Let τ cover some $t = 0 \in Z$. *Shape.* The datum has no binders, so τ ’s rigid skeleton has none. A pattern occurrence of positive arity needs bound variables in scope, so every metavariable is 0-ary. The only formula position in $t = 0$ is the root. *Root metavariable.* Then τ is a formula metavariable, with the false instance $0 = S0$. *Otherwise* $\tau = (u = v)$ with term templates u, v . u matches t , so its rigid nodes are $+$ and 0 and its metavariable occurrences are leaves; hence $\text{val}(u\theta) = \sum_j c_j \text{val}(\theta(z_j))$, c_j the multiplicity of z_j in u . v matches 0 , so v is 0 or a metavariable z . $v = 0$. Soundness requires $\sum_j c_j \text{val}(\theta(z_j)) = 0$ for all θ ; $\theta(z_j) := S0$ for all j shows $c_j = 0$. So τ is ground and covers one sentence. $v = z$, z not in u . The value of $u\theta$ does not depend on $\theta(z)$; one of $\theta(z) \in \{0, S0\}$ gives a false instance. $v = z$, z in u . Soundness

data (true unless stated)	likelihood	posterior lands on	where
<i>same theorems or instances, other axioms</i>			
instances, usage that $\mathcal{Q}$ misfits	L0, L1	root splits; depth-2 nestings	Propositions 4.10 and 4.11 and Remark 4.20
PA instances, connective-rich motives	L0	$T^* + T_{\exists}$; observed-root fragments	Remark 7.30
<i>deductively weaker</i>			
closed instances of $\forall x(0 + x = x)$	L0	closed schema $0 + z = z$	Example 4.17
instances: numerals only; finite set	L0 (L1 to $n = 512$)	R_m, m growing; $\text{Mem}(D_n)$ (L0, $n = 1024$)	Remark 4.20
PA instances, atomic motives	L0	$\mathbf{Q} + \{T_-, T_-\}$	Remark 7.30 and Proposition 7.29
narrow induction practice	two-part code	$\mathbf{Q} + T_E + T_N \subseteq \text{IS}_1$ (extrapolated)	Example 7.26
theorems, no direct induction uses	L1 ^{sch}	$\mathbf{Q} + \text{a finite library}$	Remark 7.16
<i>unsound</i>			
derived equations $t = 0$, finite class	L0	$T_{\text{esc}} = \{z = 0\}$	Example 4.19
the same	L1 ^{eq}	T_{esc} or $T_{\mathbf{Q}} \cup \{z = 0\}$ (16/16 at $p_+ = 0.25$)	Example C.6
the same, full class	L0, Dir.	memorisation- T_{esc} hybrids (conjecture)	Remark C.5
prime values of $k^2 + k + 41$	L1 ^{sch}	the false schema (counterexample absent)	Example 7.37
10% false near misses (not true)	L0, L1	verifier accepts falsehoods, 100/100	Remark 5.10
small n , rarely cited axioms	L0; L1 ^{sch}	unsound lumps (E2: unsound MAP in 7/25 seeds at $n = 8$; false acceptance in 5/25)	Remarks 5.11 and 7.32

Table 23: Kinds of misspecification and where the posterior lands; statuses are those of the cited items. Computed results hold within their pools or candidate sets. T_f : T_{Ind} with motive root f ; T_E, T_N : the two induction templates of narrow practice, on motives $\exists z(s = t)$ and $\neg(s = t)$ (Example 7.26).

requires $\sum_j c_j \text{val}(\theta(z_j)) = \text{val}(\theta(z))$ for all θ ; unit test values give $c_z = 1$ and $c_j = 0$ for $j \neq z$. So u contains z once and is otherwise ground; covering $t = 0$ forces $\theta(z) = 0$, and τ covers exactly one member of Z . $\square$

Proof of Example 4.19. By Lemma 4.18 each T_j covers at most $|T_j|$ members of Z . Since $\text{supp } P_H$ is infinite, $P_H(Z \setminus \bigcup_j \text{inst}(T_j)) > 0$, so a.s. some datum falls outside every T_j , which then has likelihood 0 under L0. $P_{T_{\text{esc}}} > 0$ on Z (full-support $\mathcal{Q}$, $z := t$). So $\pi_n(T_{\text{esc}}) = 1$ from that datum on. $T_{\text{esc}} \vdash S0 = 0$ (instance $z := S0$); $\mathbf{Q} \vdash \neg S0 = 0$ (Q1). $\mathbf{Q}$'s axioms are sentences outside Z , so a theory containing only them has likelihood 0 at the first datum. $\square$

Computed (c4, Part 2; the run reported in Example 4.19): the twenty sound theories are ground equations for the 20 most likely data, which carry mass 0.956, weighted as the human's conditional law, and share prior 0.99 against T_{esc} ; $S0 = 0$ had reached posterior ≥ 0.99 in 90.2% of runs by $n = 50$. Proposition 5.4 uses this example for the verifier.

Remark C.5 (The full class). [conjecture; computed]^{model Rem 3.5} In the full class with Dirichlet weights the KL-projection picture can fail: theories with finite KL are unsound (Lemma 4.18), with infimum KL 0 approached by hybrids of memorisation and T_{esc} ; memorisers are sound but

each has infinite KL, yet as a family they code the data with $o(n)$ extra bits. Conjecture: unsound mass $\rightarrow 1$. Computed for three hand-picked families: at $n = 10^4$ the hybrid leads memorisation by at least 7000–25000 bits (Table 24).

seed	$\lambda = 1$			$\lambda = 2$		
	11	12	13	11	12	13
distinct data at $n = 10^4$	282	325	330	282	325	330
hybrid – memorisation, $n = 10^3$	≥ 1143	≥ 1207	≥ 1065	≥ 3344	≥ 3505	≥ 3063
hybrid – memorisation, $n = 10^4$	≥ 7370	≥ 8613	≥ 8625	≥ 21703	≥ 24719	≥ 24596
escape alone – memorisation, $n = 10^4$	–4234	–2649	–2719	12871	16289	16120

Table 24: Remark C.5, computed (model check c7): $\log_2$ score (prior plus L0 marginal likelihood, $\text{Dir}(\frac{1}{2})$) relative to memorising all distinct data. Human data $t = 0$, t a $\{0, +\}$ -tree (0 with probability 0.7); model $\mathcal{Q}$ root law (0.5, 0.3, 0.1, 0.1) on $(0, S, +, \cdot)$; prior $2^{-\lambda \ell}$ under a simplified code (3 bits per symbol, 2 per metavariable occurrence, Elias γ for the number of templates), not π_λ of Definition 2.4. Hybrid: ground templates for data seen at least twice plus $z = 0$; its score is a lower bound (one labelling). Escape alone: $\{z = 0\}$. Positive favours the unsound theory. At $\lambda = 1$ the escape alone loses to memorisation by $n = 10^4$, but the hybrid does not.

Example C.6 (A derivation likelihood does not reliably rescue the axioms). $[\text{computed}]^{\text{model Ex 3.6}}$ L1^{eq} derives closed equations over $0, S, +$ (cite, refl, sym, trans, congruences), terms of size $\leq N$, $P_T = \mu_T/Z_T$ exact. Class: $T_Q := \{z+0 = z, z_1+Sz_2 = S(z_1+z_2)\}$ (Q4, Q5) and $T_Q \cup \{0+z = z\}$ (sound); T_{esc} and $T_Q \cup \{z = 0\}$ (unsound). On true equations $t = 0$ over $\{0, +\}$ the cross-entropy minimiser is unsound in all 16 settings in which $\mathcal{Q}$ gives $+$ probability 0.25, and sound in 5 of 8 when $+$ is rare under $\mathcal{Q}$ (Table 25): in grammar 1, citing $z = 0$ pays about 2 nats per $+$ through $\mathcal{Q}$, deriving from Q4, Q5 about 7. Limits: truncated, equational only, three grammars, a finite class.

Conjecture C.7. $[\text{conjecture}]^{\text{model Conj 3.7}}$ In untruncated L1^{eq} , $\ln P_{T_Q}(t = 0) = -c_{\text{der}}\#_+(t) + O(1)$ on balanced $\{0, +\}$ -terms; so T_{esc} beats T_Q on laws with many $+$ iff $\ln \frac{1}{p_+p_0} < c_{\text{der}}$, p_+, p_0 the root probabilities under $\mathcal{Q}$ (grammar 1: ≈ 2 against ≈ 7).

Details of Example C.6. We write $T_{Q0} := T_Q \cup \{0 + z = z\}$ and $T_{\text{mix}} := T_Q \cup \{z = 0\}$. Each node of the L1^{eq} grammar cites a template of T (probability α_{ax} , metavariables from $\mathcal{Q}$), or applies refl ($t \sim \mathcal{Q}$), sym, trans (valid iff the middle terms agree), congruence for S , or congruence for $+$ (a side term $u \sim \mathcal{Q}$, left or right). A tree is valid iff its steps are valid and every term lies in U_N , the closed terms of size $\leq N$. $P_T = \mu_T/Z_T$, with μ_T the least fixpoint of the grammar’s equation. These rules are complete for closed equational consequence (Birkhoff; recalled, not checked against the source (Birkhoff, 1935)). A Monte Carlo simulation agrees with the fixpoint ($Z = 0.54909$ against 0.54965 ± 0.00091 ; $|z| \leq 1.31$ on eight conclusion frequencies). Human law: $t = 0$, t a closed $\{0, +\}$ -term from a grammar choosing 0 with probability $1 - h$ and $+$ with h , conditioned on size $\leq N$, $h \in \{0.3, 0.45\}$, with and without the datum $0 = 0$.

C.7 Predictive versus deductive

Proof of Theorem 4.21. (a) is Proposition C.1(b) with the index shifted; (b) is Corollary 4.4 with $d = \infty$, and the two cases are those named after it. (c) $\mathbb{E}_{P_{T_1}} \ln \frac{P_{T_1}(X)}{P_{T_2}(X)} = \text{KL}(P_{T_1} \| P_{T_2})$ per

grammar	rule probabilities (ax, refl, sym, trans, cong _S , cong ₊)	$\mathcal{Q}$ root law (0, S , +)	minimiser	settings
1	(.35, .15, .1, .15, .1, .15), $m = 0.65$	(.55, .2, .25)	T_{esc} with $0 = 0$; T_{mix} without	8/8 unsound
2	(.5, .1, .05, .2, .05, .1), $m = 0.60$	(.55, .2, .25)	as grammar 1	8/8 unsound
3	as grammar 1	(.8995, .1, .0005)	T_{Q0} in 5; T_{mix} in 3 (all with $0 = 0$)	5/8 sound

mean $\ln P_T(t = 0)$ by size of t	1	3	5	7	9
grammar 1: T_Q	-1.88	-1.82	-8.45	-15.52	-22.89
grammar 1: T_{esc}	-0.80	-3.23	-5.22	-7.20	-9.19
grammar 3: T_Q	-1.62	-1.57	-7.91	-15.41	-23.33
grammar 3: T_{Q0}	-1.63	-1.28	-7.34	-14.55	-22.19
grammar 3: T_{esc}	-0.43	-8.58	-16.30	-24.01	-31.72

Table 25: Example C.6, computed (model check c16). Top: the cross-entropy minimiser over $\{T_Q, T_{Q0}, T_{\text{esc}}, T_{\text{mix}}\}$ in the eight settings of each grammar ($N \in \{7, 9\}$, $h \in \{0.3, 0.45\}$, with and without $0 = 0$). Bottom: per-datum log-likelihoods at $N = 9$. In grammar 1, T_Q pays about 7 nats per extra + and T_{esc} about 2; e.g. at $N = 9$, $h = 0.45$, data $t \neq 0$, the cross-entropies are 7.758 (T_Q), 7.124 (T_{Q0}), 4.951 (T_{esc}), 4.904 (T_{mix}) nats, so the odds of T_Q against T_{mix} fall by about 2.85 nats per datum. In grammar 3, T_{esc} pays $\ln \frac{1}{p+p_0} \approx 7.7$ nats per + through $\mathcal{Q}$; T_Q 's increments (6.3, 7.5, 7.9 nats between consecutive sizes) are still rising at size 9, so the large-term limit in grammar 3 is not settled.

datum, summed over the data; Pinsker's inequality $\text{TV} \leq (\text{KL}/2)^{1/2}$ (known). For the misspecified remark: $M(D_n) \geq \pi(T)P_T(D_n)$ for every T and every sequence, so the cumulative log loss of M exceeds that of T by at most $\ln(1/\pi(T))$. $\square$

Proposition C.8 (Data that come with their proofs). *[proved]^{model Prop 8.2} Let each datum be a whole valid L1 tree from $P_T(\cdot \mid \text{valid})$, showing each node's rule and conclusion and whether a citation leaf cites a theory or a logical axiom, but not which template; weights fixed. (a) $P_T(\pi) = \prod_v P_T^{\text{L0}}(\text{concl}(v)) G(\pi)/Z_T$ over the theory-citation leaves v , with $P_T^{\text{L0}} := \sum_\tau w_\tau \mathcal{Q}_\tau$, G independent of T , and Z_T a function of P_T^{L0} . (b) $P_T = P_{T'}$ on trees iff $P_T^{\text{L0}} = P_{T'}^{\text{L0}}$: the posterior concentrates on the L0 generator class of T^* , with instance union $\bigcup \text{inst}(T^*)$, and splits still tie. (c) So proofs guarantee identification of the instance union $\bigcup \text{inst}(T^*)$; conclusions guarantee only $\text{Th}(T^*)$ (though they often separate more, Proposition 4.5).*

Proof of Proposition C.8. (a) A tree's probability is the product of its nodes' probabilities. A theory-citation leaf whose template is not recorded contributes $\alpha_{\text{ax}} \sum_\tau w_\tau \mathcal{Q}_\tau(\text{concl}(v)) = \alpha_{\text{ax}} P_T^{\text{L0}}(\text{concl}(v))$, the sum over the unrecorded choices factorising over leaves (matching is unique, so $\mathcal{Q}_\tau(\text{concl}(v))$ is the sum over substitutions). No other node depends on T . Validity depends only on conclusions, so $Z_T = \Pr[\text{valid}]$ is a function of the law of theory-leaf conclusions, i.e. of P_T^{L0} . (b) "If" is (a). "Only if": the one-node tree citing s as a theory axiom has probability $\alpha_{\text{ax}} P_T^{\text{L0}}(s)/Z_T$; equality for all s makes P_T^{L0} and $P_{T'}^{\text{L0}}$ proportional, and both sum to 1. The rest is Theorem 4.2, Corollary 4.3 and Proposition 4.6(a), (b). (c) The proof-data generator class of T^* is contained in its conclusion-data class, since equal L0 laws give equal L1 laws; the latter

guarantees only $\text{Th}(T^*)$ (Corollary 4.4, L1 with parameters admissible), though conclusions can separate more (Proposition 4.5). $\square$

If the template name is recorded at each citation leaf, the argument is easier.

Remark C.9 (Near misses). [proved (the transfer); open (size of the generator class)]^{model Rem 8.3} Hänni’s “grade near misses” can be a corruption channel $K(s' \mid s)$ favouring s' near s , $P_T^K(s') := \sum_s P_T(s)K(s' \mid s)$. Each P_T^K is a fixed law with a data-independent normaliser, so Theorem 4.2, Corollary 4.4 and Theorem 5.2 (with C_d^* defined through P^K) hold when the channel is part of the well-specified model. The size principle also holds, but with a full-support channel it is vacuous ($\text{supp } P_{T^*}^K = \mathcal{S}$), and theories are separated only through KL. The generator class can grow, since K need not be injective; by how much, for natural metrics, is open. The noise mixtures of Proposition B.1 are related.

Proof of Remark C.9. P_T^K is a probability on $\mathcal{S}$ whose normaliser does not depend on the data. The proofs of Theorem 4.2, Corollary 4.4, Theorem 5.2 and Lemma 2.21 use only that each hypothesis carries a fixed law on a countable set and that the data are i.i.d. from one of them; they therefore hold with P_T^K in place of P_T when the data are i.i.d. from $P_{T^*}^K$. $\square$

D Soundness: proofs and details

Scripts named *ck* are the model track’s (`research/tracks/model/checks/`), *rk* its referee’s (`research/tracks/model/referee_code/`); the others are in `research/tracks/experiments/checks/`. Each is seeded and writes its output next to itself. Of this section’s results the model track’s referee checked Theorem 5.2, Propositions 5.5 and D.2 and the trichotomy results Propositions 2.14, 2.18 and 2.19 (now in Section 2.7). Lemma D.4, Theorems 5.6 and 5.7, Remark D.3, the mechanism of Example 5.8 and Proposition 2.17 were added after the referee report and checked by the track and while writing, not by a referee; the complete list of such results is in Appendix H.1. Nothing is checked in a proof assistant.

D.1 Fixed weights

Proof of Theorem 5.2. This is the proof of IL Thm 4.15(b) (IL App. C.9), in the class form of IL Thm 10.9; the role of shadow-measurability there is played by the definition of C_d^* . Write $L_t(T) := \prod_{i \leq n(t)} P_T(X_i) \prod_j c_j(T)$.

Step 1: C_d^* keeps the target’s likelihood. For $T \in C_d^*$, $P_T = P_{T^*}$, and every constraint has $c_j(T) = c_j(T^*) = 1$ because $\text{Th}_d(T) = \text{Th}_d(T^*)$. So $L_t(T) = L_t(T^*)$, which is positive almost surely. With $Z_t := \sum_T \pi(T) L_t(T) / L_t(T^*)$ we get $\pi_t(C_d^*) = W_d^* / Z_t$.

Step 2: constraints only help. $c_j(T^*) = 1$ and $c_j(T) \leq 1$, so pathwise $Z_t \leq Z_{n(t)}^{\mathcal{H}}$, where

$$Z_n^{\mathcal{H}} := \sum_T \pi(T) \prod_{i \leq n} \frac{P_T(X_i)}{P_{T^*}(X_i)} = \frac{M(X^n)}{P_{T^*}(X^n)}.$$

Step 3: Ville. Under P_{T^*} , $\mathbb{E}[P_T(X) / P_{T^*}(X)] = \sum_{x: P_{T^*}(x) > 0} P_T(x) \leq 1$. By independence and Tonelli, $Z^{\mathcal{H}}$ is a nonnegative supermartingale for $\sigma(X_1, \dots, X_n)$ with $Z_0^{\mathcal{H}} = 1$. By Ville’s inequality (IL Lemma C.4) the event $G := \{\sup_n Z_n^{\mathcal{H}} < 1/\delta'\}$ has probability at least $1 - \delta'$. It depends only on the data.

Step 4: conclusion. On G , in every round and for every history of queries and truthful constraints, $\pi_t(C_d^*) = W_d^* / Z_t \geq W_d^* / Z_{n(t)}^{\mathcal{H}} > W_d^* \delta' \geq \delta$. If $q \notin \text{Th}_d(T^*)$ then $q \notin \text{Th}_d(T)$ for

every $T \in C_d^*$, so $\pi_t(\{T : q \notin \text{Th}_d(T)\}) > \delta$ and q is not accepted. The prover's knowledge and randomness enter only through the schedule and the constraints, so the conclusion covers a prover that knows the future data. $\square$

Escalations. Let $Z_t := W_d^*/\pi_t(C_d^*)$, as in Step 1. Escalating a valid query q adds the constraint $1[q \in \text{Th}_d(T)]$, which keeps C_d^* and multiplies Z_t by $\pi_t(\{T : q \in \text{Th}_d(T)\}) < 1 - \delta$, because q was not accepted; data rounds multiply it by factors that form a nonnegative supermartingale from 1 (Step 3). Since $Z_t \geq W_d^*$, on the Ville event of probability $\geq 1 - \delta''$ the number E_t^+ of escalated valid queries satisfies $(1 - \delta)^{E_t^+} > W_d^* \delta''$, so $E_t^+ \leq (\ln(1/W_d^*) + \ln(1/\delta''))/\delta$ for all t (IL Thm 4.16). An invalid query escalated adds a constraint that removes the theories deriving it; IL bounds these through a reject threshold δ_r , which this protocol does not have, so their number is not bounded.

Computed (Theorem 5.2). c4, Part 1: L0, fixed weights, $\mathcal{Q}$ root law (.5, .3, .1, .1) on $(0, S, +, \cdot)$, $T^* = \{z+0 = z\}$ against $\{z+0 = z, S0+0 = S0, 0+0 = S0\}$ (weights 0.49, 0.5, 0.01), which proves $q := (0+0 = S0)$; $\pi(T^*) = 0.3$; 20 000 runs of 300 data. The frequency of $\{\exists n \leq 300 : \pi_n(T^*) \leq \delta\}$ was 0.044, 0.014, 0.004 against $\delta/\pi(T^*) = 0.10, 0.033, 0.010$. r7: nine hypotheses all proving the same invalid query, $\pi(C^*) = 0.385$, $n \leq 400$: 0.0365, 0.0125, 0.0027 against $\delta/\pi(C^*) = 0.130, 0.052, 0.013$. So the frequency of invalid acceptance stayed 2–5 times below the bound. The horizons are finite, so these check the bound, not its time-uniformity.

Remark D.1 (The constant is tight). [proved]^{model §4.1 (referee m13)} IL Prop C.5 shows that $\delta \leq W_d^* \delta'$ cannot be improved, with $R^* = \{a, b\}$ against $R' = \{a, c\}$, which generates only a and proves c . That pair is not in the template models (under L0, $\text{supp } P_{T'} \subseteq \text{supp } P_{T^*}$ implies $\text{Th}_d(T') \subseteq \text{Th}_d(T^*)$ for every d ; under L1 with parameters admissible it does so for $d = \infty$ only: $T^* = \{a, a \rightarrow b\}$ and $T' = T^* \cup \{b\}$ have the same support, but $b \in \text{Th}_{|b|}(T') \setminus \text{Th}_{|b|}(T^*)$). But $T' = \{a, c\}$ with a small fixed weight on c attains IL's acceptance probability $(1 - u)^{t^*}$ exactly, and it tends to δ' as $u, W_d^* \rightarrow 0$.

Proof of Remark D.1. IL's pair and the template models. Under L0 with full-support $\mathcal{Q}$, $\text{supp } P_T = \bigcup \text{inst}(T)$, and Th_d depends only on the instance union, so $\text{supp } P_{T'} \subseteq \text{supp } P_{T^*}$ gives $\text{Th}_d(T') \subseteq \text{Th}_d(T^*)$ for every d : no T' generates only data of T^* and still proves a sentence T^* does not. Under L1 with parameters admissible $\text{supp } P_T = \text{Th}(T)$ (Lemma 2.10(c)), so the inclusion of supports gives $\text{Th}(T') \subseteq \text{Th}(T^*)$, the case $d = \infty$ only. For finite d it fails: with ground $T^* = \{a, a \rightarrow b\}$ and $T' = T^* \cup \{b\}$, $\text{Th}(T') = \text{Th}(T^*)$, so the supports agree, but b is an axiom of T' while every derivation of b from T^* uses $a, a \rightarrow b$ and MP, so $b \in \text{Th}_{|b|}(T') \setminus \text{Th}_{|b|}(T^*)$.

The construction. Let a, b, c be ground sentences with $c \notin \text{Th}_d(T^*)$ and $c \in \text{Th}_d(\{c\})$, let $T^* = \{a, b\}$ with fixed weights $(1-u, u)$, $T' = \{a, c\}$ with fixed weights $(1-\varepsilon, \varepsilon)$, prior $(W^*, 1-W^*)$, likelihood L0 and $\delta = W^* \delta'$. Then $W_d^* = W^*$. After t data equal to a , $r_t := P_{T'}(a^t)/P_{T^*}(a^t) = ((1-\varepsilon)/(1-u))^t$, and c is accepted iff $\pi_t(T^*) = W^*/(W^* + (1-W^*)r_t) \leq \delta$, i.e. iff $r_t \geq \theta := (1-W^* \delta')/((1-W^*)\delta')$. Let t^* be the least t with $(1-u)^{-t} > \theta$ (IL Prop C.5). For $t < t^*$, $r_t < (1-u)^{-t} \leq \theta$. At t^* the inequality $(1-u)^{-t^*} > \theta$ is strict, so $r_{t^*} > \theta$ for all ε below some $\varepsilon_0 > 0$. A datum b kills T' . So the prover that waits for t^* data and then asks c (an earlier query could be escalated, and the oracle's "no" would kill T') wins iff the first t^* data are a : probability $(1-u)^{t^*}$, exactly IL's value. By IL Prop C.5 it lies in $[(1-u)/\theta, 1/\theta]$ with $1/\theta = (1-W^*)\delta'/(1-W^* \delta')$, so it tends to δ' as $u, W^* \rightarrow 0$. $\square$

Truncation (Remark 5.3). On a finite truncation $\mathcal{H}_N \ni T^*$ with the renormalised prior, the proof above runs with $\pi(\cdot)/\pi(\mathcal{H}_N)$ in place of π , so the theorem holds with W_d^* replaced by $\pi(C_d^* \cap \mathcal{H}_N)/\pi(\mathcal{H}_N) \geq \pi(T^*)$.

The computable verifier (Remark 5.3). For a finite class and $d < \infty$, $\text{Th}_d(T)$ is decidable (finitely many derivations of symbol size $\leq d$ up to renaming of parameters, axioms checked by matching), the likelihoods are computable reals (L0 by matching; L1 by Proposition 2.24(a)), and the denominator of π_t is positive. So an interval of width $\leq \varepsilon$ around the non-deriving mass can be computed; accept iff its upper end is $\leq \delta$. Acceptance then implies that the true mass is $\leq \delta$ at the same posterior, and Step 4 applies verbatim. With truncated $\mathcal{Q}$ the computation is exact.

The refuted remarks (Remark 5.3). (1) On a truncation $\mathcal{H}_N \ni T^*$ the constant $\pi(C_d^* \cap \mathcal{H}_N)/\pi(\mathcal{H}_N)$ is at least $\pi(T^*)$ but can be below $\pi(C_d^*)$: with $\mathcal{H} = \{T^*, T_2, T_3\}$, $T_2 \in C_d^*$, prior $(0.01, 0.89, 0.10)$ and $\mathcal{H}_N = \{T^*, T_3\}$ it is $0.01/0.11 \approx 0.09$, against 0.90 . (2) L2 likelihoods are finite sums only when $\mathcal{Q}$ has finite support: for $T = \{P \rightarrow b\}$, P a 0-ary formula metavariable, $P_T(b)$ at depth 1 sums $\alpha_{\text{MP}} \mu(A) \mu(A \rightarrow b)$ over the infinitely many logical-axiom instances A .

Proposition 5.4: computed. c4, Part 2, is the run of Example 4.19 (details in Appendix C.6), read with $\delta = 0.01$ and no escalation: the fraction of runs in which $S0 = 0$ had been accepted was $0.341, 0.902, 0.986, 1.000$ by $n = 10, 50, 100, 500$. r3, Part 2: against a verifier that escalates every non-accepted query, a prover asking $S0 = 0$ in every round won $11/400$ (only when the first datum was uncovered), the waiting prover $400/400$.

D.2 The cautious limit

Proposition D.2 (The limit $\delta \rightarrow 0$). *[proved]^{model Prop 4.3} Suppose each P_T has a fixed support supp_T (fixed weights; or Dirichlet weights under L0, where $P_T^{\text{Dir}}(D) > 0$ iff $D \subseteq \bigcup \text{inst}(T)$).*

- (a) $\pi_n(T) > 0$ iff $\pi(T) > 0$ and $D_n \subseteq \text{supp}_T$. So $V_{0,d}$ accepts exactly $\bigcap \{\text{Th}_d(T) : \pi(T) > 0, D_n \subseteq \text{supp}_T\}$.
- (b) $V_{0,d}$ is sound for every T^* with $\pi(T^*) > 0$ and $D_n \subseteq \text{supp}_{T^*}$, deterministically and without well-specification (IL Thm 4.14). $V_{\delta,d}$ accepts at least as much.
- (c) Under L0 with full-support $\mathcal{Q}$, if π charges every finite theory, $V_{0,d}$ accepts exactly $\text{Th}_d(D_n)$ (AS Prop 6.4, read on theorems).
- (d) Under L0, if $\text{supp } \pi = H_k(\text{DT}^\circ)$, the theories with at most k templates, the acceptance set of $V_{0,d}$ contains $\text{Th}_d(\text{Acc}_k(D_n))$, where Acc_k is the cautious k -union verifier of AS Def 2.6. In general classes the inclusion can be strict; for $H_k(\text{DT}^\circ)$ itself this is open.

Proof of Proposition D.2. (a) The likelihood of T (fixed weights, or the Dirichlet marginal under L0) is positive iff every datum lies in supp_T , and $\pi_n(T)$ is proportional to $\pi(T)$ times it. $V_{0,d}$ accepts q iff $\pi_n(\{T : q \notin \text{Th}_d(T)\}) = 0$, i.e. iff $q \in \text{Th}_d(T)$ for every T of positive posterior mass. (b) Such a T^* is one of the theories in the intersection of (a), and truthful constraints keep it; so everything accepted lies in $\text{Th}_d(T^*)$. The accepted set of $V_{\delta,d}$ grows with δ . (c) The memoriser $\text{Mem}(D_n)$, made of ground templates for the distinct data, is charged by π , its support is the set of distinct data, and $\text{Th}_d(\text{Mem}(D_n)) = \text{Th}_d(D_n)$. Every T with $D_n \subseteq \bigcup \text{inst}(T)$ has $\text{Th}_d(D_n) \subseteq \text{Th}_d(T)$, since Th_d is monotone in the axiom set. So the intersection in (a) is $\text{Th}_d(D_n)$; this is AS Prop 6.4 ($\text{Acc}_\infty(D) = D$) read on theorems. (d) A theory of positive posterior

mass lies in $H_k(\text{DT}^\circ)$ and covers D_n , so it is in the version space $\text{VS}_k(D_n)$ of AS Def 2.6, and $\bigcup \text{inst}(T) \supseteq \text{Acc}_k(D_n)$. Monotonicity of Th_d gives the inclusion. *Strictness in a general class.* Take the class $\{T_1, T_2\}$ with $T_1 = \{c, a \wedge b\}$, $T_2 = \{c, b \wedge a\}$ (ground), and $D = \{c\}$. Both theories prove a , so $V_{0,d}$ accepts a for d large enough. The instance-level intersection is $\{c\}$, which does not prove a . Whether the inclusion can be strict for $H_k(\text{DT}^\circ)$ itself is not settled (model §10, problem 10). $\square$

The two verifiers compared. Table 26 contrasts the cautious verifier of AS with the thresholded verifier of Section 5 (model §5.5, with the weight convention stated).

	cautious k -union verifier (AS)	Bayesian thresholded verifier
assumptions for soundness	realizability ($R^* \in H_k$), bound $k \geq k'$	well-specified data law; $W_d^* \geq \delta/\delta'$; with Dirichlet weights, averaging over w^* , or a threshold shrinking with n for likelihoods linear in w
soundness	deterministic; every prover, every time	probability $\geq 1 - \delta'$; every prover, every time (Theorems 5.2, 5.6 and 5.7)
without the assumptions	sound relative to $\text{cl}(R^*)$ (AS Lemma 2.8); without a bound accepts only the data (AS Prop 6.4)	can accept a false sentence with probability 1 (Proposition 5.4); at a fixed w^* a constant threshold fails (Example 5.8)
completeness	exact once an anchor is present (AS Lemma 2.8)	each theorem eventually, a.s. (Proposition 5.5; Dirichlet: a.e. w^*); no anchor, no bound; open under a shrinking threshold
splits	sound splits survive any negatives (AS Thm 6.5)	exact ties at matched fixed weights; with learned weights Occam terms decide; usage statistics can reverse (Propositions 4.6, 4.10 and 4.11)
depends on the target's length	no	yes: $\delta \leq 2^{-\lambda \ell(T^*)} \delta'$; under Theorem 5.7 also on $ T^* $

Table 26: The cautious verifier of AS against the Bayesian thresholded verifier (model §5.5, with the weight convention stated).

The time for the Bayesian verifier to accept s is set by the competitors that miss it: a split that omits a root of probability p gains a factor $(1 - p)^{-n}$ until the first datum with that root.

D.3 Dirichlet weights

Remark D.3 (Vacuity for Dirichlet weights). [proved (L0 and likelihoods affine in w)]^{model Rem 4.5; experiments §0 (M3)}

Under L0, or any likelihood affine in w (such as the experiments' chain Ch_J), read Theorem 5.2 with parameter (T, w) and prior $\pi(T)\text{Dir}_T(dw)$. Its class $\{(T, w) : P_{T,w} = P_{T^*,w^*}, \text{Th}_d(T) = \text{Th}_d(T^*)\}$ has prior mass 0 if $|T^*| \geq 2$, the component laws of T^* are linearly independent, and w^* lies outside a countable set; then $\delta \leq W_d^* \delta'$ forces $\delta = 0$. Under L1, where $P_{T,w}$ is a ratio of power series in w , the same is plausible but not proved.

Proof of Remark D.3. The class is the exact class $\{(T, w) : P_{T,w} = P_{T^*,w^*}\}$ of Remark 4.8 intersected with $\{T : \text{Th}_d(T) = \text{Th}_d(T^*)\}$, so its prior mass is at most that of the exact class. The affine-slice argument in the proof of Remark 4.8 (Appendix C.3) uses only that $w \mapsto P_{T,w}$ is affine, which holds under L0 and for the chain Ch_J (whose law is $\sum_i w_i a_i$): it gives $\text{Dir}_T(\{w : P_{T,w} = P_{T^*,w^*}\}) = 0$ for every T unless some one-component theory has law

P_{T^*, w^*} , which, by linear independence of T^* 's component laws, happens for at most one w^* per one-component theory, so for countably many w^* . Under L1, $P_{T, w} = \mu_{T, w} / Z_{T, w}$ is a ratio of power series in w , not affine, and the argument does not apply. $\square$

Lemma D.4 (Regret). *[(a), (b), (d) proved; (c) known; computed]^{model Lemma 4.6; experiments Prop X8(c)}
Let T have K templates with laws q_τ (under L0, $q_\tau = \mathcal{Q}_\tau$), $P_{T, w} := \sum_\tau w_\tau q_\tau$, weights $\text{Dir}(\alpha)$, $A := \sum_\tau \alpha_\tau$, and*

$$R_\alpha(n, K) := \min_{c \in \mathbb{N}^K, \sum_\tau c_\tau = n} \frac{\mathbb{E}_{\text{Dir}(\alpha)}[\prod_\tau w_\tau^{c_\tau}]}{\prod_\tau (c_\tau/n)^{c_\tau}} \quad (0^0 := 1).$$

(a) $P_T^{\text{Dir}}(x^n) \geq R_\alpha(n, K) P_{T, w}(x^n)$ for every w in the closed simplex and every x^n . (b) If $\alpha_\tau \leq 1$ for every τ , then $R_\alpha(n, K) \geq \frac{\Gamma(A)}{e^{(K-1)!} \prod_\tau \Gamma(\alpha_\tau)} (n+1)^{-(K-1)}$. (c) For $\alpha_\tau \equiv \frac{1}{2}$: $-\ln R_\alpha(n, K) = \frac{K-1}{2} \ln n + O(1)$ (Krichevsky and Trofimov, 1981). (d) For $\alpha_\tau \equiv \alpha$: $R_\alpha(n, K) \leq \frac{\Gamma(K\alpha)\Gamma(\alpha+n)}{\Gamma(\alpha)\Gamma(K\alpha+n)} \sim \frac{\Gamma(K\alpha)}{\Gamma(\alpha)} n^{-(K-1)\alpha}$; so at $\alpha = 1$ the exponent $K-1$ in (b) is tight.

Proof of Lemma D.4. (a) Expand over labellings $\ell : [n] \rightarrow T$ with counts $c_\tau(\ell)$:

$$P_{T, w}(x^n) = \sum_\ell \prod_\tau w_\tau^{c_\tau(\ell)} \prod_i q_{\ell_i}(x_i), \quad P_T^{\text{Dir}}(x^n) = \sum_\ell \mathbb{E}_{\text{Dir}} \left[\prod_\tau w_\tau^{c_\tau(\ell)} \right] \prod_i q_{\ell_i}(x_i).$$

For each ℓ the maximum of $\prod_\tau v_\tau^{c_\tau}$ over the simplex is attained at $v = c/n$, so

$$\mathbb{E}_{\text{Dir}} \left[\prod_\tau w_\tau^{c_\tau} \right] \geq R_\alpha(n, K) \prod_\tau (c_\tau/n)^{c_\tau} \geq R_\alpha(n, K) \prod_\tau w_\tau^{c_\tau}.$$

Sum over ℓ . This termwise argument is experiments Prop X8(c).

(b) Fix c , put $v := c/n$, $\eta := 1/(n+1)$ and $B := \{(1-\eta)v + \eta u : u \text{ in the simplex}\}$. On B , $w_\tau \geq (1-\eta)v_\tau$, so $\prod_\tau w_\tau^{c_\tau} \geq (1-\eta)^n \prod_\tau v_\tau^{c_\tau} \geq e^{-1} \prod_\tau v_\tau^{c_\tau}$, because $(1-\eta)^n = (1+1/n)^{-n} \geq e^{-1}$. The Dirichlet density $\frac{\Gamma(A)}{\prod_\tau \Gamma(\alpha_\tau)} \prod_\tau w_\tau^{\alpha_\tau-1}$ (with respect to Lebesgue measure on the first $K-1$ coordinates) is at least $\Gamma(A)/\prod_\tau \Gamma(\alpha_\tau)$ on the simplex, since $w_\tau \leq 1$ and $\alpha_\tau \leq 1$. B is the simplex scaled by η , so its volume is $\eta^{K-1}/(K-1)!$. Hence $\mathbb{E}_{\text{Dir}}[\prod w^c] \geq e^{-1} \prod v^c \cdot \frac{\Gamma(A)}{\prod_\tau \Gamma(\alpha_\tau)} \cdot \frac{\eta^{K-1}}{(K-1)!}$; divide by $\prod v^c$ and minimise over c .

(d) Take $c = (n, 0, \dots, 0)$: $\prod (c/n)^c = 1$ and $\mathbb{E}_{\text{Dir}}[w_1^n] = \Gamma(K\alpha)\Gamma(\alpha+n)/(\Gamma(\alpha)\Gamma(K\alpha+n))$; then $\Gamma(a+n)/\Gamma(b+n) \sim n^{a-b}$.

(c) is the Krichevsky–Trofimov rate (Krichevsky and Trofimov, 1981); it is not re-proved here. $\square$

Computed (Lemma D.4). c11, Part A: exact Dirichlet marginals (a Bernstein-form dynamic programme, equal to quadrature to six decimals) for $K = 2, 3$ with overlapping components on four letters, w^* interior and on the boundary, $n \leq 3000$. The largest value of regret minus bound (b) was $-2.81, -1.00, -6.45$ nats ($K = 2, \alpha = \frac{1}{2}$; $K = 2, \alpha = 1$; $K = 3, \alpha = \frac{1}{2}$). Part A2: slopes of the exact $\ln R_\alpha(n, K)$ against $\ln n$ were $-0.499, -0.993$ ($K = 2, 3$; $\alpha = \frac{1}{2}$) and $-0.993, -1.960$ ($\alpha = 1$), against $-(K-1)\max(\frac{1}{2}, \alpha)$. The experiments' `kt_regret`: $-0.499, -0.998, -1.485$ for $K = 2, 3, 4$ at $\alpha = \frac{1}{2}$.

Proof of Theorem 5.6. Steps 1, 2 and 4 of the proof of Theorem 5.2 go through with $L_t(T) := P_T^{\text{Dir}}(X^{n(t)}) \prod_j c_j(T)$ and C_d^{Dir} in place of C_d^* . For Step 3 let $M := \sum_T \pi(T) P_T^{\text{Dir}}$ and $Z_n :=$

$M(X^n)/P_{T^*}^{\text{Dir}}(X^n)$. Each P_T^{Dir} is a probability on sequences, consistent in n , so M is too, and under $P_{T^*}^{\text{Dir}}$

$$\mathbb{E}[Z_{n+1} \mid X^n] = \sum_{x: P_{T^*}^{\text{Dir}}(X^n x) > 0} \frac{M(X^n x)}{P_{T^*}^{\text{Dir}}(X^n)} \leq \frac{M(X^n)}{P_{T^*}^{\text{Dir}}(X^n)} = Z_n.$$

Ville's inequality gives $G := \{\sup_n Z_n < 1/\delta'\}$ with $P_{T^*}^{\text{Dir}}(G) \geq 1 - \delta'$. For the consequence, $f(w^*) := P_{T^*, w^*}^\infty(G^c)$ has $\int f d\text{Dir}_{T^*} = P_{T^*}^{\text{Dir}}(G^c) \leq \delta'$, since $P_{T^*}^{\text{Dir}}$ on infinite sequences is the mixture $\int P_{T^*, w}^\infty \text{Dir}_{T^*}(dw)$; apply Markov's inequality.

Pool version (experiments Prop X8(a), (b); no escalation). Let F be the countable family, $C := \sum_{T \in F} 2^{-\text{bits}(T)} \leq 1$ (Kraft for the experiments' code, Remark A.1), $M_T := P_T^{\text{Dir}}$, and $Z_n := \sum_{T \in F} (2^{-\text{bits}(T)}/C) M_T(D_n)/M_{T^*}(D_n)$, a nonnegative supermartingale under M_{T^*} with $Z_0 = 1$ by the computation above. For every $R_n \subseteq F$ with $T^* \in R_n$, however chosen, pathwise

$$\pi_R(T^* \mid D_n) = \frac{2^{-\text{bits}(T^*)} M_{T^*}(D_n)}{\sum_{T \in R_n} 2^{-\text{bits}(T)} M_T(D_n)} \geq \frac{2^{-\text{bits}(T^*)}}{C Z_n} \geq \frac{2^{-\text{bits}(T^*)}}{Z_n}.$$

Off the Ville event this exceeds $2^{-\text{bits}(T^*)}\delta' \geq \delta$. Accepting s with $T^* \not\prec_J s$ needs mass $\geq 1 - \delta$ on theories deriving s , hence $\pi_R(T^* \mid D_n) \leq \delta$. If F is a fixed pool H plus all memorisers whose codes lie outside the template code, $C \leq 1$ may fail; but $C \leq \sum_{T \in H} 2^{-\text{bits}(T)} + 1$, because the memorisers' codes form a prefix-free set, and the argument runs with $2^{-\text{bits}(T^*)}/C$ in place of $2^{-\text{bits}(T^*)}$. $\square$

Proof of Theorem 5.7. $Z'_n := M(X^n)/P_{T^*, w^*}(X^n)$ is a nonnegative supermartingale under the i.i.d. law P_{T^*, w^*} with $Z'_0 = 1$, by the computation above with P_{T^*, w^*} as reference law (M is a consistent probability on sequences). Let $G := \{\sup_n Z'_n < 1/\delta'\}$, so $P(G) \geq 1 - \delta'$. Members of C_d^{Dir} have T^* 's marginal likelihood and pass every constraint, and the other constraints are ≤ 1 , so

$$\pi_t(C_d^{\text{Dir}}) \geq \pi(C_d^{\text{Dir}}) \frac{P_{T^*}^{\text{Dir}}(X^n)}{M(X^n)} = \pi(C_d^{\text{Dir}}) \frac{P_{T^*}^{\text{Dir}}(X^n)/P_{T^*, w^*}(X^n)}{Z'_n} \geq \frac{\pi(C_d^{\text{Dir}}) R_\alpha(n, K)}{Z'_n},$$

with $n = n(t)$, by Lemma D.4(a), which needs the likelihood to be linear in w . On G this exceeds $\pi(C_d^{\text{Dir}}) R_\alpha(n, K) \delta' \geq \delta_n$. No member of C_d^{Dir} derives an invalid q , so its non-deriving mass exceeds δ_n .

Pool version (experiments Prop X8(c)). With F , $C \leq 1$ and M_T as in the pool version of Theorem 5.6, $Z'_n := \sum_{T \in F} (2^{-\text{bits}(T)}/C) M_T(D_n)/P_{T^*, w^*}(D_n)$ is a nonnegative supermartingale under P_{T^*, w^*} with $Z'_0 = 1$, so $P[\sup_n Z'_n \geq 1/\delta'] \leq \delta'$. For every pool $R_n \subseteq F$ with $T^* \in R_n$, pathwise and by Lemma D.4(a),

$$\pi_R(T^* \mid D_n) \geq \frac{2^{-\text{bits}(T^*)} M_{T^*}(D_n)}{C Z'_n P_{T^*, w^*}(D_n)} \geq \frac{2^{-\text{bits}(T^*)} R_\alpha(n, K)}{Z'_n},$$

which off the Ville event exceeds $2^{-\text{bits}(T^*)} R_\alpha(n, K) \delta' \geq \delta_n$. Conclude as in the pool version of Theorem 5.6. $\square$

Example 5.8: details. *Why $T^* \not\prec q$.* In the term model in which $+$ returns its first argument on $(a, 0)$ exactly when a has root 0 or S (and returns the formal term $a + b$ otherwise), every instance of $0 + 0 = 0$ and $(Sz) + 0 = Sz$ holds, but $(0 \cdot 0) + 0$ denotes the term $(0 \cdot 0) + 0 \neq 0 \cdot 0$. *Why $P_{T'} = (1 - \varepsilon)P_{T^*, w^*}$ on the data's support.* A datum is $t + 0 = t$ with $t = 0$ or $t = St'$.

Under T' it has probability $\mathcal{Q}(t)$, i.e. 0.5 or $(0.5 - \varepsilon)\mathcal{Q}(t')$; under P_{T^*, w^*} it has $w_1^* = 0.5/(1 - \varepsilon)$ or $w_2^*\mathcal{Q}(t') = (0.5 - \varepsilon)\mathcal{Q}(t')/(1 - \varepsilon)$. *The decomposition.* T^* 's instance sets are disjoint, so $\ln P_{T^*, w^*}(D) - \ln P_{T^*, \hat{w}}(D) = \sum_f n_f \ln(w_f^*/\hat{w}_f) = -n\text{KL}(\hat{w} \| w^*)$, with $\hat{w}$ the empirical weights. With the previous point,

$$\ln \frac{P_{T'}(D)}{P_{T^*}^{\text{Dir}}(D)} = n \ln(1 - \varepsilon) + \left[\ln P_{T^*, \hat{w}}(D) - \ln P_{T^*}^{\text{Dir}}(D) \right] - n \text{KL}(\hat{w} \| w^*).$$

The bracket is the regret of Lemma D.4(c) at $K = 2$, also Proposition 4.10(b), and $2n\text{KL}(\hat{w} \| w^*) \rightarrow \chi_1^2$ in law. *Not written out:* a version of that χ^2 limit uniform over the window $1 \ll n \ll 1/\varepsilon$, which the crossing argument needs. While $\varepsilon n \ll 1$ the log-odds of T' grow like $\frac{1}{2} \ln n$ and cross $\ln 99$ near $n \approx 8000$. *Moderate ε .* The referee's runs (**r3**, geometric grid, $n \leq 8000$ and $n \leq 80000$) found no acceptance at $\varepsilon = 10^{-3}$ and 10^{-4} , at the fixed w^* (median of the largest log-ratio 2.81 and 3.96, below $\ln 99 = 4.60$) and with w^* from the prior. *The shrinking threshold* of the model track's run is $\delta_n = \pi(T^*) \delta' \frac{1}{e\pi} (n + 1)^{-1}$ with $\delta' = 0.02$, the bound of Lemma D.4(b) at $K = 2$, $\alpha = \frac{1}{2} (1/(e\pi) = 0.1171)$.

implementation	ε	fixed w^* , constant δ	$w^* \sim \text{Dir}(\frac{1}{2}, \frac{1}{2})$, constant δ	fixed w^* , shrinking δ_n
model (c11 , every $n \leq 2 \cdot 10^6$, 300 runs)	10^{-5}	0.997	0.007	0/300
	10^{-6}	1.000	0.003	0/300
experiments (check_fixed_weight , same)	10^{-5}	0.977 (median first n 8500)	0.003	0/300
	10^{-6}	1.000 (median first n 7525)	0.007	0/300
model's referee (r3 , geometric grid of n)	10^{-5}	≥ 0.980	≥ 0.008	–
	10^{-6}	≥ 1.000	≥ 0.010	–

Table 27: Example 5.8: probability that the underivable q is ever accepted; nominal bound $\delta/\pi(T^*) = 0.02$. The referee checked a geometric grid of n , so its numbers are lower bounds. The differences between implementations are Monte Carlo and grid effects.

check_fixed_weight also matched its closed form against the posterior code (ln Bayes factors to $4.1 \cdot 10^{-13}$) and checked Lemma D.4(b) at $K = 2$ for $n \leq 2000$ (margin ≥ 2.1 nats).

E Time penalties: proofs and details

Scripts cited here are in the folders **checks/** (the model track's) and **referee_code/** (its referee's) of **research/tracks/model/**; each is seeded and writes its output next to itself. Sizes are symbol counts in closure-normal form with de Bruijn indices: a quantifier, a connective, a function or relation symbol, a constant, an index and a parameter each count one.

E.1 The two inductors

Proposition E.1 (Why consistent assigners). *[proved]^{model Prop 6.0} Let R be a unary predicate, $w_1, w_2, \dots$ distinct closed terms, and φ_i be $R(w_i)$ or $\neg R(w_i)$ by independent fair coins, all labelled true (a consistent labelling). FI_{all} 's log loss is at most a constant (the program “always accept”); the expected log loss of AI and of FI_{cons} is at least n bits.*

This confirms Hänni's informal separation, and is why the fair comparison is with FI_{cons} .

Proof of Proposition E.1. FI_{all} : the program “always accept” has some length c and labels every φ_i true, so $W_{\text{FI}_{\text{all}}}(D_n) \geq 2^{-c}$; with $W_{\text{FI}_{\text{all}}}(\emptyset) \leq 1$ (Kraft, for a prefix machine) the log loss $-\log_2 W(D_n) + \log_2 W(\emptyset)$ is at most c . AI and FI_{cons} : let D be the data so far and $\psi := R(w_i)$ for the next index. A hypothesis compatible with $D + (\psi, 1)$ and with $D + (\neg\psi, 1)$ would prove, or accept, both ψ and $\neg\psi$, against consistency. So the two sets of compatible hypotheses are disjoint subsets of those compatible with D , and the predictive probabilities $a := W(D + (\psi, 1))/W(D)$ and $a' := W(D + (\neg\psi, 1))/W(D)$ satisfy $a + a' \leq 1$. The coin shows ψ or $\neg\psi$ with probability $\frac{1}{2}$ each, independently of D , so the expected step loss is $\frac{1}{2}(-\log_2 a - \log_2 a') \geq -\log_2 \frac{a+a'}{2} \geq 1$ bit, by convexity of $-\log_2$. Summing over the n steps gives at least n bits. $\square$

Proof of Theorem 6.3. $W_{\text{FI}} \geq 2^{-c'} W_{\text{AI}}$. For a decider p let f_p be the program: on input φ , dovetail searches for proofs of φ and of $\neg\varphi$ from $B \cup A_p$ (both sets are decidable, so the theorems are recursively enumerable), and output acc or rej for the first proof found. Its code is a fixed wrapper followed by p , so $|f_p| \leq |p| + c'$ and $p \mapsto f_p$ is injective. Let p be compatible with D . Then $B \cup A_p$ is consistent, so at most one of $\varphi, \neg\varphi$ is provable, and f_p labels each φ_i with b_i . Also $\Gamma_{f_p} \subseteq \text{Th}(B \cup A_p)$, so $B \cup \Gamma_{f_p}$ is consistent and f_p is a compatible FI_{cons} hypothesis. Summing $2^{-|f_p|} \geq 2^{-c'} 2^{-|p|}$ over compatible p gives the claim.

$W_{\text{AI}} \geq 2^{-c''} W_{\text{FI}}$ (*Craig’s trick*). For an FI_{cons} hypothesis f define A_f^C as in Theorem 6.3. Its decider p_f : given χ , try each of the at most $|\chi|$ decompositions $\chi = \psi^{\wedge m}$; for each, run f for $m - 1$ steps on ψ and check whether it accepts in exactly $m - 1$ steps, and, if $\psi = \neg\varphi$, run f on φ and check whether it rejects in exactly $m - 1$ steps. Every run is bounded, so p_f halts on every input, even when f is partial. Each member $\varphi^{\wedge(k+1)}$ is logically equivalent to $\varphi \in \Gamma_f$ (likewise for $\neg\varphi$), and each member of Γ_f has such a power in A_f^C , with k the halting time. So $\text{Cn}(B \cup A_f^C) = \text{Cn}(B \cup \Gamma_f)$, which is consistent, and $B \cup A_f^C$ decides every φ_i as f does. Hence p_f is compatible whenever f is. Its code is a fixed wrapper followed by f : $|p_f| \leq |f| + c''$, and $f \mapsto p_f$ is injective.

Losses. Put $c := \max(c', c'')$. The semimeasure log loss of an inducer on D_n is $-\log_2 W(D_n) + \log_2 W(\emptyset)$; each of the two terms differs by at most c between AI and FI_{cons} , and both depend on D_n only as a set. If one weight is 0, so is the other, and both losses are infinite. $\square$

Remark E.2 (The convention, and “what is a sequence”). [proved]^{model §6.1} Renormalising at each step, $P(b \mid \varphi, D) := W(D + (\varphi, b)) / \sum_{b'} W(D + (\varphi, b'))$, the argument of Theorem 6.3 gives only $2c$ bits *per step*. The constant equivalence is proved only in the semimeasure convention, which answers Hänni’s “wtf is a sequence here?” (any reveal order, any labelling). His “same spec length as the axiom system plus a const” holds in both directions, and B needs no arithmetic. His route through the schema does need arithmetic; it works with two sorts, or in one sort for sound assigners, and Proposition 6.5 shows that it fails for some consistent unsound ones (Section 6.2).

Per-step renormalisation. With $P_X(b \mid \varphi, D) := W_X(D + (\varphi, b)) / (W_X(D + (\varphi, 0)) + W_X(D + (\varphi, 1)))$, the numerator and the denominator each change by a factor in $[2^{-c}, 2^c]$ between the inducers, so each step’s ratio lies in $[2^{-2c}, 2^{2c}]$; over n steps the bound is $2cn$ bits, and the argument gives nothing better.

E.2 Hänni’s schema

Proof of Proposition 6.4. Consistency. Take the structure whose arithmetic sort is the standard $\mathbb{N}$ and whose L -sort is a model M of Γ_f . In $\mathbb{N}$, $\text{Accept}_f(\ulcorner \varphi \urcorner)$ is true iff f accepts φ , and then $\varphi \in \Gamma_f$ holds in M ; likewise for rejections. So every member of A_f holds, and so does $\mathbf{Q}$. $\supseteq$. If

$f(\varphi) = \text{acc}$, then $\text{Accept}_f(\ulcorner \varphi \urcorner)$ is a true Σ_1 sentence, hence provable in $\mathbf{Q}$ (Σ_1 -completeness for $<$ -free Σ_1 sentences, which applies as Accept_f is written without $<$; known, e.g. Boolos et al., 2007, chapter on $\mathbf{Q}$, not checked); MP with its member of A_f gives φ . Rejections are similar. So $\Gamma_f \subseteq \text{Cn}(B \cup A_f)$. $\subseteq$. An L -sentence provable from $B \cup A_f$ holds in every structure $\mathbb{N} \oplus M$ with $M \models \Gamma_f$; its truth there depends only on M . So it holds in every model of Γ_f , and by completeness it lies in $\text{Cn}(\Gamma_f)$. $\square$

Proof of Proposition 6.5. Γ_f . Since $\text{Con}(\text{PA})$ holds, the search on $\ulcorner \text{Con}(\text{PA}) \urcorner$ never ends, so $\Gamma_f = \{\neg \text{Con}(\text{PA})\}$. $\text{PA} + \neg \text{Con}(\text{PA})$ is consistent by Gödel's second incompleteness theorem. $\neg \text{Con}(\text{PA})$. $\text{Accept}_f(\ulcorner \neg \text{Con}(\text{PA}) \urcorner)$ is a true Σ_1 sentence, so PA proves it; with its member of A_f , $\text{PA} \cup A_f \vdash \neg \text{Con}(\text{PA})$. $\text{Con}(\text{PA})$. $\text{PA} \vdash \neg \text{Con}(\text{PA}) \rightarrow \text{Accept}_f(\ulcorner \text{Con}(\text{PA}) \urcorner)$: if a proof of $\perp$ exists there is a least one, and PA proves that the search halts on it with output acc . This formalisation step is standard and already provable in IS_1 . With the member $\text{Accept}_f(\ulcorner \text{Con}(\text{PA}) \urcorner) \rightarrow \text{Con}(\text{PA})$ of A_f , $\text{PA} \cup A_f \vdash \neg \text{Con}(\text{PA}) \rightarrow \text{Con}(\text{PA})$, so $\text{PA} \cup A_f \vdash \text{Con}(\text{PA})$. Both together are a contradiction. $\square$

E.3 Templates

Proof of Proposition 6.6. Let $\tau \in \text{DT}^\circ$ with $\text{inst}(\tau) \subseteq C_f$, and suppose τ has a metavariable M with an occurrence at position q . Constant bodies exist for every type with two different root symbols: $\lambda \bar{z}.0$ and $\lambda \bar{z}.(0 + 0)$ for terms, $\lambda \bar{z}.(0 = 0)$ and $\lambda \bar{z}.\neg(0 = 0)$ for formulas. By the preservation lemma (AS Lemma 2.4), for a constant body the subtree of $\tau\theta$ at q is that constant, whatever the occurrence's arguments, and the rigid symbols of τ above q are the same in every instance. Every member of C_f has the form $\text{Accept}_f(N) \rightarrow \varphi$, where N is the numeral $\ulcorner \varphi \urcorner$ substituted for the free variable of the fixed formula $\text{Accept}_f(x)$. Since $\text{inst}(\tau) \neq \emptyset$, the path from the root to q is a path of such a member.

Case 1: q is the root or lies in the antecedent. (i) If q lies inside an occurrence of the numeral N (its root included), every member of C_f carries S or 0 at q . Put $\theta(M) := \lambda \bar{z}.(0 + 0)$ and any bodies elsewhere. Then $\tau\theta$ has $+$ at q . If $\tau\theta$ were in C_f , q would again lie inside a numeral, since the path to q is the same; so $\tau\theta \notin C_f$. (ii) Otherwise q is the root or a position of the fixed skeleton of $\text{Accept}_f(x)$. All members of C_f carry the same symbol there ($\rightarrow$ at the root). Choose the constant body of the right type whose root symbol differs from it; then $\tau\theta \notin C_f$. Both contradict $\text{inst}(\tau) \subseteq C_f$.

Case 2: no metavariable occurs at the root or in the antecedent. Then the antecedent of every instance is one fixed sentence $\text{Accept}_f(N_0)$, and the only member of C_f with that antecedent is $\text{Accept}_f(\ulcorner \varphi_0 \urcorner) \rightarrow \varphi_0$ for the single φ_0 with $\ulcorner \varphi_0 \urcorner = N_0$. So τ has at most one instance. But a template with a metavariable has infinitely many instances: each type has infinitely many bodies, and $\theta \mapsto \tau\theta$ is injective (AS Thm 2.5(a)). Contradiction.

So τ is ground. A ground template has one instance; C_f is infinite, so no finite set of DT° templates has instance union C_f , and such a set with instances in C_f covers at most as many members as it has templates.

Reflection schema (pa Prop 4.3). With $\text{Prv}_T(x)$ in place of $\text{Accept}_f(x)$ the members are $\text{Prv}_T(\ulcorner \varphi \urcorner) \rightarrow \varphi$, with unary numerals, and the proof goes through word for word. $\square$

Implicit in the notes. Case 1(i) uses that “inside the numeral” is determined by the path to q , which is rigid in τ ; we have made this explicit. The notes, and the statement, treat the acceptance part C_f . The rejection part, and templates whose instances mix the two parts, are not treated: at a skeleton position where Accept_f and Reject_f carry different symbols, Case 1(ii) would need a constant body whose root differs from both (Appendix H.5).

Proof of Proposition 6.7. (a) Let $f(\varphi) = \text{acc}$ with $\varphi \in \Sigma_n$. Then $\text{PA} \vdash \text{Accept}_f(\ulcorner \varphi \urcorner)$ (a true Σ_1 sentence) and $\text{PA} \vdash \text{Sent}_{\Sigma_n}(\ulcorner \varphi \urcorner)$ (a true Δ_1 fact). So $\text{PA} + \rho_{f,n} \vdash \text{Tr}_n(\ulcorner \varphi \urcorner)$, and $\text{PA} \vdash \text{Tr}_n(\ulcorner \varphi \urcorner) \rightarrow \varphi$. For rejections, $\text{PA} + \rho_{f,n} \vdash \neg \text{Tr}_n(\ulcorner \varphi \urcorner)$ and $\text{PA} \vdash \varphi \rightarrow \text{Tr}_n(\ulcorner \varphi \urcorner)$, so $\neg \varphi$ follows. (b) In $\mathbb{N}$, Accept_f and Reject_f hold exactly of f 's actual decisions, and Tr_n is Σ_n -truth, so both conjuncts of $\rho_{f,n}$ are true when f is Σ_n -sound. *Length.* f enters $\rho_{f,n}$ only through its index, written as a binary numeral of length $O(|f|)$ inside Accept_f and Reject_f ; the rest of the sentence depends on n only. $\square$

E.4 Time penalties

Proof of Proposition 6.8. (a) Parsing χ as an implication with antecedent $\text{Accept}_f(t)$ or $\text{Reject}_f(t)$ is a comparison with a fixed formula, linear in $|\chi|$. Computing the Gödel number of the consequent (or of the formula under its negation) is polynomial in $|\chi|$, and so is writing it as a binary numeral and comparing it with t . With unary numerals, t itself has length $\ulcorner \varphi \urcorner \leq |\chi|$, so writing out the numeral is again polynomial in $|\chi|$. No step runs f . (b) $\chi = \psi^{\wedge m}$ forces $m \leq |\chi|$, and there are at most $|\chi|$ candidate decompositions. For each, f is simulated for $m - 1 < |\chi|$ steps on ψ (and on φ if $\psi = \neg \varphi$), in time polynomial in $|\chi|$ for fixed f . $\square$

Proof of Theorem 6.9. Algorithm. Given w of length m , compute φ_w and the time-constructible bound $\ell(m)$, guess a K-derivation of symbol size at most $\ell(m)$, and check it: each line is an instance of a logical axiom (A1–A3, A5 and the equality axioms by matching; A4 by first-order matching, which is polynomial), a member of T (time $O(\ell^e)$), or follows by MP or Gen. Accept iff the derivation is valid and ends in φ_w . All checks are polynomial in $\ell(m)$, of a degree that depends only on e and on the degree d_K of K's line checks; so the machine runs in time $O(\ell(m)^{c_e})$. *Completeness.* If $w \in X$, such a derivation exists by hypothesis. *Soundness.* If $w \notin X$, then $\neg \varphi_w \in \Gamma_{f_X}$, so $T \vdash \varphi_w$ would make $T \cup \Gamma_{f_X}$ inconsistent. No derivation exists, and the machine rejects. $\square$

Proof of Corollary 6.10. Uniformity. The constant c_e of Theorem 6.9 depends on T only through e . Membership in a finite DT° theory is matching against each template, in time $O(|T| \cdot |\chi|)$ (AS Thm 2.5), so $e = 1$ for all of them, with a T -dependent constant factor that does not change the class. *The hierarchy theorem* (known; exact form not checked against Cook, 1972; Seiferas et al., 1978; Žák, 1983): if t_2 is time-constructible and $t_1(m+1) = o(t_2(m))$, then $\text{NTIME}(t_2) \setminus \text{NTIME}(t_1) \neq \emptyset$. (a) Apply it with $t_1 := s^{c_1}$ and $t_2 := s^{c_1+1}$ to get a decidable $X \in \text{NTIME}(s^{c_1+1}) \setminus \text{NTIME}(s^{c_1})$. Suppose some finite DT° theory T with $T \cup \Gamma_{f_X}$ consistent had, for all but finitely many m , a derivation of size at most $s(m)$ of every φ_w with $w \in X$, $|w| = m$. Put the finitely many words of the exceptional lengths in a table. The machine of Theorem 6.9 with $\ell = s$, plus the table, decides X in $\text{NTIME}(s^{c_1})$: a contradiction. (b) The same with c_e in place of c_1 , for each e . *A single X for all e* (proof sketch). Take $X \in \text{NTIME}(s^{\log_2 s}) \setminus \text{NTIME}(s^{(\log_2 s)/2})$, when these bounds are time-constructible and satisfy the gap condition. Every s^{c_e} is eventually below $s^{(\log_2 s)/2}$, so the argument of (a) applies to every e . The time-constructibility and gap conditions for these bounds are not checked. $\square$

Remark E.3 (Upper bounds). [proof sketch (A_f^C , two-sorted A_f); not established for $\rho_{f,n}$]^{model §6.4} The collapse constructions through A_f^C and the two-sorted A_f pay at most a polynomial of the assigner's *deterministic* running time k in derivation size: cite $\varphi^{\wedge(k+1)}$ and eliminate $\wedge$; or prove the true Σ_1 sentence $\text{Accept}_f(\ulcorner \varphi \urcorner)$ in $\mathbb{Q}$, in size polynomial in the computation (recalled from

Pudlák, 1998, not checked), then apply MP. For $\rho_{f,n}$ the same route also needs a PA-proof of the Tarski biconditional for φ , whose size is not bounded here.

Upper bounds (Remark E.3); proof sketch. From A_f^C , a derivation of φ cites $\varphi^{\wedge(k+1)}$, $k = \text{time}_f(\varphi)$, and applies $\wedge$ -elimination, in size polynomial in $k|\varphi|$. From A_f in two sorts, a derivation is a Q-proof of the true Σ_1 sentence $\text{Accept}_f(\ulcorner \varphi \urcorner)$ followed by MP; the size of such a Q-proof is polynomial in the length of the computation (recalled from Pudlák, 1998, not checked). From $\rho_{f,n}$, the same plus a PA-proof of the Tarski biconditional $\text{Tr}_n(\ulcorner \varphi \urcorner) \rightarrow \varphi$, whose size is not bounded in the notes.

E.5 What the derivation likelihoods charge

The counterexample to the pre-referee claim (Remark 6.11; referee M2). Let $\varphi^{(0)} := \forall x (x = x)$, the logical axiom of reflexivity (size 4). Round $j = 1, \dots, k$: apply $\forall E$ with the term $p + p$ (p the first parameter), then Gen on p ; this doubles the number of occurrences of the bound variable and keeps one $\forall$ at the root. A final $\forall E$ with $p + p$ gives φ_k , an equation whose two sides are balanced sums of 2^{k+1} copies of p ; its size is $2(2^{k+2} - 1) + 1 = 2^{k+3} - 1$. The tree has one citation node; each $\forall E$ contributes its own node and the 5 grammar nodes of $p + p$ (the $+$ node and two parameters, each with a one-node index chain), each Gen its node and a one-node chain: $\nu = 1 + 6 + 8k = 7 + 8k$. In the referee's grammar ($\alpha_{\text{ax}} = \alpha_{\text{lg}} = \alpha_{\text{MP}} = 0.2$, $\alpha_{\text{Gen}} = \alpha_{\forall E} = 0.1$, so $m = 0.6$; seven logical schemas cited uniformly; terms 0:.4, S :.2, $+$:.1, $\cdot$:.1, parameter:.2; parameter index geometric with ratio $\frac{1}{2}$) a round costs $-\ln(0.1 \cdot 0.1 \cdot (0.2 \cdot 0.5)^2) - \ln(0.1 \cdot 0.5) = 12.206$ nats and the citation plus final $\forall E$ 12.77 nats. Every derivation of φ_k contains φ_k , so $\ell_{\min}(\varphi_k) \geq 2^{k+3} - 1$, and $-\ln P_T(\varphi_k) \leq -\ln \Pr(\text{tree})$ because $Z_T \leq 1$. For every $\kappa > 0$ and C the claimed inequality fails for large k ; Table 28 gives the numbers.

One-node example. For $B_m(x) := (x + \dots + x = 0)$ with m summands and $t := S^m 0$, the A4 instance $\forall x B_m(x) \rightarrow B_m(t)$ has size quadratic in m but probability $e^{-O(m)}$: the referee's script gives sizes 145, 1765, 6725 symbols at costs at most 99.4, 369.0, 728.5 nats for $m = 10, 40, 80$ (r2_l1_size.out).

k	$\nu = 7 + 8k$	$ \varphi_k = 2^{k+3} - 1$	$-\ln \Pr(\text{tree})$ (nats)	$\ln \varphi_k / \nu$
0	7	7	12.77	0.278
1	15	15	24.97	0.181
3	31	63	49.38	0.134
10	87	8191	134.83	0.104
14	119	131071	183.65	0.099
30	247	$8.6 \cdot 10^9$	378.95	0.093

Table 28: The counterexample to the pre-referee claim of Remark 6.11. Sizes were built and checked by the referee's script for $k \leq 14$ (r2_l1_size.out; formula beyond) and independently by c12_l1_size.out, Part A, for $k \leq 12$, which also counts $\nu = 7 + 8k$. Costs are the referee's, for the grammar given in the text. $\ln |\varphi_k| / \nu \rightarrow \ln 2/8 \approx 0.087$, below the $1/e \approx 0.368$ that Lemma E.4 allows.

Lemma E.4 (Symbol size against grammar choices). *[proved; computed]^{model Lemma 6.10} Let π be a valid L1 tree from a finite T with ν grammar nodes, $a_T \geq 2$ a bound on the sizes of the templates of T and of K 's logical templates, and $c_T := 2a_T^2$. Every conclusion in π has symbol size at most $(c_T + 1)\nu^2 e^{\nu/e}$.*

Proof of Lemma E.4. For a node v let $G(v)$ be the number of Gen nodes in its subtree and $\Pi(v) := \prod_u |t_u|$ over the $\forall E$ nodes u in its subtree, t_u the substituted term ($\Pi = 1$ if there are none). We show $|\text{concl}(v)| \leq (c_T \nu^2 + G(v)) \Pi(v)$ by induction on the tree. *Citation of a DT° template* (of T , or A1–A3, A5, equality; substitution θ). An occurrence $M(\bar{t})$ becomes $\theta(M)[\bar{t}/\bar{z}]$. The body $\theta(M)$ has at most ν nodes and each argument t_i , a subterm of the template, has size at most a_T , so the piece has size at most νa_T . There are at most a_T occurrences and at most a_T rigid symbols, so $|\tau\theta| \leq a_T + a_T^2 \nu \leq c_T \nu \leq c_T \nu^2$. *Citation of A4* ($\forall x B \rightarrow B[t/x]$, not a template; Lemma A.5). B and t have at most ν nodes each, so the size is at most $1 + (|B| + 1) + |B| |t| \leq 3\nu^2 \leq c_T \nu^2$ (such a tree has $\nu \geq 3$). *$\forall E$ with term t .* The conclusion $\psi[t/x]$ replaces each occurrence of the bound variable by t , which contains no bound indices, so no index shifting is needed: $|\psi[t/x]| \leq |\psi| |t| \leq |\text{premise}| |t|$, and Π gains the factor $|t|$. *Gen.* Replacing a parameter by an index keeps the size and the new $\forall$ adds one; G grows by one and $\Pi \geq 1$. *MP.* The conclusion is a proper subformula of the major premise, and G and Π only grow from a child to its parent. Finally $G(v) \leq \nu$, and the substituted terms of different $\forall E$ nodes are disjoint sets of grammar nodes, so $\sum_u |t_u| \leq \nu$. For k terms, AM–GM gives $\prod_u |t_u| \leq (\nu/k)^k \leq \sup_{y>0} (\nu/y)^y = e^{\nu/e}$. Hence $|\text{concl}(v)| \leq (c_T \nu^2 + \nu) e^{\nu/e} \leq (c_T + 1) \nu^2 e^{\nu/e}$. $\square$

Computed (c12_l1_size.out, Part B). Chains without A4 citations satisfy the stronger bound $(c_T + 1) \nu e^{\nu/e}$ (the citation case gives $c_T \nu$). This is what the script tests: on 9865 conclusions of 3000 random valid $\forall E$ /Gen chains, with terms from a PCFG and cited sentences with up to three quantifiers and repeated variables, neither the inductive bound nor the final bound was violated; the largest $\ln(\text{size})/\nu$ was 0.220. The first version of the lemma had $\nu^{\nu+1}$ in place of $e^{\nu/e}$, and ν in place of ν^2 ; the A4 case (the referee’s one-node example) forced ν^2 (model §12.4).

Proposition E.5 (L1 charges its own code length exponentially). *[proved; computed]^{model Prop 6.11}* Assume L1 with $m < 1$, $\mathcal{Q}$ uniformly subcritical (Definition 2.6), T finite and a geometric parameter law. There are $\zeta, C > 0$, depending on the grammar and on T , with $P_T(\varphi) \leq C e^{-\zeta \nu_{\min}(\varphi)} / Z_T$ for every φ .

Proof of Proposition E.5. The whole tree (derivation nodes, $\mathcal{Q}$ nodes, parameter-chain nodes) is a multi-type branching process with boundedly many children per node, each node’s children drawn independently given its type. Give derivation nodes weight 1, $\mathcal{Q}$ nodes weight $\eta h_{\mathcal{Q}}$ ($h_{\mathcal{Q}}$ from Definition 2.6) and chain nodes weight η . A derivation node has expected weighted offspring at most $m + \eta(\nu_T h_{\mathcal{Q}, \max} + 1)$, with ν_T bounding the number of metavariables of a template of T or of K ; this is below 1 for small η , since $m < 1$. $\mathcal{Q}$ nodes satisfy the condition with $\rho_{\mathcal{Q}}$, and a chain node continues with probability $g_{\text{par}} < 1$. So the uniform condition holds with $\rho_{\text{br}} := \max(m + \eta(\nu_T h_{\mathcal{Q}, \max} + 1), \rho_{\mathcal{Q}}, g_{\text{par}}) < 1$, and Lemma A.2(b),(d) gives $\mathbb{E}[e^{\zeta \nu}] \leq C$ for some $\zeta > 0$. A valid tree concluding φ has $\nu \geq \nu_{\min}(\varphi)$, so $P_T(\varphi) = \mu_T(\varphi) / Z_T \leq \Pr[\nu \geq \nu_{\min}(\varphi)] / Z_T \leq C e^{-\zeta \nu_{\min}(\varphi)} / Z_T$ by Markov’s inequality. $\square$

Computed (c12_l1_size.out, Part C). For the joint process with $m = 0.55$ and the grammar $\mathcal{Q}_A$ of the model appendix, 10^5 simulated trees: mean $\nu = 26.9$; $\Pr(\nu \geq L) = 0.362, 0.137, 0.0288, 0.00238, 3.0 \cdot 10^{-5}$ for $L = 25, 50, 100, 200, 400$; the log-tail falls by 0.0249 nats per node between $L = 100$ and 200.

Theorem E.6 (Code length bounds the assigner’s time, logarithmically). *[proved]^{model Thm 6.12}* In the setting of Theorem 6.9, let T be a finite DT° theory with $T \cup \Gamma_{f_X}$ consistent, and let every φ_w , $w \in X$, have a valid L1 tree from T with at most $\ell(|w|)$ grammar nodes, $\ell(m) \geq m$ time-constructible. Then $X \in \text{NTIME}(2^{(d+1)\ell(m)})$, where d , the polynomial degree of the syntactic operations used in checking, depends neither on T nor on ℓ .

Proof of Theorem E.6. On input w , put $\ell := \ell(|w|)$ and guess a tree with at most ℓ grammar nodes. Each node is a choice from a finite list (rule, template of T , logical schema, production of $\mathcal{Q}$) or one of at most $\ell + O(1)$ variables, so the guess has $O(\ell \ln \ell)$ bits. Compute every node's conclusion explicitly, check validity (MP needs equality of formulas, $\forall E$ a root $\forall$) and compare the root's conclusion with φ_w . By Lemma E.4 every formula has size at most $S := (c_T + 1)\ell^2 e^{\ell/e}$, and all operations are polynomial of a fixed degree d in that size. So the time is at most $\ell \cdot S^d$, which is at most $2^{(d+1)\ell}$ for $\ell \geq \ell_0(T)$, since $e^{d/e} < 2^d$ and the remaining factor is polynomial in ℓ ; smaller inputs go in a table. Completeness and soundness are as in Theorem 6.9: citations name a template of T , so no membership test is needed, and a valid tree is a K-derivation from $\bigcup \text{inst}(T)$ (each $\forall E$ is an A4 instance plus MP). $\square$

Proof of Corollary 6.12. Apply the hierarchy theorem with $t_1 := 2^{(d+1)s}$ and $t_2 := 2^{s^2}$; the condition $t_1(m+1) = o(t_2(m))$ is $s(m)^2 - (d+1)s(m+1) \rightarrow \infty$. This gives a decidable $X \in \text{NTIME}(2^{s^2}) \setminus \text{NTIME}(2^{(d+1)s})$. If, for some T and all but finitely many m , every $w \in X$ of length m had $\nu_{\min}(\varphi_w) \leq s(m)$, Theorem E.6 with $\ell = s$ and a table (for the exceptional lengths and for $\ell < \ell_0(T)$) would put X in $\text{NTIME}(2^{(d+1)s})$. Since d does not depend on T , one X serves every T . The likelihood bound is Proposition E.5 with $\nu_{\min}(\varphi_w) > s(m)$. $\square$

Proof of Proposition 6.13. (a) Summing over valid trees π concluding φ ,

$$\mu_T^\sigma(\varphi) = \sum_{\pi} \Pr(\pi) e^{-\kappa|\pi|} \leq e^{-\kappa \ell_{\min}^\sigma(\varphi)} \sum_{\pi} \Pr(\pi) = e^{-\kappa \ell_{\min}^\sigma(\varphi)} \mu_T(\varphi) \leq e^{-\kappa \ell_{\min}^\sigma(\varphi)} Z_T;$$

divide by Z_T^σ . (b) List the conclusions of all nodes of the tree in post-order (total size ℓ). For each $\forall E$ node v with premise u , insert the A4 instance $\text{concl}(u) \rightarrow \text{concl}(v)$ before v , of size $|\text{concl}(u)| + |\text{concl}(v)| + 1$; the substituted term has no bound variables, so it is free for x . Then $\text{concl}(v)$ follows by MP; Gen nodes are K's Gen, and citations are axioms of K or members of $\bigcup \text{inst}(T)$. Each node is the premise of at most one node and there are at most ℓ nodes, so the insertions total at most 3ℓ and the derivation has size at most 4ℓ . If $\ell_{\min}^\sigma(\varphi_w) \leq s(m)/4$, there would be a K-derivation of size $\leq s(m)$; by Corollary 6.10(a) this fails for some $w \in X$ of length m , for infinitely many m . On those data (a) gives $-\ln P_T^\sigma(\varphi_w) \geq \kappa \ell_{\min}^\sigma(\varphi_w) - \ln(Z_T/Z_T^\sigma) > \kappa s(m)/4 - \ln(Z_T/Z_T^\sigma)$. (c) An explicit K-derivation written with at least one bit per symbol has bit length at least its symbol size, which exceeds $s(m)$ on the same data. So $g(\ell_T(\varphi_w)) = 2^{-\kappa \ell_T(\varphi_w)} < 2^{-\kappa s(m)}$, and after normalisation $-\log_2 P_T^g(\varphi_w) = \kappa \ell_T(\varphi_w) + \log_2 Z_T > \kappa s(m) + \log_2 Z_T$. $\square$

Implicit in the notes. Part (c) needs the explicit derivations of Remark 2.23 to be derivations in K (or derivations of a calculus that translates into K with linear overhead, which changes the constant) and the code to spend at least one bit per written symbol; the notes say “bits, which is at least their symbol count”. We state both assumptions.

The route to Conjecture 6.14, and what is missing. Represent conclusions with sharing (formula DAGs, grammar-compressed terms), so that $\forall E$ need not copy t into every occurrence; for the doubling family above a DAG of size $O(k)$ suffices. That every tree with ν grammar nodes has such representations of size polynomial in ν is part of what has to be shown. MP needs equality of two compressed formulas; equality of compressed strings and of grammar-compressed trees is decidable in polynomial time (recalled by the track, not checked: Plandowski 1994; Busatto, Lohrey and Maneth 2008; Schmidt-Schauß 2005). The missing step is Gen and de Bruijn indices under sharing: abstracting a parameter turns it into an index whose value depends on the binder depth at each occurrence, so a shared subterm can need different indices in different contexts.

With that step, the proof of Theorem E.6 would give $\text{NTIME}(\ell^c)$. Model §10 lists this as problem 12, together with the derivation-penalised equivalence (problem 13), a single-sorted repair of Hänni’s schema for consistent but unsound assigners (problem 14), and for which f the set A_f^C , or a set with its consequences and polynomial membership, is a finite union of DT° templates (problem 15).

Remark E.7 (Links to the other sections). [(i) proved; (ii) and the argument of (iv): proof sketch; the numbers of (iv): computed]^{universal Prop U2t, §8; experiments E7; pa §3.8, F9} (i) No penalty monotone in derivation length or time reverses Theorem 3.4: in $\mathbf{C}_{\min}$ a chain weight non-increasing in the number of eliminations keeps $P_{H_v}^f(s) \leq P_{H_{\text{sch}}}^f(s)$ on every datum $s \neq \forall x\varphi$ (Proposition B.1, U2t). This counts steps, not symbols. (ii) In the equational fragment $\{\text{Q4}, \text{Q5}\}$, a Kt-style penalty on generating $0 + S^k 0 = S^k 0$ charges $O(\ln k)$, less than the derivation charge of Proposition B.15 (at least $k+1$ axiom leaves). (iii) The experiments’ membership-time factor is a small extra simplicity penalty by construction (Remark 6.15). (iv) A time-bounded evaluation of the likelihood sums over the derivations found within the bound, so it lower-bounds $\mu_T(s)$ and favours adopting theorems with long proofs as axioms; a two-part code is the extreme case. Under L1^{sch} with its default axiom prior, memorising already beats deriving at the first occurrence for 11-line proofs (Proposition 7.15, Table 4). (ii) and (iv) agree: a derivation likelihood favours the theory that makes the data cheap to state, not the logically minimal one. A PA comparison under a derivation-time penalty ($\text{I}\Sigma_n$ as Q plus one partial-truth sentence, against the schema) was not quantified.

F Proofs and computations for Section 7

Proofs follow track pa’s final notes (and track experiments’ for Props X9–X11); steps they leave out are named. Track pa’s referee re-checked every derivation of Propositions 7.3, 7.8 and 7.14 and of Remark 7.9 with an independent checker, recomputed the cost tables, the toy tower and the Euler example by independent code, and read the proofs of Propositions 7.11, 7.20, 7.21, 7.23, 7.31, 7.35 and 7.36, Remark F.1 and Proposition 7.24(a), finding them correct for the stated models after the corrections recorded in Appendix H. Added in the revision and therefore not refereed: the proof of Proposition 7.6, Proposition 7.12, Section 7.4 (Propositions 7.15 and 7.17 to 7.19), the full proof of Proposition 7.22, Lemma F.3, Proposition 7.25, Remarks F.4 and F.7, and experiments’ Propositions 7.29 and F.6. They were checked by the tracks and again while writing this appendix. Nothing is checked in a proof assistant.

F.1 The checker and the reproduction

`checks/nd.py` checks natural-deduction derivations line by line. Rules: hyp (open a hypothesis, to be discharged), ax (an instance of an axiom of the theory), tc (tautological consequence of earlier lines, by truth tables over their prime subformulas), $\rightarrow\text{I}$, $\forall\text{E}$ (with a capture check), $\forall\text{I}$ and $\exists\text{E}$ (with eigenvariable conditions), $\exists\text{I}$, refl and subst (equality). A derivation must end with no open hypotheses. Schematic derivations are written in a unary predicate symbol P ; substituting a formula for P , after renaming the derivation’s bound and eigenvariables away from it, maps derivations to derivations (the substitution theorem for predicate symbols; Shoenfield, 1967, location not verified). `test_nd.py` replays the four derivations of Proposition 7.3 on 200 random concrete motives with a parameter (seed 20261008): 200/200 check. Negative tests: the checker rejects all of 10 unsound steps (a $\forall\text{I}$ eigenvariable free in a hypothesis, a tc step that is not a tautology, an $\exists\text{E}$ eigenvariable that is not fresh in the conclusion or in a hypothesis, a

mismatched subst, capture in $\forall E$, a wrong goal, an open hypothesis, tc through a quantifier, a wrong $\exists I$ witness). The referee’s independent re-checker (own substitution, α -normal form, truth tables, rule semantics and statements of all axioms and schemas) accepts all 20 derivations of the first version and rejects all 173 corrupted variants (a negated conclusion on some line, or a bad eigenvariable).

A script matched every number quoted in track pa’s notes against the outputs (87 quotes, 0 failures; three planted errors were caught). All twelve outputs were byte-identical on a second run. Seeds: c1 7, c2 `random.Random(n)` per n (as in u7), c3 1, c4 11 and 12, c5 5 and 6, c6 41 and 42, c8 31, 33, 11–13.

F.2 Equivalent axiomatisations

Proof of Proposition 7.3. Write “ φ is progressive” for $\forall x(\theta_\varphi \rightarrow \varphi)$. (A) Assume $\varphi(0)$ and $\forall x(\varphi \rightarrow \varphi(Sx))$; we show progressiveness. Let $\theta_\varphi(x)$. By Q3, $x = 0$, and then $\varphi(x)$; or $x = Sw$. In the second case $w + S0 = S(w + 0) = Sw$ by Q5 and Q4, so $w < x$ by Dlt; hence $\varphi(w)$, hence $\varphi(Sw)$, i.e. $\varphi(x)$. CVI(φ) gives $\forall x\varphi$. (B) Assume φ progressive and apply Ind to θ_φ . Base: $y < 0$ gives $y + Sz = 0$ for some z (Dlt), so $S(y + z) = 0$ by Q5, contradicting Q1; so $\theta_\varphi(0)$. Step: assume $\theta_\varphi(x)$ and $y < Sx$. Then $y + Sz = Sx$ for some z , so $S(y + z) = Sx$ (Q5) and $y + z = x$ (Q2). By Q3, $z = 0$ or $z = Sv$. If $z = 0$, then $y = x$ (Q4) and $\varphi(x)$ follows from $\theta_\varphi(x)$ by progressiveness. If $z = Sv$, then $y + Sv = x$, so $y < x$ and $\varphi(y)$ by $\theta_\varphi(x)$. So $\theta_\varphi(Sx)$, and $\forall x\theta_\varphi(x)$; progressiveness gives $\forall x\varphi$. (C1) If LNP(φ) fails, $\exists x\varphi$ holds and every x with $\varphi(x)$ has some $y < x$ with $\varphi(y)$. Then $\neg\varphi$ is progressive, so CVI($\neg\varphi$) gives $\forall x\neg\varphi$, a contradiction. (C2) Let φ be progressive and suppose $\exists x\neg\varphi$. LNP($\neg\varphi$) gives an x with $\neg\varphi(x)$ and $\forall y < x\varphi(y)$; progressiveness gives $\varphi(x)$, a contradiction. The base axioms cited by the checked derivations are exactly those named: (A) CVI, Dlt, Q3–Q5; (B) Dlt, Ind, Q1–Q5; (C1), (C2) only the schema. $\square$

For restricted motive classes the classical equivalences ($I\Sigma_n \Leftrightarrow III_n \Leftrightarrow L\Sigma_n \Leftrightarrow L\Pi_n$; Kaye, 1991; Hájek and Pudlák, 1993, theorem numbers not checked) need bounded collection, because θ_φ adds a bounded quantifier, and some fail for parameter-free schemes (Kaye et al., 1988); DT° templates cannot say “motive in Σ_n ”, so only the full-schema equivalence matters here.

Proof of Proposition 7.5. $CL(PA_S; D_n) = \text{prior}_S + \sum_{i \leq n} (c_S(f_i) + L_Q(\varphi_i))$. The prior is a constant, and the L_Q terms are the same under every S . The differences $c_S(f_i) - c_{S'}(f_i)$ are i.i.d. and bounded, so the strong law gives the limit. The posterior odds are $2^{-\Delta^{CL}}$, which tend to 0 or ∞ exponentially when the limit is nonzero. $\square$

Proof of Proposition 7.6. For CVI(P); Ind(P) is the same with its 11-symbol antecedent (Ind(P) fails in the structure $\{0, a\}$ with $S0 = 0$, $Sa = a$, $P = \{0\}$, and it is not an instance of CVI, whose antecedent has root $\forall$). Both theories have 9 axioms, so a citation costs $\log_2 10 + \log_2 9 + 2\beta$ (it writes $P(x)$). The term positions of both forms lie under binders or are 0, so no $\forall E$ step can have produced them from a shorter variable without capture; formulas on the chain below have full size. Let π derive CVI(P) from PA_{Ind} . 1. π cites an axiom. CVI(P) is not valid: read $<$ as the full relation and P as empty; then $\forall y(y < x \rightarrow P(y))$ fails for every x (take $y = x$), so the antecedent holds, while $\forall xP(x)$ fails. So some line is a citation, costing at least $\log_2 10 + \log_2 9$. 2. Another line writes at least 13 symbols. Follow the premises back from the last line through $\forall E$ (premise $\forall v A$ with $A[t/v]$ the current formula), $\forall I$ (premise the matrix) and $\exists E$ (premise with the same conclusion). Every formula on this chain is CVI(P) with a quantifier prefix and terms in place of variables, which keeps the connective skeleton. The chain ends at a line of another

rule. It is not ax: the Q axioms and Dlt do not contain P ; an Ind instance is an implication whose antecedent is a conjunction, while every chain formula, prefix removed, has an antecedent with root $\forall$; and no instance has a $\forall$ -prefix. It is not refl or $\exists$ I (wrong root). A hyp, tc or subst line writes a formula with $\text{CVI}(P)$'s skeleton, at least 18 symbols (13 skeleton symbols and one at each of 5 term positions); a hyp line still writes it, even if discharged later. A $\rightarrow$ I line writes its discharged antecedent, which has the skeleton of $\text{CVI}(P)$'s antecedent: at least 13 symbols. So some non-citation line costs at least $\log_2 10 + 13\beta$. 3. Premise references cost ≥ 0 . So $\text{bits}(\pi) \geq 2\log_2 10 + \log_2 9 + 13\beta$, which exceeds the citation by $\log_2 10 + 11\beta$. $\square$

Proposition 7.8. Each law is proved by one induction: in T_{left} , $x + 0 = x$ on x ($0 + 0 = 0$; $Sx + 0 = S(x + 0) = Sx$) and $\forall y(x + Sy = S(x + y))$ on x ; in T_{right} , $0 + x = x$ on x and $Sp + x = S(p + x)$ on x with parameter p . The four derivations (11, 22, 11 and 21 lines) are checked; they cite only T_{Ind} and the other convention's two laws.

derivation (checked)	lines	symbols	writes of P	overhead	template prior
Sep from $\text{Repl}_J(\lambda xy. \varphi(x) \wedge y = x)$	36	207	19	1209.3	72 (Sep)
Repl_{sp} from Coll	15	108	12	574.1	185 (Repl_{sp})
Repl_J from Coll and two Sep instances	43	237	21	1432.0	167 (Repl_J)
Found from $\text{EInd}(\neg x \in S)$	21	138	0	758.1	109 (Found)

Table 29: ZF: overheads in bits per use ($\text{L1}^{\text{sch}}, \beta = \log_2 23$). Sep is $\forall X \exists Y \forall u(u \in Y \leftrightarrow u \in X \wedge \varphi)$; Repl_J is the image form $\forall x \forall y \forall z(\psi \wedge \psi[z/y] \rightarrow y = z) \rightarrow \forall X \exists Y \forall y(y \in Y \leftrightarrow \exists x(x \in X \wedge \psi))$; Repl_{sp} the bounding form with $\exists!$ spelled out; Coll, EInd, Found as in AS Table 6. The checker caught a capture error in a first version of the Sep constructor. Computed; upper bounds (pa §1.4).

Further numbers for Remark 7.7. On a 0.1 grid of $(p_{\text{Ind}}, p_{\text{CVI}})$, PA_{Ind} is the best minimal theory for $p_{\text{Ind}} \geq 0.7$, PA_{CVI} in a broad middle region and PA_{LNP} for large p_{LNP} ; PA_{all} , if allowed, wins every mixed cell. It pays $81 + 86$ prior bits and $\log_2(11/9) \approx 0.29$ bits per citation more than PA_{Ind} , and is ahead in expectation after 12, 2, 1, 8 schema uses for $(p_{\text{CVI}}, p_{\text{LNP}}) = (0.01, 0)$, $(0.05, 0)$, $(0.10, 0.05)$, $(0, 0.01)$; one CVI use alone saves 1468 bits. Under L1^{naive} the overhead grows by about $\beta(K - 1)$ bits per motive symbol, K the number of written copies of P : CVI from PA_{Ind} ($K = 14$) costs 1621, 1915, 2515, 3832 bits at $|\varphi| = 5, 10, 20, 40$ (schematic: 1468), and Ind from PA_{CVI} ($K = 18$) 1088, 1487, 2264, 4030 (schematic: 880); mean of 5 random motives per size, seed 7.

F.3 The MDL finding

Proof of Proposition 7.11. For an induction datum $\text{Ind}(\varphi)$ whose motive has root f , $P_{H_F} = w\mathcal{Q}(f)\mathcal{Q}(\varphi | f) = w\mathcal{Q}(\varphi) = P_{T^*}$; ground data have the same weight under both. $\square$

Proof of Proposition 7.12. (a) The total code is the prior, plus the KT code of the citation index, plus a sum over contexts of the KT code of the symbols coded there. Sequential KT in a context equals the Dirichlet($\frac{1}{2}$) marginal of that context's counts, whatever their order (known; it is the Pólya-urn form of the Dirichlet-multinomial; Krichevsky and Trofimov, 1981). By hypothesis the two hypotheses have the same count vectors in every context

outside C , so those terms cancel; H' has no counts in C . The index and prior terms remain. (b) $\text{KT}(k; A) \ln 2 = \ln \Gamma(N + \frac{A}{2}) - \ln \Gamma(\frac{A}{2}) - \sum_{a:k_a > 0} [\ln \Gamma(k_a + \frac{1}{2}) - \ln \Gamma(\frac{1}{2})]$. Stirling, $\ln \Gamma(m+a) = (m+a-\frac{1}{2}) \ln m - m + \frac{1}{2} \ln 2\pi + o(1)$, applied to the $j+1$ growing arguments (j the number of nonzero counts) gives $N \ln N - \sum_a k_a \ln k_a + \frac{A-1}{2} \ln N + \frac{1-j}{2} \ln 2\pi + j \ln \Gamma(\frac{1}{2}) - \ln \Gamma(\frac{A}{2}) + o(1)$; the first two terms are $N\hat{H}(k)$ in nats. (c) Let τ_0 be the template of H whose data visit c . H' 's counts are those of H for the other templates and $m_{c,a}$ for the templates that correspond to (τ_0, a) . Hence $n\hat{H}(n') = n\hat{H}(n) + m_c\hat{H}(m_c)$ (chain rule for empirical entropies), and the entropy terms of (a) cancel exactly. The $\log_2$ terms give $\frac{|H'|-1}{2} - \frac{|H|-1}{2} - \frac{|A_c|-1}{2}$ times $\log_2 n$, since m_c is linear in n ; the rest is $O(1)$. For a split of T_{Ind} into the roots F , $|H'| - |H| = |F| - 1$ and $|A_c| = 9$. $\square$

Corrections and reproduction. The first version, like u7 itself, charged prior bits only for templates that some datum uses, while the KT index ranged over all seven (referee m1). Under G1 nothing changes; under G2 (no $\vee, \forall, \exists$ at the root) 355 prior bits were missing and under G3 470. Every template is now charged; H_{used} is AS's split H_F with the index restricted to F . With u7's own bookkeeping the script reproduces AS Table 12 exactly at $n = 1000$ (G1: $-223, +2325, +2701$; G2: $-915, +1141, +1428$); the G2 rows of AS Table 12 carry the same undercount, which changes no conclusion there. A first SDPC run ($+803$ to $+12270$ bits) was an encoding artefact and is refuted.

law, split	n	NAIVE	PC	DPC	SDPC	RDPC	CF
G1, H_{root}	1000	-223.0	+2325.3	+2700.6	+770.0	+1676.6	+709.8
	16000	-16466.1	-1198.5	-3035.6	+765.8	+3724.6	-301.8
	256000	-274655.9	-94088.5	-163027.5	+761.9	+5985.9	-17241.7
G2, H_{used}	1000	-926.0	+1129.8	+1417.1	+403.7	+870.1	+236.5
	16000	-21484.6	+1035.8	+2805.3	+393.7	+1566.1	-2341.2
	256000	-350773.1	-12391.6	+4289.5	+383.7	+2346.1	-45211.8
G3, H_{used}	1000	-1300.8	+772.5	+543.0	+284.6	+931.3	+270.0
	16000	-24985.6	-3058.9	-9258.6	+272.8	+1736.6	-246.7
	256000	-404445.7	-75061.9	-186306.5	+260.8	+2564.9	-7814.9

Table 30: $\text{CL}(\text{split}) - \text{CL}(T^*)$ in bits ($T^* = \mathbf{Q} + T_{\text{Ind}}$; positive: T_{Ind} stays whole). Prior parts $+780$ (G1), $+425$ (G2), $+310$ (G3). SDPC for H_{root} under G2 and G3: $+769.9 \rightarrow +761.8$ and $+769.7 \rightarrow +761.9$. CF on G1 at $n = 256000$: -0.067 bits per datum. Computed (pa §2.2; c2_md1.out, c2b_cf.out).

Proof of Proposition 7.14. (a) Each wrapper $w_f(P)$ ($\neg\neg P, P \wedge P, P \vee P, (0 = 0) \rightarrow P, \forall zP, \exists zP$, with z not free in P) is logically equivalent to P and has root f , so $\text{Ind}(w_f(P)) \in \text{inst}(T_f)$, and $\text{Ind}(P)$ follows by Lemma F.3 (a special case of Proposition 7.25(a)). The six checked derivations have 17–19 lines; overheads over citing $\text{Ind}(P)$: $\neg 409.9, \wedge 441.6, \vee 441.6, \rightarrow 488.9, \forall 369.2, \exists 425.6$ bits. No wrapper exists for the root $=$. (b) The instances of $T_{=}$ are induction for equations, so $\mathbf{Q} + T_{=} \subseteq \text{IOpen}$; IOpen does not prove that $\sqrt{2}$ is irrational (Shepherdson, 1964), and PA does. $\square$

F.4 Theorem data

Proofs of Propositions 7.15 and 7.17. Proposition 7.15: adding s adds $\beta|s|$ prior bits; each occurrence of s costs ℓ_{cite} instead of $D_T(s)$; the other data change only through the index. Proposition 7.17: let $Z := \sum_s 2^{-\ell(s)} \leq 1$ and $q := 2^{-\ell}/Z$. Then $\mathbb{E}_P[\ell + \log_2 P] = \sum_s P(s) \log_2 \frac{P(s)}{Zq(s)} =$

$\text{KL}(P\|q) - \log_2 Z \geq 0$, with equality iff $P = q$ and $Z = 1$. Adopting a new datum costs at least $\ell(s)$ prior bits, so memorisation can gain only on repeated data. $\square$

Proof of Proposition 7.18. $\text{Th}(\mathbf{Q} \cup S) \subseteq \text{Th}(\mathbf{PA})$, and equality would make $\mathbf{PA}$ finitely axiomatisable, which it is not (Ryll-Nardzewski, 1952). So some $\sigma \in \text{Th}(\mathbf{PA}) \setminus \text{Th}(\mathbf{Q} \cup S)$ has $P_{\mathbf{PA}}(\sigma) > 0$ (Lemma 2.10, parameters admissible) and $P_{\mathbf{Q} \cup S}(\sigma) = 0$; it occurs a.s. after finitely many data. There are countably many such S . The second claim is Theorem 4.2. For $\mathbf{PA} \cup S$: for generic weights its law differs from $P_{\mathbf{PA}}$, so it loses by Theorem 4.2; that the decay is polynomial, as for a spare slot (Proposition 4.14), is a proof sketch, and whether some special weight gives $P_{\mathbf{PA} \cup S} = P_{\mathbf{PA}}$ was not checked. $\square$

Proof of Proposition 7.19. (a) is the definition of the score. (b) and (c) compare prior bits among score-1 theories: F_0 has every sentence as an instance, and an inconsistent theory derives every sentence, so either has score 1 unless refuted within d . The example in (b): $\beta(7 + 13) = 31.7 + 58.8 = 90.5 > 72$ bits. $\square$

The theorems in other axiomatisations. The six theorems of Table 4 cost 1148.4–4163.7 bits more in $\mathbf{PA}_{\text{CVI}}$ and 2241.2–8117.4 bits more in $\mathbf{PA}_{\text{LNP}}$ than in $\mathbf{Q} + T_{\text{Ind}}$, with $\beta^* = 143$ –501 and 240–861 bits per symbol (upper bounds: Ind is re-derived at each use; a direct $\mathbf{PA}_{\text{CVI}}$ derivation of $\forall x 0 + x = x$ by the referee costs 652 bits more instead of 1161). The four recursion-convention theorems of Proposition 7.8 give $\beta^* = 30.5$ –45.6. Decodable-code costs are 3.6–7.5% higher. All 19 derivations are checked.

Compressible data and streams (Remark 7.16). Per datum $f(\varphi)$ (φ a random motive, seed 41) the options cost: derive in $\mathbf{PA}_{\text{Ind}}$, memorise ($\beta|f(\varphi)| + \ell_{\text{cite}}$), or adopt $f(P)$ ($\ell_{\text{cite}} + \beta|\varphi|$, plus $\beta|f(P)|$ once). Ind, $|\varphi| = 21$: 101.5 (a citation) against 431.7. CVI, $|\varphi| = 20$: derive 1565.0, memorise 332.2, adopt 97.0 (+81); $|\varphi| = 418$: 3365.4, 5733.3, 1897.3. LNP, $|\varphi| = 21$: 2335.5, 350.3, 101.5 (+86). Streams (seed 42; bits relative to $\mathbf{Q} + T_{\text{Ind}}$, $n = 1000$): $u = 0$: $\mathbf{Q} + T_{\text{Ind}}$ plus library -561857 , $\mathbf{Q}$ plus library -562036 (MAP); $u = 0.1$: -513380 (MAP) against -489898 ; $u = 0.5$: -285950 (MAP) against -147800 .

F.5 Data from $\text{Th}(\mathbb{N})$, the chain and narrow practice

Proof of Proposition 7.20. If T proves a false φ , then $\neg\varphi \in \text{Th}(\mathbb{N})$ and $T \not\vdash \neg\varphi$ by consistency. Otherwise T 's theorems are true, and since they are r.e. while $\text{Th}(\mathbb{N})$ is not, some true σ has $T \not\vdash \sigma$. In both cases some s has $\mu(s) > 0 = P_T(s)$, and the probability that it is absent from n draws is $(1 - \mu(s))^n \rightarrow 0$. $\square$

Proof of Proposition 7.21. An inconsistent theory derives every sentence, so its 0/1 likelihood is never 0 without a rule. Under $\mathbf{R}_\infty$ it derives the negation of the first datum. Under $\mathbf{R}_{d_n}$ its shortest refutation has some finite size d , and it is refuted once $d_n \geq d$. Consistency of r.e. theories is Π_1 -complete (known), so no computable rule refutes exactly the inconsistent theories. (AS Thm 6.15 is a related but different statement about DTRC-type accept sets.) $\square$

Proof of Proposition 7.22 (adapted from the proof of Proposition 6.6). Suppose τ has a metavariable M with an occurrence at position q . Constant bodies of each sort with two different root symbols exist: $\lambda\bar{z}.0$ and $\lambda\bar{z}.(0 + 0)$ for terms, $\lambda\bar{z}.(0 = 0)$ and $\lambda\bar{z}.\neg(0 = 0)$ for formulas; for

a constant body the subtree of $\tau\theta$ at q is that constant (AS Lemma 2.4), and the symbols of τ above q are kept. Give every other metavariable any body. *Case 1: q is the root or lies in the antecedent.* If q lies inside the numeral argument of Prv_T , take $\lambda\bar{z}.(0+0)$: the argument is then not a numeral, so $\tau\theta \notin \text{Refl}_T$. Otherwise q is the root or a position of Prv_T 's fixed skeleton, where all members of Refl_T carry the same symbol; choose the constant body with a different root symbol. *Case 2: no metavariable occurs at the root or in the antecedent.* Then the antecedent is a fixed $\text{Prv}_T(N_0)$, and every member of Refl_T with this antecedent is $\text{Prv}_T(\ulcorner\varphi_0\urcorner) \rightarrow \varphi_0$ for the one φ_0 with $\ulcorner\varphi_0\urcorner = N_0$; so $\text{inst}(\tau)$ has at most one element. But a template with a metavariable has infinitely many instances, since each type has infinitely many bodies and $\theta \mapsto \tau\theta$ is injective (AS Thm 2.5(a)). So τ has no metavariable. $\square$

Remark F.1 (Pointwise limits are trivial). [proved]

^{pa} Rem 4.5 Under 0/1 support every sentence is eventually decided correctly by memorisation alone (with inconsistent theories present, given R_{d_n}). Whether the mass of not-yet-refuted inconsistent theories deriving a sentence tends to 0 is open.

Proofs of Proposition 7.23 and Remark F.1. $P_{\text{mix}}(D_n) \geq \pi(T) \prod_s L_\varepsilon(T; s)$; for $s \notin E$ that T derives, $L_\varepsilon(T; s) \geq (1 - \varepsilon)P_T(s)$ (if T refutes s the bound is $+\infty$); for $s \in E$, $L_\varepsilon(T; s) \geq \varepsilon\mu_0(s)$. Take $-\log_2$. With $\varepsilon = 0$, apply the same to $T \cup E$, whose likelihood is positive on all of D_n . For Remark F.1: a true s occurs eventually, after which every surviving theory derives it; for a false s , $\neg s$ occurs eventually, after which no surviving consistent theory derives s ; with inconsistent theories present this needs R_{d_n} (Proposition 7.21). Their remaining mass can be large (Proposition 7.36); whether it tends to 0 is open. $\square$

Remark F.2 (Toy drift). [computed]^{pa} §4 Along $T_0 = \text{PA}$, $T_{k+1} = T_k + \text{Con}(T_k)$, with Geometric($\frac{1}{2}$) data levels and prior $2^{-50(k+1)}$, the MAP is the largest level seen (16 at $n = 10^5$) and the regret grows like 50 bits per level (798.9 bits at $n = 10^5$; linear, 0.2637 bits per datum, if the theories' level law is misspecified). The toy covers only the provable part of the data: under a full-support law a positive fraction of the data is unprovable in every T_k . Each step adds a ground sentence, since one reflection sentence suffices (Proposition 6.7) and the reflection schema is not a template (Proposition 7.22).

Details (seed 1; data levels Geometric(ρ), each T_k generating its theorems with a truncated geometric level law of rate s): for $\rho = s = \frac{1}{2}$ the MAP level and regret are 2, 98.1 bits at $n = 10$, 9, 448.6 at 10^3 and 16, 798.9 at 10^5 , and the probability that the next datum is unprovable is $7.6 \cdot 10^{-6}$ at 10^5 ; with L_ε , $\varepsilon = 0.01$, the MAP lags the record (11 at 10^5) and the regret is 674.0; for $\rho = s = 0.8$ the regret is 2598.9 bits at 10^5 (MAP 52); for $\rho = \frac{1}{2}$, $s = 0.3$ it is 0.2637 bits per datum at 10^5 . The referee's independent re-implementation gives identical numbers.

Proof of Proposition 7.24. (a) $I\Sigma_n$ is finitely axiomatisable for $n \geq 1$ (known). $I\Sigma_{n+1} \vdash \text{Con}(I\Sigma_n)$ (known; Hájek and Pudlák, 1993, traced by the referee to Cor. I.4.34 through a secondary source; not verified), so $\text{PA} \vdash \text{Con}(I\Sigma_n)$, while $I\Sigma_n \not\vdash \text{Con}(I\Sigma_n)$ by Gödel's second theorem, $I\Sigma_n$ being consistent (true). So $P_{\text{PA}}(\text{Con}(I\Sigma_n)) > 0 = P_{I\Sigma_n}(\text{Con}(I\Sigma_n))$, and $\text{Con}(I\Sigma_n)$ occurs a.s. Eliminating each $I\Sigma_n$ is not concentration, since the tail $\{I\Sigma_m : m > N\}$ could hold mass; Theorem 4.2 gives $\pi_n(C_{\text{PA}}^*) \rightarrow 1$, and no $I\Sigma_n$ is in C_{PA}^* since its law has a different support. The waiting time is about $1/P_{\text{PA}}(\text{Con}(I\Sigma_n))$. (b) *Sketch.* Deriving $\text{Ind}(\varphi)$, $\varphi \in \Sigma_n$, from σ_n goes through the Tarski biconditional $\varphi \leftrightarrow \text{Tr}_n(\ulcorner\varphi\urcorner)$ (the predicate of Proposition 6.7), whose derivation grows with $|\varphi|$, while citing T_{Ind} costs $L_Q(\varphi)$ plus a constant. Not formalised; the growth rate was not checked. (c) Refuted by Examples 7.26 and 7.27. $\square$

Lemma F.3 (Equivalent motives). *[proved]*

pa Lemma 4.7 If $B \vdash \forall \bar{p} \forall x (\chi \leftrightarrow \psi)$ then $B + \text{Ind}(\chi) \vdash \text{Ind}(\psi)$.

Proof of Lemma F.3. From $\forall x (\chi \leftrightarrow \psi)$ at 0, x and Sx : $\psi(0)$ gives $\chi(0)$, and $\psi(x) \rightarrow \psi(Sx)$ gives $\chi(x) \rightarrow \chi(Sx)$. So the antecedent of $\text{Ind}(\psi)$ gives that of $\text{Ind}(\chi)$, which gives $\forall x \chi$, hence $\forall x \psi$. Generalise over the parameters. $\square$

Proof of Proposition 7.25. (a) Fix the path from C 's root to one occurrence $F(x, \bar{z})$. Instantiate every other atom as a constant: $\top := (0 = 0)$ ($A, B := \lambda.0$), $\perp := (S0 = 0)$ ($A := \lambda.S0$, $B := \lambda.0$), other formula metavariables $\lambda.(0 = 0)$ or $\lambda.(S0 = 0)$; Q1 refutes $S0 = 0$. Since each metavariable occurs once, these choices are independent, so each sibling on the path can be made $\top$ or $\perp$ as needed: $\top$ beside $\wedge$ and $\leftrightarrow$, $\perp$ beside $\vee$, $\top$ as the antecedent of $\rightarrow$ when the path goes to the consequent, and $\perp$ as the consequent when it goes to the antecedent (that step then acts as a negation, as $\neg$ does). Set $F := \lambda x \bar{z}. \psi'(x)$, which ignores $\bar{z}$, so the quantifiers on the path become vacuous. By propositional logic, Q1 and the vacuous-quantifier laws, $\text{Q1} \vdash \forall x (C\theta \leftrightarrow \psi')$ or $\text{Q1} \vdash \forall x (C\theta \leftrightarrow \neg \psi')$, depending only on the path. Take $\psi' := \psi$ in the first case and $\psi' := \neg \psi$ in the second. Then $\text{Q1} \vdash \forall x (C\theta \leftrightarrow \psi)$, and Lemma F.3 applied to the instance $\tau\theta = \text{Ind}(C\theta)$ gives $\text{Ind}(\psi)$. (b) Term metavariables are replaced by terms, which contain no logical symbols, so every instance has C 's skeleton. A formula with a fixed skeleton is logically equivalent to a prenex formula of fixed complexity Σ_k , and Lemma F.3 turns the instances into Σ_k induction. $\text{I}\Sigma_k \subsetneq \text{PA}$ because $\text{PA} \vdash \text{Con}(\text{I}\Sigma_k)$ (known, as in Proposition 7.24(a)) while $\text{I}\Sigma_k \not\vdash \text{Con}(\text{I}\Sigma_k)$. $\square$

Remark F.4 (Read-once is not necessary). *[proved; computed]*

pa Rem 4.9 $T_{\text{Ind}}[P := \lambda x. x = 0 \vee F(x)]$ is not read-once, and every instance's motive holds at 0; yet $\text{Q1}, \text{Q2} + \text{Ind}(x = 0 \vee \exists y (x = Sy \wedge P(y))) \vdash \text{Ind}(P)$ (checked, 47 lines). Which non-read-once templates give full induction is open.

The checked derivation has 227 written symbols, costs 1403.3 bits against a 15.4-bit citation, and cites only Ind , Q1 , Q2 . The motive χ is not equivalent to P , so Lemma F.3 does not apply.

Details of Examples 7.26 and 7.27. Narrow practice (the referee's counterexample r8, rechecked by the track): $|T_{\text{Ind}}| = 14$, $|T_E| = 39$, $|T_N| = 31$ (`dtlib`), prior difference 280 bits; coverage checked on 3000 data in seeds 31 and 33; the code, the identity (three contexts: motive root and the two “=” positions) and its expansion give $+230.8 / +230.8 / +232.0$ at $n = 100$ and $+174.6 / +174.6 / +174.6$ at 3000. G1: motives of depth 2 quantifying only atoms give $1 + 7 + 3 \cdot 49 + 2 = 157$ skeletons, all found among 200000 motives (seed 12); 112 occur in the first 3000 data. Each motive is a Boolean combination of atoms and single quantifiers over atoms, so it is equivalent to a Σ_2 formula. The theory of the 112 seen skeletons (119 templates) is $+42905.6$ and $+42838.5$ bits behind at $n = 1000, 3000$; T^* uses 19 formula contexts. With all 157 skeletons (164 templates) the slope is $+2.0$ bits per doubling. The entropy terms cancel only asymptotically: G1 draws each connective independently given its context, so their difference is $O_P(1)$ by Wilks's theorem for nested multinomial models (cited from memory; proof sketch).

F.6 Root fragments and the Q axioms

Track experiments' $T^* = \text{Q1}.. \text{Q7} + T_{\text{Ind}}$ has one-hole motives without parameters; its language has $<$ with no Q axiom about it. T_f for $f \in \{=, <\}$ has motive $a(x) f b(x)$ with 1-ary term metavariables; for $\neg$, $\neg P_1(x)$; for $\wedge, \vee, \rightarrow, \leftrightarrow$, $P_1(x) f P_2(x)$; for $\forall, \exists$, $f y P_1(x, y)$.

Proposition F.5 (Root fragments). *[proved]^{experiments Prop X9} $\text{inst}(\{T_f : 9 \text{ roots}\}) = \text{inst}(T_{\text{Ind}})$; for each connective root f and any Q^- , $Q^- \cup \{T_f\}$ and $Q^- \cup \{T_{\text{Ind}}\}$ have the same consequences; the argument fails for $=$ and $<$.*

Proof of Proposition F.5. (a) Every motive has one of the 9 roots, and an instance of T_{Ind} with root f is the instance of T_f whose sub-bodies are read off the motive (for $\forall, \exists$ the bound variable becomes P_1 's second argument); conversely $\text{inst}(T_f) \subseteq \text{inst}(T_{\text{Ind}})$. (b) One direction is (a). For the other, given φ choose: $P_1 = P_2 = \varphi$ for $\wedge, \vee$; $P_1 = \neg\varphi$ for $\neg$; $P_1 = (0 = 0)$, $P_2 = \varphi$ for $\rightarrow, \leftrightarrow$; $P_1(x, y) = \varphi(x)$ for $\forall, \exists$ (nonempty domains). The chosen motive ψ satisfies $\vdash \forall x(\psi \leftrightarrow \varphi)$, so $\text{Ind}(\psi) \vdash \text{Ind}(\varphi)$. (c) No such body exists for an atomic root; Proposition 7.29 shows the atomic fragments are weaker. $\square$

Proof of Proposition 7.29. T^* proves every induction instance with parameters. For $\varphi(x, \bar{w})$ take the parameter-free motive $\psi(x) := \forall \bar{w}[(\varphi(0, \bar{w}) \wedge \forall y(\varphi(y, \bar{w}) \rightarrow \varphi(Sy, \bar{w}))) \rightarrow \varphi(x, \bar{w})]$. $\psi(0)$ is valid; if $\psi(x)$ and the base and step hold for $\bar{w}$, then $\varphi(x, \bar{w})$ and so $\varphi(Sx, \bar{w})$; $\psi(Sx)$. T_{Ind} gives $\forall x\psi$, which is induction for φ . So $T^* \supseteq \text{PA}$ and $T^* \vdash \sigma$, as $\text{PA} \vdash \sigma$ (known). $Q + \{T_=: T_<\} \not\vdash \sigma$. By Shepherdson (1964) (secondary sources only) there is a model M of Q plus open induction with parameters in which $x \cdot x = 2 \cdot y \cdot y$ has a solution with $y \neq 0$; M is the non-negative part of a discretely ordered ring, with $Sx := x + 1$, so Q1–Q7 hold (Q3 by discreteness). Instances of $T_=:$ are open induction, so they hold. Read $<$ as the empty relation: every instance of $T_<$ has the false conjunct $a(0) < b(0)$ in its antecedent, so it holds; σ does not mention $<$. So $M \models Q + \{T_=: T_<\}$ and $M \not\models \sigma$. $\square$

The same holds for any theory whose induction-type templates are all term-only: by Proposition 7.25(b) it lies in some $\text{IS}_k \subsetneq \text{PA}$.

Proposition F.6 (Redundant Q axioms). *[proved; computed]^{experiments Prop X11} $Q1, Q2, Q4..Q7 + T_{\text{Ind}} \vdash Q3$; no other Q_i is redundant given full induction.*

Proof of Proposition F.6. T_{Ind} cites $\text{Ind}(\psi)$ for $\psi(x) := x = 0 \vee \exists y(x = Sy)$; $\psi(0)$ holds and $\psi(Sx)$ holds with $y := x$, so $\forall x\psi$, which gives Q3. For the other axioms: in a structure in which every element is the value of a numeral, every induction instance holds (induction on k in the metatheory). Each structure below has this property, satisfies the Q_j , $j \neq i$, and falsifies Q_i : $i = 1$: $\{0\}$, all operations 0; $i = 2$: $\{0, 1\}$, $S0 = S1 = 1$, $x + 0 = x$, $x + 1 = 1$, $x \cdot 0 = 0$, $x \cdot 1 = x$; $i = 4$: $\mathbb{N}$, $x \oplus y := x + y + 1$, $x \otimes y := y(x + 1)$; $i = 5$: $\mathbb{N}$, $x \oplus y := x$, $x \otimes y := 0$; $i = 6$: $\mathbb{N}$, standard $+$, $x \otimes y := xy + 1$; $i = 7$: $\mathbb{N}$, standard $+$, $x \otimes y := 0$. Each verification is a line of arithmetic (e.g. $i = 4$: $x \oplus Sy = x + y + 2 = S(x \oplus y)$, $x \otimes Sy = (y + 1)(x + 1) = (x \otimes y) \oplus x$); `check_models.out` evaluates all seven axioms in each structure on a finite range. $\square$

F.7 The Bayesian DTRC, the $\forall$ -merge and Euler's polynomial

Proof of Proposition 7.31. The Dirichlet($\frac{1}{2}$) marginal of counts $(n_1, \dots, n_K, 0)$ over $K + 1$ categories divided by that of $(n_1, \dots, n_K)$ over K is $\frac{\Gamma((K+1)/2)\Gamma(n+K/2)}{\Gamma(K/2)\Gamma(n+(K+1)/2)}$, and $\Gamma(n+a+\frac{1}{2})/\Gamma(n+a) \sim n^{1/2}$. Add $\beta|\tau|$ prior bits. $\square$

Proofs of Propositions 7.35 and 7.36. Proposition 7.35: the likelihood is the constant $\mu_0(D)$ for every compatible theory and 0 otherwise. Proposition 7.36: every T -derivation is a T' -derivation, and each of its citations loses a factor $1 - w$ under the rescaled weights; the best

T' -derivation is at least as probable. With Dirichlet weights and σ unused, Proposition 7.31 applies. $\square$

Remark F.7 (Overlapping templates). [proof sketch]

pa Rem 5.4 For the lump and F_0 the code assigns each datum to its cheapest covering template, a two-part bound; if the greedy assignment is optimal (not proved), the exact marginal is at most about n bits lower, and the lump stays ≥ 219 bits behind at $n = 100$.

Sketch for Remark F.7. The exact marginal sums over all assignments of data to covering templates. Its code length is at least the body code length under the best assignment, because the index probabilities of all assignments sum to at most 1. If the best assignment is the greedy one (induction data on T_{Ind} , whose motive is coded once, rather than on F_0 , which codes the whole sentence), the exact code is lower than the two-part code by at most the greedy index cost, about n bits for two covering templates. At $n = 100$ the lump would then still be at least $319.3 - 100 \approx 219$ bits behind. Optimality of the greedy assignment is not proved.

Full dtrc run. Table 5 at $n = 300, 1000$: root split +771.4, +769.9; spare +18.2, +19.0; memorise +41418.2, +133845.4; over-general +8964.3, +27876.8; lump +1048.9, +3205.5; bare F_0 +14712.6, +46251.0 bits (the first version had eight entries that did not match the output; referee M1). Without negatives the lump has mass 1.000 at $n = 10$ and $7.679 \cdot 10^{-97}$ at $n = 100$; from $n = 100$ the remainder beside $Q + T_{\text{Ind}}$ is almost all on the spare ($5.808 \cdot 10^{-6}$). The second negative is $(0 = 0) \wedge \forall x(0 = S0 \rightarrow 0 = S0) \rightarrow \forall x(0 = S0)$. The theories without $0 = S0$ as an instance have mass $2.6 \cdot 10^{-53}$ and $2.2 \cdot 10^{-13}$ at $n = 10, 30$, hence the acceptance by $V_{\delta,0}$ for $\delta \geq 3 \cdot 10^{-13}$.

candidate	$n = 10$	30	100	300	1000	3000
split by head of t (sound)	+115.5	+109.1	+101.0	+93.3	+84.6	+76.7
memorise instances	+519.5	+1263.8	+3749.9	+12204.1	+34023.6	+83197.3
over-general $z_1 + 0 = z_2$	+145.5	+351.5	+927.2	+2618.8	+7975.5	+22702.2
$\forall x(x + 0 = x)$, fixed rule code	+38.2	+104.7	+337.2	+1001.6	+3326.9	+9970.8

Table 31: $\forall x\varphi$ as a sound merge: $\text{CL} - \text{CL}(z + 0 = z)$ in bits on data $t + 0 = t$ (t random closed terms, seed 12). The over-general template is refuted by the negative $0 + 0 = S0$. With Q3 in the theory the head split would give $\forall x(x + 0 = x)$ by cases; here the theories are the templates alone. A first run in which the head split overtook the schema at $n = 1000$ was an artefact of coding split bodies from a root context (refuted). Computed (pa §5.5).

Euler’s polynomial (Example 7.37). $\text{Prime}(m) := \neg m = 0 \wedge \neg m = S0 \wedge \forall a \forall b(a \cdot b = m \rightarrow a = S0 \vee b = S0)$ has a 27-symbol frame and three occurrences of $m = k \cdot k + k + 41$ (the first version counted two; referee m9), so $|\text{datum}(k)| = 27 + 3(3(k + 1) + 45)$ with unary numerals; the template has 171 symbols. Under the unconditioned geometric law the false instances have mass 0.0049 ($\rho = 0.9$) and 0.0727 ($\rho = 0.97$). Code length of memorisation minus the false schema (bits), flat / depth-indexed grammar, and the schema’s predictive probability of $k = 40$: $\rho = 0.9$, $n = 100$: 51445/51373, $1.9 \cdot 10^{-3}/4.1 \cdot 10^{-4}$; $n = 10^4$: 161030/160816, $1.4 \cdot 10^{-3}/4.8 \cdot 10^{-5}$; $n = 10^6$: 314225/320382, $1.4 \cdot 10^{-3}/5.0 \cdot 10^{-7}$; $\rho = 0.97$, $n = 10^6$: 1820112/1920893, $8.6 \cdot 10^{-3}/5.0 \cdot 10^{-7}$ (seeds 5, 6; primality by trial division; the referee’s closed-form recomputation is identical). *Covering (sketch).* A template that covers infinitely many data must have a metavariable at or above a

n	$R = 10$ (rules of ND)			$R = 2$ ($C_{\min}$: cite, $\forall E$)		
	fixed	learned, shared	learned, by depth	fixed	shared	by depth
10	33.2	23.4	12.5	10.0	20.0	2.5
10^3	3321.9	2004.0	41.1	1000.0	2000.0	5.8
10^6	3321928.1	2000004.0	85.9	1000000.0	2000000.0	10.8

Table 32: $CL(H_V) - CL(H_{sch})$ in bits on pure instance data under three rule codes (term costs cancel). The depth-indexed code gives $\frac{R-1}{2} \log_2 n + O(1)$ ($4.31 \log_2 n$ at $n = 10^6$ for $R = 10$, tending to 4.5). In a mixed practice with a fraction f of φ -instances the depth-indexed difference is $h(f)$ bits per datum at $n = 10^6$ (0.4691 at $f = 0.9$ and 0.1; 1.0001 at $f = \frac{1}{2}$), the shared one 1.896, 1.377, 0.483 bits per datum at $f = 0.9, 0.5, 0.1$. Computed in closed form; matches the referee’s r3 (pa §5.5).

position of the numeral k (otherwise k is fixed), so some instance puts there a term that makes the sentence false: the numeral $k = 41j$ at the positions of k ($41 \mid k^2 + k + 41$), or a composite such as $41 \cdot 41$ at a position above them. So every finite unguarded covering union is false unless it memorises. *Eventual behaviour (sketch)*. Under the flat grammar the schema pays a linear KL term (about 0.007 bits per datum at $\rho = 0.9$) for the false k it predicts, while memorisation’s prior grows like $(\log n)^2$, so memorisation eventually wins, beyond $n = 10^6$; under the depth-indexed grammar both costs grow like $(\log n)^2$ and the false schema stays ahead. Constants estimated, not computed. A refutation datum ($40^2 + 40 + 41 = 41^2$) removes the schema.

G Experiments: proofs and details

G.1 Grammar, prior code and pools

Q. A term node is 0, S , $+$, $\cdot$ with weights .45, .35, .10, .10, a variable with total weight .3 split uniformly over the variables in scope (holes and bound variables alike), or p with weight .15 where the guard allows it. A formula node is $=$, $<$, $\neg$, $\wedge$, $\vee$, $\rightarrow$, $\leftrightarrow$, $\forall$, $\exists$ with weights .35, .20, .10, .10, .05, .10, .02, .05, .03. Weights are renormalised over the options available. The mean number of same-sort children is 0.75 for closed terms, at most 0.75 for other terms and 0.72 for formulas, so bodies are finite almost surely; the referee’s exact size distributions agree.

Prior code. $\text{bits}(\tau)$ is $-\log_2$ of a subcritical stochastic grammar over templates (formula and term nodes, metavariable occurrences, arities and guards); a theory of m distinct components costs $|\gamma(m)| + \sum_i \text{bits}(\tau_i) - \log_2 m!$ with γ the Elias code. The referee’s independent implementation gives the same lengths to four decimals; Kraft’s inequality holds (Remark A.1).

Pools. Theories are deduplicated at each n by their components (canonical template plus guards); if $\text{Mem}(D_n)$ equals a pool theory, that theory keeps its mass and Mem becomes an alias without mass (the first version counted it twice, which affected E6 at $n = 1, 2$ only). Under L0 $\text{Mem}(D_n)$ is the best memoriser covering D_n : a further ground sentence costs prior bits and the factor of Proposition 4.14(a) and cannot raise the likelihood. Build points: E1 $b \in \{1, 2, \dots, 64\}$ (powers of 2); E3(a) $\{8, 32, 64\}$; E2, E3(b), E7 $\{8, 16, 32, 64\}$; E4 and E6 have none. In E2 a theory of Q axioms and induction-type templates is tagged equivalent to T^* iff it contains Q1, Q2, Q4–Q7 and a full-induction template (T_{Ind} , IndSwap or a connective fragment); other tags come from a budgeted refuter, and an unrefuted theory is “unknown”. The false probes are $0 = S0$, $\forall x (x + 0 = 0)$, induction with a wrong base, $\forall x (Sx = x)$ and $\forall x \forall y (y = x)$; held-out induction instances are drawn outside the training stream.

$\mathbf{L1}_{\text{cit}}^{\text{sel}}$. The class probabilities c_i are exact for components $\forall^a \chi$ with χ quantifier-free, term metavariables only, every leading variable used and no MP components (a finite Markov chain on the number of leading $\forall$ and the presence of p); other components are left out of $\mathbf{L1}_{\text{cit}}^{\text{sel}}$ pools. Monte Carlo agrees on the eight E6 forms (40000 chains each; largest $|z| = 2.17$).

G.2 Proof of Proposition 8.1

(a) Bodies from $\mathcal{Q}$ contain no parameter other than p , so $\tau\theta$ is already in closure-normal form and the event “the datum is s ” is the event $\tau\theta = s$. By AS Thm 2.5 at most one θ satisfies $\tau\theta = s$, so $P(\tau\theta = s) = \sum_{\theta: \tau\theta=s} \prod_M \mathcal{Q}(\theta(M))$ has at most one term, $\prod_M \mathcal{Q}(\theta_s(M))$. A body containing p has probability 0 under a closed guard. $\square$

(b) Expanding, $\prod_j \sum_i w_i a_{ij} = \sum_z \prod_i w_i^{n_i(z)} \prod_j a_{zjj}$, and the Dirichlet moments $\mathbb{E} \prod_i w_i^{n_i} = \frac{\Gamma(K\alpha)}{\Gamma(K\alpha+n)} \prod_i \frac{\Gamma(\alpha+n_i)}{\Gamma(\alpha)}$ give the formula (Lemma 2.8). *The programme.* The summand depends on z only through $\prod_j a_{zjj}$ and the counts. A datum covered by one component contributes a fixed factor and one count. For the others, with $S_j := \{i : a_{ij} > 0\}$, the sum over their assignments is the polynomial $\prod_j \sum_{i \in S_j} a_{ij} x_i$ evaluated by the linear functional $x^c \mapsto \prod_i \Gamma(\alpha + c_i + c_i^0) / \Gamma(\alpha)$ (c^0 the fixed counts), times $\Gamma(K\alpha) / \Gamma(K\alpha + n)$. A dense array (at most three overlapping components) or variable elimination over count vectors computes the coefficients exactly. In the latter a component’s factor is applied, and its count dropped, once no later group contains it; this is valid because the functional is a product over components. *The bounds.* All terms are nonnegative, so any single term is a lower bound. For the upper bound, $\mathbb{E}_w L(w) \leq \max_w L(w)$ with $L(w) = \prod_j \sum_i w_i a_{ij}$, and $f := \ln L$ is concave on the simplex. For an interior w with $L(w) > 0$ and any w° in the simplex, $f(w^\circ) \leq f(w) + \nabla f(w) \cdot (w^\circ - w) \leq f(w) + \max_i \partial_i f(w) - n$, since $\nabla f(w) \cdot w^\circ \leq \max_i \partial_i f(w)$ and $w \cdot \nabla f(w) = \sum_j \sum_i w_i a_{ij} / \sum_k w_k a_{kj} = n$. $\square$

(c) At most J rules are applied. Each step chooses among finitely many outcomes (stop, or a rule that applies) with probabilities summing to 1; each rule’s input (a term from $\mathcal{Q}_e$, bodies from $\mathcal{Q}$, a uniform choice among the applicable MP components) has a probability distribution ($\mathcal{Q}$ is proper). A rule is applied only when it applies, so no run fails; the runs have total probability 1, and the datum is a function of the run. $\square$

(d) Let $G_k(s)_i$ be the probability that the chain reaches step k at s after citing component i , divided by w_i ; so $G_0(s)_i$ is the L0 coefficient of (a). For $1 \leq k \leq J$ the chain leaves s' only if a rule applies to it, then takes rule r with probability $(1 - c_{\text{stop}})c_r / \sum_{r'} \text{applies to } s' c_{r'}$ and input ξ with probability $P(\xi)$. Hence

$$G_k(s) = \sum_{(s', r, \xi): r(s', \xi)=s} G_{k-1}(s') (1 - c_{\text{stop}}) \frac{c_r}{\sum_{r'} \text{applies to } s' c_{r'}} P(\xi), \quad a_i(s) = \sum_{k \leq J} G_k(s)_i \text{stop}_k(s),$$

with $\text{stop}_k(s) = 1$ if $k = J$ or no rule applies to s , and c_{stop} otherwise. The recursion is exact iff every predecessor s' with $G_{k-1}(s') > 0$ is enumerated. The following give complete predecessor sets.

(d1) $\forall E$ guided by templates. Let $U = \forall x B$ be a component whose derived template $D := B[u/x]$ (u a fresh 0-ary term metavariable) is in DT° . Then x is no argument of a metavariable occurrence (else an argument would contain u), so x occurs only at rigid positions, bodies never contain x , and $(B\theta)[t/x] = D(\theta \cup \{u \mapsto t\})$. So the pairs (θ, t) with $\forall E(U\theta, t) = s$ correspond to the matchers of D against s , of which there is at most one (AS Thm 2.5). If x does not occur in B , the predecessor is $\forall x s$, with total weight $\sum_t \mathcal{Q}_e(t) = 1$.

- (d2) $\forall E$ from P or $\forall x P(x)$ at step 0. For a parameter-free s the predecessors are $\forall x \psi_S$, with ψ_S the sentence s with a nonempty set S of occurrences of one closed term t replaced by x . Since the PCFG probability is a product over nodes and every node of ψ_S lies in a context with a variable, $\mathcal{Q}(\forall x \psi_S) = \mathcal{Q}(\forall x s) \prod_{o \in S} r_o$ with $r_o := (p_{\text{var}}/\nu_o)/\mathcal{Q}_v(t)$; here p_{var} is the normalised variable weight, ν_o is 1 plus the binder depth at o , and $\mathcal{Q}_v(t)$ is the probability of t in a context with a variable. So $\sum_{S \neq \emptyset} \mathcal{Q}(\forall x \psi_S) = \mathcal{Q}(\forall x s) (\prod_o (1 + r_o) - 1)$; for $\forall x P(x)$ the same holds up to the factor of the root $\forall$. The code uses this closed form only in theories without MP components, where these parameter-free predecessors admit only $\forall E$. If s contains p and the guard is closed, t contains p and S is all occurrences of t .
- (d3) *Gen*, then $\forall E$ ($J = 2$). A step-1 Gen output is $\text{gen}(s_0) = \forall x s_0[x/p]$ for a cited $s_0 = T\theta_0$ containing p , and $\forall E(\text{gen}(s_0), t) = s_0[t/p]$. Let D be the component with its rigid occurrences of p replaced by a fresh 0-ary metavariable u , and suppose $D \in \text{DT}^\circ$ (no rigid p inside an argument; the track's notes leave this hypothesis implicit). Then $s_0[t/p] = D\theta$ with $\theta(M) = \theta_0(M)[t/p]$ and $\theta(u) = t$, so θ is the unique matcher of D against s . Closed-guard bodies satisfy $\theta_0(M) = \theta(M)$; open-guard bodies $\theta_0(M)$ arise from $\theta(M)$ by replacing some occurrences of t by p ; without a rigid p , t occurs in some open-guard body. The code enumerates exactly these candidates and keeps those with $\forall E(\text{gen}(s_0), t) = s$.
- (d4) *Gen* and *MP*. $\text{gen}^{-1}(\forall x \chi) = \chi[p/x]$ if χ uses x and s has no parameter. For an MP component $A \rightarrow B$, $B \in \text{DT}^\circ$ contains A 's metavariables, so the matcher θ of B against s is unique and the predecessor is $A\theta$.
- (d5) *Pruning*. A candidate is skipped only if its ($\forall$ -spine, parameter flag) is outside the abstract set of step $k - 1$, computed forward from the components; by induction on k every step- k sentence lies in that set. $\square$

The code admits into L1 pools only theories whose $\forall$ -rooted components have derived templates in DT° (twice derived at $J = 2$) and whose formula-metavariable components are P or $\forall x P(x)$ at $J = 1$ (none at $J = 2$). A term with more than 12 occurrences in a datum stops the enumeration and raises a counter; it was 0 in every run. The support check does not test the hypothesis of (d3). It holds for every component without a rigid p , which includes every hand theory of E1 and E6; for data-derived theories at $J = 2$ (E1 *allq2*) exactness rests on the computed checks: the guided recursion equals a brute-force recursion over all subsets of all occurrences to 10^{-9} on samples from 10 theories at $J = 1, 2$ (same class, so not independent), and the referee's own recursion agrees on 904 triples at $J = 1, 2$ to $2.1 \cdot 10^{-14}$.

G.3 Proof of Proposition 8.2

(a) A_{xy} and A_{yx} differ by the order of two universal quantifiers. M_x^{open} cites $\forall x (x + p = p + x)$ (body $z := p$), which under the closure reading is $\forall w \forall x (x + w = w + x)$, i.e. A_{yx} ; and each of its instances $\forall x (x + b = b + x)$, b a term with at most the parameter p , follows from A_{xy} by $\forall E$ (and Gen on p under the closure reading). Likewise M_y^{open} cites $\forall y (p + y = y + p)$, i.e. A_{xy} .

(b) Take the domain $\mathbb{N} \sqcup \{\alpha, \beta\}$, standard on $\mathbb{N}$, with $<$ arbitrary, $S\alpha = \alpha$, $S\beta = \beta$, and for $a \in \{\alpha, \beta\}$, $n \in \mathbb{N}$: $a + n = n + a = a$; $a \cdot n = n \cdot a = a$ if $n \neq 0$ and 0 if $n = 0$; $\alpha + \alpha = \alpha \cdot \alpha = \alpha$, $\beta + \beta = \beta \cdot \beta = \beta$, $\alpha + \beta = \alpha \cdot \beta = \alpha$, $\beta + \alpha = \beta \cdot \alpha = \beta$. Each $\mathbb{N} \cup \{a\}$ is closed under S , $+$, $\cdot$ and $+$ is commutative on it. Closed terms denote naturals, so the closed instances of S_{ab} hold. An instance $\forall x (x + t = t + x)$ of M_x , t closed of value n , holds for $x \in \mathbb{N}$ by arithmetic and for $x = a$ since $a + n = a = n + a$; likewise for M_y . An instance $\forall w (f(w) + g(w) = g(w) + f(w))$ of S_{ab}^{open} (one parameter) holds because for $w \in \mathbb{N}$ both values lie in $\mathbb{N}$ and for $w = a$ both lie in

$\mathbb{N} \cup \{a\}$. But $\alpha + \beta = \alpha \neq \beta = \beta + \alpha$, so A_{xy} fails and, by soundness, is not derivable. With two parameters, $p_1 + p_2 = p_2 + p_1$ would be an instance of S_{ab}^{open} ; the convention excludes it. $\square$
 Computed (`checks/check_models.out`): the 285 closed terms with at most 7 nodes denote naturals; the instances of S_{ab} , M_x , M_y hold for $x \in \{0, \dots, 14, \alpha, \beta\}$, and those of S_{ab}^{open} for 154 terms with at most 5 nodes and $w \in \{0, \dots, 3, \alpha, \beta\}$; A_{xy} and the instance $\forall x (x + p = p + x)$ of M_x^{open} fail at (α, β) .

G.4 Bounds, counters and further tables

Exact L1 pools. E1’s support check left out 65 (*open*) and 43 (*allq2*) data-derived theories for $\varphi = \neg Sx = 0$, summed over seeds, mostly skeleton clusters, and nothing else. *Dirichlet bounds* replaced the exact sum, with the largest posterior mass the theory could have had: in E1 for some skeleton theories and $\text{Mem}(D_n)$ under L1, at most $8 \cdot 10^{-14}$; in E2 for `skel6@32` and `skel6@64` in 14 cases over 25 seeds and both pools, 0 to double precision; in E6 for $\text{Mem}(D_n)$, 0; in E3 and E4(B, C), in no case checked. *Fallbacks.* The likelihood and derivability-oracle fallback counters were 0 in every run of E1, E3(a) and E4; the referee re-ran the 75 E1 jobs with the oracle instrumented (5896 queries, no fallback). *E4 queries.* $\forall x (x + 0 = x)$, $1 + 0 = 0$, $0 + 0 = 1$, $0 = S0$, $2 + 0 = 3$, $7 + 0 = 8$, $(1 + 1) + 0 = S(1 + 1)$, $0 = 0$; two are true. The code’s list has nine entries because it writes $1 + 0 = 0$ twice. *E5(a2).* Under $\{\varphi(z), \varphi(Sz)\}$ a datum $\varphi(t)$ has probability $Q(t)(1 - w_2 + w_2/q_S)$ if t is S -rooted and $Q(t)(1 - w_2)$ otherwise ($q_S = 0.35$), so $\text{BF} = \mathbb{E}[(1 - w_2 + w_2/q_S)^{n_S} (1 - w_2)^{n - n_S}]$, $w_2 \sim \text{Beta}(\frac{1}{2}, \frac{1}{2})$. Quadrature: substitution $w_2 = u^2$, $2 \cdot 10^5$ -point trapezoid rule; at $n = 500$ it agrees with the exact programme to 10^{-4} bits, and the referee’s exact Beta-moment sum agrees with it to $3 \cdot 10^{-4}$ bits up to $n = 10^6$.

n	Q seen	SeenQ(D_n)		CL(T) – CL(T^*), bits					acc. seeds	
		mass	vs Q-l.	frag-c.	$+T_\wedge$	$+\forall x(0+x=x)$	$+0=S0$	Q-lumped	legacy	causal
8	2.5	0.744	–16.8	696.9	87.6	13.8	5.1	–100.0	18	2
16	4.0	0.803	–38.1	698.2	87.9	14.2	5.4	–41.0	3	3
32	5.7	0.923	–87.6	700.0	88.2	14.6	5.9	51.8	2	0
64	6.7	0.991	–255.0	703.6	88.5	15.1	6.3	248.2	0	0
512	7.0	0.996	–2928.5	716.3	89.2	16.6	7.8	2928.5	–	0

Table 33: E2, 25 seeds. Left: the sound competitor $\text{SeenQ}(D_n) = \text{Trim}(T^*, D_n)$; its code length minus that of **Q-lumped** ranges over seeds from -49.1 to $+8.9$ at $n = 8$ and from -92.8 to $+12.7$ at 16. At $n \leq 32$ it is usually in the pool under the name of a data-derived theory; from $n = 64$ it is usually T^* or $T^* - Q_i$. Middle: code lengths relative to T^* (means; positive means worse); the over-general induction templates are 147 and 538 bits behind at $n = 8$, 8303 and 32624 at 512. Right: seeds in which $V_{0.05,0}$ accepts a false probe, with the first version’s legacy pool (data-derived theories from the first 16, 40, 64 data, used at every n ; no trimmed theories) and with the causal pool; at some $n \leq 64$, 19 against 5 seeds. Mean unsound mass, legacy against causal: 0.720/0.256, 0.147/0.197, 0.081/0.077, 0.009/0.009 at $n = 8, 16, 32, 64$. The legacy counts reproduce the referee’s. The referee’s variant, legacy plus SeenQ, accepts in seeds 1, 4, 11 ($n = 16$) and 17 ($n = 8$); the causal pool accepts in these and in seed 14 at $n = 8$, where $\text{Trim}(\text{Q-lumped})$ is the MAP with mass 1.000, so it is not uniformly kinder to soundness.

n	<i>heavy data</i>			<i>numerals data</i>			
	C_1	C_2	R_4	R_9	R_{11}	R_{12}	R_{16}
128	−0.2	−11.5	−162.0	−86.6	−3.0	+46.6	+300.2
512	−76.3	−204.4	−978.8	−1270.0	−1248.5	−1220.7	−1015.0
1024	−165.5	−447.0	−2063.1	−2835.1	−2894.6	−2895.1	−2751.7

Table 34: E3(a): code length relative to H_{sch} on the extended families (L0, mean over 5 seeds, bits; negative means preferred). The best R_m moves to larger m as n grows and is interior; C_2 pulls away from C_1 linearly. The first version’s pool stopped at R_4 and C_1 , which it named as the winners.

(A) tight construction								
u	w^*	δ'	t^*	exact $P(\text{win})$	MC (10^5)	pipeline	selected	bound
0.05	0.01	0.01	150	0.00046	0.00046	0/400	20/20	0.01
0.01	0.001	0.05	680	0.00108	0.00123	not run	20/20	0.05
0.5	0.01	0.01	10	0.00098	0.00095	2/400	20/20	0.01
0.1	0.1	0.1	47	0.00707	0.00694	5/400	20/20	0.1
0.2	0.05	0.2	17	0.02252	0.02298	6/400	20/20	0.2
(B, C) pools; L0 / L1; bound 0.05								
setting	prover wins			largest invalid mass		MAP at $n = 256$		
B well specified	0/100, 0/100			0.452, 0.533		$H_{\text{sch}} \times 100$		
C1 10% near misses $t + 0 = St$	100/100, 100/100			1.000, 1.000		$H_{\text{sch}} \cup \{z + 0 = Sz\} \times 100$		
C2 heavy-tailed terms	0/100, 0/100			0.452, 0.536		$C_1 \times 96, H_{\text{sch}} \times 4$		
C3 only $\varphi(0)$; decoy in pool	0/100, 0/100			0.107, 0.112		Mem $\times 100$		

Table 35: E4. (A) “Pipeline”: wins of the full posterior–derivability–threshold pipeline, which equals the closed-form event on all 400 streams in the two settings checked; “selected data”: only $\varphi(0)$ shown. (B) The largest invalid mass sits on $\forall x (x + 0 = x)$ at $n \leq 6$ ($n \leq 3$ in 184 of 200 streams), carried by H_{open} (mean 0.23 under L0, 0.21 under L1), by $H_{\forall}$ under L1 (0.15) and by a generalisation of $\varphi(z)$. (C1) First acceptance ($0 + 0 = 1$ or $2 + 0 = 3$) at the first mistaken datum in 90 of 100 streams per likelihood, 2–5 data later in the rest.

G.5 Reproduction and checks

All commands run from `code/` (Python 3.11, numpy 2.4, scipy 1.17). Results are in `results/*.md` and `/*.json`, the first version’s in `results/v1/`. AS’s `dtrc` code is read, not modified. `python3 -m pytest -q tests` runs the 21 unit tests (18s); `experiments/ek_*.py` runs E_k ; `sh run_all.sh` runs everything (about 50 minutes on 4 shared cores; E3 10 min, E4 32 min, the rest at most 3 min each). Seeds: E1 0–4, E2 and E7 0–24, E3 and E6 0–4, E4 streams 0–399 (A) and 0–99 (B, C), E5 and E8 0–9; the random subsets for Min have seed $(c + \text{seed}) \cdot 1000 + b$ with $c = 1000$ (E1), 500 (E2), 77 (E3). After the batch run E1–E3 were re-run with their final reporting code; all posteriors are identical, and only derived labels changed. Two L0 entries of the first version’s E1 table were wrong on its own results and are corrected in Table 7. The unit tests compare each core computation with brute force or Monte Carlo, and with the referee’s independent `ref_core.py` (within 10^{-9}). Nothing is checked in a proof assistant. The check scripts (`research/tracks/experiments/checks/`, outputs next to them) give: `check_x6`, E1’s L1 odds within $1.1 \cdot 10^{-11}$ bits of prior $+n$ with prior differences (0.0899, 0.0899, 1.4404 bits) from the referee’s code, and the `L1selcit` tie to $1.0 \cdot 10^{-13}$; `check_fixed_weight`, Example 5.8 in this code (every $n \leq 2 \cdot 10^6$, 300 runs: 0.977, 1.000 at fixed w^* ; 0.003, 0.007 with w^* from the

spare added to $\{\varphi(z)\}$	mean $\log_2$ BF at $n = 1, 16, 256, 4096$	slope	prior bits
false sentence $0 = S0$ (never cited)	-1.00, -2.84, -4.83, -6.83	-0.500	8.4
disjoint schema $z \cdot 0 = 0$ (never matched)	-1.00, -2.84, -4.83, -6.83	-0.500	14.5
nested $\varphi(Sz)$	+0.17, -1.05, -2.85, -3.97	-0.281	21.3
nested $\varphi(Sz)$, quadrature	$n = 10^2 \dots 10^8$: see caption	-0.251 ± 0.002	21.3

Table 36: E5 (10 seeds; Bayes factors without the prior; slopes in $\log_2$ BF per doubling of n , on $n = 256 \dots 4096$ except for the quadrature row, $10^2 \dots 10^8$). (a2) The nested spare by quadrature: mean $\log_2$ BF -2.196, -3.028, -3.867, -4.658, -5.575, -6.332, -7.200 at $n = 10^2, \dots, 10^8$ (s.e. 0.028–0.031); slope -0.251 ± 0.002 on $10^2 \dots 10^8$ and -0.251 ± 0.003 on $10^4 \dots 10^8$; per decade -0.250, -0.253, -0.238, -0.276, -0.228, -0.261; 1000 draws of $n_S \sim \text{Bin}(n, 0.35)$ per n , numpy seed 0; the first version reported -0.259 from 200 draws, and the referee’s independent sum -0.233 on $10^2 \dots 10^7$ with 40 draws. (b) L_5 data: L_5 has mass $2 \cdot 10^{-6}$ at $n = 64$, 0.66 at 128, 1.000 from 256 (10/10 seeds); L_∞ data: L_∞ has 1.000 from $n = 16$. $\mathcal{Q}_{\text{num}}(S^j 0) = 2^{-(j+1)}$; prior bits L_1 10.9, L_5 89.4, L_8 187.9, L_∞ 17.1; per datum L_5 gains $3 - \log_2 5 \approx 0.68$ bits, which repays the 72-bit prior difference and L_5 ’s Dirichlet cost of about $2 \log_2 n$ bits between $n = 64$ and 128. (c) Stage lengths 1, 24, 81, 115, 104, 83, 68, 55, 45, 39, 33, 33, 32, 27 (740 data); they shorten after $i = 4$ because L_∞ ’s mean cost on round-robin data, $1 + (i - 1)/2$ bits, outgrows L_i ’s $\log_2 i$.

data from	generator’s mass at $n = 2, 4, 8, 16, 256$	first n (≥ 0.99), per seed	$P(\text{proves } A_{xy})$ at $n = 4; 256$
A_{xy}	0.34, 0.90, 1.000, 1.000, 1.000	8, 8, 8, 4, 2	1.000; 1.000
M_x	0.60, 0.80, 0.90, 0.99, 1.000	32, 16, 8, 16, 16	0.205; $7 \cdot 10^{-49}$
S_{ab}	0.28, 0.56, 0.95, 1.000, 1.000	16 in every seed	0.060; $1 \cdot 10^{-11}$

Table 37: E6 under L1 (means over 5 seeds). Under $L1_{\text{cit}}^{\text{sel}}$ at $n = 256$, for all three sources: A_{xy} 0.044, A_{yx} 0.044, S_{ab} 0.128, M_x 0.392, M_y 0.392, open forms $\leq 6 \cdot 10^{-9}$, so $P(\text{proves } A_{xy}) = 0.088$. Prior bits: A_{xy}, A_{yx} 31.6; $S_{ab}, S_{ab}^{\text{open}}$ 30.0; the four M forms 28.4. Provability of A_{xy} is read from Proposition 8.2, not from $\vdash_J$.

prior; 0 with the shrinking threshold) and $R_\alpha(n, 2) \geq 1/(e\pi(n + 1))$ for $n \leq 2000$; `kt_regret`, slopes -0.499, -0.998, -1.485 of $\log_2 R_\alpha(n, K)$ for $K = 2, 3, 4$; `check_models`, the structures of Propositions 8.2 and F.6; `check_e4_tight` and `check_e4_where`, E4 as in Table 35; `check_e1_v1`, E1’s generator masses unchanged against the first version to three decimals; `check_bounded` and `check_classprob_open`, as in Appendices G.1 and G.4. The referee’s own code reproduced the first version’s E1–E7 bit for bit, and the final code reproduces the referee’s legacy E2 counts, E3 extensions and atomic-motive run.

n	<i>wellspec</i> , L1	<i>wellspec</i> , L0	<i>skew</i> , L1	<i>skew</i> , L0
16	-1.7 (-3.4, 0.8)	-2.2 (-4.5, 1.2)	1.7 (-1.5, 8.3)	-0.2 (-2.3, 5.6)
64	-3.5 (-5.5, 1.2)	-2.8 (-5.2, 0.1)	7.7 (-0.4, 14.3)	1.2 (-2.9, 5.7)
256	-7.3 (-8.3, -6.1)	-6.8 (-7.8, -5.6)	48.8 (37.9, 72.6)	17.4 (6.4, 34.2)
1024	-9.8 (-11.6, -6.5)	-9.3 (-11.4, -6.2)	196.5 (165.4, 232.8)	72.1 (41.9, 98.0)
4096	-13.0 (-14.6, -11.3)	-12.9 (-14.5, -11.4)	817.0 (738.9, 912.3)	299.6 (210.7, 351.5)

Table 38: E8: $\log_2$ Bayes factor split : whole without the prior (mean over 10 seeds; range).

(λ, τ)	(1, 0)	(1, 1)	(1, 4)	(2, 0)	$(\frac{1}{2}, 0)$
T^* at $n = 32$	0.197	0.197	0.197	0.040	0.186
T^* at $n = 64$	0.751	0.751	0.752	0.594	0.719
T^* at $n = 128$	0.991	0.992	0.993	1.000	0.961
T^* at $n = 512$	0.996	0.996	0.996	1.000	0.980
unsound at $n = 8$	0.256	0.293	0.372	0.860	$3 \cdot 10^{-4}$
unsound at $n = 16$	0.197	0.217	0.262	0.792	$7 \cdot 10^{-5}$
unsound at $n = 32$	0.077	0.084	0.103	0.737	0.014
unsound at $n = 64$	0.009	0.009	0.008	0.206	0.039
unsound at $n = 512$	0.004	0.004	0.004	$2 \cdot 10^{-4}$	0.019
acc. at $n = 8, 16, 32, 64$	2, 3, 0, 0	2, 3, 0, 0	5, 4, 1, 0	18, 19, 18, 5	0, 0, 0, 0
MAP at $n = 8$: e/w/u	0/18/7	0/16/9	0/16/9	0/3/22	0/25/0

Table 39: E7 (E2’s data and pools; means over 25 seeds). “Acc.”: seeds in which $V_{0.05,0}$ accepts a false probe; no seed accepts from $n = 128$. Last column: MAP tags equivalent / weaker / unsound at $n = 8$. T^* has mass below 10^{-3} at $n \leq 16$ in every row. $\lambda = \frac{1}{2}$ is proper only on finite pools. The first version (5 seeds, legacy pool) reported a largest time-factor effect of 0.002 and 0.016; the referee’s re-run of that pool with SeenQ gave an unsound MAP at $n = 8$ in 19 of 25 seeds at $\lambda = 2$, the same direction as here.

H How the results were established

H.1 Process

Roles. A research brief (`research/00-brief.md`) quoted the question and set out hypotheses H1–H7 to be confirmed, refined or refuted (Table 1). Four research tracks then worked in parallel: *model* (the general theory), *universal* (the case $\forall x\varphi$), *pa* (PA and ZF) and *experiments* (the package `code/bai` and the experiments E1–E8). Each track wrote notes (`notes.md`) with seeded check scripts. An adversarial referee for each track read the notes, re-derived the proofs, wrote independent code (`referee_code/`) and filed a report (`referee.md`). The track revised its notes into `notes-final.md`, which ends with a verification log mapping every referee issue to its resolution. The referees found no fatal error. They raised, by track: *model*, 5 major and 16 minor issues and 6 questions; *universal*, 3 major and 11 minor issues and 7 questions; *pa*, 5 major and 11 minor issues and 5 questions the track had missed; *experiments*, 5 major and 14 minor issues. A planning editor then wrote an outline, a notation file and a ledger of every claim with its status and source, and reconciled 43 cross-track conflicts (Appendix H.4). Section writers wrote each section from the final notes only, re-reading the proofs as they went and recording what they found (Table 41). A framing editor wrote the introduction, the discussion and this appendix, and ran the build.

What this means for the reader. All roles were separate Claude instances (Anthropic). No human has checked the mathematics. Nothing is checked in a proof assistant. Track *pa*’s natural-deduction derivations are checked line by line by a small checker, `nd.py`, which rejects ten planted unsound steps and the referee’s 173 mutants (Appendix F.1); its referee re-checked the derivations with an independent checker. Results added in the revisions were checked by their track and re-read by the section writers, but not by a referee; among them are (numbered as in the tracks’ notes, which the grey source notes give) *model* Thms 4.7, 4.8, Lemma 4.6, Props 5.5(d), 5.6, 6.11, 6.14, 7.6, 8.2, Lemma 6.10, Thm 6.12, Cor 6.13 and Ex 3.6; *universal* Lemma N0, Props N1–N3, B2, U2n, U2t and U10(e); *pa* Props 1.4, 2.3, 3.1–3.4, 4.8 and Lemma 4.7; *experiments* Props X10, X12, X13 and E8. The appendices name these where they give the proofs (e.g. Appendices C and F).

H.2 Corrections made during verification

Table 40 lists every claim of the first versions that was refuted or weakened, with the reason and where the paper states what holds instead. “Found by” gives the referee’s issue number (M major, m minor) or “track” when the track found it in its own revision. Refuted claims that matter to the reader are also kept in the text, labelled refuted.

Table 40: Claims refuted or weakened during verification.

item; found by	claim of the first version	reason; what holds now	where
<i>model track</i> Def 1.5; M5	“every type has mean offspring below 1” makes $\mathcal{Q}$ subcritical	no formula type of L_A satisfies it; with infinitely many types a chain is infinite with probability $\prod_k (1 - 1/(k+2)^2) = \frac{1}{2}$; the domination step fails. Refuted; replaced by uniform weighted subcriticality	Remark A.3 and Definition 2.6

item; found by	claim of the first version	reason; what holds now	where
Rem 1.10; m3	a stronger theory has a larger Z_T , “hence a smaller $P_T(s)$ on each datum”	$T = \{a, a \rightarrow b\}$, $T' = T \cup \{b\}$: $P_T(b) = 0.032$, $P_{T'}(b) = 0.346$. Refuted; true for data whose shortest derivation is unchanged	Remark 2.23
Thm 5.1(b); M4	the exact generator class gets posterior mass $\rightarrow 1$ for a.e. w^*	its mass is 0 at every n ($ T^* \geq 2$, independent components, w^* off a countable set). Refuted; replaced by identification of the instance union	Remark 4.8 and Theorem 4.7
Prop 6.9; M2	plain L1 charges derivation symbol size	a conclusion of size $2^{k+3} - 1$ from a tree with $7 + 8k$ grammar nodes. Refuted; replaced by Lemma 6.10–Prop 6.14 and Conj 6.15	Remark 6.11
§7 (c5); m4	renormalisation is coherent essentially when every theory is complete	coherent in 199 of 300 posteriors, 2 of them with all theories complete. Refuted; the 50/50 rule characterised instead	Remark A.9 and Proposition 2.17
§4; M1	Thm 4.1 applies to the Dirichlet-weight model	vacuous there; a constant threshold accepts with probability 0.977–1.000 against nominal 0.02. Refuted; averaged and shrinking-threshold versions proved	Remark D.3, Example 5.8, and Theorems 5.6 and 5.7
§5.3; M3	“a derivation likelihood does not change the MDL finding” [proved]	proved only for matched fixed weights. Withdrawn; Prop 5.6 with (d) a conjecture	Proposition 4.11 and Remark 4.13
§4.1; m8	“ W^* only grows under truncation”; “L2 likelihoods are finite sums”	the truncated constant can fall to 0.09 against $\pi(C_d^*) = 0.90$; with full-support $\mathcal{Q}$ the sums are infinite. Refuted	Remark 5.3
brief H6; track	a collapse schema “requires running T to check membership”, so a Kt penalty prices it	membership is polynomial without running T . Refuted	Proposition 6.8
Hänni’s note; track	his schema is consistent for every consistent assigner	single-sorted over PA it can be inconsistent. Refuted in that reading; his conclusion holds via Craig	Proposition 6.5 and Theorem 6.3
weakened; m1, m2, m6, m9–m12, m14–m16	—	Cor 6.8’s constant uniform only for template theories (m1); Prop 4.2’s prover replaced by a waiting prover, as an every-round prover wins only 11/400 against escalation (m2); Prop 2.5’s numbers corrected to 0.16–0.35 (m6); consistency hypothesis added to Prop 1.9(b) (m9); Prop 2.6(c) restricted to first-order templates with 0-ary term metavariables (m10); $\alpha_{ax} + \alpha_{lg} > 0$ (m11) and parameters admissible (m12) added to Lemma 1.7; split ties only at matched fixed weights (m14); Prop 2.5’s limit 0 or ∞ only if $T^* \in \{T_1, T_2\}$ (m15); Ex 3.4, Rem 3.5 are L0 only, and L1 was computed in Ex 3.6 (m16)	the corresponding items of Sections 2 and 4 to 6
<i>universal track</i>			
U11(e); M1	without a guard H_{open} wins on closed data and the posterior takes the ω -step	$R_1 = \{\varphi(0), \varphi(Sz)\}$ beats H_{open} by $(1 - q)\mathcal{W}_o$ per datum; $P(T \vdash \forall x \varphi \mid D_n) = 0.000$ from $n = 1000$. Refuted under L1; holds for $\{H_{\forall}, H_{\text{open}}\}$ and under L1 ^{sel}	Remark 3.11 and Proposition 3.12

item; found by	claim of the first version	reason; what holds now	where
remark after U3; m8	$\wedge$ -detours cannot break the factor c	in $C_{\wedge}$ the ratio is $0.15553 > c = 0.15$. Refuted; whether the direction can reverse is open	Remark 3.5
U16(b), (c); M2	"= 1 over $\mathbf{Q}$ " in the sentence-only class	true only in a three-hypothesis class; with depth ≤ 4 memorisers win, with depth ≤ 40 splits win to $n = 10^{12}$; the referee's " $\rightarrow 0$ over $\mathbf{Q}$ " holds only for bounded depth. Restated, limit a conjecture	Proposition B.16, Appendix B.7
U10(c); m3	$\vdash_C$ and $\vdash$ conflated in the faithful regime	split into (c1), (c2) with the extra hypothesis	Theorem 3.9
weakened; M3, m1, m2, m4–m7, m9–m11	—	U10(c3) restricted to $C_{\min}$ -like calculi (M3); memoriser universe named, as the prior was improper (m1); Σ_1 -completeness of $\mathbf{Q}$ restricted to $<$ -free sentences (m2); learned weights are outside U7, Prop B2 added (m4); U12(b) recomputed with equal priors (m5); U14(b) does not apply at $q = \frac{1}{2}$, U14(c) is a sufficient condition (m6); U10(d) for quantified φ a sketch in general (m7); inconsistent theories reported as Inc and the verifier moved to Bel_{cons} (m9); wording (m10); the probability qualifier of U6 restored (m11)	the corresponding items of Section 3
<i>pa track</i>			
Prop 3.5(c), F8; M4	with Σ_n motives every covering DT° theory has PA-strength induction, so the posterior lands on the L_∞ side	a theory inside $\text{I}\Sigma_1$ covers a Σ_1 /open practice and overtakes $\mathbf{Q} + T_{\text{Ind}}$ near $n \approx 2^{26.7}$; a term-only theory inside $\text{I}\Sigma_2$ covers G1. Refuted	Proposition 7.24 and Examples 7.26 and 7.27
§2.2; m2	the SDPC margin of the root split is constant	it falls by 1 bit per doubling. Refuted	Proposition 7.12, Section 7.3
§2.2 first run; track	under a shared grammar the root split loses by a growing margin (+803...+12270 bits)	an encoding artefact (bound variables coded as holes). Refuted	Section 7.3
§4.5 first run; track	the head-symbol split of $x + 0 = x$ overtakes the schema at $n = 1000$	an artefact of root-context coding; the split closes in at 5 bits per doubling. Refuted	Remark 7.34
§4.5; M5	under L1 the odds of $\forall x(x + 0 = x)$ fall like $2^{-3.3n}$, "as H4 predicts"	true for a fixed rule code only; $\frac{R-1}{2} \log_2 n$ under a learned depth-indexed code. Refuted as a general rate	Remark 7.34
§7; M2–M4	"it robustly gets a theory deductively equivalent to PA"	true only within the hand-picked candidate sets and for axiom-instance data. Refuted as stated	Section 7.7.1
§1.2; m5	a derivation of $\text{CVI}(P)$ costs more "because it has two lines"	a citation also writes $P(x)$. Argument wrong; conclusion re-proved	Proposition 7.6
numbers; M1, m1, m9	eight entries of the DTRC table; the G2/G3 margins; the Euler lead	table regenerated from the output and every quote checked by script (M1); every template charged (m1); lead $2.1 \cdot 10^5 \rightarrow 3.1 \cdot 10^5$ bits at $n = 10^6$ (m9)	Tables 5 and 30 and Example 7.37

item; found by	claim of the first version	reason; what holds now	where
weakened; m3, m4, m6–m8	—	break-even ratios are code-specific (m3); a decodable variant of the code added (m4); one refutation rule R_d (m6); eliminating each $I\Sigma_n$ is not concentration, Doob’s theorem added (m7); a depth theorem of AS cited too loosely, replaced by Π_1 -completeness of consistency (m8)	the corresponding items of Section 7
<i>experiments track</i>			
E2; M1	in 3 of 5 seeds an unsound lump has mass ≥ 0.997 at some $n \leq 32$ and a $\delta = 0.05$ verifier accepts a false sentence	the pool used unseen data and lacked SeenQ(D_n); with causal pools 5 of 25 seeds. Refuted as a frequency	Remarks 5.11 and 8.4
E2; M1	the MAP list at $n = 8$	built from data 9–16, unseen at $n = 8$. Refuted	Remark 8.4
E6; M2	the five closed-guard forms of commutativity are logically equivalent	S_{ab} , M_x , M_y are strictly weaker. Refuted	Proposition 8.2
§0; M3	soundness at a constant threshold for any well-specified data	fails for multi-component T^* at fixed weights. Refuted	Example 5.8
§5; M4	PA: “robust at the level of theorems, not of axioms”	atomic motives lead to $Q + \{T_=: T_<\}$, strictly weaker. Refuted	Proposition 7.29 and Remark 7.30
E1, E2; track	$P(T \vdash \varphi(t^*) \mid D) = 1.000$ from $n = 8$ in every E1 configuration; no false probe accepted on E2 data	open data under L0 give 0 at $n = 4, 8$; the mean over seeds hid seeds 1, 2 and 4. Refuted	Remark 8.4
weakened; m2, m4–m8, m10–m12; track	—	Prop X5’s almost-everywhere qualification (m2, M3); Mem(D_n) is the best memoriser only under L0 (m4); held-out PA instances drawn outside the stream (m5); every n checked in E4(B, C) (m6); E7’s time-factor result holds by construction (m7); $L1_{\text{cit}}^{\text{sel}}$ is per-citation rejection (m8); Gold’s text only if every stage ends (m10); E3(a) gains compared within L0 (m11); E3(a) families extended, so the MAP is interior (m12); Mem(D_n) double-counted in E6 at $n = 1, 2$, $T^* - Q3$ mis-tagged, two wrong L0 entries in the first E1 table (track)	Section 8, Appendix G

Track against referee. In five places the final notes disagree with the referee, with reasons, and the paper follows the final notes. (1) Pa, m11: AS Thm 5.16 holds in closure-normal form with parameters; the real gap is that it concerns a single covering template (Proposition 7.24). (2) Pa, M4: the law G1 does not force a formula metavariable; its 157 skeletons are covered by a term-only theory inside $I\Sigma_2$, and the posterior keeps PA through the Occam balance (Example 7.27). (3) Experiments, M2: open guards do reach $\forall x \forall y$ under the single-parameter convention (Proposition 8.2(a)). (4) Universal, M2: the referee’s “ $\rightarrow 0$ over Q ” holds only with bounded split depth (Proposition B.16). (5) Model, m13: the tightness of the soundness constant holds exactly, not only as a supremum (Remark D.1).

H.3 Corrections made while writing the paper

The section writers re-read the final notes against their proofs and outputs. Table 41 lists what they changed or qualified. None of these changes a headline result; several narrow a scope or lower a status.

Table 41: Corrections and qualifications made while writing.

section	what was found, and what the paper does	where
model	Kraft’s inequality for the prior needs theories taken up to renaming of metavariables; added to the definition. The number of templates is coded as $\gamma(k+1)$, not $\gamma(k)$, so that the empty theory has a code.	Definition 2.1, Appendix A.1
model	Lemma 1.7(b), $Z_T \geq \alpha_{ax} + \alpha_{lg}$, fails for $T = \emptyset$; $Z_\emptyset \geq \alpha_{lg}$ suffices where it is used. Prop 1.9(c) uses only $\text{supp } P_T \subseteq \text{Th}(T)$, which needs no hypothesis.	Appendices A.4 and A.5
model	The notation file’s “ $\leq c$ ” for the C_{U3} factor bounds only derivations that cite the added axiom. The direction (factor at most 1) is not proved for $C_\wedge$; the table caption says for which calculi it is. Some “size principle” marks in Table 12 are the writer’s applications of Lemma 2.21 and Theorem B.4, flagged in its caption. Prop 2.8 needs $\mathcal{Q}$ symmetric in the holes.	Tables 11 and 12 and Proposition A.8
universal	The proof of Lemma S1 omitted a case (a summand that is a variable always instantiated by 0); added, conclusion unchanged. Thm U2’s equality under Hänni’s scores needs $\forall x \varphi$ consistent. In c5 Part A, H_{both} carries a uniform prior on its weight. U5’s “survival 0.503” is a per-datum estimate of $p_S = \frac{1}{2}$, so only the gain per datum is reported. The notes’ citations of the superseded first E3 result and of “only derivation length prices computation” are replaced (conflicts C9, C11).	Lemma B.17 and Tables 14 and 15
ident	“A false spare sentence makes $T^* \cup \{\sigma\}$ inconsistent” holds only when T^* refutes σ . “Well specified: Occam loss” under L1 is a conjecture; only a bounded gain is proved. Spares in the span of T^* ’s components never vanish, contrary to a summary sentence of the model notes. Check c7: the escape template alone loses to memorisation at $\lambda = 1$ by $n = 10^4$ (−2649 to −4234 bits); only the hybrid beats memorisation. Check c2 Part B: the misspecified Stirling prediction is off at $n = 100$ (−1.50 against −0.06).	Remarks 4.13 and 4.15, Proposition 4.14, and Tables 20 and 24
sound	The laws of Bel, Dis, Indep (model Props 7.1(a), (c), (d), 7.2) need unbounded derivability, since $\text{Th}_d(T)$ is not closed under consequence; stated for $d = \infty$. Check c5 records only 2- and 3-monotonicity; 4-monotonicity was checked by the referee on two atoms. The computed support of Thm 4.1 ($n \leq 300, 400$) checks the bound, not time-uniformity.	Section 2.7, Appendices A.7 and D.1
time	The closing statement’s upper bound rests on a proof sketch, so its status is lowered. Prop 6.14(b), (c) depend on the hierarchy theorem. Prop 6.14(c) also needs the explicit derivations to be K-derivations (or to translate at linear cost) and a code with at least one bit per written symbol. The size of the PA-proof of the Tarski biconditional in the upper bound for the reflection sentence is not bounded in the notes. Prop 6.4 is proved for the acceptance part of Hänni’s schema only.	Remarks 6.1 and E.3 and Propositions 6.6 and 6.13
pa	Rem 4.9’s “no instance is logically equivalent to $\text{Ind}(P)$ ” is not proved; a weaker true statement is used. Prop 5.3 is proved for the two-part likelihood, not for the normalised L1, and is stated under $L1^{\text{max}}$. The “astronomically large” waiting times of Prop 4.6(a) are a heuristic. Props 3.3 and 4.6(a) need fixed weights (setting W); that $\text{PA} \cup S$ decays polynomially is a proof sketch for generic weights. “The redundant union wins after 1–12 uses” counts schema uses in expectation; one use of CVI alone saves 1468 bits. “E3(b) keeps the theorems, not the axioms” holds for two of three connective-rich laws.	Remarks 7.7, 7.30 and F.4 and Propositions 7.18, 7.24 and 7.36

section	what was found, and what the paper does	where
experiments	The proof of Prop X4(c) (Gen, then $\forall E$, at $J = 2$) treats a derived component as a DT° template without saying so, and the code’s support check does not test it; the hypothesis is added, and for data-derived theories at $J = 2$ exactness rests on the computed checks. At $\lambda = 2$ the verifier also accepts a false probe in 5 of 25 seeds at $n = 64$ (unsound mass 0.206), which the notes do not report.	Proposition 8.1, Appendix G.2 and Table 39

H.4 Cross-track conflicts

The tracks used different calculi, priors and names, and some cited each other’s superseded results. The ledger records 43 conflicts; Table 42 gives each with its resolution.

Table 42: Cross-track conflicts and their resolutions.

	conflict	resolution
C1	the per-datum $\forall E$ factor differs between calculi	each factor stated with its calculus and normalisation (Table 11); the direction is shared
C2	two selection-aware likelihoods	$L1^{\text{sel}}$ (stream) for Prop B2; $L1_{\text{cit}}^{\text{sel}}$ (per citation) for E1, E6 (Remarks 3.7 and A.6)
C3	prior codes differ, so prior shares differ	every share names its code (Table 16)
C4	three verifier constants	one theorem with classes C_d^* , C_d^{Dir} , $C_\forall^*$
C5	Thm 4.1 used for Dirichlet weights	never; Theorems 5.6 and 5.7 instead
C6	Ex 4.9 numbers differ between implementations	ranges with all sources (Table 27)
C7	pa cites E2’s superseded “3 of 5”	5 of 25 (causal pools); pa §5.4 is one run
C8	E6 cited as identification among deductively equivalent forms	logically and merely instance-equivalent forms stated separately
C9	universal cites the superseded first E3(a)	final E3(a) only
C10	nested spare-slot slopes differ	E5(a2) -0.251 ± 0.002 primary; the rate stays a proof sketch
C11	“only derivation length prices computation”	“derivation length in written symbols”
C12, C31	pa’s two-part code called L1	named $L1^{\text{sch}}$; its numbers are not L1 results
C13	sign of the Occam drift (fixed vs learned grammar)	both with scopes; Proposition 7.12 reconciles
C14	rate of the $\forall E$ penalty	depends on the rule law and normalisation; the direction does not
C15	guard learning rate (exponential vs logarithmic)	both, in their models (ρ fixed or learned)
C16	memorisation verdicts	each with its setting; H4(a) refuted for memorisation in general
C17	misspecification gives weaker or unsound theories	classified by kind of data (Table 23)
C18–C23	clashing names and symbols	the notation of Section 2
C24	numbering of Robinson’s axioms	the same in all tracks
C25, C26	symbol counts; Dirichlet α	bit counts name their code; tables state α
C27	parameter admissibility	stated wherever $\text{supp } P_T = \text{Th}(T)$ is used
C28	fixed weight vectors in the experiments	a yardstick, not an instance of a theorem (Remark C.3)
C29	E8 per-doubling rates	quoted with their windows
C30	“PA-equivalents from $n = 100$ ” vs “SeenQ first”	different data laws and candidate sets; both with scopes
C32	Hänni’s equivalence: argument vs conclusion	argument fails single-sorted over PA; conclusion holds (Proposition 6.5 and Theorem 6.3)

	conflict	resolution
C33	instance-only generators vs E1	no conflict; Ch_1 is $\text{C}_{\min}$ -like
C34	three spare-slot formulas	one statement, three sources (Proposition 4.14)
C35	model Prop 5.6(b) vs E8	E8 not cited for (b)
C36	U11(e) vs AS Prop 4.5	the Bayesian counterpart, with the refuted claim labelled
C37	over-Q limit for splits	class-dependent (Propositions 3.12 and B.16)
C38	track-vs-referee disagreements	the final notes, with reasons (Appendix H.2)
C39	three verifier definitions	the Bel_{cons} verifier inherits soundness (Remark 5.9)
C40	units of d in Th_d	stated at each use
C41	“statements given without proof are easily derivable” in two tracks	the same message, different pairs; no conflict
C42	cost of computing L1	the wording of Proposition 2.24
C43	Thm B under L0^{cl} vs E1 under L0	different likelihoods

H.5 Conjectures, proof sketches and open items

Conjectures. Theorem 4.7(c) (every w^*); Proposition 4.11(d) (Occam rate under L1); Remark C.5 (unsound mass in the full class); Conjecture C.7; Proposition B.6(iii) (memoriser rate); the limits in Proposition 3.12(d) and Proposition B.16(d); Conjecture 6.14; Conjecture 6.16; Conjecture 7.28; the reverse overheads in Remark 7.9.

Proof sketches. Lemma A.5(b); Theorem 4.7(b’); Proposition 4.14(b), (c), (d2); the mechanism of Example 5.8; Theorem 3.9(d2) in general; Proposition 3.12(c); Proposition B.14(b) for $f_q > 0$ and (c); Proposition B.16(a); the spine formula of Remark 3.5; Corollary 6.10(b) for a single X ; Remark E.3 and the upper bound in Remark 6.1; Remark E.7(ii) and the argument of (iv); Proposition 7.24(b); the last part of Proposition 7.18; the slope in Example 7.27; the covering argument in Example 7.37; Remark F.7. The waiting times in Proposition 7.24(a) are a heuristic.

Open items found while writing. These are not resolved in the paper. (1) Model Lemma 1.2(b) has only a proof sketch; the missing step is named in Appendix A.3. (2) The direction of Theorem 3.4 is not proved in $\text{C}_\wedge$. (3) In Proposition B.15 the full-sum comparison at $q = \frac{1}{2}$ is open. (4) The ident results added after the referee report (Theorem 4.7(b) with its new proof, Proposition 4.11, Proposition 4.14(d), Example C.6, Proposition C.8, Remark C.9) were not refereed. (5) In Example 5.8 a χ^2 limit uniform over the window $1 \ll n \ll 1/\varepsilon$ is not written out; completeness under the shrinking threshold, and the strictness of Proposition D.2(d) for $H_k(\text{DT}^\circ)$, are open. (6) In Remark E.3 the Q-proof size bound is recalled from Pudlák (1998) and the size of the Tarski biconditional’s proof is not bounded; for Corollary 6.10 with a single X , the time-constructibility and gap conditions are not checked; the route to Conjecture 6.14 lacks de Bruijn indices and Gen under sharing, and a polynomial bound on the size of shared representations; Proposition 6.6 is not proved for the rejection part of the schema; the $\text{I}\Sigma_n$ comparison under a derivation-time penalty proposed by pa’s referee was not quantified. (7) Pa: the stronger form of Rem 4.9; Proposition 7.36 under the normalised L1; Proposition 7.18 and Proposition 7.24(a) with Dirichlet weights. (8) Experiments: exactness of L1 for data-derived theories at $J = 2$ rests on computed checks (brute force at $J = 1, 2$ and the referee’s recursion on 904 triples), not on Proposition 8.1(d).

H.6 References not verified

The bibliography files record, for each entry, what was checked. The following were not verified against their sources, and the text hedges each claim that rests on them. *Content or theorem not checked:* Hutter (2007) (bibliographic data and abstract only); Leike and Hutter (2015) (abstract); Gaifman (1964) (condition cited from memory; secondary sources use constants); Berk (1966) and Kleijn and van der Vaart (2006) (abstracts; conditions not read); Ghosal and van der Vaart (2017) (theorem number only via citing papers); Horning (1969) and Angluin (1988) (existence only); Rousseau and Mengersen (2011) (issue number not re-confirmed; not applied); Craig (1953), Seiferas et al. (1978) and Žák (1983) (full text not read); Cook (1972) (exact form of the theorem not checked; pages omitted); Pudlák (1998) (the Q-proof bound not checked in the chapter); Boolos et al. (2007) (chapter not checked); Shepherdson (1964) (secondary sources only); Zarach (1996) (editor and publisher of the volume not checked); Gitman et al. (2016) and Kaye and Wong (2007) (abstracts); Kaye et al. (1988) (existence). *Locations or numbering not checked:* Mendelson (1997) (system K, numbering); Athreya and Ney (1972) (chapter); Hájek and Pudlák (1993) (Cor. I.4.34 traced by a referee through a secondary source) and Kaye (1991) (chapter numbers); Shoenfield (1967) (location; publisher recalled). *Recalled from memory:* Church (1936), Turing (1936), Birkhoff (1935), Schwarz (1978), Clarke and Barron (1990), Hoeffding (1963), Shafer (1976) (cited only for the framework), Ville (1939) (data copied from IL’s bibliography) and Schwartz (1965) (data checked by the model track’s referee only). *Not opened:* Parikh (1973), Baaz and Pudlák (1993) (editors and pages not recorded) and Hrubeš (2007). *Cited without an entry, from memory:* Wilks’s theorem, the Paris–Kirby results on fragments, the sequential form of the KT estimator, Levin’s Kt and the compressed-string algorithms behind the route to Conjecture 6.14.

H.7 Reproduction

Every computed number comes from a seeded script whose output is saved next to it.

- *Track checks.* `research/tracks/<track>/checks/` holds each track’s scripts and their `.out` files (model c1–c16; universal c1–c10; pa `test_nd`, c1–c10; experiments’ check scripts), and `referee_code/` the referees’ independent scripts and outputs. Universal re-ran all ten of its scripts and the referee’s seven, and pa all twelve of its scripts, in a scratch directory, with byte-identical outputs; pa’s `c10_quotes.py` checks 87 quoted numbers of its notes against the outputs.
- *Code.* From `code/`: `python3 -m pytest -q tests` (21 unit tests, including a comparison with the referee’s independent implementation); `sh run_all.sh` (tests and E1–E8, about 50 minutes on four shared cores); seeds and per-experiment commands in Appendix G.5. Results are in `code/results/*.md` and `*.json`, the first version’s in `code/results/v1/`.
- *Paper.* From `paper/`: `./build.sh` merges the bibliographies and runs pdfL^AT_EX and BibT_EX; `./test-section.sh <section> <appendix>` builds one section alone.